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Longbing Cao · Philip S. Yu · Chengqi Zhang · Yanchang Zhao

Domain Driven Data Mining

Springer

Longbing Cao
University of Technology,
Sydney
Fac. Engineering & Information Tech.
Centre for Quantum Computation
and Intelligent Systems
Broadway NSW 2007
Australia
lbcao@it.uts.edu.au

Philip S. Yu
Department of Computer Science
University of Illinois at Chicago
851 S. Morgan St.
Chicago IL 60607-7053
USA
psyu@cs.uic.edu

Chengqi Zhang
University of Technology,
Sydney
Fac. Engineering & Information Tech.
Centre for Quantum Computation
and Intelligent Systems
Broadway NSW 2007
Australia
chengqi@it.uts.edu.au

Yanchang Zhao
University of Technology,
Sydney
Fac. Engineering & lnformation Tech.
Centre for Quantum Computation
and Intelligent Systems
Broadway NSW 2007
Australia
yczhao@it.uts.edu.au

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To Sabrina Yue Cao and Bobby Yu Cao for their time, the peace and understanding they have given during writing this book.

Preface

Data mining has emerged as one of the most active areas in information and communication technologies (ICT). With the booming of the global economy, and ubiquitous computing and networking across every sector and business, data and its deep analysis becomes a particularly important issue for enhancing the soft power of an organization, its production systems, decision-making and performance. The last ten years have seen ever-increasing applications of data mining in business, government, social networks and the like.

However, a crucial problem that prevents data mining from playing a strategic decision-support role in ICT is its usually limited decision-support power in the real world. Typical concerns include its actionability, workability, transferability, and the trustworthy, dependable, repeatable, operable and explainable capabilities of data mining algorithms, tools and outputs.

This monograph, *Domain Driven Data Mining*, is motivated by the real-world challenges to and complexities of the current KDD methodologies and techniques, which are critical issues faced by data mining, as well as the findings, thoughts and lessons learned in conducting several large-scale real-world data mining business applications. The aim and objective of domain driven data mining is to study effective and efficient methodologies, techniques, tools, and applications that can discover and deliver actionable knowledge that can be passed on to business people for direct decision-making and action-taking.

In deploying current data mining algorithms and techniques into real-world problem-solving and decision-making, we have faced the crucial need to bridge the gap between academia and business, as well as addressing the gap between technical evaluation systems and real business needs. We have been confronted by the extreme imbalance between the large number of algorithms published versus the very few that are deployed in a business setting; the large number of patterns mined versus the few that satisfy business interests and needs; and many patterns identified versus the lack of recommended decision-support actions.

To bridge the above-mentioned gaps, and to narrow the extreme imbalance, it is crucial to amplify the decision-support power of data mining. Most importantly, it is

critical to enhance the actionability of the identified patterns, and to deliver findings that can support decision-making. These are the drivers of this book.

Our purpose is to explore the directions and possibilities for enhancing the decision-support power of data mining and knowledge discovery. The book is organized as follows. In Chapter one, we summarize the main challenges and issues surrounding the traditional data mining methodologies and techniques, and the trends and opportunities for promoting a paradigm shift from data-centered hidden pattern mining to domain-driven actionable knowledge delivery. Chapter two presents the domain-driven data mining methodologies. From Chapters three to five, we mainly extend the discussions about domain-driven data mining methodologies. In Chapter three, ubiquitous intelligence surrounding enterprise data mining is considered. Chapter four discusses knowledge actionability, while Chapter five summarizes several types of system frameworks for actionable knowledge delivery. Chapters six to eight present several techniques supporting domain-driven data mining. Chapter six introduces the concept of combined mining, leading to combined patterns that can be more informative and actionable. In Chapter seven, we discuss agent-driven data mining, which can enhance the power of mining complex data. Chapter eight summarizes the technique of post mining for enhancing knowledge power through post-processing of identified patterns. Chapters nine and ten illustrate the use of domain driven data mining in the real world. In Chapter nine, domain-driven data mining is used to identify actionable trading strategies and actionable market microstructure behavior patterns in capital markets. Chapter ten utilizes domain-driven data mining in identifying actionable combined associations and combined patterns in social security data. Chapter eleven lists some of the open issues and discusses trends in domain-driven data mining research and development. Chapter twelve lists materials and references about domain-driven data mining.

A typical trend in real-world data mining applications is to treat a data mining system as a problem-solving system within a certain environment. Looking at the problem-solving from the domain-driven perspective, many open issues and opportunities arise, indicating the need for next-generation data mining and knowledge discovery far beyond the data mining algorithms themselves. We realize that we are not at the stage for covering every aspect of these open issues and opportunities. Rather, it is our intention to raise them in this book for wider, deeper and more substantial investigation by the community.

We would like to convey our appreciation to all contributors, including Ms. Melissa Fearon and Ms. Jennifer Maurer from Springer US, for their kind support and great effort in bringing the book to fruition.

Chicago, USA
July 2009 *Philip S Yu*

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We appreciate all colleagues, contributors and reviewers for their kind contributions to the professional activities related to domain driven data mining (DDDM or D^3M), including the DDDM workshop series and special issues.

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Chapter 1
Challenges and Trends

1.1 Introduction

This chapter introduces the challenges and trends of data mining and knowledge discovery from data (KDD). We emphasize the issues surrounding real-world data mining, distinguish data-centered data mining from domain driven data mining, and propose the trends from data-centered hidden pattern discovery to domain driven actionable knowledge delivery (AKD) as a new KDD paradigm shift.

The goals of this chapter consist of the following aspects.

- Briefly looking retrospectively at the history and evolution of data mining and knowledge discovery;
- Summarizing the main issues frustrating traditional data mining methodologies and approaches in mining real-world applications;
- Discussing the factors and need to discover actionable knowledge in real-world data mining applications; and
- Exploring the paradigm shift from data-centered hidden pattern mining to domain driven actionable knowledge delivery.

These goals are addressed through interpreting the following contents.

- Section 1.2 briefs the evolution history of data mining and knowledge discovery from the viewpoint of the data mining process, objectives, and performance in deployment;
- In Section 1.3, we state the challenges and technical issues facing existing data mining and knowledge discovery methodologies and systems when they are deployed in dealing with real-world data mining applications;
- Based on the retrospection on traditional methodologies and approaches which feature data-centered hidden pattern mining, in Section 1.4, we advocate the need of domain driven actionable knowledge discovery by scrutinizing the major dimensional requirements including domain-oriented problems, KDD context, interestingness systems, expectations on the resulting KDD deliverables, and infrastructure to support real-world data mining;

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- Finally, with all the above thoughts in mind and the alleged general shift from data mining to knowledge discovery, Section 1.5 foresees a specific paradigm shift from data-centered hidden knowledge discovery to domain driven actionable knowledge delivery. To support such a revolution, the idea of domain driven data mining (*DDDM* or D^3M for short) presents thinkings and hints from the system, economic, organizational and social science perspectives.

1.2 KDD Evolution

In the last decade, data mining [117, 118], has become an active research and development area in information technology fields. In particular, data mining is gaining rapid development in various aspects such as the data mined, knowledge discovered, techniques developed, and applications involved. Table 1.1 illustrates such key research and development progress in KDD.

Table 1.1 Data mining development.

Dimension	Key Research Progress
Data mined	- Relational, data warehouse, transactional, object-relational, active, spatial, time-series, heterogeneous, legacy, WWW - Stream, spatio-temporal, multi-media, ontology, event, activity, links, graph, text, etc.
Knowledge discovered	- Characters, associations, classes, clusters, discrimination, trend, deviation, outliers, etc. - Multiple and integrated functions, mining at multiple levels, mining exceptions, etc.
Techniques developed	- Database-oriented, association and frequent pattern analysis, multi-dimensional and OLAP analysis methods, classification, cluster analysis, outlier detection, machine learning, statistics, visualization, etc. - Scalable data mining, stream data mining, spatio-temporal data and multimedia data mining, biological data mining, text and Web mining, privacy-preserving data mining, event mining, link mining, ontology mining, etc.
Application involved	- Engineering, retail market, telecommunication, banking, fraud detection, intrusion detection, stock market, etc. - Specific task-oriented mining - Biological, social network analysis, intelligence and security, etc. - Enterprise data mining, cross-organization mining, etc.

A typical feature of traditional data mining is that KDD is presumed as a predefined and automated process. It targets the production of pre-defined and automatic algorithms and tools. As a result, algorithms and tools developed have no capability to adapt to external environment constraints. Millions of patterns and al-

gorithms have been published in literature, but unfortunately very few of them have been transferred into real business.

Many researchers and developers have realized the limitation of traditional data mining methodologies, and the gap between business expectation and academic attention. The research on challenges of KDD and innovative and workable KDD methodologies and techniques has actually become a significant and productive direction of KDD. In the panel discussions of SIGKDD 2002 and 2003 [9, 102], a couple of grand challenges for extant and future data mining were identified. Actionable knowledge discovery is one of the key focuses among them, because it can not only afford important grounds to business decision makers for performing appropriate actions, but also deliver expected outcomes to business. However, it is not a trivial task to extract actionable knowledge utilizing traditional KDD methodologies. This situation partly results from the scenario that the extant data mining is a data-driven trial-and-error process [9], in which data mining algorithms extract patterns from the converted data through predefined models.

To bridge the gap between business and academia, it is important to understand the difference of objectives and goals of data mining in research and in the real world. Real-world data mining applications place extra constraints and expectations on the mined results; for instance, financial data mining and crime pattern mining are highly constraint-based [17, 102]. The differences involve key aspects such as the problem concerned, KDD context mined, patterns interested, processes of mining, interestingness cared, and infrastructure supporting data mining.

To handle the above differences, real-world experience [30, 32] and lessons learned in data mining in capital markets [138] show the significance of involving domain factors. Domain factors consist of the involvement of domain knowledge [227] and experts, the consideration of constraints, and the development of in-depth patterns, which are essential for filtering subtle concerns while capturing incisive issues. In combining these, a sleek data mining methodology is necessary to find the distilled core of a problem. They form the grounds of domain driven data mining.

1.3 Challenges and Issues

In this section, we introduce the challenges and issues facing current data mining and knowledge discovery. Our point of view results from the lessons and experiences in conducting knowledge discovery on enterprise data in areas such as social security, governmental services, capital markets and telecommunication.

Several KDD-related mainstream forums have discussed the actualities and future of KDD, for instance, the panel discussions in SIGKDD and ICDM. In respect of the progress and prospect of existing and future data mining, many great challenges, for instance, link analysis, multi-data sources, and complex data structure, have been identified for future research effort on knowledge discovery. Mining actionable knowledge and involving domain intelligence are two such challenges.

They are more generally significant issues of existing and future KDD. They hinder the shift from data mining to knowledge discovery, in particular, blocking the shift from hidden pattern mining to actionable knowledge discovery. The wide acceptance and deployment of data mining in solving complex enterprise applications is thus further restrained. Moreover, they are closely related and to some extent create a cause-effect relation, that is the involvement of domain intelligence contributing to actionable knowledge delivery.

We explore the challenges and issues from the following aspects:

- Organizational and social factors surrounding data mining applications;
- Human involvement and preferences in the data mining process;
- Domain knowledge and intelligence making data mining close to business needs;
- Actionable knowledge discovery supporting decision-making actions;
- Decision-support knowledge delivery facilitating corresponding decision-making, and
- Consolidation of the relevant aspects for decision-support.

1.3.1 Issues of Traditional Data Mining Studies

If we look at traditional data mining, including methodologies, techniques, algorithms, tools and case studies, we might have listened to or seen comments and issues much divided in academia and the business world, for instance:

- Data miner: 'I find something interesting!' 'Many patterns are found!' 'They satisfy my technical metric thresholds very well!'
- Business people: 'So what?', 'They are just commonsense.' 'I don't care about them.' 'I don't understand them.' 'How can I use them?'

In fact, we have seen some extreme imbalance existing in the current data mining community from the workability or performance perspective, for instance:

- Algorithm imbalance: Many published algorithms vs. several really workable in the business environment,
- Pattern imbalance: Many patterns mined vs. a small portion or none of them satisfying business expectations,
- Decision imbalance: Many patterns identified vs. effectively very few of them can be taken over for business use.

We have reached the conclusion that there is a serious problem for the decision-support power of existing data mining methodologies and techniques, from aspects such as:

- Dependability of the identified patterns and knowledge,
- Repeatability of the proposed algorithms and methods,
- Trust of the proposed algorithms and methods, as well as identified patterns and knowledge,

- Explainability and interpretability of the identified patterns and knowledge,
- Deliverability and transferability of the identified patterns and knowledge from data miners to business people.

In summary, we say the algorithms, models and resulting patterns and knowledge are short of workable, actionable and operable capabilities.

As a result, we often see big gaps in many aspects, for instance:

- A gap between a converted research issue and its actual business problem,
- A gap between academic objectives and business goals,
- A gap between technical significance and business interest,
- A gap between identified patterns and business expected deliverables.

How do these gaps arise? We try to analyze this from two perspectives: macro-level and micro-level. From the macro-level, we focus on methodological issues bothering traditional data mining. There is a gap between academia and practitioners. In academia, some things are of concern but not others, for instance:

- Innovative algorithms and patterns,
- Only checking technical significance,
- Do not really catch the needs of business people,
- Do not take the business environment into account, or
- Over-simplified data, surroundings and problem definition.

However, practitioners and business analysts value something else, for example,

- Can it solve my business problem?
- Has it considered surrounding social, environmental and organizational factors?
- Can I interpret it in my business language, experience and knowledge?
- Can I adjust it as I need by following my business rules and processes?
- Can it make my job more efficient rather than causing new issues?
- Can it be integrated into my business rules, operational systems and workflow?
- What could be the impact on business if I use it? Is that manageable?

Another angle to scrutinize the gap is more micro-level, from the technical and engineering perspective. There are issues that have been overlooked in traditional data mining, for instance:

- Problem dynamics and interaction in a system: an example in stock data mining is overlooking market dynamics, for instance, interaction within hidden groups,
- Problem environment: for instance, a model is not differentiated in developed and undeveloped markets,
- Business processes, organizational factors, and constraints: for instance, a trading pattern may only apply for market orders rather than limit orders,
- Human involvement: for instance, is a trading pattern applicable for individual or firm-based investors?

1.3.2 Related Efforts on Tackling Traditional Data Mining Issues

Existing efforts related to tackling challenges and issues in traditional data mining are multi-fold. They can be categorized into three major efforts:

- developing more effective interestingness metrics,
- converting and summarizing learned rules through post-analysis and post-mining [232], and
- combining multiple relevant techniques.

The main efforts of developing effective interestingness metrics are on *objective technical* interestingness metrics ($t_o()$) [106, 120]. They aim to capture the *complexities of pattern structure and statistical significance*. Other work appreciates *subjective technical* measures ($t_s()$) [140, 158, 184], which recognize *to what extent a pattern is of interest to particular user preferences*. For example, *probability-based belief* is used to describe user confidence of unexpected rules [158]. There is very limited research on developing business-oriented interestingness, for instance, profit mining [210].

The related work on developing alternative interestingness measures focuses on technical interestingness only [155]. In this aspect, complexities of pattern structures and statistical significance are mainly emphasized. What has often been missing is the reflection of general user/business preferences and expectations in evaluating the identified patterns. Emerging research on general business-oriented interestingness is isolated from technical significance. A question to be asked is, "what makes interesting patterns actionable in the real world?" For that, knowledge actionability needs to pay equal attention to both *technical* and *business-oriented* interestingness from both *objective* and *subjective* perspectives [41].

To promote the transfer from data mining to knowledge discovery [101], post-analysis and post-mining have been the main approach. It is used to filter/prune rules and summarize learned rules [142], reduce redundancy [134], or match against expected patterns by similarity/difference [139]. A recent highlight is to *extract actions from learned rules* by splitting attributes into 'hard/soft' [223] or 'stable/flexible' [204] to extract actions that may improve the loyalty or profitability of customers. However, most of the existing post-analysis and post-mining work focuses on specific methods such as frequent pattern mining, especially association rules or its combination with specific methods. Specific strategies for post-analysis and post-mining are mainly developed. This limits the actionability of learned actions and the generalization of the proposed approaches, and leads to a limited systematic problem-solving capability.

In recent years, the combination of relevant algorithms has emerged as a powerful tool for identifying more effective patterns. Typical work consists of a combination of two or more methods. For instance, *class association rules* (or *associative classifier*) build classifiers on association rules [124]. In [160], clustering is used to reduce the number of learned association rules. In [61], a comprehensive overview is drawn on combined mining, including approaches of combining multiple data

sources, multiple features, and multiple methods for more informative combined patterns.

In summary, issues surrounding traditional data mining studies can be summarized into the following key points.

- Real-world business problems are often buried in complicated environment and factors. The environmental elements are often filtered or largely simplified in traditional data mining research. As a result, there is a big gap between a syntactic system and its actual target problem. The identified patterns cannot be used for problem-solving.
- Even though good data mining algorithms are important, any real-world data mining is a problem-solving process and system. It involves many other businesses such as catering for user interaction, environmental factors, connected systems and deliverables to business decision-makers.
- Existing work often stops at pattern discovery which is based on technical significance and interestingness. Business concerns are not considered in assessing the identified patterns. Consequently, the identified patterns are only of technical interest.
- There are *often many* patterns mined but they are not informative and transparent to business people. They cannot easily obtain *truly interesting and operable* patterns for their businesses.
- A large proportion of the identified patterns may be either *commonsense* or *of no particular interest* to business needs. Business people feel confused as to*why* and *how* they should care about those findings.
- Actions extracted or summarized through post-analysis and post-processing without considering business concerns do not reflect the genuine expectations of business needs. Therefore, they cannot support smart decision-making.
- Business people often do not know, and are also not informed about, *how to interpret and use* them and *what straightforward actions can be taken* to engage them in business operational systems and decision-making.

These aspects greatly contribute to the significant gap between data mining research and applications, the weak capabibility of existing KDD approaches for AKD, and the resulting limitation of widespread deployment of data mining.

1.3.3 Overlooking Ubiquitous Intelligence

Traditional data mining methodologies and approaches overlook or largely simplify the involvement of ubiquitous intelligence. In real-world data mining, ubiquitous intelligence has to be involved because of both environmental factors surrounding the problems such as constraints, and the need to solve problems in an appropriately relevant way to satisfy both effectiveness and actionability at the level expected by business. Ubiquitous intelligence can be categorized into the following aspects.

- In-depth data intelligence: deep analysis of data intelligence besides the commonly mentioned hidden patterns on transactional or demographic data; for instance, dealing with business appearance data, business performance data and business behavioral data, and combining patterns identified in individual aspects to form more informative patterns.
- Domain intelligence: involving domain knowledge, prior knowledge, constraints, expert knowledge etc. that are of importance in enhancing data mining decision-making capabilities and performance.
- Organizational and social intelligence: involving organizational and social factors in the data mining environment: for instance, business processes, business rules and things like social networks.
- Human intelligence: involving human-related factors such as human roles, expert knowledge and user preferences in data mining.
- Network and web intelligence: for particular applications such as distributed and online businesses, network and web factors such as application network structures may be necessarily considered, and the strategy design for merging patterns from multiple data sources may need to cater for factors related to data allocation.

In general, it is not necessary, and on most occasions is actually very difficult, to cater for all types of intelligence and respective aspects. In practice, even though some of them can be considered and involved, there is a need to consolidate them in a systematic way. A methodological means is *intelligence metasynthesis* [31, 74, 75, 76, 77, 79, 168, 169, 170, 171, 172, 173, 174, 175], which synthesizes ubiquitous intelligence for problem-solving.

Chapter 3 presents more details and thinking with regard to the definition, illustration, utilization and respective issues related to the above types of intelligence.

1.3.4 Organizational and Social Factors

In data mining, problems and applications are normally surrounded by many organizational and social factors in businesses and organizations. Such organizational and social factors form the environment of real-world data mining. Catering for such factors is necessary since they reflect the real user and business need, and form the 'future' living environment of identified knowledge after deployment. Other reasons include (1) reflecting real business and environmental requirements, (2) constraining the knowledge discovery process and model design, and (3) determining the delivery manner of the discovered knowledge.

The consideration of organizational and social factors in domain driven data mining also differentiates it from the classic data mining methodologies. In D^3M, we believe data mining and knowledge discovery is a systematic and social process, involving data mining systems and their respective environments. It can make data mining outcomes very dependable, trustworthy, practical, effective and beneficial. However, it is challenging to represent and involve organizational and social factors into data mining processes and models. In Chapter 3, we will further discuss

organizational and social factors, under the concept of organizational and social intelligence.

1.3.5 Human Involvement

The role of human beings in data mining has been recognized in many aspects, for instance, interactive data mining, human-centered data mining, and catering for user preferences. In mining complex data mining applications and problems, it is critically important to involve domain experts and to cater for their preferences. This may be due to varying reasons. For instance, many issues such as imaginary reasoning cannot be handled well by current computer and computerized systems; many aspects cannot be reflected into data mining models easily; however, some factors related to human beings can greatly enhance data mining capabilities or results.

Even though it is widely recognized that human involvement and intelligence can make a difference in promoting actionable knowledge discovery, it is not a trivial thing to involve humans in the data mining process, in particular dynamic data mining. It is also challenging to represent human intelligence in data mining models and systems. We will further discuss human intelligence in Chapter 3.

1.3.6 Domain Factors

Enterprise data mining always needs to involve domain knowledge and intelligence. Domain knowledge and intelligence is important for actionable knowledge discovery and operationalizable outcome delivery for many reasons. Domain knowledge and intelligence can play significant roles in filling the business knowledge shortage of data miners if they are not business experts. Without the care of such information, data mining models may generate outcomes totally of no interest to business. This is because the modeling process is highly abstract and filters many domain factors that are critical for bridging the gap between academic research-based findings and industrial problem-solving-oriented solutions.

While there is a common understanding shared by both data mining researchers and practitioners, there is very limited related work on defining, extracting, representing, integrating and utilizing domain knowledge in data mining modeling and its process. In fact, besides commonly agreed domain knowledge, domain intelligence could be explored in a broad manner. This then would address critical issues such as definition, acquisition and the use of domain intelligence in data mining process and model design. We will discuss these issues in Chapter 3.

1.3.7 Knowledge Decision Power

The decision-making power of identified knowledge is largely overlooked in traditional data mining, even though discovering actionable knowledge has been taken as the essential goal of KDD. However, even now, it is still one of the great challenges to existing and future KDD, as pointed out by the panelists and retrospective literature [9]. This situation partly results from the limitations of traditional data mining methodologies, which view KDD as a data-driven trial-and-error process targeting automated hidden knowledge discovery. The methodologies do not take into consideration the constrained and dynamic environment and ubiquitous factors surrounding KDD, which naturally excludes humans and problem domains in the loop. As a result, data mining research has been over-simplified by the development and demonstration of specific algorithms, while it runs off the rails in failing to produce actionable knowledge that is the key purpose in fulfilling specific user needs.

To revert to the core objectives of KDD, the following key points have recently been highlighted or need to be catered for: comprehensive organizational factors around the problem [17], domain knowledge and intelligence [227], human role and intelligence, and the deliverables in the form of business-friendly and operationalizable manners for straightforward decision-making in the process and environment of real-world KDD. A proper consideration and consolidation of these aspects in the KDD process has been reported as making KDD promising in being able to dig out actionable knowledge that satisfies real-life dynamics and requests, even though it is a very difficult issue. This pushes us to think of what knowledge actionability is, and how to make actionable knowledge discovery.

Domain driven data mining has been proposed to engage the relevant aspects in actionable knowledge discovery. On top of the data-driven framework, domain driven data mining aims to develop proper methodologies and techniques for integrating domain knowledge, human roles and interaction, as well as actionability measures into the KDD process to discover and deliver actionable knowledge in the constrained environment. Furthermore, it aims to ensure the delivery of actionable and dependable knowledge that can be taken over by business people for direct decision-making and business operation purposes.

1.3.8 Decision-Support Knowledge Delivery

Data mining currently stops at the delivery of identified patterns, while such patterns are not necessarily operationalizable after being taken over by business people. If this gap cannot be narrowed or bridged, data mining will not be widely accepted by businesses or be converted into enterprise production. For this purpose, there is a need for decision-support knowledge delivery, namely delivering or converting the identified knowledge into business-friendly and actionable means that can be

easily interpreted and seamlessly deployed by business people into the enterprise operation.

One way of doing this is to convert the identified actionable knowledge into operationalizable business rules. In this sense, we know that domain driven data mining can facilitate a knowledge delivery process from data to generally interesting patterns, further to actionable knowledge, and eventually into operationalizable business rules.

From the viewpoint of systems, domain driven actionable knowledge delivery needs to be supported by a complex intelligent system. Such a system is guided by a methodology, namely domain driven data mining methodology, for process and project management. The system needs to engage facilities and services for generating decision-support knowledge and to assist in the deployment of such deliverables into business applications seamlessly and effectively.

1.4 KDD Paradigm Shift

One of the fundamental objectives of KDD is to discover knowledge that is of key interest to real business needs and user preferences. This creates a big challenge to existing and future data mining research and applications. To better understand this challenge, let's review traditional data-centered data mining methodologies and the expectation of real-world KDD.

1.4.1 Data-Centered Interesting Pattern Mining

Traditionally, data mining has been emphasized as a process of data-centered interesting pattern mining. There is nothing wrong with it from the conceptual perspective. After all, data mining targets useful information hidden in data. However, if pattern mining has only, or mainly, paid attention to data itself, problems could emerge when it migrates to handling real-world applications. In fact, the dominant research and development in traditional data mining has been notable forthe corresponding research scope, methodologies and research interests concentrating on data itself.

To better understand the major periodical characteristics of traditional data mining, we scrutinize them from the following multiple dimensions: aims and objectives, objects and dataset mined, models and methods, process and system, and performance and evaluation.

- The main *Aim* of data mining is to develop innovative approaches. As a result of this motivation and follow-up trend, the majority of papers accepted by prestigious journals and conferences in this area talk about new approaches or performance revolution;

- Correspondingly, the *Objective* of data mining is to develop or update and demonstrate new algorithms or new performance on a very nicely shaped data set;
- *Object mined*: data is the object being mined, which is expected to tell the whole story of a concern;
- *Datasets* mined are mainly artificial data or are greatly abstracted or refined from real problems and data. Mining is not directly conducted on real data from businesses;
- *Models and methods* in data mining systems are usually predefined. It is the data mining researcher rather than a user that can interpret and deploy the algorithms and outcomes;
- The *Process* of data mining stops at delivering identified patterns, and does not care whethe or how business people can take it over for business decision-support.
- The *System* is usually packed with multiple-step modules customized for analysts to go through one by one with particular knowledge about them; while the algorithm part is normally automated, in which a user is not necessary, and in fact cannot do much on the modeling and algorithms;
- In general, the *Performance* of an algorithm is mainly measured from the technical side, and specific focus has been on things like accuracy and computational sides; no particular interestingness measures from the business side are defined or considered;
- The *Evaluation* of the mined results is fundamentally based on and stops at checking technical metrics; if they beat certain thresholds presumed by data mining researchers then an algorithm is promising; the performance from the business perspective is usually not assessed or even considered.

In summary, traditional KDD is a *data-centered* and *technically dominated* process targeting automated hidden pattern mining [9, 33]. The main goal of traditional data mining research is to let data create/verify research innovation, pursue the high performance of algorithms, and demonstrate novel algorithms. As a result, the mining process stops at discovering knowledge that is mainly of interest to academic or technical people.

1.4.2 From Data Mining to Knowledge Discovery

In the history of data mining evolution as a discipline, there has been a distinct effort advocated by [101] highlighting the need for a paradigm shift *from Data Mining to Knowledge Discovery*. Fayaad et al. believe data mining is a particular step of knowledge discovery, which is the application of specific algorithms for extracting patterns from data. KDD focuses on the overall process of identifying valid (on new data), novel (to the system or even the user), potentially useful (leading to some benefit to the user or task), and ultimately understandable (if not immediately then after some postprocessing) patterns in data, from data preparation, data

selection, data cleaning, incorporation of appropriate prior knowledge, data mining, and proper interpretation of the results of mining, all repeated in multiple iterations. Their constructive point is to view KDD as an intersection of research fields such as machine learning, pattern recognition, databases, statistics, artificial intelligence, knowledge acquisition, data visualization, and high-performance computing, man-machine interaction. The notion *interestingness* is usually taken as an overall measure of pattern value in terms of objective (such as accuracy and utility) or subjective (such as novelty or understandability) either explicitly or implicitly (such as ordering). Based on the infrastructure constructed for KDD, a pattern is considered to be knowledge if it exceeds some interestingness threshold. The knowledge identified should be purely user-oriented, domain specific and is determined by whatever functions and thresholds the user chooses.

1.4.3 Multi-Dimensional Requirements on Actionable Knowledge Delivery

Actionable Knowledge Delivery (AKD) is important because of the multiple dimensions of requirements on both macro-level and micro-level from real-world applications.

On the macro-level, issues are related to methodological and fundamental aspects. For instance, an intrinsic difference exists in academic thinking and business deliverable expectation. An example is that researchers usually are interested in innovative pattern types, while practitioners care about getting a problem solved. A strategic position needs to be taken on whether to focus on a hidden pattern mining process centered by data, or an AKD-based problem-solving system as the deliverable. Some of typical macro-level issues need to be addressed.

- Environment: Refer to any factors surrounding data mining models and systems, for instance, domain factors, constraints, expert group, organizational factors, social factors, business processes, and workflow. They are inevitable and important for AKD. Some factors such as constraints have been considered in the current data mining research, but many others have not. It is essential to represent, model and involve them in AKD systems and processes.
- Human role: To handle many complex problems, human-centered and human-mining-cooperated AKD is crucial. Critical problems related to this include how to involve domain experts and expert groups in the mining process, and how to allocate the roles between human and mining systems.
- Process: Real-world problem-solving has to cater for dynamic and iterative involvement of environmental elements and domain experts along the way.
- Infrastructure: The engagement of environmental elements and humans at run time in a dynamic and interactive way requires an open system with closed-loop interaction and feedback. AKD infrastructure should provide facilities to support such scenarios.

- Dynamics: To deal with the dynamics in data distribution from training to testing and from one domain to another, in domain and organizational factors, in human cognition and knowledge, in the expectation of deliverables, and in business processes and systems.
- Evaluation: Interestingness needs to be balanced between technical and business perspectives, special attention needs to be paid to deliverable formats, and to its actionability and generalizable capability.
- Risk: Risk needs to be measured in terms of its presence and then size if any in conducting an AKD project and system.
- Policy: Data mining tasks often involve policy issues such as security, privacy, trust existing not only in data and environment, but also in the use and management of data mining findings in an organization's environment.
- Delivery: What is the right form of delivery and presentation of AKD models and findings so that end users can easily interpret, execute, utilize and manage the resulting models and findings, and integrate them into business processes and production systems.

At the micro-level, issues related to technical and engineering aspects need to be sorted out. For instance, if we take the position that data mining is an AKD-based problem-solving system, we then need to develop facilities for involving system dynamics, system environment, and interaction in a system. If environmental elements are useful for AKD, how can we engage business processes, organizational factors and constraints? The following lists a few dimensions that can address these concerns.

- Architecture: AKD system architectures need to be flexible for incorporating specific environmental elements, AKD processes, and evaluation systems.
- Interaction: To cater for interaction with business people along the way of AKD processes, it requires appropriate user interfaces, user modeling and servicing to support individuals and group interaction.
- Dynamics and adaptation: Data, environmental elements and business expectations change all the time. AKD systems, models and evaluation metrics are required to be adaptive.
- Actionability: What do we mean by actionability? How should we measure it? What is the tradeoff between technical and business sides? Do subjective and objective perspectives matter? This requires essential metrics to be developed.
- Findings delivery: End users certainly feel more comfortable if patterns can be presented in a business-friendly way and combined with business operational systems and business rules. In this sense, AKD deliverables are required to be easily interpretable, convertible into business rules or to be presented as business rules, and to be linked to decision-making systems.

1.4.4 From Data-Centered Hidden Knowledge Discovery to Domain Driven Actionable Knowledge Delivery

While the above distinction draws an ideal roadmap of knowledge discovery for real-life applications, it is widely recognized that it would be a long-term effort and a disciplinary challenge to achieve the objectives of knowledge discovery and to satisfy user and business needs. In addition, many issues that have not been paid enough attention previously or have emerged with the need to mine complex data for complex knowledge in real-world data mining applications, for instance, human roles and human intelligence, organizational and social factors existing in complex problem domains.

To deal with the disciplinary challenge and emergent complexities, and to make the outcomes workable for business decision-making in real-life knowledge discovery, we believe there is a need for an explicit paradigm shift, namely *from data-centered hidden knowledge discovery to domain driven actionable knowledge delivery*.

The following reasons argue the case for the above explicit paradigm shift.

- Knowledge discovery is a process that needs to be supported by a system; from the system viewpoint, KDD needs to care about the system environment and factors related to data mining in the environment;
- Factors in the knowledge discovery environment involve aspects related to human, domain, organization and society, and network and web where applicable. Such factors may play crucial roles in discovering and delivering actionable knowledge;
- It is challenging to properly define, extract, represent and involve relevant factors in the knowledge discovery process. Methodologies, techniques and tools have to be studied for facilitating corresponding factors;
- Besides the understandability of identified knowledge, more effort is necessary to measure business expectations and actionable capabilities. Business expectations include interestingness from the business perspective (we call this *business interestingness*; correspondingly, *technical interestingness* refers to interestingness measured mainly from the technical perspective), and actionable capabilities (we call this *knowledge actionability*) reflect its immediate problem-solving capabilities by taking corresponding decision-making actions indicated by the knowledge.

D^3M is proposed as a methodology and a collection of techniques targeting *domain driven actionable knowledge delivery* to drive KDD toward enhanced problem-solving infrastructure and capabilities in real business situations. This book addresses the methodologies, frameworks, factors, approaches and applications of D^3M. There are still many open issues in D^3M, and we believe that effective and efficient approaches and techniques will be proposed further by other colleagues. One of the main motivations of this book is to draw attention to those unavoidable issues and aspects in handling complex enterprise applications, and mining complex data for complex knowledge.

1.4.5 D^3M: Domain Driven Actionable Knowledge Delivery

In the real world, discovering and delivering knowledge that is actionable in solving business problems has been viewed as the essence of KDD. However, the existing data mining is mainly *data-centered* and *technically dominated*, and stops at hidden pattern mining favoring technical concerns and expectation, while many other factors surrounding business problems have not been systematically or comprehensively considered and balanced. We believe this kind of environment is likely to continue for a long time. It will be one of the great challenges to the existing and future KDD community, as discussed in retrospective literature, to shift data mining into operationalizable soft power.

The problem with the classic data mining research environment partly results from the periodical interest and focus of traditional data mining methodologies. Their main objectives and tasks have been to break through this new scientific field by setting up theoretical underpinnings and approaches. Real-world complexities have not been given much consideration due to other more urgent and foundational research issues. In this process, external factors surrounding problem domains and problem-solving systems. As a result, very often data mining research mainly aims at developing, demonstrating and pushing the use of specific algorithms. As a result, it runs off the rails in producing knowledge that caters for user and business needs, and in supporting straightforward decision-making action-taking on business problems.

In the wave of rethinking the original objectives of KDD and the subsequent development, the following key points have recently been reviewed: comprehensive constraints around a problem [17], domain knowledge and human role [9, 32, 115] in the KDD process and environment, dependability and actionability, and decision-support knowledge delivery. A proper engagement of these aspects in the KDD process has been reported as making KDD promising in digging out actionable knowledge to satisfy real life dynamics and requests, even though this is a very tough issue. Driven by such rethinking, a comprehensive and systematic retrospection and innovation may be necessary to effect a paradigm shift from *data-centered* and *technically dominated* hidden pattern mining to *problem-solving-oriented* capabilities and deliverables.

Aiming at complementing the shortcomings of traditional data mining, in particular, strengthening the *problem-solving-oriented* capabilities and deliverables in enterprise data mining, we propose a practical methodology, called *Domain Driven Actionable Knowledge Delivery*, or *Domain Driven Data Mining* (D^3M) by following the widely accepted terminology 'Data Mining'. The basic idea of D^3M is as follows. On top of the data-centered framework, it aims to develop proper methodologies and techniques for integrating *domain knowledge*, *human role and interaction*, *organizational and social factors*, as well as *capabilities and deliverables* toward delivering actionable knowledge and supporting business decision-making action-taking in the KDD process.

D^3M targets the discovery of actionable knowledge in the real business environment. Such research and development is very important for developing the next-

generation data mining methodologies and infrastructures [9, 33]. Most importantly, D^3M highlights the crucial roles of ubiquitous intelligence, including in-depth data intelligence, domain intelligence and human intelligence, and their consolidation, by working together to tell hidden stories in businesses, exposing actionable and operationalizable knowledge to satisfy real user needs and business operation decision-making. End users hold the right to say "good" or "bad" to the mined results.

To be aligned with traditional data mining, Table 1.2 compares major dimensions of the research driven by traditional data-centered data mining and domain driven data mining. The dimensions consist of Aims, Goals, Objects mined, Datasets, Process, Methods and models, Performance and Evaluation, as well as Deliverables.

Table 1.2 Data-Centered vs. Domain Driven Data Mining.

Dimensions	Traditional Data-Centered	Domain Driven
Aims	Developing innovative approaches	Solving business problems
Goals	Let data create/verify research innovation;	Let data and domain knowledge
	Demonstrate and push the use of novel algorithms discovering knowledge of interest technically	Tell hidden story in business; discovering actionable knowledge to satisfy real user needs
Objectives	Algorithms are the focus	Problem-solving is the target
Object mined	Data tells the story	Data and domain-oriented factors tell the story
Datasets	Mining abstract and refined datasets	Mining constrained real-life data
Models and methods	Predefined	Ad-hoc, running-time and personalized model customization
Process	Data mining is an automated process	Human is in the circle of data mining process
Performance	Technical sides such as accuracy and computational aspects	Technical and business aspects such as actionability
Evaluation	Evaluation based on technical metrics	Business says "yes" or "no"
Deliverables	Hidden patterns	Operationalizable business rules

1.5 Towards Domain Driven Data Mining

Data mining research and development is challenged and boosted by increasingly recognized and emerging issues in the real world, the need to deal with more and more sophisticated problems, and the expectation of delivering more sophisticated but straightforward and immediate knowledge for supporting complex decision-making. These form the fundamental driving forces of the KDD evolution. For instance, some typical recent progress that has been made in data mining includes stream data mining handling stream data, and link mining studying linkage across entities. Challenges and prospects coming from the real world force us to rethink some key aspects in data mining. This includes problem understanding and defini-

tion, KDD context, patterns mined, mining process, interestingness system, and infrastructure supports. The outcome of this retrospection and rethinking is a paradigm enhancement or shift from a traditional data-focused and research-oriented environment to a domain driven and problem-solving-targeted era.

1.5.1 The D^3M Methodology

As a result of developing the next-generation methodologies, techniques and tools for handling ubiquitous intelligence surrounding real-world data mining, we propose the concept of *Domain Driven Actionable Knowledge Delivery*, or D^3M by following the widely accepted notion 'data mining'.

Definition 1.1. Domain Driven Data Mining. Domain driven data mining refers to the set of methodologies, frameworks, approaches, techniques, tools and systems that cater for human, domain, organizational and social, and network and web factors in the environment, for the discovery and delivery of actionable knowledge.

Definition 1.2. Actionable Knowledge. Actionable knowledge is business-friendly and understandable, reflects user preferences and business needs, and can be seamlessly taken over by business people for decision-making action-taking.

To support domain driven data mining for actionable knowledge delivery, we highlight the following key components:

- *Problem understanding and definition* is domain-specific, and must consider and involve ubiquitous intelligence surrounding the domain problem;
- *Ubiquitous intelligence* needs to be cared and utilized in the whole KDD process;
- *Human roles, intelligence and involvement* contribute importantly to the problem-solving and decision-support capabilities;
- *Data mining* has to cater for a constrained (with constraints), dynamic (dynamic data, environment and deliverables), distributed (data, domain knowledge or decision-making crossing multiple nodes), heterogeneous (mixed and multiple data sources), interactive (human-mining interaction), social (involving groups and communities, organizational structures and relationships, and factors such as reputation and policies) and/or networked (resources, actors and modules interconnected) context;
- *Patterns* target in-depth ones, which differentiate themselves from generally interesting patterns, but disclose deep knowledge and inside principles about businesses that cannot be exposed by single or direct use of traditional methods and strategies; typical strategies such as *combined mining* [61] are helpful for finding more sophisticated knowledge in complex data;
- *Actionable knowledge discovery* is a loop-closed and iterative refinement process, multiple feedbacks, iterations and refinement are involved in the understanding of data, resources, the roles and utilization of relevant intelligence, the presentation of patterns, the delivery specification, and knowledge validation;

during the process and iterations, understanding and deliverables are progressively improved and enhanced toward the final deliverables satisfying user and business needs and supporting direct decision-making action-taking;

- *Performance evaluation* needs to consider not only generally recognized technical interestingness, but more importantly business interestingness and expectations from both subjective and objective perspectives;
- *KDD infrastructure* needs to be greatly enhanced to facilitate domain driven actionable knowledge delivery by developing system support for dynamically involving humans, engaging ubiquitous intelligence, adapting to business and requirement dynamics and movement, plugging and playing resources, actors and algorithms, suitable for visual KDD process management, visualizing modeling, supporting dynamic parameter and feedback tuning, and presenting outcomes in a personalizable and customizable manner.

In this domain driven framework, computer-based KDD systems and domain experts complement each other in terms of in-depth granularity through interactive interfaces. The involvement of domain experts and their knowledge, in particular, qualitative intelligence, can assist in developing highly effective domain-specific data mining techniques and reduce the complexity of the knowledge producing process in the real world. In-depth pattern mining discovers more interesting and actionable patterns from a domain-specific perspective. A system following this framework can embed effective supports for domain knowledge and experts' feedback, and refines the lifecycle of data mining in an iterative manner.

1.5.2 Problem: Domain-Free vs. Domain-Specific

In traditional data mining, researchers spend the majority of their time on searching and constructing research problems, while real-world data mining research issues come from real challenges. Traditionally in a research-focused situation, it is typical that, even though a problem may come from a real scenario, it is always abstracted and pruned into a very general and brilliant research issue to fulfil innovation and significance requirements. Such a research issue is usually domain-free, which means it does not necessarily involve specific domain intelligence. Undoubtedly, this is important for developing the science of KDD.

On the other hand, in real-world scenarios, challenges always come from both general and domain-specific problems. Therefore, the objectives and goals of applying KDD are basically problem-solving and the satisfaction of real user needs. Problem-solving and satisfying real user needs present strongly usable requirements. Requirements mainly come from a specific domain and involve concrete functional and non-functional concerns. The analysis and modeling of these requirements call for domain intelligence, in particular domain background knowledge and the involvement of domain experts. Therefore, real-world data mining is more likely to mix both general and domain-specific issues. However, domain-specific data mining does not necessarily focus on the specific domain-problem only. Here *domain*

can refer to either a big industrial sector, for instance, telecommunication or banking, or a specific category of business, such as customer relationship management for banking.

Domain intelligence can play a significant role in real-world data mining. Domain knowledge in the business field takes the form of either precise knowledge, concepts, beliefs, and relations, or vague preference and bias. For instance, in financial data mining, traders often take 'beating market' as a personal preference to judge an identified trading rule's reliability. The key to taking advantage of domain knowledge in the KDD process is knowledge and intelligence integration, which involves understanding how it can be represented and injected into the knowledge discovery process. Ontology-based domain knowledge representation, transformation and mapping [35, 107, 108] between a business and data mining system is one of proper approaches to modeling domain knowledge. Ontology-based specifications can represent domain knowledge in terms of ontological items and semantic relationships.

Through ontology-based representation and transformation, business terms are mapped to the data mining system's internal ontologies in a uniform manner. We can build an internal *data mining ontological domain* to represent and generalize the KDD process and systems in a general way. In addition, constructing standard domain-specific terminology is helpful for interaction amongst specific applications crossing the particular domain and the exchange of discovered knowledge across the domain. To match items and relationships between two specific domain problems and reduce and aggregate synonymous concepts and relationships in each situation, ontological rules, logical connectors and cardinality constraints can support ontological transformation from one domain to another, and semantic aggregations of semantic relationships and ontological items intra or inter domains.

1.5.3 KDD Context: Unconstrained vs. Constrained

The law, business rules and regulations are common forms of constraints in human society. Similarly, actionable knowledge delivery in the real world can only be conducted well in a constrained rather than unconstrained context. Constraints involve technical, economic and social aspects in the process of developing and deploying actionable knowledge. For instance, constraints can involve aspects such as environmental reality and expectations on data format, knowledge representation, and outcome delivery in the mining process. Other aspects of domain constraints include the domain and characteristics of a problem, domain terminology, specific business process, policies and regulations, particular user profiling and favorite deliverables. In particular, we highlight following types of constraints - *domain constraint*, *data constraint*, *interestingness constraint* and *deployment constraint*.

Real world business problems and requirements are often tightly embedded in domain-specific business process and business rules (*domain constraint*). Potential matters to satisfy or react to domain constraints may consist of building domain

models, domain metadata, semantics and ontologies [35], supporting human involvement, human-machine interaction, qualitative and quantitative hypotheses and conditions, merging with business processes and enterprise information infrastructure, fitting regulatory measures, conducting user profile analysis and modeling, etc. Relevant hot research areas include interactive mining, guided mining, and knowledge and human involvement.

Patterns that are actionable to business are often hidden in large quantities of data with complex data structures, quality, dynamics and source distribution (*data constraint*). Constraints on particular data may be embodied in terms of aspects such as very large volume, ill-structure, multimedia, diversity, high dimensions, high frequency and density, distribution and privacy, etc. Data constraints seriously affect the development of, and performance requirements on, mining algorithms and systems, and constitute a big challenge to data mining. As a result, some popular research work on data constraint-oriented issues are emerging such as stream data mining, link mining, multi-relational mining, structure-based mining, privacy mining, multimedia mining and temporal mining.

Often, mined patterns are not actionable to business even though they are sensible to research. There may be significant interestingness conflicts or gaps between academia and business (*interestingness constraint*). What makes one rule, pattern and finding more interesting than aother? In the real world, simply emphasizing technical interestingness such as objective statistical measures of validity and surprise is not adequate. Social and economic interestingness (we refer to Business Interestingness) such as user preferences and domain knowledge should be considered in assessing whether a pattern is actionable or not. Business interestingness may be instantiated in specific social and economic measures in terms of a problem domain. For instance, metrics such as *profit*, *return* and *return on investment* are usually used by traders to judge whether a trading rule is sufficiently interesting or not.

Furthermore, interesting patterns often cannot be deployed to real life if they are not integrated with business rules and processes (*deployment constraint*). The delivery of an interesting pattern must be integrated with the domain environment such as business rules, process, information flow, presentation, etc. In addition, many other realistic issues must be considered. For instance, a software infrastructure may be established to support the full lifecycle of data mining; the infrastructure needs to be integrated with the existing enterprise information systems and workflow; parallel KDD [198] may be involved with parallel supports on multiple sources, parallel I/O, parallel algorithms, and memory storage; visualization, privacy and security should receive much-deserved attention; false alarming should be minimized.

Some other types of constraints include knowledge type constraint, dimension/level constraint and rule constraint [115]. Several types of constraints play significant roles in effectively discovering knowledge actionable to the business world. In practice, many other aspects such as scalability and efficiency of algorithms may be enumerated. They consist of domain-specific, functional, nonfunctional and environmental constraints. These ubiquitous constraints form a constraint-based context for actionable knowledge delivery. All the above constraints must, to varying de-

grees, be considered in relevant phases of real-world data mining. For this reason, it is also called *constraint-based data mining* [17, 115].

1.5.4 Interestingness: Technical vs. Business

Traditionally, interestingness mainly refers to technical significance such as statistical significance metrics. This situation is still the dominant way to measure pattern significance in mainstream research and development. Technical significance refers to interestingness measures developed corresponding to the particular technical modeling methods used. For instance, if a method is driven by statistics, relevant statistic measures are then defined to measure the method's performance. In general, data mining methods and algorithms come from the bodies of knowledge of statistics, pattern recognition and machine learning, and databases. Corresponding technical measures involve aspects of statistical performance (for instance, *support* and *confidence*), computational performance (e.g., *response time*), and system performance (e.g., *accuracy*).

The technical interestingness family has been expanded throughout the evolution of KDD. This includes major efforts on:

- Developing *enhanced interestingness measures* for the same method. A typical example is *association rule mining*, although many interestingness measures have been proposed;
- Developing *subjective interestingness measures* for the same method. Measures such as *understandability* and *actionability* are typical examples.

We believe there is a need to develop *business interestingness* for the following reasons:

- The need for*satisfying and correspondingly measuring business and user needs* in the paradigm shift from data mining to knowledge discovery, and in particular from data-centered hidden knowledge discovery to domain driven actionable knowledge delivery; business interestingness metrics should be developed in terms of corresponding business needs and user preferences.
- The need for*measuring the significance of identified knowledge from the business perspective* besides what have been justified from the technical end when patterns are filtered. In general, patterns are pruned and filtered in terms of particular technical interestingness metrics developed to measure the technical significance and performance of the discovered knowledge. Business interestingness is necessary and complementary for measuring business significance and performance of the identified patterns. By complementing technical interestingness systems, business interestingness systems can filter outcomes to make them not only of technical significance and interest but also of business significance and interest.

The KDD interestingness system is therefore expected to be enhanced and expanded by adding business interestingness. Further, with the emergence of business interestingness, there is the added need to develop integrated interestingness

system to combine and balance technical significance with business in knowledge discovery, and in measuring the overall performance of knowledge significance in problem-solving. We refer to such overall performance as *knowledge actionability*, meaning the extent to which an identified pattern can be significant and trustful from both technical and business sides, while also supporting decision-making action-taking. In Chapter 4 on Knowledge Actionability, we further discuss the definition, and measurement of business interestingness, the relationships between technical and business interestingness, and the concept and measurement of knowledge actionability.

1.5.5 Pattern: General vs. Actionable

Many mined patterns are more interesting to data miners than to business persons. Generally interesting patterns are thus because they satisfy a technical interestingness measurement. We call them *general patterns* or technically interesting patterns.

However, general patterns are not necessarily useful for solving business problems. To improve this situation, we advocate in-depth pattern mining which aims to develop patterns that are actionable in the business world. It targets the discovery of actionable patterns to support smart and effective decision-making, namely a pattern P (composed of itemset x) must satisfy

$$\forall P : x.tech_int(P) \wedge x.biz_int(P) \longrightarrow x.act(P). \tag{1.1}$$

Therefore, in-depth patterns may be delivered through improving either technical interestingness *tech_int*() or business interestingness *biz_int*(). However, actionable patterns are those satisfying both *tech_int*() and *biz_int*(). In domain driven actionable knowledge discovery, both technical and business interestingness measures must be satisfied from both objective and subjective perspectives.

Technically, it could be through enhancing or generating more effective interestingness measures [155]. For instance, a series of research projects have been done on designing right interestingness measures for association rule mining [196]. It may also be achieved through developing alternative models for discovering deeper patterns. Some other solutions include post mining actionable patterns on an initially discovered pattern set. Additionally, techniques can be developed to deeply understand, analyze, select and refine the target data set in order to find in-depth patterns.

Actionable patterns in most cases can be created through rule reduction, model refinement or parameter tuning by optimizing generic patterns. In this case, actionable patterns are a revised optimal version of generic patterns, which capture deeper characteristics and understanding of the business. Of course, such patterns can also be directly discovered from the data set with sufficient consideration of business constraints.

On the other hand, for those generic patterns identified based on technical measures, their business interestingness needs to be checked so that business requirements and user preference can be given proper consideration. Domain intelligence, including business requirements, objectives, domain knowledge and qualitative intelligence of domain experts, can play roles in enhancing pattern actionability. This can be achieved through selecting and adding business features, involving domain knowledge, supporting interaction with users, tuning parameters and data sets by domain experts, optimizing models and parameters, adding factors to technical interestingness measures or building business measures, and improving the result evaluation mechanism through embedding domain knowledge and human involvement.

1.5.6 Infrastructure: Automated vs. Human-Mining-Cooperated

Traditional data mining is an automated trial and error process. Deliverables are presumed as predefined automated algorithms and tools. It is arguable that such automated methodology has both strengths and weaknesses. The positive side is that it makes the user's life easy. However, it confronts challenges in aspects such as lacking capability in involving domain intelligence and adapting to dynamic situations in the business world. In particular, automated data mining encounters big trouble in handling the dynamics and ad-hoc requests widely seen in enterprise data mining applications.

Actionable knowledge discovery in a constrained context determines that real-world data mining is more likely to be human involved rather than automated. Human involvement is embodied through the cooperation between an individual or a group of people (including users and business analysts, mainly domain experts) and a data mining system. This is achieved through the complementation of human qualitative intelligence such as domain knowledge and field supervision, and mining quantitative intelligence like computational capability. Therefore, real-world data mining usually presents as a human-mining-cooperated interactive knowledge delivery process.

The human role can be embodied in the full period of data mining from business understanding, data understanding, problem definition, data integration and sampling, feature selection, hypothesis proposal, business modeling and learning, to the evaluation, refinement, interpretation and delivery of algorithms and resulting outcomes. For instance, the experience, metaknowledge and imaginary thinking of domain experts can guide and assist with the selection of features and models, adding business factors into modeling, creating high quality hypotheses, designing interestingness measures by injecting business concerns, and quickly evaluating mining results. This assistance can largely improve the effectiveness and efficiency of mining actionable knowledge.

Usually, humans serve on feature selection and result evaluation. Humans can play roles at a specific stage or the full stages of data mining. Humans can be an essential constituent or the center of a data mining system. The complexity of dis-

covering actionable knowledge in a constraint-based context decides to what extent and how a human must be involved. As a result, human-mining cooperation presents, to varying degrees, as human-centered, human-guided mining [9, 102], or human-assisted mining.

1.6 Summary

In this chapter, we have explored the evolution of data mining and knowledge discovery research and development, and pointed out the major challenges and issues troubling traditional data mining methodologies and techniques, and further proposed the methodologies of domain driven data mining.

The research on domain driven data mining is very important for developing the next-generation data mining methodology and infrastructure. It can assist in a paradigm shift from *data-driven hidden pattern mining* to *domain driven actionable knowledge delivery*, and provides support for KDD to tackle the real business situations as widely expected.

Our conclusions from this chapter are as follows:

- There is a big gap between traditional data mining research and business expectations on identified knowledge;
- To narrow the gap, domain driven data mining aims to provide corresponding methodologies and techniques;
- Domain driven data mining aims at the development of a series of methodologies, techniques, tools and applications for domain driven actionable knowledge discovery and delivery;
- Domain driven data mining aims at the contribution of a paradigm shift from data-centered hidden pattern mining to domain driven actionable knowledge discovery and delivery.

In Chapter 2, we introduce the methodology of domain driven data mining.

Chapter 2
D^3M Methodology

2.1 Introduction

On the basis of the discussions and retrospection on existing data mining methodologies and techniques in Chapter 1, this chapter presents an overall picture of domain driven data mining (D^3M). We focus on the high level of architecture and concepts of the D^3M methodology.

The goals of this chapter consist of the following aspects:

- An overview of the D^3M methodology;
- The main components of the D^3M methodology; and
- The methodological framework of the D^3M methodology.

Correspondingly, this chapter will introduce the following content.

- Section 2.2 presents a concept map of the D^3M methodology, which is composed of the structure, major components, and their relationships.
- In Section 2.3, we outline key methodological components consisting of the D^3M methodology. While some of these can be found in current data mining systems, they will be re-interpreted or revisited under the umbrella of D^3M. We also highlight some elements that are ignored or weakly addressed in classic methodologies and approaches.
- Finally, in Section 2.4, the theoretical underpinnings and process model of the D^3M methodology are introduced. This presents some high-level ideas of how D^3M is built on and what the D^3M process looks like.

2.2 D^3M Methodology Concept Map

Fig. 2.1 illustrates a high-level concept map of D^3M methodology. The concept map consists of the following layers from the outer most layer to the central core.

L. Cao et al., *Domain Driven Data Mining*, DOI 10.1007/978-1-4419-5737-5_2,
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- Specific domain problems: In general, this can apply to any domain problems from retail to government to social network, from either a sector or specific business problem perspective. However, since D^3M mainly targets complex knowledge from complex data, we do not concern ourselves with those problems and businesses that have been or can be well-handled by existing data mining and knowledge discovery techniques.
- Fundamental research issues: Driven by specific domain problems and business needs, we here extract the main fundamental research issues emerging from them. We emphasize the following two aspects:
  - Infrastructure capability: We are concerned with key issues including: whether a data mining system can generally handle ubiquitous intelligence surrounding a domain problem or not, how it consolidates the relevant intelligence through what infrastructure, and what will be presented to support decision-making action-taking by end users.
  - Decision-support power: We are concerned with key issues reflecting and enhancing the decision-making power of identified knowledge and deliverables, in terms of key performances such as adaptability, dynamics, actionability, workability, operability, dependability, repeatability, trust, explainability, transferability, and usability.
- D^3M theoretical foundations: D^3M supporting techniques need foundational support from many relevant areas, from the information sciences to social sciences. In particular, we see a strong need to create new scientific fields, such as data sciences, web sciences and service sciences, targeting the establishment of a family of scientific foundations, techniques and tools for dealing with increasingly emergent complexities and challenges in the corresponding areas.
- D^3M supporting techniques: To engage and consolidate the fundamental issues surrounding domain driven actionable knowledge delivery, we need to develop corresponding techniques and tools for involving and utilizing ubiquitous intelligence, supporting knowledge representation and deliverables, catering for project and process management, and implementing decision-making pursuant to the findings.

2.3 D^3M Key Components

The D^3M methodology consists of the following key components.

- Constrained knowledge delivery environment
- Considering ubiquitous intelligence
- Cooperation between human and KDD systems
- Interactive and parallel KDD support
- Mining in-depth patterns
- Enhancing knowledge actionability

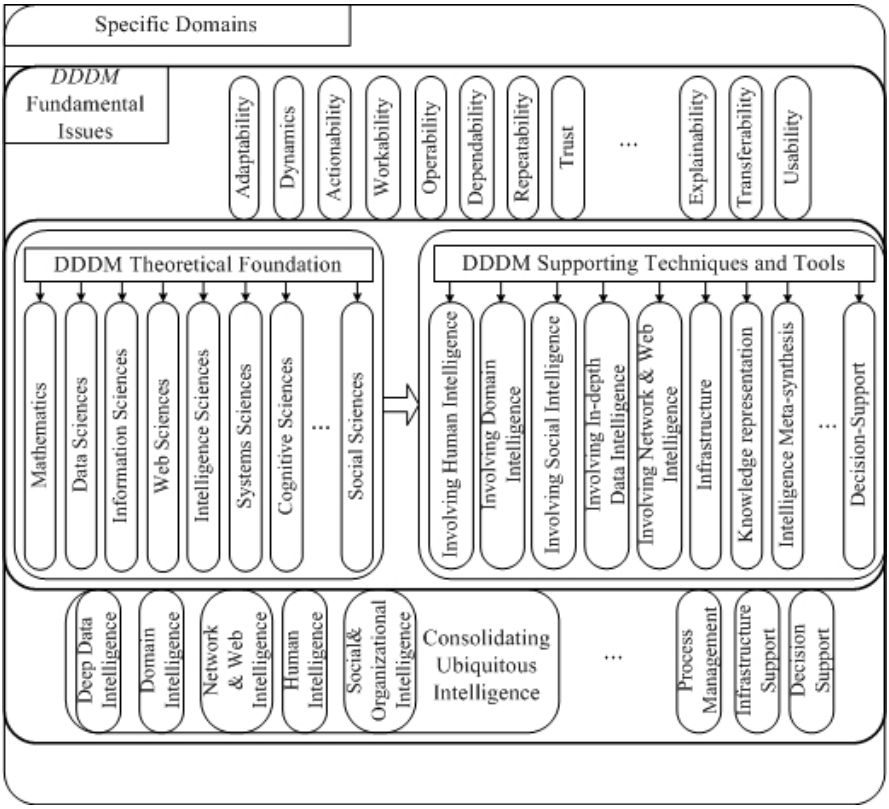

Fig. 2.1 D^3M concept map

- Reference model
- Qualitative research
- Closed-loop and iterative refinement

The nomination of the above key components is based on the support needed to cater for relevant factors in the domain and for actionable knowledge delivery. They have potential for re-shaping KDD processes, modeling and outcomes toward discovering and delivering knowledge that can be seamlessly used for decision-making action-taking, if they are appropriately considered and supported from technical, procedural and business perspectives.

2.3.1 Constrained Knowledge Delivery Environment

In human society, everyone is constrained by either social regulations or personal situations. Similarly, actionable knowledge is discovered in a constraint-based con-

text mixing environmental reality, expectations and constraints in the pattern mining process. Specifically, [33] list several types of constraints which play significant roles in a process which effectively discovers knowledge actionable to business. These include *domain constraints*, *data constraints*, *interestingness constraints*, and *deliverable constraints*.

Some major aspects of *domain constraints* include the domain and characteristics of a problem, domain terminology, specific business process, policies and regulations, particular user profiling and favorite deliverables. Potential matters to satisfy or react on domain constraints consist of building domain models, domain metadata, semantics [181] and ontologies [35, 107, 108], supporting human involvement, human-mining interaction, qualitative and quantitative hypotheses and conditions, merging with business processes and enterprise information infrastructure, fitting regulatory measures, conducting user profile analysis and modeling, etc. Relevant hot research areas include interactive mining, guided mining, and knowledge and human involvement.

Constraints on particular data, namely *data constraints*, may be embodied in terms of aspects such as very large volume, ill-structure, multimedia, diversity, high dimensionality, high frequency and density, distribution and privacy, dynamics and changes. Data constraints seriously affect the development of and performance requirements on data mining algorithms and systems, and constitute some grand challenges to real-world data mining. As a result, popular researches on data constraint-oriented issues are emerging such as stream data mining, link mining, multi-relational mining, structure-based mining, privacy mining, multimedia mining, temporal mining, dynamic data mining, and change and difference mining.

What makes a rule, pattern and finding more interesting than the other? This involves *interestingness constraints*. In the real world, simply emphasizing technical interestingness, such as objective statistical measures of *validity* and *surprise*, is not adequate. Social and economic interestingness (we refer to *Business Interestingness*) such as benefit-cost for operating and implementing the identified knowledge, which embodies user preferences and domain knowledge, should be considered in assessing whether a pattern is actionable or not. Business interestingness would be instantiated into specific social and economic measures in terms of the problem domain. For instance, *profit*, *return*, *return on investment*, or *cost-benefit ratio* are usually used by traders to measure the economic performance of a trading rule, and to judge whether a trading rule is in their interest or not.

Furthermore, the delivery of an interesting pattern needs to consider operationalization factors, which consequently involves *deliverable constraints*. Deliverables such as business rules, processes, information flow, presentation, etc. may need to be integrated into the domain environment.In addition, many other realistic issues must be considered. For instance, a software infrastructure may be needed to support the full lifecycle of data mining; the infrastructure needs to integrate with the existing enterprise information systems and workflow; parallel KDD may be involved with parallel support on multiple sources, parallel I/O, parallel algorithms, and memory storage; visualization, privacy and security should receive much-deserved attention; false alarming needs to be minimized.

In practice, constraints on real-world data mining may also be embodied in many other aspects, such as the frequency and density of data, and the scalability and efficiency of algorithms, which have to be facilitated. These plus the above-mentioned constraints form the conditions and environment of data mining from process, operational, functional and nonfunctional aspects. These ubiquitous constraints form a constraint-based context for actionable knowledge discovery and delivery. All the above constraints must, to varying degrees, be considered in the relevant phases of real-world data mining. Thus it is also called constraint-based data mining [17, 115].

In summary, actionable knowledge discovery and delivery will not be a trivial task and should be put into a constraint-based environment. On the other hand, tricks may not only include how to find a right pattern with a right algorithm in a right manner, but may also involve a suitable process-centric and domain-oriented process management methodology and infrastructure support.

2.3.2 Considering Ubiquitous Intelligence

Traditionally, data mining only pays attention to and relies on data to disclose possible stories wrapping a problem. We call such finding *data intelligence* disclosed from data. Driven by this strategic idea, data mining focuses on developing methodologies and methods in terms of data-centered aspects, particularly the following issues:

- Data type such as numeric, categorical, XML, multimedia, composite
- Data timing such as temporal, time-series, sequential and real-time data
- Data spacing such as spatial and temporal-spatial
- Data speed such as data stream, rare and loose occurrences
- Data frequency such as high frequency data
- Data dimension such as multi-dimensional data
- Data relation such as multi-relational data, linkage, casual data
- Data quality such as missing, noisy, uncertain and incomplete data

On the other hand, domain factors consisting of qualitative and quantitative aspects hide intelligence for problem-solving. Both qualitative and quantitative intelligence is instantiated in terms of domain knowledge, constraints, actors/domain experts and environment. They are further instantiated into specific bodies. For instance, constraints may include domain constraints, data constraints, interestingness constraints, deployment constraints and deliverable constraints. To deal with constraints, various strategies and methods may be undertaken; for instance, interestingness constraints are modeled in terms of interestingness measures and factors, such as objective interestingness and subjective interestingness.

In a summary, we list ubiquitous intelligence hidden and explicitly existing in domain problems in terms of the following major aspects.

(1) Domain knowledge aspect

- Including domain knowledge, background and prior information,

(2) Human aspect

- Referring to direct or indirect involvement of humans, imaginary thinking, brainstorming, etc.
- Empirical knowledge
- Belief, request, expectation, etc.

(3) Constraint aspect

- Including constraints from system, business process, data, knowledge, deployment, etc.
- Privacy
- Security

(4) Organizational aspect

- Organizational factors
- Business process, workflow, project management
- Business rules, law, trust

(5) Environmental aspect

- Surrounding business processes, workflow
- Linkage systems
- Surrounding situations and scenarios

(6) Evaluation aspect

- Technical interestingness corresponding to a specific approach
- Profit, benefit, return, etc.
- Cost, risk, etc.
- Business expectation and interestingness

(7) Deliverable and deployment aspect

- Delivery manners
- Embedding into business system and process

Correspondingly, a series of issues needs to be studied in order to involve and utilize such ubiquitous intelligence in the actionable knowledge delivery system and process. The involvement of ubiquitous intelligence forms a key element and characteristic of domain driven data mining towards domain driven actionable knowledge delivery. For instance, the following are some such tasks for involving and utilizing ubiquitous intelligence.

- Definition of ubiquitous intelligence
- Representation of domain knowledge
- Ontological and semantic representation of ubiquitous intelligence
- Ubiquitous intelligence transformation between business and data mining
- Human role, modeling and interaction

- Theoretical problems in involving ubiquitous intelligence in KDD
- Metasynthesis of ubiquitous intelligence in knowledge discovery
- Human-cooperated data mining
- Constraint-based data mining
- Privacy, sensitivity and security in data mining
- Open environment in data mining
- In-depth data intelligence
- Knowledge actionability
- Objective and subjective interestingness
- Gap resolution between statistical significance and business expectation
- Domain-oriented knowledge discovery process model
- Profit, benefit/cost, risk, impact of mined patterns

2.3.3 Cooperation between Human and KDD Systems

The real-life requirements for discovering actionable knowledge in a constraint-based environment determine that real-world data mining is more likely to follow man-machine-cooperated mode, namely human-mining-cooperated rather than automated. Human involvement is embodied through the cooperation between humans (including users and business analysts, mainly domain experts) and a data mining system. This is because of the complementation between human qualitative intelligence such as domain knowledge and field supervision, and the quantitative intelligence of KDD systems like computational capabilities. Therefore, real-world complex data mining presents as a human-mining-cooperated interactive knowledge discovery and delivery process.

The role of humans in AKD may be embodied in the full period of data mining from business and data understanding, problem definition, data integration and sampling, feature selection, hypothesis proposal, business modeling and learning to the evaluation, refinement and interpretation of algorithms and resulting outcomes. For instance, the experience, meta-knowledge and imaginary thinking of domain experts can guide or assist with the selection of features and models, add business factors into the modeling, create high quality hypotheses, design interestingness measures by injecting business concerns, and quickly evaluate mining results. This assistance can largely improve the effectiveness and efficiency of identifying actionable knowledge.

In existing data mining applications, humans often take part in feature selection and results evaluation. In fact, a human may play a critical role in a specific stage or during the full stages of data mining on demand. In some cases, for example, mining a complex social community, a human is an essential constituent or the center of a data mining system. The complexity of discovering actionable knowledge in a constraint-based environment determines to what extent humans must be involved. As a result, the human-mining cooperation could be, to various degrees, human-centered mining, human-guided mining, or human-supported or -assisted mining.

To support human involvement, human-mining interaction, otherwise known as interactive mining [6, 9], is absolutely necessary. Interaction often takes explicit forms, for instance, setting up direct interaction interfaces to fine tune parameters. Interaction interfaces may take various forms as well, such as visual interfaces, virtual reality, multi-modal, mobile agents, etc. On the other hand, it could also go through implicit mechanisms, for example accessing a knowledge base or communicating with a user assistant agent. Interaction communication may be message-based, model-based, or event-based. Interaction quality relies on performance such as user-friendliness, flexibility, run-time capability, representability and even understandability.

2.3.4 Interactive and Parallel KDD Support

To support domain driven data mining, it is important to develop interactive mining support for involving domain experts, and human-mining interaction. Interactive facilities are also useful for evaluating data mining findings by involving domain experts in a closed-loop manner. On the other hand, parallel mining support is often necessary for dealing with concurrent applications, distributed and multiple data sources. In cases with intensive computation requests, parallel mining can greatly upgrade the real-world data mining performance.

For interactive mining support, intelligent agents [214] and service-oriented computing [186] are good technologies. They can create flexible, business-friendly and user-oriented human-mining interaction through building facilities for user modeling, user knowledge acquisition, domain knowledge modeling, personalized user services and recommendation, run-time support, and mediation and management of user roles, interaction, security and cooperation.

Parallel KDD [122] is good at parallel computing and management support for dealing with multiple sources, parallel I/O, parallel algorithms and memory storage. For instance, to tackle cross-organization transactions, we can design efficient parallel KDD computing and systems to wrap the data mining algorithms. This can be through developing parallel genetic algorithms and proper processor-cache memory techniques. Multiple master-client process-based genetic algorithms and caching techniques can be tested on different CPU and memory configurations to find good parallel computing strategies.

The facilities supporting interactive and parallel mining can largely improve the performance of real-world data mining in aspects such as human-mining interaction and cooperation, user modeling, domain knowledge acquisition and involvement in KDD, and reducing computation complexity. They are essential parts of the next-generation KDD infrastructure for dealing with complex enterprise data for complex knowledge.

Based on our experience in building agent service-based stock trading and mining system F-Trade [1], agent service and ontological engineering techniques can be used for building interactive and parallel mining facilities. User agents, knowledge management agents, ontology services [35] and run-time interfaces can be built to support interaction with users, take users' requests and to manage information from users in terms of ontologies. Ontologies can represent domain knowledge and user preferences, and further map them to a data mining domain to support a seamless mapping crossing user/business terminology, data mining ontologies, and underlying data source fields. Subsequently, a universal and transparent one-stop portal may be feasible for domain experts and business users to assist with training, supervising and tuning feature selection, modeling and refinement, as well as evaluating the outcomes. This can help avoid the requirements imposed on domain experts and business users for learning technical detail and jargon, and enable them to concentrate on their familiar environment and language, as well as their interests and responsibilities.

2.3.5 Mining In-Depth Patterns

In-Depth Patterns indicate patterns that

- uncover not only appearance dynamics and rules but also inside driving forces; for instance, in stock data mining, not only price movement trends but also the interior driving forces of such movements,
- reflect not only technical concerns but also business expectations, and
- disclose not only generic knowledge but also something that can support straightforward decision-making actions.

Greater effort is essential to uncover in-depth patterns in data. 'In-depth patterns' (or 'deep patterns') are not straightforward such as frequency-based, but can only be discovered through more powerful models following thorough data and business understanding and effectively involving domain intelligence or expert guidance. An example is to mine for insider trading patterns in capital markets. Without deep understanding of the business and data, a naive approach is to analyze the price movement change in data partitions of pre-event, event and post-event. A deeper pattern analysis on such price difference analysis may be considered by involving domain factors such as considering market or limit orders, market impact, and checking the performance of *potential abnormal return*, *liquidity*, *volatility* and *correlation*.

However, in general, the modeling of data mining is only concerned with the technical significance. Technical significance is usually defined in a straightforward manner by reflecting the significance of the findings in terms of the utilized techniques. Consequently, pattern interestingness is measured in terms of such technical metrics. When they are delivered to business people, business analysts either cannot understand them very well or cannot justify their significance from the business

[1] www.F-TRADE.info

end. In many cases, business people may just find them unconvincing, unjustifiable, unacceptable, impractical and inoperable. Such situations have hindered the deployment and adoption of data mining in real applications. Therefore it is essentially critical to develop pattern interestingness catering for business concerns, preferences and expectations. The resulting patterns (P) satisfy both technical and business interestingness ($\forall P, x.tech_int(P) \wedge x.biz_int(P) \rightarrow x.act(P)$). As a result, it is more likely that they reflect the genuine needs of business and can support smarter and more effective decision-making.

In-depth pattern mining needs to check both technical ($tech_int()$) and business ($biz_int()$) interestingness in a constraint-based environment. Technically, this could be through enhancing or generating more effective interestingness measures [155]. For instance, a series of interestingness measures have been proposed to evaluate associations more properly in association rule mining. It could also be through developing alternative models by involving domain factors and business interestingness for discovering patterns of business interest. Some other solutions include further mining actionable patterns on the initially discovered pattern set. Additionally, techniques can be developed to deeply understand, analyze, select and refine the target data set and feature set in order to find in-depth patterns.

More attention should be paid to business requirements, objectives, domain knowledge and qualitative intelligence of domain experts for their impact on mining deep patterns. Consequently, business interestingness needs to be developed to reflect such business reality, user preferences, needs and expectations. This can be through selecting and adding business features, involving domain knowledge in modeling pattern significance and impact, supporting interaction with users, tuning the parameters and data set by domain experts, optimizing models and parameters, adding organizational factors into technical interestingness measures or building domain-specific business measures, improving the results evaluation mechanism through embedding domain knowledge and expert guidance.

2.3.6 Enhancing Knowledge Actionability

Patterns which are interesting to data miners may not necessarily lead to business benefits if deployed. For instance, a large number of association rules are often found, while most of them are workable in business. These rules are generic patterns satisfying technical interestingness, while they are not measured and evaluated in the business sense.

In traditional data mining, or when data mining methods are used in applications, a common scenario is that many mined patterns are more interesting to data miners than to business people. Business people have difficulties in understanding and taking them over for one of the following reasons:

- they reflect commonsense in business,
- they are uninterpretable, thus incomprehensible by following general business thinking,

- they are too many and are indistinguishable, not indicating which are truly useful and more important to business,
- their significance is not justifiable from a business perspective, and
- their presentation is usually far from that of the legacy or existing business operation systems, and thus they cannot be integrated into the systems directly or there is no advice on how they are to be combined into the business working system.

Any findings falling into one of the above scenarios would have difficulty in supporting business decision-making.

To boost the actionable capability of identified patterns, techniques for further actionability enhancement are necessary for generating actionable patterns useful to business. This may be conducted from several perspectives.

First, the appropriate measurement of pattern actionability needs to be defined. Technical and business interestingness measures should be defined and satisfied from both objective and subjective perspectives. For those generic patterns identified based on technical measures only, business interest and performance need to be further checked so that business requirements and user preferences can be given proper consideration.

Second, actionable patterns in many cases can be created through rule reduction, model refinement or parameter tuning by optimizing and filtering generic patterns. If this is the case, actionable patterns are a revised optimal version of generic patterns, which capture deeper characteristics and understanding of the business, consequently present as in-depth or optimized patterns.

Third, pattern actionability is also reflected in the delivery manner. Some forms of patterns are easily understandable by business people, while others are elusive. To support business operation, it is necessary for the deliverables to be converted into forms that can be easily or even seamlessly fed into business rules, processes and operation systems. For this, one option is to convert patterns into business rules following the business terminology system and the existing business rule specifications.

Finally, more direct efforts are necessary to enhance the KDD modeling and mining process targeting the output of actionable knowledge directly discovered from data set by considering the ubiquitous intelligence surrounding the problem.

2.3.7 Reference Model

Reference models such as those in CRISP-DM [2] are very helpful for guiding and managing the knowledge discovery process. It is recommended that these reference models be respected in domain driven actionable knowledge delivery. However, actions and entities for domain driven data mining, such as considering constraints, and integrating domain knowledge, should be paid more attention in the corresponding modeling and procedures.

[2] http://www.crisp-dm.org/

On the other hand, new reference models are essential for supporting components such as in-depth modeling and actionability enhancement.

For instance, Fig. 2.2 illustrates the reference model for actionability enhancement in domain driven data mining.

2.3.8 Qualitative Research

In developing real-world data mining applications, qualitative research methodology [88] is very helpful for capturing business requirements, constraints, requests from organization and management, risk and contingency plans, expected representation of the deliverables, etc. For instance, a questionnaire can assist with uncovering human concerns and business specific requests.

During the process of conducting real-world data mining, a questionnaire can be used to collect feedback on business requirements, constraints, interest, expectations, and requests from domain experts and business users. The information collected may involve aspects, factors, elements, measures and critical values about the organization, operation, management, business processes, workflow, business logics, existing business rules, risk and contingency plans, expected representation of the deliverables, and so on.

It is recommended that questionnaires be designed for every procedure in the domain driven actionable knowledge delivery process. Analytical and contingency reports are then developed for every procedure. Follow-up interviews and discussions may be necessary. Data and records must be collected, analyzed, clarified and finally documented in a knowledge management system as the evidence and guidance for business and data understanding, feature selection, parameter tuning, result evaluation, and deliverable presentation. The findings will also guide the design of user interfaces and user modeling, as well as working mechanisms for involving domain knowledge and experts' roles in the process.

In addition, reference models are helpful for guiding the implementation and for managing the actionable knowledge discovery and delivery process and project. For instance, Fig. 2.2 illustrates the reference model for actionability enhancement.

2.3.9 Closed-Loop and Iterative Refinement

Actionable knowledge discovery in a constraint-based context is more likely to be a closed-loop rather than open process. A closed-loop process indicates that the outputs of data mining are fed back to change relevant parameters or factors in particular stages. The feedback and change effect may be embodied through analyzing and adjusting the relationships between outputs and particular parameters and factors, and eventually tuning the parameters and factors accordingly.

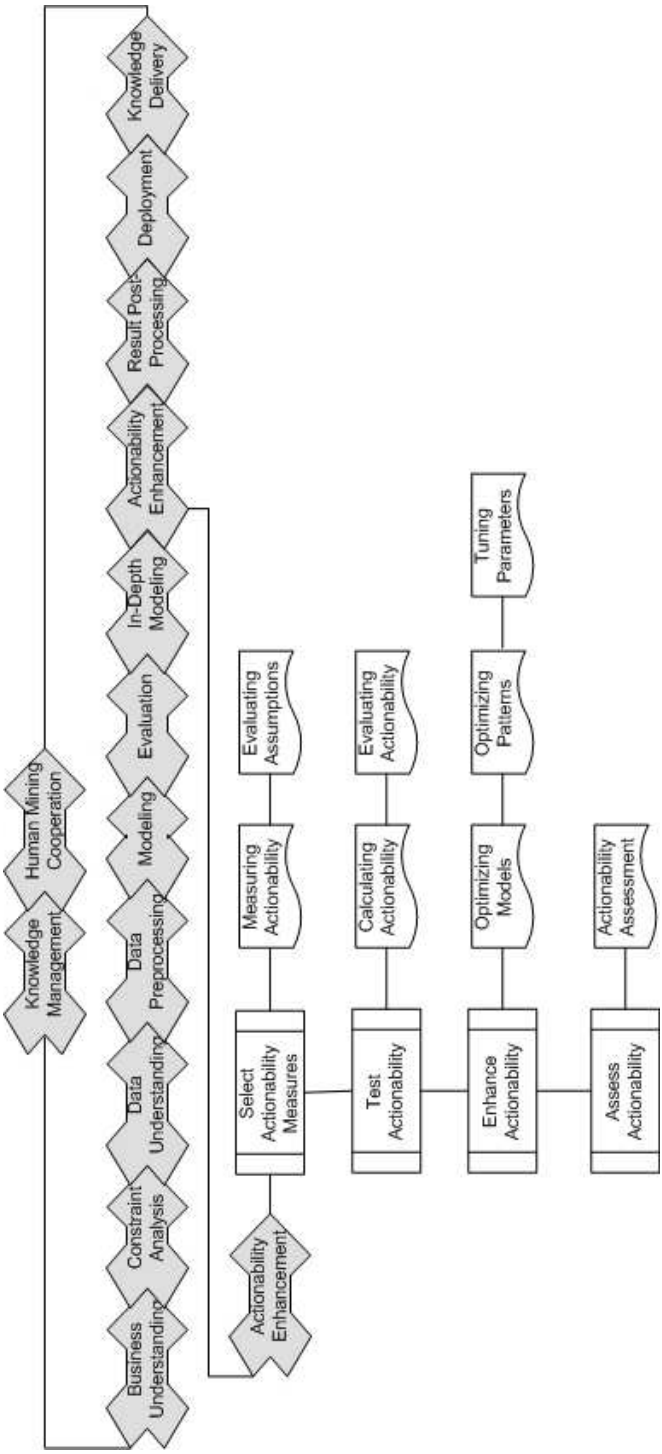

Fig. 2.2 Knowledge actionability enhancement in D^3M

The real-world data mining process is likely iterative because the evaluation and refinement of features, models and outcomes cannot be completed in a one-off way. Iterative interaction may be conducted to varying stages such as sampling, hypothesis, feature selection, modeling, evaluation and/or interpretation before reaching the final stage of knowledge and decision-support report delivery.

Consequently, real-world data mining cannot be undertaken just using an algorithm. Rather, It is necessary to build a proper data mining infrastructure to discover actionable knowledge from constraint-based scenarios in a closed-loop iterative manner. To this end, an agent-based data mining infrastructure [23, 49] can provide good facilities for interaction and message passing amongst different modules, and support user modeling and user agent interaction toward autonomous, semi-autonomous or human-mining-cooperated problem-solving.

2.4 D^3M Methodological Framework

2.4.1 Theoretical Underpinnings

Research and development, and the effective use of D^3M involves multiple disciplines. Its theoretical underpinnings involve analytical, computational and social sciences. We interpret the theoretical infrastructure for D^3M from the perspectives of methodological support, fundamental technologies, and supporting techniques and tools.

From the methodological support perspective, D^3M needs the support of multiple fields, including the information sciences, intelligence sciences, system sciences, cognitive sciences, organizational sciences, and social sciences. *Information and intelligence sciences* provide support for intelligent information processing and systems. *System sciences* furnish methodologies and techniques for domain factor modeling and simulation, closed-loop system design and analysis, and feedback mechanism design. *Cognitive sciences* incorporate principles and methods for understanding human qualitative intelligence such as imaginary thinking, empirical knowledge, belief and intention which is important for understanding and analyzing complex domain problems. *Social sciences* supply foundations for conceiving organizational and social factors and business processes surrounding problem domain.

In particular, we highlight the need to involve a few new scientific fields: data sciences, knowledge sciences, web and network sciences, service sciences, and complexity sciences. We need them because they are critical for handling the increasingly emergent issues and complexities, as well as increasingly for the breadth and depth of their involvement in our business, data and environment.

- Data sciences, on top of the current efforts on data engineering, offer a systematic and fundamental understanding and exploration of the ever-increasing data, which certainly forms one of the essential foundations for deep data understanding, exploration and analysis in D^3M;

- Knowledge sciences, on top of the current efforts on knowledge engineering, entails the systematic and fundamental understanding and exploration of the ever-increasing atock of knowledge, from both prior, empirical and human knowledge, to emerging knowledge from discovery, interaction and computing; it certainly forms one of the essential foundations for knowledge representation, transformation, reasoning, emergence, transferring and use in D^3M;
- Web sciences and network sciences, as an attempt to understand and explore the ever-growing phenomenon of the World Wide Web and increasingly emerging networks, contribute to the understanding, identification, facilitation and involvement of networks and networking in D^3M.
- Service sciences, which are a melding of technologies for an understanding of business processes and organizations, and services systems, contribute to the infrastructure establishment and knowledge delivery from data mining systems to business operations.
- Complexity sciences is a discipline studying complex systems, which can provide methodologies and techniques for involving and managing surrounding factors in understanding and analyzing complex data.

In addition, areas and knowledge bodies such as optimization theory, risk analysis, economics and finance are also important for understanding and measuring business impact and the interestingness of identified patterns.

Besides the mainstream KDD techniques, fundamental technologies needed also involve user modeling, formal methods, logics, representation, knowledge engineering, ontological engineering, semantic web, and cognitive engineering. The modeling of pattern impact and business interestingness may refer to the relevant technologies such as statistical significance, impact analysis, benefit-cost analysis, risk management and analysis, and performance measurements in economics and finance. To understand domain-specific ubiquitous intelligence, and the evolution of user modeling and group thinking, we refer to techniques and tools in fields like systems simulation, communication, artificial social system, open complex systems, swarm intelligence, social network analysis, reasoning and learning. The deliverable presentation may involve means in knowledge representation, business rule presentation, visualization and graph theory.

2.4.2 Process Model

The existing data mining methodology, for instance CRISP-DM, generally supports autonomous pattern discovery from data. By contrast, the idea of domain driven knowledge discovery is to involve ubiquitous intelligence into data mining. The D^3M highlights a process that discovers in-depth patterns from a constraint-based environment with the involvement of domain experts and their knowledge. Its objective is to maximally accommodate both naive users as well as experienced analysts, and to satisfy business goals. The patterns discovered are expected to be integrated into business systems and to be aligned with existing business rules. To make do-

main driven data mining effective, user guides and intelligent human-machine interaction interfaces are essential through incorporating both human qualitative intelligence and machine quantitative intelligence. In addition, appropriate mechanisms are required for dealing with multiform constraints and domain knowledge.

The main functional components of the D^3M process model are shown in Fig. 2.3, in which we highlight those processes specific to D^3M in thick boxes. The lifecycle of the D^3M process is as follows, but be aware that the sequence is not rigid, some phases may be bypassed or moved back and forth in dealing with a real problem. Every step of the D^3M process may involve ubiquitous intelligence and interaction with business users and/or domain experts.

P_1 . Problem understanding (to identify and define the problems, including its scope and challenges etc.);
P_2 . Constraints analysis (to identify constraints surround the above problems, from data, domain, interestingness and delivery perspectives);
P_3 . Definition of analytical objectives, and feature construction (to define the goal of data mining, and accordingly features selected or constructed to achieve the objectives);
P_4 . Data preprocessing (data extraction, transformation and loading, in particular, the data preparation such as processing missing and privacy data);
P_5 . Method selection and modeling (to select the appropriate models and methods for achieving the above objectives); or
P_5' . In-depth modeling (to apply deep modeling either by using more effective models disclosing the very core of the problem, or by using multi-step mining or combined mining);
P_6 . Initial generic results analysis and evaluation (to analyze/assess the initial findings);
P_7 . It is quite possible that each phase from P_1 may be iteratively reviewed through analyzing constraints and interaction with domain experts in a back-and-forth manner; or
P_7' . In-depth mining on the initial generic results where applicable;
P_8 . Actionability measurement and enhancement (to check the interestingness from both technical and business perspectives, and to enhance the performance by applying more effective methods etc.);
P_9 . Back and forth between P_7 and P_8;
P_{10}. Results post-processing (to post-analyse or post-mine the initial resulting data);
P_{11}. Reviewing phases from P_1 may be required;
P_{12}. Deployment (to deploy the results into business lines);
P_{13}. Knowledge delivery and report synthesis for smart decision making (to synthesize the eventual findings into decision-making report to be delivered to business people).

The D^3M process highlights the following aspects that are critical for the success of a data mining in the real world. They are

- context and environment (including the factors from data, domain, organizational social aspects),

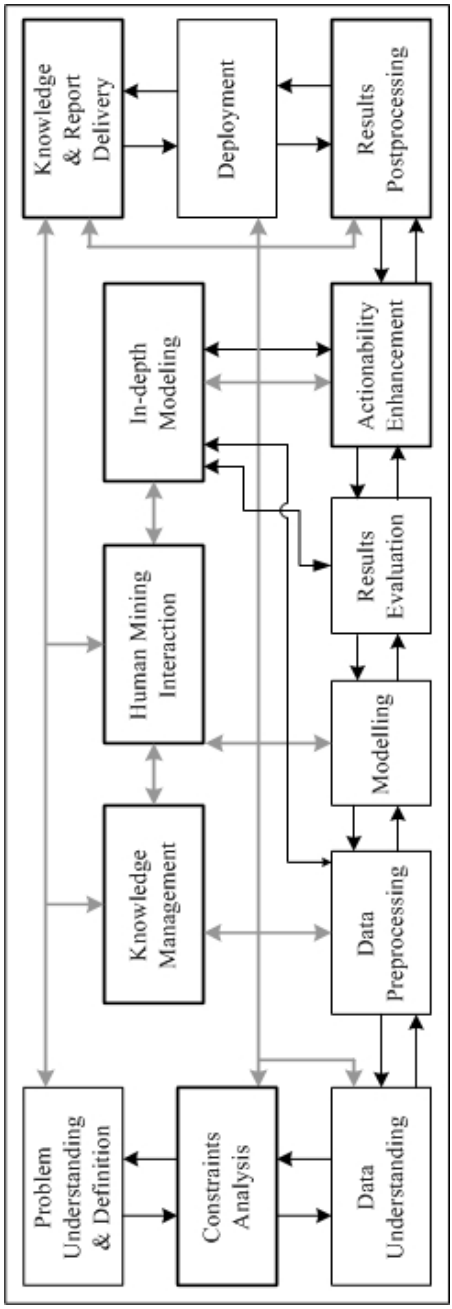

Fig. 2.3 D^3M process model

- constraints (including constraints from data, domain, interestingness and deliverable perspectives),
- domain knowledge (including domain expert knowledge and knowledge from business systems),
- organizational and social factors (including factors from aspects such as organizational rules, relationships, social networks),
- human qualitative intelligence and roles (including human empirical knowledge, imaginary thinking etc.),
- user preferences (including user expectations and needs),
- interaction and interfaces (including the tools and interfaces for interaction between a user and data mining systems),
- cooperation between humans and the data mining system (reflecting the allocation of tasks between human and a data mining system),
- in-depth pattern mining (to discovery deep patterns reflecting geniune interest of business owners and decision-makers),
- parallel support (including the support for mining patterns in parallel),
- business impact and interestingness (reflecting business concerns from objective and subjective perspectives),
- knowledge actionability (including concerns from technical and business aspects),
- feedback (from business modelers, as well as business owners and decision-making), and
- iterative refinement (as needed, back to corresponding steps).

These aspects are consistent with the key components in D^3M, as we discussed in Section 2.3.

2.4.3 D^3M Evaluation System

The D^3M evaluation system caters for significance and interestingness ($Int(p)$) of a pattern (p) from both technical and business perspectives. $Int(p)$ is measured in terms of *technical interestingness* ($t_i(p)$) and *business interestingness* ($b_i(p)$) [41].

$$Int(p) = I(t_i(p), b_i(p)) \tag{2.1}$$

where $I(.)$ is the function for aggregating the contributions of all particular aspects of interestingness.

Further, $Int(p)$ is described in terms of *objective* (o) and *subjective* (s) factors from both *technical* (t) and *business* (b) perspectives.

$$Int(p) = I(t_o(), t_s(), b_o(), b_s()) \tag{2.2}$$

where $t_o()$ is objective technical interestingness, $t_s()$ is subjective technical interestingness, $b_o()$ is objective business interestingness, and $b_s()$ is subjective business interestingness.

We say p is truly *actionable* (i.e., $\widetilde{p}$) to both academia and business if it satisfies the following condition:

$$Int(p) = t_o(\mathbf{x}, \widetilde{p}) \wedge t_s(\mathbf{x}, \widetilde{p}) \wedge b_o(\mathbf{x}, \widetilde{p}) \wedge b_s(\mathbf{x}, \widetilde{p}) \tag{2.3}$$

where '$\wedge$' indicates the interestingness 'aggregation'.

In general, $t_o()$, $t_s()$, $b_o()$ and $b_s()$ of practical applications can be regarded as independent of each other. With their normalization (expressed by ˆ), we can get:

$$\begin{aligned} Int(p) &\rightarrow \hat{I}(\hat{t_o}(), \hat{t_s}(), \hat{b_o}(), \hat{b_s}()) \\ &= \alpha\hat{t_o}() + \beta\hat{t_s}() + \gamma\hat{b_o}() + \delta\hat{b_s}() \end{aligned} \tag{2.4}$$

The AKD optimization problem in D^3M can be expressed as follows:

$$\begin{aligned} AKD^{e,\tau,m\in M} &\longrightarrow O_{p\in P}(Int(p)) \\ &\rightarrow O(\alpha\hat{t_o}()) + O(\beta\hat{t_s}()) + \\ &\quad O(\gamma\hat{b_o}()) + O(\delta\hat{b_s}()) \end{aligned} \tag{2.5}$$

The *actionability* of a pattern p is measured by $act(p)$:

$$\begin{aligned} act(p) &= O_{p\in P}(Int(p)) \\ &\rightarrow O(\alpha\hat{t_o}(p)) + O(\beta\hat{t_s}(p)) + \\ &\quad O(\gamma\hat{b_o}(p)) + O(\delta\hat{b_s}(p)) \\ &\rightarrow t_o^{act} + t_s^{act} + b_o^{act} + b_s^{act} \\ &\rightarrow t_i^{act} + b_i^{act} \end{aligned} \tag{2.6}$$

where t_o^{act}, t_s^{act}, b_o^{act} and b_s^{act} measure the respective actionable performance in terms of each aspect.

Due to the inconsistency often existing at different aspects, we often find that the identified patterns only fit in one of the following sub-sets:

$$\begin{aligned} Int(p) \rightarrow \{&\{t_i^{act}, b_i^{act}\}, \{\neg t_i^{act}, b_i^{act}\}, \\ &\{t_i^{act}, \neg b_i^{act}\}, \{\neg t_i^{act}, \neg b_i^{act}\}\} \end{aligned} \tag{2.7}$$

where '$\neg$' indicates the corresponding element is not satisfactory. Ideally, we look for actionable patterns p that can satisfy the following condition:

IF

$$\begin{aligned} \forall p \in \widetilde{P}, \exists \mathbf{x} : t_o(\mathbf{x}, p) \wedge t_s(\mathbf{x}, p) \wedge b_o(\mathbf{x}, p) \\ \wedge b_s(\mathbf{x}, p) \rightarrow act(p) \end{aligned} \tag{2.8}$$

THEN:

$$p \rightarrow \widetilde{p}. \tag{2.9}$$

In the real-world data mining, it is often very challenging to find the most actionable patterns that are associated with both 'optimal' t_i^{act} and 'optimal' b_i^{act}. Clearly, D^3M favors patterns confirming the relationship $\{t_i^{act}, b_i^{act}\}$. There is a need to deal with possible conflict and uncertainty amongst respective interestingness elements. Technically, there is an opportunity to develop techniques to balance and combine all types of interestingness metrics to generate uniform, balanced and interpretable mechanisms for measuring knowledge deliverability. Under sophisticated situations, domain experts from both computation and business areas need to interact with each other, ideally through an m-space with intelligence meta-synthesis facilities such as letting one run models with quantitative outcomes to support discussions with other experts. If $t_i()$ and $b_i()$ are inconsistent, experts argue and compromise with each other through m-interactions in the m-space, like what happens in a board meeting, but with substantial online resources, models and services.

2.4.4 D^3M Delivery System

Well experienced data mining professionals attribute the weak executable capability of existing data mining findings to the lack of proper tools and mechanisms for implementing the deployment of the resulting models and algorithms ideally by business users rather than analysts. In fact, the barrier and gap comes from the weak, if not none, capability of existing data mining deployment systems, existing in presentation, deliverable and execution aspects. They form the D^3M delivery system, which is much beyond the identified patterns and models themselves.

- Presentation: studies how to present data mining findings that can be easily recognized, interpreted and taken over as they need;
- Deliverable: studies how to deliver data mining findings and systems to business users so that the findings are handy to be re-formated, transformed, or cut and pasted into their own business systems and presentation on demand, and the systems can be understood and taken over by end users; and
- Execution: studies how to integrate data mining findings and systems into production systems, and how the findings to be executed easily and seamlessly in an operational environment.

Supporting techniques need to be developed for AKD presentation, deliverable and execution. For instance, the following lists some of techniques.

- Presentation: typical tools such as visualization techniques are essentially helpful, visual mining could support the whole data mining process in a visual manner;
- Deliverable: business rules are widely used in business organizations, one of the methods for delivering patterns is to convert them into business rules; for this

we can develop a tool with underlying ontologies and semantics to support the transfer from pattern to business rules;
- Execution: tools to make deliverables executable in an organization's environment need to be developed, one of the efforts is to generate PMML to convert models to executables so that the models can be integrated into production systems, and run on a regular basis to provide cases for business management.

2.5 Summary

This chapter has presented the basic concept and overall picture of domain driven data mining methodology. We have discussed the concept map of D^3M, key methodological components consisting of D^3M, and its theoretical underpinnings.

Conclusions from this chapter consist of:

- Driven by challenges and complexities from specific domain problems, domain driven data mining provides a systematic solution and guideline from identifying fundamental research issues, to developing corresponding techniques and tools;
- Major methodological components of D^3M reflect the corresponding problem-solving solutions to tackle key challenges and issues existing in traditional data mining;
- The identified key components within D^3M exhibit tremendous new opportunities for us to explore in the data mining area, by engaging knowledge and lessons from many other disciplines, including traditional ones such as information sciences, as well as new fields such as data sciences, web sciences, service sciences, knowledge sciences and complexity sciences;
- It is far from mature as a new research area, and there are many great opportunities and prospects for us to further investigate in domain driven data mining.

In Chapter 3, we specifically explore ubiquitous intelligence surrounding and contributing to domain driven data mining.

Goal Generation from Possibilistic Beliefs Based on Trust and Distrust

Célia da Costa Pereira and Andrea G.B. Tettamanzi

Università degli Studi di Milano
Dipartimento di Tecnologie dell'Informazione
via Bramante 65, I-26013 Crema, Italy
{celia.pereira,andrea.tettamanzi}@unimi.it

Abstract. The extent to which a rational agent changes its beliefs may depend on several factors like the trustworthiness of the source of new information, the agent's competence in judging the truth of new information, the mental spirit of the agent (optimistic, pessimistic, pragmatic, etc), the agent's attitude towards information coming from unknown sources, or sources the agent knows as being malicious, or sources the agent knows as providers of usually correct information, and so on.

We propose and discuss three different agent's belief behaviors to be used in a goal (desire) generation and adoption framework. The originality of the proposals is that the trustworthiness of a source depends not only on the degree of trust but also on an *independent* degree of distrust. Explicitly taking distrust into account allows us to mark a clear difference between the distinct notions of *negative* trust and *insufficient* trust. More precisely, it is possible, unlike in approaches where only trust is accounted for, to "weigh" differently information from *helpful*, *malicious*, *unknown*, or *neutral* sources.

1 Introduction and Motivation

The goals to be adopted by a BDI agent [22] in a given situation may depend on the agent's beliefs, desires, and obligations [23]. Most of the existing works on goal generation, like for example the one proposed in [11], consider the notion of belief as an all-or-nothing concept: either the agent believes something, or it does not. However, as pointed out by Hansson in [16] for example, believing is a matter of degree. He underlines two notions of degree of belief. One is the static concept of degree of confidence. In this sense, the more an agent's degree of belief in a sentence is higher, the more confidently it entertains that belief. The other notion is the dynamic concept of degree of resistance to change. In that sense, the higher an agent's degree of belief in a sentence is, the more difficult it is to change that belief.

In classical logic, beliefs are considered to be certainly true, and the negation of beliefs to be certainly false. This assumption does not cover the intermediate cases pointed out by Hansson, that is, the cases in which beliefs are neither fully believed nor fully disbelieved. In the possibility theory setting [12, 13], the notion of graded belief is captured in terms of two measures: necessity and possibility. A first degree of belief in a proposition, computed thanks to the necessity measure, is valued on a

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unipolar scale, where 0 means an absence of belief rather than believing the opposite (Paul does not believe p does not mean that Paul believes $\neg p$). A second degree, valued on a different unipolar scale and attached to propositions, expresses plausibility and is computed thanks to the possibility measure. If the plausibility of a proposition is 0, it means certainty of falseness (If Paul thinks that it is fully impossible that it will rain tomorrow, it means that Paul is certain that tomorow it will not rain), whereas 1 just reflects possibility, not certainty (If Paul thinks that it is fully possible that it will rain tomorrow, it does not mean that Paul is certain that tomorow it will rain). Thanks to the duality between necessity and possibility measure, the set of disbelieved proposition can then be inferred from the set of believed ones.

Recently, we have proposed [7] a belief change operator for goal generation in line with Hansson's considerations. In that approach, the sources of information may also be partially trusted and, as a consequence, information coming from such sources may be partially believed.

The main lack in that approach is the way the concept of *distrust* is implicitly considered, that is, as the complement of *trust* ($trust = 1 - distrust$). However, trust and distrust may derive from different kinds of information (or from different sides of the personality) and, therefore, can coexist without being complementary [15, 9, 20]. For instance, one may not trust a source because of lack of positive evidence, but this does not necessarily mean (s)he distrusts it. Distrust can play an important role in an agent's goal generation reasoning complementing trust. In particular, in a framework of goal generation based on beliefs and desires[1], taking distrust explicitly into account allows an agent, e.g., to avoid dropping a goal just because favorable information comes from an unknown source (neither trusted nor distrusted) — the absence of trust does not erroneously mean full distrust.

We propose a way to take these facts into consideration by using possibility theory for representing degrees of beliefs and by concentrating on the influence of new information in the agent's beliefs. The latter point supposes that a source may be trusted and/or distrusted to a certain extent. This means that we explicitly consider not only the trust degree in the source but also the distrust degree. To this aim, the trustworthiness of a source is represented as a (trust, distrust) pair, and intuitionistic fuzzy logic [1] is used to represent the uncertainty on the trust degree introduced by the explicit presence of distrust. On that basis, we propose three belief change operators to model the attitude of an agent towards information from trusted, malicious, neutral (trusted and malicious to same extent) or unknown sources. Besides, such operators allow us to account for the fact that the acceptance or rejection of new information depends on several factors like the agent's competence in judging the truth of incoming information, the agent's state of spirit, the correctness of information provided by the source in the past, and so on.

The paper is organized as follows: Section 2 provides minimal background on fuzzy sets, possibility theory, and intuitionistic fuzzy logic; Section 3 motivates and discusses a bipolar view of trust and distrust; Section 4 introduces an abstract beliefs-desires-goals agent model; Section 5 presents a trust-based belief change operator adapted from previous work and proposes three extensions thereof for dealing with explicitly given trust and distrust; Section 6 describes the goal generation process; and Section 7 concludes.

[1] Here, for sake of simplicity, we will not consider intentions.

2 Basic Considerations

2.1 Fuzzy Sets

Fuzzy sets [26] allow the representation of imprecise information. Information is imprecise when the value of the variable to which it refers cannot be completely determined within a given universe of discourse. For example, among the existing fruits, it is easy to define the set of apples. Instead, it is not so easy to define in a clear cut-way the set of ripe apples because ripeness is a gradual notion. A fuzzy set is appropriate to represent this kind of situation.

Fuzzy sets are a generalization of classical sets obtained by replacing the characteristic function of a set A, χ_A, which takes up values in $\{0, 1\}$ ($\chi_A(x) = 1$ iff $x \in A$, $\chi_A(x) = 0$ otherwise) with a *membership function* μ_A, which can take up any value in $[0, 1]$. The value $\mu_A(x)$ or, more simply, $A(x)$ is the membership degree of element x in A, i.e., the degree to which x belongs in A. A fuzzy set is then completely defined by its membership function.

2.2 Possibility Theory and Possibility Distribution

The membership function of a fuzzy set describes the more or less possible and mutually exclusive values of one (or more) variable(s). Such a function can then be seen as a possibility distribution [27]. Indeed, if F designates the fuzzy set of possible values of a variable X, $\pi_X = \mu_F$ is called the possibility distribution associated to X. The identity $\mu_F(u) = \pi_X(u)$ means that the membership degree of u to F is equal to the possibility degree of X being equal to u when all we know about X is that its value is in F. A possibility distribution for which there exists a completely possible value ($\exists u_0; \pi(u_0) = 1$) is said to be *normalized*.

Possibility and Necessity Measures. A possibility distribution π induces a *possibility measure* and its dual *necessity measure*, denoted by Π and N respectively. Both measures apply to a crisp set A and are defined as follows:

$$\Pi(A) \equiv \sup_{s \in A} \pi(s); \tag{1}$$

$$N(A) \equiv 1 - \pi(\bar{A}) = \inf_{s \in \bar{A}} \{1 - \Pi(s)\}. \tag{2}$$

In words, the possibility measure of set A corresponds to the greatest of the possibilities associated to its elements; conversely, the necessity measure of A is equivalent to the impossibility of its complement $\bar{A}$.

A few properties of possibility and necessity measures induced by a normalized possibility distribution on a finite universe of discourse U are the following, for all subsets $A, B \subseteq U$:

1. $\Pi(A \cup B) = \max\{\Pi(A), \Pi(B)\}$;
2. $\Pi(\emptyset) = 0, \Pi(U) = 1$;
3. $N(A \cap B) = \min\{N(A), N(B)\}$;
4. $N(\emptyset) = 0, N(U) = 1$;

5. $\Pi(A) = 1 - N(\bar{A})$ (duality);
6. $\Pi(A) \geq N(A)$;
7. $N(A) > 0$ implies $\Pi(A) = 1$;
8. $\Pi(A) < 1$ implies $N(A) = 0$.

An immediate consequence of these properties is that either $\Pi(A) = 1$ or $\Pi(\bar{A}) = 1$. Both a set A and its complement having a possibility of 1 is the case of complete ignorance on A.

2.3 Intuitionistic Fuzzy Logic

Fuzzy set theory has been extended to intuitionistic fuzzy set (IFS for short) theory [1]. In fuzzy set theory, it is implicitly assumed that the fact that an element x "belongs" with a degree $\mu_A(x)$ in a fuzzy set A implies that x should "not belong" to A to the extent $1 - \mu_A(x)$. An intuitionistic fuzzy set S, instead, explicitly assigns to each element x of the considered universe of discourse both a degree of membership $\mu_S(x) \in [0, 1]$ and one of non-membership $\nu_S(x) \in [0, 1]$ which are such that $\mu_S(x) + \nu_S(x) \leq 1$. Obviously, when $\mu_S(x) + \nu_S(x) = 1$ for all the elements of the universe, the traditional fuzzy set concept is recovered.

Deschrijver and Kerr showed [10] that IFS theory is formally equivalent to interval-valued fuzzy set (IVFS) theory, which is another extension of fuzzy set theory in which the membership degrees are subintervals of $[0, 1]$ [24]. The IFS pair $(\mu_S(x), \nu_S(x))$ corresponds to the IVFS interval $[\mu_S(x), 1 - \nu_S(x)]$, indicating that the degree to which x "belongs" in S can range from $\mu_S(x)$ to $1 - \nu_S(x)$. The same authors define the *hesitation degree*, $h \in [0, 1]$, as the length of such interval, $h = 1 - \mu_S(x) - \nu_S(x)$. Hesitation h represents the uncertainty about the actual membership degree of x.

IFS is suitable to representing gender, for example. Indeed, according to the Intersex Society of North America [21], approximately 1 in 2000 children are born with a condition of "ambiguous" external genitalia – it is not clear if they are female or male. Let M be a fuzzy set representing males, and F be a fuzzy set representing females. A newborn x can be identified as a male if $\mu_M(x) = 1$, $\nu_M(x) = 0$, and $h = 0$; or as a female if $\mu_F(x) = 1$, $\nu_F(x) = 0$, and $h = 0$; or as "ambiguous" if $\mu_M(x)(\mu_F(x)) = \alpha$, $\nu_M(x)(\nu_F(x)) = \beta$, and $h = 1 - \alpha - \beta > 0$.

3 Representing Trust and Distrust

Most existing computational models usually deal with trust in a binary way: they assume that a source is to be trusted or not, and they compute the probability that the source can be trusted. However, sources can not always be divided into trustworthy and untrustworthy in a clear-cut way. Some sources may be trusted to a certain extent. To take this fact into account, we represent trust and distrust as fuzzy degrees. A direct consequence of this choice is that facts may be believed to a degree and desires and goals may be adopted to a given extent.

It must be stressed that our aim is not to compute degrees of trust and distrust of sources; like in [18], we are just interested in how these degrees influence the agent's beliefs, and, by way of a deliberative process, desires and goals.

Approaches to the problem of deriving/assigning degrees of trust (and distrust) to information sources can be found, for example, in [5], [8], [2].

We propose to define the *trustworthiness* score of a source for an agent as follows:

Definition 1 (Trustworthiness of a Source). *Let $\tau_s \in [0,1]$ be the degree to which an agent trusts source s, and $\delta_s \in [0,1]$ the degree to which it distrusts s, with $\tau_s + \delta_s \leq 1$. The trustworthiness score of s for the agent is the pair (τ_s, δ_s).*

Following Deschrijver and Kerr's viewpoint, the trustworthiness (τ, δ) of a source corresponds to the interval $[\tau, 1-\delta]$, indicating that the trust degree can range from τ to $1-\delta$. Therefore, the hesitation degree $h = 1 - \tau - \delta$ represents the uncertainty, or doubt, about the actual trust value. E.g., if a source has trustworthiness $(0.2, 0)$, this means that the agent trusts the source to degree 0.2, but possibly more, because there is much room for doubt ($h = 0.8$). More precisely, it means that the agent may trust the source to a degree varying from 0.2 to 1. Instead, if the trustworthiness is $(0.6, 0.4)$, the agent trusts the source to degree 0.6 but not more ($h = 0$).

Thanks to these considerations, we can represent the trustworthiness score of a source more faithfully than in existing approaches. In particular, we can explicitly represent the following special cases:

$(0,1)$: the agent has reasons to fully distrust the source, hence it has no hesitation ($h = 0$);
$(0,0)$: the agent has no information about the source and hence no reason to trust the source, but also no reason to distrust it; therefore, it fully hesitates in trusting it ($h = 1$);
$(1,0)$: the agent has reasons to fully trust the source, hence it has no hesitation ($h = 0$).

As we can see, by considering both the (not necessarily related) concepts of trust and distrust, it is possible to differentiate between absence of trust caused by presence of distrust (e.g., information provided by a malicious source) versus by lack of knowledge (e.g., as towards an unknown source).

Let us consider the following example. Paul's son needs urgent treatment. John, Robert and Jimmy claim they can treat Paul's son. In Situation 1 (S_1), John is the first who meets Paul and his son. In Situation 2 (S_2), the first to meet them is Robert; and in Situation 3 (S_3), the first to meet them is Jimmy.

S_1. Paul has reason to think that John is an excellent physician, therefore he trusts John;
S_2. Paul knows that Robert is an ex-physician. Paul has motivations to think that the reason why Robert does not practice medicine anymore is he failed several times when treating people in a situation similar to that of his son. In this case, the rational behavior would be to distrust Robert;
S_3. Paul does not know Jimmy who introduces himself as a physician. In this case trusting or distrusting Jimmy depends on Paul's internal factors like his competence in judging the truth of incoming information, his state of spirit (optimistic, pessimistic, or pragmatic), etc. For example, if Paul is wary and pessimistic, he will distrust Jimmy; instead, if Paul is optimistic and does not perceive malicious purposes from somebody he does not know, he will accept Jimmy's help.

We can notice that in S_3, Paul has lack of positive and negative evidence. The fact is that Paul neither trusts nor distrusts Jimmy. He just does not know him.

This case shows that trust is not always the complement of distrust. Indeed, here we have $trust\ degree = 0$; $distrust\ degree = 0$, and we do not have $trust \neq 1 - distrust$. In this particular case the doubt (hesitation degree) is maximal, i.e., 1. It means that Paul behavior can depend on uncertain trust values going from completely trusting Jimmy to completely distrusting Jimmy.

We can distinguish information sources thanks to the extent of negative and positive evidence we dispose of for each source. Let (τ, δ) be the trustworthiness of a source for an agent.

Definition 2. *A source is said to be* helpful *if* $\tau > \delta$, malicious *if* $\tau < \delta$, neutral *if* $\tau = \delta \neq 0$. *When* $\tau = \delta = 0$, *the source is said to be* unknown.

Definition 3 (Comparing Sources). *Let* s_1 *and* s_2 *be two sources with trustworthiness scores of* (τ_1, δ_1) *and* (τ_2, δ_2) *respectively. Source* s_1 *is said to be more trustworthy than source* s_2, *noted* $s_1 \succeq s_2$, *if and only if* $\tau_1 \geq \tau_2$ *and* $\delta_1 \leq \delta_2$.

This is a partial order in the sense that it is not always possible to compare two trustworthiness scores.

Working Example. John thinks his house has become too small for his growing family and would like to buy a larger one. Of course, John wants to spend as little money as possible. A friend who works in the real estate industry tells John prices are poised to go down. Then John reads in a newspaper that the real estate market is weak and prices are expected to go down. Therefore, John's desire is to wait for prices to lower before buying. However, John later meets a real estate agent who has an interesting house on sale, and the agent tells him to hurry up, because prices are soaring. On the way home, John hears a guy on the bus saying his cousin told him prices of real estate are going up.

To sum up, John got information from four sources with different scores. The first source is friendly and competent; therefore, its score is $(1, 0)$. The second is supposedly competent and hopefully independent: therefore, its score might be something like $(\frac{1}{2}, \frac{1}{4})$. The third source is unknown, but has an obvious conflict of interest; therefore John assigns it a score of $(0, 1)$. Finally, the guy on the bus is a complete stranger reporting the opinion of another complete stranger. Therefore, its score cannot be other than $(0, 0)$.

We all have a basic intuition, suggested by common sense, of how information from these sources should be accounted for to generate a goal (buy now vs. wait for prices to go down) and planning actions accordingly. Below, we provide a formalization of the kinds of deliberations a rational agent is expected to make in order to deal with information scored with trust and distrust degrees.

4 Representing Graded Beliefs, Desires and Goals

An agent's belief is a piece of information that the agent believes in. An agents's desire is something (not always material) that the agent would like to possess or perform.

Here, we do not distinguish between what is positively desired and what is not rejected by explicitly considering positive and negative desires like it is done by Casali and colleagues [3]. This is an interesting point that we will consider in a future work. Nevertheless, our "desires" can be regarded as positive desires.

Desires (or motivations) are necessary but not sufficient conditions for action. When a desire is met by other conditions that make it possible for an agent to act, that desire becomes a *goal*. Therefore, given this technical definition of a desire, all goals are desires, but not all desires are goals. The main distinction we made here between desires and goals is in line with the one made by Thomason [23] and other authors: goals are required to be consistent whereas desires need not be.

4.1 Basic Notions

In this section, we present the main aspects of the adopted formalism.

Definition 4 (Language). *Let $\mathcal{A}$ be a set of atomic propositions and let $\mathcal{L}$ be the propositional language such that $\mathcal{A} \cup \{\top, \bot\} \subseteq \mathcal{L}$, and, $\forall \phi, \psi \in \mathcal{L}$, $\neg\phi \in \mathcal{L}$, $\phi \wedge \psi \in \mathcal{L}$, $\phi \vee \psi \in \mathcal{L}$.*

$\Omega = \{0,1\}^{\mathcal{A}}$ is the set of all possible interpretations on $\mathcal{A}$. An interpretation $\mathcal{I} \in \Omega$ is a function $\mathcal{I} : \mathcal{A} \rightarrow \{0,1\}$ assigning a truth value to every atomic proposition and, by extension, to all formulas in $\mathcal{L}$.

Representing Graded Beliefs. As convincingly argued by Dubois and Prade [12, 13], a *belief* can be regarded as a necessity degree. A cognitive state of an agent can then be modeled by a normalized possibility distribution π:

$$\pi : \Omega \rightarrow [0,1]. \tag{3}$$

$\pi(\mathcal{I})$ is the possibility degree of interpretation $\mathcal{I}$. It represents the plausibility order of the possible word situation represented by interpretation $\mathcal{I}$. If the agent deems more plausible the world $\mathcal{I}_1$ than $\mathcal{I}_2$, then $\pi(\mathcal{I}_1) \geq \pi(\mathcal{I}_2)$.
The notation $[\phi]$ denotes the set of all models of a formula $\phi \in \mathcal{L}$:

$$[\phi] = \{\mathcal{I} \in \Omega : \mathcal{I} \models \phi\}.$$

Definition 5 (Possibility of a world situation). *The extent to which the agent considers ϕ as possible, $\Pi([\phi])$, is given by:*

$$\Pi([\phi]) = \max_{\mathcal{I} \in [\phi]} \{\pi(\mathcal{I})\}. \tag{4}$$

A desire-generation rule is defined as follows:

Definition 6 (Desire-Generation Rule). *A desire-generation rule R is an expression of the form $\beta_R, \psi_R \Rightarrow_D^+ d$, where $\beta_R, \psi_R \in \mathcal{L}$, and $d \in \{a, \neg a\}$ with $a \in \mathcal{A}$. The unconditional counterpart of this rule is $\alpha \Rightarrow_D^+ d$, which means that the agent (unconditionally) desires d to degree α.*

Intuitively this means: "an agent desires d as much as it believes β_R and desires ψ_R.

Given a desire-generation rule R, we shall denote $\mathrm{rhs}(R)$ the literal on the right-hand side of R.

Example (continued). John's attitude towards buying a larger house may be described as follows:

$$\begin{aligned} R_1 &: \quad \text{need_larger_house}, \top \Rightarrow_D^+ \text{buy_house}, \\ R_2 &: \neg\text{prices_down}, \text{buy_house} \Rightarrow_D^+ \neg\text{wait}. \\ R_3 &: \quad \text{prices_down}, \text{buy_house} \Rightarrow_D^+ \text{wait}. \end{aligned}$$

4.2 Agent's State

The state of an agent is completely described by a triple $\mathcal{S} = \langle \mathcal{B}, \mathcal{R}_J, \mathcal{J} \rangle$, where

- $\mathcal{B}$ is the agent's belief set induced by a possibility distribution π;
- $\mathcal{R}_J$ is a set of desire-generation rules, such that, for each desire d, $\mathcal{R}_J$ contains at most one rule of the form $\alpha \Rightarrow_D^+ d$;
- $\mathcal{J}$ is a fuzzy set of literals.

$\mathcal{B}$ is the agent's belief set induced by a possibility distribution π (Equation 3); the the degree to which an agent believes ϕ is given by Equation 5. $\mathcal{R}_J$ contains the rules which generate desires from beliefs and other desires (subdesires). $\mathcal{J}$ contains all literals (positive and negative form of atoms in $\mathcal{A}$) representing desires which may be deduced from the agent's desire-generation rules. We suppose that an agent can have inconsistent desires, i.e., for each desire d we can have $\mathcal{J}(d) + \mathcal{J}(\neg d) > 1$.

4.3 Semantics of Belief and Desire Formulas

The semantics of belief and desire formulas in $\mathcal{L}$ are the following.

Definition 7 (Graded belief and desires formulas). *Let $\mathcal{S} = \langle \mathcal{B}, \mathcal{J}, \mathcal{R}_{\mathcal{J}} \rangle$ be the state of the agent, and ϕ be a formula, the degree to which the agent believes ϕ is given by:*

$$\mathcal{B}(\phi) = N([\phi]) = 1 - \Pi([\neg\phi]). \tag{5}$$

Straightforward consequences of the properties of possibility and necessity measures are that $\mathcal{B}(\phi) > 0 \Rightarrow \mathcal{B}(\neg\phi) = 0$ and

$$\mathcal{B}(\top) = 1, \tag{6}$$
$$\mathcal{B}(\bot) = 0, \tag{7}$$
$$\mathcal{B}(\phi \wedge \psi) = \min\{\mathcal{B}(\phi), \mathcal{B}(\psi)\}, \tag{8}$$
$$\mathcal{B}(\phi \vee \psi) \geq \max\{\mathcal{B}(\phi), \mathcal{B}(\psi)\}. \tag{9}$$

If ϕ is a literal, $\mathcal{J}(\phi)$ is directly given by the state of the agent. Instead, the desire degree of non-literal propositions is given by:

$$\mathcal{J}(\neg\phi) = 1 - \mathcal{J}(\phi), \tag{10}$$
$$\mathcal{J}(\phi \wedge \psi) = \min\{\mathcal{J}(\phi), \mathcal{J}(\psi)\}, \tag{11}$$
$$\mathcal{J}(\phi \vee \psi) = \max\{\mathcal{J}(\phi), \mathcal{J}(\psi)\}. \tag{12}$$

Note that since $\mathcal{J}$ needs not be consistent, the De Morgan laws do not hold, in general, for desire formulas.

Definition 8 (Degree of Activation of a Rule). *Let R be a desire-generation rule. The degree af activation of R, $\mathrm{Deg}(R)$, is given by*

$$\mathrm{Deg}(R) = \min(\mathcal{B}(\beta_R), \mathcal{J}(\psi_R))$$

and for its unconditional counterpart $R = \alpha \Rightarrow_D^+ d$: $\mathrm{Deg}(R) = \alpha$.

Definition 9 (Degree of Justification). *The degree of justification of desire d is defined as*

$$\mathcal{J}(d) = \max_{R \in \mathcal{R}_{\mathcal{J}} : \mathrm{rhs}(R) = d} \mathrm{Deg}(R).$$

This represents how rational it is for an agent to desire d.

Example (continued). John's initial state may be described by a possibility distribution π on

$$\Omega = \left\{ \begin{array}{l} \mathcal{I}_0 = \{\text{need_larger_house} \mapsto 0, \text{prices_down} \mapsto 0\}, \\ \mathcal{I}_1 = \{\text{need_larger_house} \mapsto 0, \text{prices_down} \mapsto 1\}, \\ \mathcal{I}_2 = \{\text{need_larger_house} \mapsto 1, \text{prices_down} \mapsto 0\}, \\ \mathcal{I}_3 = \{\text{need_larger_house} \mapsto 1, \text{prices_down} \mapsto 1\} \end{array} \right\},$$

such that

$$\pi(\mathcal{I}_0) = 0, \quad \pi(\mathcal{I}_1) = 0, \quad \pi(\mathcal{I}_2) = 1, \quad \pi(\mathcal{I}_3) = 1,$$

whereby

$$\begin{aligned} \mathcal{B}(\text{need_larger_house}) &= 1 - \max\{\pi(\mathcal{I}_0), \pi(\mathcal{I}_1)\} = 1, \\ \mathcal{B}(\neg\text{need_larger_house}) &= 1 - \max\{\pi(\mathcal{I}_2), \pi(\mathcal{I}_3)\} = 0, \\ \mathcal{B}(\text{prices_down}) &= 1 - \max\{\pi(\mathcal{I}_0), \pi(\mathcal{I}_2)\} = 0, \\ \mathcal{B}(\neg\text{prices_down}) &= 1 - \max\{\pi(\mathcal{I}_1), \pi(\mathcal{I}_3)\} = 0. \end{aligned}$$

Therefore, the set of John's justified desires will be

$$\begin{aligned} \mathcal{J}(\text{buy_house}) &= 1 \quad \text{(because of Rule } R_1\text{)}, \\ \mathcal{J}(\neg\text{buy_house}) &= 0 \quad \text{(no justifying rule)}, \\ \mathcal{J}(\text{wait}) &= 0 \quad \text{(because of Rule } R_3\text{)}, \\ \mathcal{J}(\neg\text{wait}) &= 0 \quad \text{(because of Rule } R_2\text{)}. \end{aligned}$$

5 Belief Change

Here, we discuss and compare three possible extensions of a trusted-based operator akin to the one proposed in [7] to deal with bipolar trust and distrust degrees. To begin with, we define a basic belief change operator based on trust.

5.1 Trust-Based Belief Change Operator

Here, we suppose that a source of information may be considered trusted to a certain extent. This means that its membership degree to the fuzzy set of trusted sources is $\tau \in [0, 1]$. Let $\phi \in \mathcal{L}$ be incoming information from a source trusted to degree τ. The belief change operator is defined as follows:

Definition 10 (Belief Change Operator *). *The possibility distribution π' which induces the new belief set $\mathcal{B}'$ after receiving information ϕ is computed from possibility distribution π relevant to the previous belief set $\mathcal{B}$ ($\mathcal{B}' = \mathcal{B} * \frac{\tau}{\phi}$) as follows: for all interpretation $\mathcal{I}$,*

$$\pi'(\mathcal{I}) = \bar{\pi}'(\mathcal{I}) / \max_{\mathcal{I}}\{\bar{\pi}'(\mathcal{I})\}, \tag{13}$$

where

$$\bar{\pi}'(\mathcal{I}) = \begin{cases} \pi(\mathcal{I}), & \text{if } \mathcal{I} \models \phi \text{ and } \mathcal{B}(\neg\phi) < 1; \\ \tau, & \text{if } \mathcal{I} \models \phi \text{ and } \mathcal{B}(\neg\phi) = 1; \\ \pi(\mathcal{I}) \cdot (1-\tau), & \text{if } \mathcal{I} \not\models \phi. \end{cases} \tag{14}$$

Notice that Equation 13 guarantees that π' is a normalized possibility distribution. The condition $\mathcal{B}(\neg\phi) < 1$ in Equation 14 is equivalent to $\exists \mathcal{I}' : \mathcal{I}' \models \phi \Rightarrow \pi(\mathcal{I}') > 0$, i.e., $\Pi([\phi]) > 0$; likewise, the condition $\mathcal{B}(\neg\phi) = 1$ is equivalent to $\Pi([\phi]) = 0$, which implies $\pi(\mathcal{I}) = 0$. Therefore, the second case in Equation 14 provides for the *revision* of beliefs that are in contradiction with new information ϕ. In general, the operator treats new information ϕ in the negative sense: being told ϕ denies the possibility of world situations where ϕ is false (third case of Equation 14). The possibility of world situations where ϕ is true may only increase due to normalization (Equation 13) or revision (second case of Equation 14).

In the following, we present three alternatives for extending the belief change operator $*$. In such extensions, we suppose that a source of information may be considered trusted to a degree $\tau \in [0, 1]$ and/or distrusted to a degree $\delta \in [0, 1]$; τ and δ are respectively the source's membership degrees in the fuzzy sets of trusted and distrusted sources.

5.2 Open-Minded Belief Change Operator

This operator represents the changes in the beliefs of an agent which is both optimistic and does not perceive malicious purposes from neutral sources ("An optimistic agent discerns (him)herself as luckier" [17]). The proposed operator provides a formal representation of how an agent which gives the benefit of the doubt to the sources could change its beliefs when new information is received. More precisely, the following definition illustrates the attitude of an open-minded agent when choosing which among the possible trust degrees in $[\tau, 1-\delta]$ to consider as its trust degree.

Definition 11 (Open-Minded Operator). *Let ϕ be the incoming information with the trustworthiness score (τ, δ). An open-minded belief change operator $*_m$ can be defined as follows:*

$$\mathcal{B} *_m \frac{(\tau, \delta)}{\phi} = \mathcal{B} * \frac{\tau + (h/2)}{\phi}. \tag{15}$$

As we can see, such an agent chooses a degree of trust which is proportional to the degree of hesitation. Due to its optimism, the greater the hesitation, the higher the adopted trust degree.

Observation 1. *If $h = 0$, then, by applying the $*_m$ operator, information ϕ coming from any source with trustworthiness degree (τ, δ), is perceived as trusted with degree τ: $\mathcal{B} *_m \frac{(\tau,\delta)}{\phi} = \mathcal{B} * \frac{\tau}{\phi}$.*

Observation 2. *By applying the $*_m$ operator, information coming from a neutral source is considered as half-trusted.*

Indeed, $\forall\, \alpha \in [0, 1/2]^2$ we have: $\mathcal{B} *_m \frac{(\alpha,\alpha)}{\phi} = \mathcal{B} * \frac{1/2}{\phi}$. In particular, when a piece of information comes from a completely unknown source, (i.e., with a trustworthiness score equal to $(0, 0)$), the agent considers that information anyway by giving it a half degree of trust.

Observation 3. *By applying the $*_m$ operator, information is not considered at all only in the cases in which it comes from a completely distrusted source, i.e., a source with a trustworthiness score equal to $(0, 1)$. Benefit of the doubt is given in all the other cases.*

Example (continued). By using the open-minded change operator, John's initial beliefs would change to

$$\begin{aligned}\mathcal{B}' &= \mathcal{B} *_m \frac{(1,0)}{\phi} *_m \frac{(1/2,1/4)}{\phi} *_m \frac{(0,1)}{\neg\phi} *_m \frac{(0,0)}{\neg\phi} \\ &= \mathcal{B} * \frac{1}{\phi} * \frac{5/8}{\phi} * \frac{0}{\neg\phi} * \frac{1/2}{\neg\phi},\end{aligned}$$

where $\phi =$ prices_down. This is how the possibility distribution over belief interpretations changes, given that $\mathcal{I}_1, \mathcal{I}_3 \models \phi$ and $\mathcal{I}_0, \mathcal{I}_2 \models \neg\phi$:

	$\pi(\mathcal{I}_0)$	$\pi(\mathcal{I}_1)$	$\pi(\mathcal{I}_2)$	$\pi(\mathcal{I}_3)$
initial	0	0	1	1
$*\frac{1}{\phi}$	0	0	0	1
$*\frac{5/8}{\phi}$	0	0	0	1
$*\frac{0}{\neg\phi}$	0	0	0	1
$*\frac{1/2}{\neg\phi}$	1	0	1	1

This yields: $\mathcal{B}'(\text{prices_down}) = \mathcal{B}'(\neg\text{prices_down}) = 0$. Therefore, an open-minded John would not change the justification degree of his desires, that is, his justification to buy a house would still be full, while he would be unresolved whether to wait or not, for neither desire would be justified.

5.3 Wary Belief Change Operator

Here, we present a belief change operator which illustrates the attitude of a conservative (pessimistic) agent which does not give the benefit of the doubt to unknown sources and perceives information coming from malicious sources as false (to some extent, depending on the degree of distrust).“A pessimistic agent discerns (him)herself as unluckier”.

Definition 12 (Effective Trust/Distrust). *Let (τ, δ) be the trustworthiness score of a helpful source (malicious source). The degre of effective trust τ_e (effective distrust δ_e) is given by $\tau_e = \tau - \delta \in (0, 1]$ ($\delta_e = \delta - \tau \in (0, 1]$).*

[2] by definition, i.e. because $\tau + \delta \leq 1$, the highest value τ and δ can take up in case of equality is $1/2$.

Definition 13 (Wary Operator). *Let ϕ be incoming information with trustworthiness score (τ, δ). A wary belief change operator $*_w$ can be defined as follows:*

$$\mathcal{B} *_w \frac{(\tau,\delta)}{\phi} = \begin{cases} \mathcal{B} * \frac{\tau_e}{\phi} & \text{if } \tau_e > 0 \text{ and } h \neq 0; \\ \mathcal{B} * \frac{\delta_e}{\neg\phi} & \text{if } \delta_e > 0 \text{ and } h \neq 0; \\ \mathcal{B} & \text{if } \tau = \delta. \end{cases} \tag{16}$$

Observation 4. *Whereas in the previous generalization proposal the special case that falls back to operator $*$ is when $\delta = 1 - \tau$, here we start from the assumption that operator $*$ applies to the special case where $\delta = 0$.*

Observation 5. *If the agent has motivations for (effectively) distrusting, with degree α, the source of information from which comes the new piece of information ϕ, by applying the operator $*_w$, it will trust its opposite $\neg\phi$, with degree α.*

Observation 6. *In both cases of unknown sources (i.e., trustworthiness score equal to $(0,0)$), or neutral sources (i.e., trustworthiness score of the form (α,α)), applying $*_w$ will not change at all the agent degrees of beliefs.*

Example (continued). By using the wary change operator, John's initial beliefs would change to

$$\begin{aligned} \mathcal{B}' &= \mathcal{B} *_w \frac{(1,0)}{\phi} *_w \frac{(1/2,1/4)}{\phi} *_w \frac{(0,1)}{\neg\phi} *_w \frac{(0,0)}{\neg\phi} \\ &= \mathcal{B} * \frac{1}{\phi} * \frac{1/4}{\phi} * \frac{1}{\phi} * \frac{0}{\neg\phi}, \end{aligned}$$

where $\phi =$ prices_down. This is how the possibility distribution over belief interpretations changes, given that $\mathcal{I}_1, \mathcal{I}_3 \models \phi$ and $\mathcal{I}_0, \mathcal{I}_2 \models \neg\phi$:

	$\pi(\mathcal{I}_0)$	$\pi(\mathcal{I}_1)$	$\pi(\mathcal{I}_2)$	$\pi(\mathcal{I}_3)$
initial	0	0	1	1
$*\frac{1}{\phi}$	0	0	0	1
$*\frac{1/4}{\phi}$	0	0	0	1
$*\frac{1}{\phi}$	0	0	0	1
$*\frac{0}{\neg\phi}$	0	0	0	1

This yields $\mathcal{B}'(\text{prices_down}) = 1$ and $\mathcal{B}'(\neg\text{prices_down}) = 0$. Therefore, a wary John would increase the justification degree of the desire to wait for prices to go down before buying. The justification of waiting is now equal to that of buying a house.

5.4 Content-Based Belief Change Operator

Neither of the previous two proposed operator extensions attempts to consider the content of incoming information as well as the agent competence in judging its truth.

Some experiments have shown, however, that this consideration can help when evaluating the truth of new information. Fullam and Barber [14] showed that integrating

both policies for information valuation based on characteristics of the information and the sources providing it yields significant improvement in belief accuracy and precision over no-policy or single-policy belief revision.

Experiments on human trust and distrust of information in the context of a military sensemaking task have concluded that people tend to trust information from an unknown source (whose prototypical case is $\tau = \delta = 0$) to the extent that it does not contradict their previous beliefs [19]. The basic rationale for this behavior appears to be that people trust themselves, if anybody.

The third proposed extension of the belief change operator intends to model this type of behavior.

Definition 14 (Content-Based Operator). *Let ϕ be incoming information with trustworthiness score (τ, δ). A content-based belief change operator $*_c$ can be defined as follows:*

$$\mathcal{B} *_c \frac{(\tau, \delta)}{\phi} = \mathcal{B} * \frac{\tau + h \cdot \mathcal{B}(\phi)}{\phi}. \tag{17}$$

Observation 7. *By applying operator $*_c$, the only case in which information will be completely rejected is when the source is fully distrusted.*

Observation 8. *By applying operator $*_c$, the only case in which the information content does not influence the agent's beliefs is when $h = 0$.*

Example (continued). By using the content-based change operator, John's initial beliefs would change to

$$\begin{aligned}\mathcal{B}' &= \mathcal{B} *_c \frac{(1,0)}{\phi} *_c \frac{(1/2, 1/4)}{\phi} *_c \frac{(0,1)}{\neg\phi} *_c \frac{(0,0)}{\neg\phi} \\ &= \mathcal{B} * \frac{1}{\phi} * \frac{3/4}{\phi} * \frac{0}{\neg\phi} * \frac{0}{\neg\phi},\end{aligned}$$

where $\phi =$ prices_down. This is how the possibility distribution over belief interpretations changes, given that $\mathcal{I}_1, \mathcal{I}_3 \models \phi$ and $\mathcal{I}_0, \mathcal{I}_2 \models \neg\phi$:

	$\pi(\mathcal{I}_0)$	$\pi(\mathcal{I}_1)$	$\pi(\mathcal{I}_2)$	$\pi(\mathcal{I}_3)$
initial	0	0	1	1
$*\frac{1}{\phi}$	0	0	0	1
$*\frac{3/4}{\phi}$	0	0	0	1
$*\frac{0}{\neg\phi}$	0	0	0	1
$*\frac{0}{\neg\phi}$	0	0	0	1

This yields: $\mathcal{B}'(\text{prices_down}) = 1$ and $\mathcal{B}'(\neg\text{prices_down}) = 0$. Therefore, in this case, John would behave like a wary John.

5.5 Further Observations

Observation 9. *By applying the operators $*_m$ and $*_c$, a completely distrusted piece of information ($\delta = 1$, $\tau = 0$ and $h = 0$) is not considered at all.*

When there is no doubt about the trust degree, both the open-minded and content-based agents behave like a trust-based agent.

Observation 10. *If there is no hesitation, i.e., if* $h = 0$,

$$\mathcal{B} *_m \frac{(\tau, \delta)}{\phi} = \mathcal{B} *_c \frac{(\tau, \delta)}{\phi} = \mathcal{B} * \frac{\tau}{\phi}.$$

Observation 11. *By using the open minded operator or the content-based operator, the more different not fully distrusted sources confirm formula* ϕ*, the more* ϕ *is believed.*

Observation 12. *By using a wary operator, the more different and:*

- *helpful sources confirm a belief by providing the same information, the more believed is the information provided;*
- *malicious sources confirm a belief by providing the same information, the less believed is the information provided.*

6 Desire and Goal Change

Belief change may induce changes in the justification degree of some desires/goals. Actually, this is the only way to modify the agent goals from the outside (external reason) [4]. As we have seen in the previous sections, different kinds of agent may have different behaviors (*open-minded*, *wary*, or *content-based*, for example) when changing their beliefs on the basis of new incoming information. This may result in different belief behaviours for different agents in the same situation and, as a consequence, in different desire/goal sets. To account for the changes in the desire set caused by belief change, one has to recursively [7]:

(i) calculate for each rule $R \in \mathcal{R}_{\mathcal{J}}$ its new activation degree by considering $\mathcal{B}'$ and
(ii) update the justification degree of all desires in its right-hand side ($\mathrm{rhs}(R)$).

Desires may also change for internal reasons. This is represented by the insertion of a new desire-generation rule in $\mathcal{R}_{\mathcal{J}}$ or the retraction of an existing rule. The new fuzzy set of justified desires, $\mathcal{J}'$, is computed as follow:

(i) calculate for each rule $R \in \mathcal{R}_{\mathcal{J}}$ its new activation degree by considering the fact that a rule is inserted or retracted and
(ii) update the justification degree of all desires in the right-hand side of the rules in $\mathcal{R}_{\mathcal{J}}$.

Goals serve a dual role in the deliberation process, capturing aspects of both *intentions* [6] and *desires* [25]. The main point about desires is that we expect a rational agent to try and manipulate its surrounding environment to fulfill them. In general, considering a problem $\mathcal{P}$ to solve, not all generated desires can be adopted at the same time, especially when they are not feasible at the same time. We assume we dispose of a $\mathcal{P}$-dependent function $\mathcal{F}_{\mathcal{P}}$ wich, given a possibility distribution inducing a set of graded beliefs $\mathcal{B}$ and a fuzzy set of desires $\mathcal{J}$, returns a degree γ which corresponds to the certainty degree of the most certain feasible solution found. We may call γ the *degree of feasibility* of $\mathcal{J}$ given $\mathcal{B}$, i.e., $\mathcal{F}_{\mathcal{P}}(\mathcal{J}|\mathcal{B}) = \gamma$.

Definition 15 (γ-Goal Set). *A γ-goal set, with $\gamma \in [0,1]$, in state $\mathcal{S}$ is a fuzzy set of desires $\mathcal{G}$ such that:*

1. *$\mathcal{G}$ is justified: $\mathcal{G} \subseteq \mathcal{J}$, i.e., $\forall d \in \{a, \neg a\}$, $a \in \mathcal{A}$, $\mathcal{G}(d) \leq \mathcal{J}(d)$;*
2. *$\mathcal{G}$ is γ-feasible: $\mathcal{F}_{\mathcal{P}}(\mathcal{G}|\mathcal{B}) \geq \gamma$;*
3. *$\mathcal{G}$ is consistent: $\forall d \in \{a, \neg a\}$, $a \in \mathcal{A}$, $\mathcal{G}(d) + \mathcal{G}(\neg d) \leq 1$.*

In general, given a fuzzy set of desires $\mathcal{J}$, there may be more than one possible γ-goal sets $\mathcal{G}$. However, a rational agent in state $\mathcal{S} = \langle \mathcal{B}, \mathcal{J}, \mathcal{R}_{\mathcal{J}} \rangle$, for practical reasons, may need to elect one precise set of goals, $\mathcal{G}^*$, to pursue, which depends on $\mathcal{S}$.

In that case, a goal election function should be defined. Let us call G_γ the function which maps a state $\mathcal{S}$ into the γ-goal set elected by a rational agent in state $\mathcal{S}$: $\mathcal{G}^* = G_\gamma(\mathcal{S})$.

The choice of one goal set over the others may be based on a preference relation on desire sets. Therefore, G_γ must be such that:

- $(\mathbf{G1})$ $\forall \mathcal{S}, G_\gamma(\mathcal{S})$ is a γ-goal set;
- $(\mathbf{G2})$ $\forall \mathcal{S}$, if $\mathcal{G}$ is a γ-goal set, then $G_\gamma(\mathcal{S})$ is preferred at least as $\mathcal{G}$, or more.

(G1) requires that a goal election function G_γ does indeed return a γ-goal set; while **(G2)** requires that the γ-goal set returned by function G_γ be "optimal", i.e., that a rational agent always selects one of the most preferrable γ-goal sets.

7 Conclusion

The issue of how to deal with independently and explicitly given trust and distrust degrees of information sources within the context of goal generation has been approached by generalizing a trusted-based belief change operator.

The three proposed alternative extensions have different scopes. The open-minded operator makes sense in a collaborative environment, where all sources of information intend to be helpful, except that, perhaps, some of them may lack the knowledge needed to help. The wary operator is well suited to contexts where competition is the main theme and the agents are utility-driven participants in a zero-sum game, where a gain for an agent is a loss for its counterparts. The content-based operator is aimed at mimicking the usual way people change their beliefs.

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Chapter 3
Ubiquitous Intelligence

3.1 Introduction

In Chapters 1 and 2, we have stated the importance of involving and consolidating relevant ubiquitous intelligence surrounding data mining applications for actionable knowledge discovery and delivery.

Ubiquitous intelligence surrounds a real-world data mining problem. D^3M identifies and categories ubiquitous intelligence into the following types.

- Data intelligence,
- Human intelligence,
- Domain intelligence,
- Network and web intelligence, and
- Organizational and social intelligence.

For the success of actionable knowledge delivery based on D^3M, it is necessary to not only involve individual types of intelligence, but also to consolidate the relevant ubiquitous intelligence into the modeling, evaluation and the whole process and systems. In this chapter, we discuss the concepts and aims of involving corresponding intelligence, the techniques and corresponding case studies for involving the intelligence into D^3M.

3.2 Data Intelligence

3.2.1 What is data intelligence

Definition 3.1. (Data Intelligence) reveals interesting stories and/or indicators hidden in data about a business problem. The intelligence of data emerges in the form of interesting patterns and actionable knowledge.

There are two levels of data intelligence:

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- *General level of data intelligence*: refers to the patterns identified from explicit data, presenting general knowledge about a business problem, and
- *In-depth level of data intelligence*: refers to the patterns identified in more complex data, using more advanced techniques, disclosing much deeper information and knowledge about a problem.

Taking association rule mining as an example, a general level of data intelligence is frequent patterns identified in basket transactions, while *associative classifiers* reflect deeper levels of data intelligence.

3.2.2 Aims of involving data intelligence

D^3M aims to disclose data intelligence from multiple perspectives. One of the angles from which to observe data intelligence is the data explicitness or implicitness.

- *Explicit data intelligence* refers to the level of data intelligence disclosing explicit characteristics or exhibited explicitly. An example of explicit data intelligence is the trend of a stock market index or the dynamics of a stock price.
- *Implicit data intelligence* refers to the level of data intelligence disclosing implicit characteristics or exhibited implicitly. In stock markets, an example of implicit data intelligence is the trading behavior patterns of a hidden group in which investors are associated with each other.

Both explicit data intelligence and implicit data intelligence may present intelligence at either a general or in-depth level.

Another angle of scrutinizing data intelligence is from either a syntactic or a semantic perspective.

- *Syntactic data intelligence*, refers to the kind of data intelligence disclosing syntactic characteristics. An example of syntactic data intelligence is itemset associations.
- *Semantic data intelligence*, refers to the kind of data intelligence disclosing semantic characteristics. An example of semantic data intelligence is temporal trading behavior embedding temporal logic relationship amongst trading behaviors.

Similarly, both syntactic data intelligence and semantic data intelligence may present intelligence at either a general or in-depth level.

3.2.3 Aspects of data intelligence

Even though mainstream data mining focuses on the substantial investigation of varying data for hidden interesting patterns or knowledge, the real-world data and its surroundings are usually much more complicated. The following list identifies aspects that may be associated with data intelligence.

- Data type such as numeric, categorical, XML, multimedia and composite data
- Data timing such as temporal and sequential
- Data spacing such as spatial and temporal-spatial
- Data speed and mobility such as high frequency, high density, dynamic data and mobile data
- Data dimension such as multi-dimensional, high-dimensional data, and multiple sequences
- Data relation such as multi-relational, linkage record
- Data quality such as missing data, noise, uncertainty, and incompleteness
- Data sensitivity like mixing with sensitive information

Deeper and wider analysis is required to mine for in-depth data intelligence in complex data. Two kinds of aspects: data engineering and data mining, need to be further developed for processing and analyzing real-world data complexities such as multi-dimensional data, high-dimensional data, mixed data, distributed data, and processing and mining unbalanced, noisy, uncertain, incomplete, dynamic, and stream data.

3.2.4 Techniques disclosing data intelligence

Since the emergence of data mining, people have mainly focused on intelligence understanding and disclosure from the data perspective. This is still a current feature of mainstream KDD research. Techniques for disclosing data intelligence are embodied through major data preparation, feature extraction and selection, and the data mining methods. If we categorize them, fundamental underpinnings for dealing with real-life data complexities present the following multi-dimensional view.

- *data quality enhancement* to enhance data quality and readiness for pattern mining,
- *data matching and integration* to match/integrate data from multiple and/or heterogeneous data sources,
- *information coordination* to access multiple data sources through coordination techniques such as multi-agent coordination,
- *feature extraction* such as extracting and representing features mixing semi-structured, ill-structured with structured data,
- *parallel computing* for processing multiple sources of high frequency data in parallel,
- *collective intelligence* of data such as through aggregating intelligence identifying in an individual data source,
- *dimension reduction* to reduce the number of dimensions to a handleable level,
- *space mapping* such as mapping an input space to a feature hypersphere,
- *computational complexity* of data such as engaging quantum computing for more efficient computation,

Table 3.1 Transactional Data Mixing Ordered and Unordered Data

Customer ID	Policies/Activities	Debt
1	$(c_1, c_2) / (a_1 - a_2)$	Y
2	$(c_2, c_4, c_5) / (a_1 - a_3)$	N
2	$(c_2) / (a_2 - a_3 - a_4)$	Y
2	$(c_1, c_3, c_4) / (a_1 - a_3 - a_4)$	N
3	$(c_1, c_3) / (a_2 - a_3)$	Y
3	$(c_1, c_2, c_4) / (a_1 - a_3)$	Y
4	$(c_2, c_4) / (a_1 - a_2 - a_5)$	N
4	$(c_1, c_2, c_4) / (a_1 - a_3)$	N
4	$(c_1, c_2, c_5) / (a_1 - a_3 - a_4)$	Y
4	$(c_2, c_3) / (a_2 - a_4)$	N

- *data privacy and security* for mining patterns while protecting sensitive information from disclosure, and
- *pattern mining algorithms* such as frequent pattern mining, classification, combined pattern mining methods.

3.2.5 An example

In the real world, data may consist of many things such as demographics, unordered and ordered transactions and business impact or outcomes. An example is data from the social security area such as the e-government allowance service-related data in Centrelink[1], Australia. Suppose there are two datasets, a transactional dataset with unordered policies and corresponding ordered activities (Table 6.2), and a customer demographic dataset with customer basic information (Table 6.1). In this example, we also consider the impact of customers on e-government service objectives, namely whether a customer incurs debts or not (represented by '*Y*' if yes or '*N*' for no). Table 6.2 shows whether a customer has a debt or not under various policies and associated activities. For instance, Centrelink has policies that customers should report their income fortnightly or irregularly, depending on different allowances. Various activities are also conducted with customers, e.g., reviewing from Centrelink, reminder letters sent to customers, and so on.

Traditionally, such data is mined individually; the transactional data mixing unordered transactions with ordered ones is partitioned into unordered and ordered datasets respectively for further frequent pattern mining or classification. For instance, when association mining is used to mine frequent associations, the rules that

[1] www.centrelink.gov.au

Table 3.2 Customer Demographic Data

Customer ID	Gender	...
1	F	
2	F	
3	M	
4	M	

Table 3.3 Traditional Association Rules

Rules	Supp	Conf	Lift
$c_1 \rightarrow Y$	4/10	4/6	1.3
$c_1 \rightarrow N$	2/10	2/6	0.7
$c_2 \rightarrow Y$	4/10	4/8	1
$c_2 \rightarrow N$	4/10	4/8	1
$c_3 \rightarrow Y$	1/10	1/3	0.7
$c_3 \rightarrow N$	2/10	2/3	1.3
...	...	...	...

Table 3.4 Traditional Sequential Patterns

Rules	Supp	Conf	Lift
$a_1 \rightarrow Y$	3/10	3/7	0.9
$a_1 \rightarrow N$	4/10	4/7	1.1
$a_2 \rightarrow Y$	3/10	3/5	1.2
$a_2 \rightarrow N$	2/10	2/5	0.8
$a_1 - a_2 \rightarrow Y$	1/10	1/2	1
$a_1 - a_2 \rightarrow N$	1/10	1/2	1
$a_1 - a_3 \rightarrow Y$	2/10	2/5	0.8
$a_1 - a_3 \rightarrow Y$	3/10	3/5	1.2
...	...	...	...

can be discovered from the unordered transactional dataset are shown in Table 6.3. Similarly, we can identify frequent sequential patterns as shown in Table 6.4.

However, such frequent patterns are not informative for business decision-makers, because they do not include information of customer demographics. They are simplified and separated from real business scenarios in which unordered and ordered activities are mixed. As a result, the identified patterns do not reflect the reality of business data, and thus are not informative. Their actionable capability is not powerful enough to support business needs.

In the following, we illustrate the concept of in-depth data intelligence. We use combined mining [61, 232, 233, 236] to handle the above problem to produce more informative and actionable patterns.

Table 3.5 Combined Association Rules

Rules	Supp	Conf	Lift	*Cont*	I_{rule}
$F \wedge c_1 \rightarrow N$	2/10	1/2	1	1	1.4
$F \wedge c_2 \rightarrow Y$	2/10	2/3	1.3	1.3	1.3
$M \wedge c_2 \rightarrow N$	3/10	3/5	1.2	1.2	1.2
$M \wedge c_2 \rightarrow Y$	2/10	2/5	0.8	0.8	0.8
...	...	...	...	...	...

Table 3.6 Combined Association Rule Pairs

Pairs	Combined Rules	I_{pair}
$\mathscr{P}_1$	$M \wedge c_3 \rightarrow Y$ $M \wedge c_2 \rightarrow N$	0.55
$\mathscr{P}_2$	$F \wedge c_2 \rightarrow Y$ $M \wedge c_2 \rightarrow N$	0.63
...	... ...	...

First, the whole population is partitioned into two groups, male and female, based on the demographic data in Table 6.1, and then the demographic and transactional data of the two groups is mined separately, as partially shown in Table 6.5, where *Cont* denotes the contribution of the transactional data, and I_{rule} reflects the interestingness of the combined rules. (The definitions of *Cont* and I_{rule} will be given in Section 6.5.1). We can see from Table 6.5 that (1) rules are more informative than those in Table 6.3, and (2) more rules with high *confidence* and *lift* can be found by combining the rules from two separate datasets.

Second, it would be more interesting to organize the rules into *contrast pairs* as shown in Table 6.6, where I_{pair} is the interestingness of a pair rule. For instance, $\mathscr{P}_1$ is a rule pair for the male group. It shows that c_3 is associated with debt but c_2 is not. $\mathscr{P}_1$ is actionable in that it suggests c_2 is a preferred policy to replace c_3 and to avoid debt raised on male customers. Moreover, male customers should be excluded when initiating policy c_3. $\mathscr{P}_2$ is a pair rule with the same policy but different demographics. With the same policy c_2, male customers have no debts while females tend to have debts. It suggests that c_2 is a preferable policy for male customers but an undesirable policy for female customers.

A simple way to find the rules in Table 6.5 is to join Tables 6.2 and 6.1 in a pre-processing stage and then apply traditional association rule mining to the derived table. Unfortunately, it is often infeasible to do so in many applications where datasets contain hundreds of thousands of records or more.

Third, frequent patterns combining unordered and ordered frequent patterns can be identified as shown in Table 6.7. From Table 6.5, we know that male customers under policy c_2 do not tend to have debts. However, in Table 6.7 we can see that if activities $a_1 - a_3$ are taken, male customers under policy c_2 are very likely to have

Table 3.7 Combined Frequent Patterns with Both Unordered and Ordered Itemsets

Rules	Supp	Conf	Lift	*Cont*	I_{rule}
$M \wedge c_2 \wedge a_1 - a_3 \rightarrow Y$	2/10	2/3	1.3	1.6	2
...	...	...	...	...	...

Table 3.8 Classification on the Transactional Data

Customer ID	Policies/Activities	Prediction
1	(c_1, c_2) / $(a_1 - a_3)$	Y
2	$(c_2, c_4$ / (a_1)	N
...	...	...

debt since the interestingness I_{pair} of the pattern is as high as 2. Obviously, the ordered activity dataset provides much richer information to make a more reasonable decision.

Fourth, classification can be conducted on the identified frequent pattern sets. Table 6.8 shows some examples of the frequent patterns which can be used for classification.Unlike the features in conventional classification on the demographic data, in frequent pattern based classification, both ordered and unordered transactional data can be used to make the decision so that the performance of classification can be greatly improved.

The above analysis and examples show the following points:

- Patterns identified by traditional methods to target the discovery from a single dataset or using a single method can only disclose a general level of intelligence. They do not reflect the full picture of business scenarios, and only indicate limited information for decision-making;
- The idea of combined mining is very useful, general and flexible for identifying in-depth patterns through combining itemsets from multiple datasets or using multiple methods;
- The identified combined patterns disclose in-depth data intelligence, which are more informative and actionable for business decision-making.

In Chapter 6, we further introduce the architecture, frameworks and techniques for conducting combined mining for disclosing in-depth data intelligence.

3.3 Domain Intelligence

3.3.1 What is domain intelligence

Definition 3.2. (Domain Intelligence) refers to the intelligence that emerges from the involvement of domain factors and resources in pattern mining, which wrap not

only a problem but its target data and environment. The intelligence of domain is embodied through the involvement into KDD process, modeling and systems.

Domain intelligence involves qualitative and quantitative aspects. They are instantiated in terms of aspects such as domain knowledge, background information, prior knowledge, expert knowledge, constraints, organizational factors, business process, workflow, as well as environmental aspects, business expectation and interestingness.

3.3.2 Aims of involving domain intelligence

D^3M aims at involving multiple types of domain intelligence.

- *Qualitative domain intelligence* refers to the type of domain intelligence that discloses qualitative characteristics or involves qualitative aspects. Taking stock data mining as an example, fund managers have qualitative domain intelligence such as 'beating the market', when they evaluate the value of a trading pattern.
- *Quantitative domain intelligence* refers to the type of domain intelligence that discloses quantitative characteristics or involves quantitative aspects. An example of quantitative domain intelligence in stock data mining is whether a trading pattern can 'beat VWAP[2]' or not.

The roles of involving domain intelligence in actionable knowledge delivery are multi-form.

- Assisting in the modeling and evaluation of the problem. An example is "my trading pattern can beat the market index return" when domain intelligence of 'beat market index return' is applied to evaluate a trading pattern.
- Making mining realistic and business-friendly. By considering domain knowledge, we are able to work on an actual business problem rather than an artificial one abstracted from an actual problem.

3.3.3 Aspects of domain intelligence

In the mainstream of data mining, the consideration of domain intelligence is mainly embodied through involving domain knowledge, prior knowledge, and mining the process and/or workflow associated with a business problem. In a specific data mining exercise, domain intelligence may be presented in multiple aspects, for instance, some of the following:

- Domain knowledge,

[2] VWAP is a trading acronym for Volume-Weighted Average Price, the ratio of the value traded to total volume traded over a particular time horizon.

- Background and prior information,
- Meta-knowledge and meta-data,
- Constraints,
- Business process,
- Workflow,
- Benchmarking and criteria definition, and
- Business expectation and interest.

3.3.4 Techniques involving domain intelligence

D^3M highlights the role of domain intelligence in actionable knowledge discovery and delivery. To incorporate domain intelligence into data mining, the following theoretical underpinnings are essential:

- *representation and involvement of domain knowledge* into the data mining system,
- *formal modeling* of domain factors and resources,
- *interaction design* to provide interfaces and interaction channels for domain expert-data mining interaction at the running time,
- *involving domain factors in the data mining model and process* to study techniques and tools for incorporating domain factors into data mining models and process,
- *catering for business process and workflow* in the mining process and interaction between a domain expert and a data mining system,
- *benchmarking* to develop benchmarks and criteria for pattern evaluation, and
- *business interestingness* to measure the pattern importance of business interest, and the tradeoff strategies for filtering patterns of both technical and business importance.

In particular, domain knowledge in business fields often takes the form of precise knowledge, concepts, beliefs, relations, or vague preference and bias. The integration of domain knowledge subjects to how it can be represented and filled into the knowledge discovery process. Ontological engineering and semantic web can be used for representing such forms from both syntactic and semantic perspectives. For example, ontology-based specifications can be developed to build a business ontological domain representing domain knowledge that can be mapped to a low-level domain for a mining system.

3.3.5 Ontology-Based Domain Knowledge Involvement

It has gradually become accepted that domain knowledge can play a significant role in real-world data mining. For instance, in cross-market mining, traders often take

'beating market', as a personal preference to judge an identified trading rule's actionability. In this case, a stock data mining system needs to embed the formulas calculating *market return* and *rule return*, and set an interface for traders to specify a favorite threshold and comparison relationship between the two returns in the evaluation process. Subsequently, the key is to take advantage of domain knowledge in the KDD process.

The integration of domain knowledge subjects to how it can be represented and filled into the knowledge discovery and delivery process. One promising approach is ontological engineering [35, 107, 108] and semantic web [181]. They can be used to represent, transform, match, map, discover and dispatch domain knowledge, meta-knowledge, as well as the identified knowledge for improving knowledge discovery and delivery. Agent-based cooperation mechanisms can further be used to support ontology-represented domain knowledge in the AKD process.

Domain knowledge in business fields often takes forms of precise knowledge, concepts, beliefs, relations, or vague preference and bias. Using ontology-based specifications, we can build a business ontological domain to represent domain knowledge in terms of ontological items and semantic relationships. For instance, in identifying trading patterns in stock markets, we can develop ontological items to represent return-related items such as *return*, *market return*, *rule return*, etc. There is class_of relationship between *return* and *market return*, while *market return* is associated with *rule return* in some form of user-specified logic connectors, say 'beating the market' if *rule return* is larger ($>$) than *market return* by a preset threshold. Corresponding ontological representations can then be developed to manage the above items and relationships.

Further, business ontological items are mapped to data mining system's internal ontologies. A data mining ontological domain is built for KDD system collecting standard domain-specific ontologies and discovered knowledge. To match items and relationships between two domains, and reduce and aggregate synonyous concepts and relationships in each domain, ontological rules, logical connectors and cardinality constraints need to be studied to support the ontological transformation from one domain to another, and the semantic aggregation of semantic relationships and ontological items intra or inter domains. For instance, the following rule transforms ontological items from the business domain to the data mining domain. Given input item A from a user, if it is associated with B by is_a relationship, then the output is B from the mining domain:

$$\forall(AANDB), \exists B ::= is_a(A,B) \Rightarrow B, \tag{3.1}$$

the resulting output is B.

For rough and vague knowledge, we can even fuzzify and map them to precise terms and relationships. For the aggregation of fuzzy ontologies, fuzzy aggregation and defuzzification mechanisms can help sort out proper output ontologies.

3.4 Network Intelligence

3.4.1 What is network intelligence

Definition 3.3. (Network Intelligence) refers to the intelligence that emerges from both web and broad-based network information, facilities, services and processing surrounding a data mining problem and system.

Network Intelligence involves both *web intelligence* and *broad-based network intelligence* such as information and resources distribution, linkages amongst distributed objects, hidden communities and groups, web service techniques, messaging techniques, mobile and personal assistant agents for decision-support, information and resources from network, and in particular the web, information retrieval, searching and structuralization from distributed and textual data. The information and facilities from the networks surrounding the target business problem either consist of the problem constituents, or can contribute to useful information for actionable knowledge discovery and delivery. Therefore, they should be catered for in domain driven AKD.

3.4.2 Aims of involving network intelligence

The aims of involving network intelligence into D^3M include multiple aspects, for example,

- to mine web data,
- to mine network data,
- to support pattern mining,
- to support decision-making on top of mined patterns, and
- to support social data mining by providing facilities for social interaction in an expert team.

In particular, we care about

- Discovering the business intelligence in networked data related to a business problem: for instance, discovering market manipulation patterns in cross-markets.
- Discovering networks and communities existing in a business problem and its data: for instance, discovering hidden communities in a market investor population.
- Involving networked constituent information in pattern mining on target data: for example, mining blog opinion for verifying market abnormal trading.
- Utilizing networking facilities to pursue information and tools for actionable knowledge discovery: for example, involving mobile agents to support distributed and peer-to-peer mining.

3.4.3 Aspects of network intelligence

In saying network intelligence, on one hand, we expect to fulfill the power of web and network information and facilities for data mining in terms of many aspects, for instance,

- Information and resource distribution
- Linkages amongst distributed objects
- Hidden communities and groups
- Information and resources from the network and in particular the web
- Information retrieval
- Structuralization and abstraction from distributed textual (blog) data
- Distributed computing
- Web network communication techniques
- Web-based decision-support techniques
- Dynamics of networks and the web
- Multiagent-based messaging and mining

On the other hand, we focus on mining web and network intelligence. In this regard, there are many emergent topics to be studied. We list a few here.

- Social network mining
- Hidden group and community mining
- Context-based web mining
- Opinion formation and evolution dynamics
- Distributed and multiple source mining
- Mining changes and dynamics of network
- Multiagent-based distribute data mining

3.4.4 Techniques for involving network intelligence

To incorporate the web and networked information and facilities discussed above into data mining, fundamental underpinnings such as the following are important:

- *application integration* to link applications for data collection and integration,
- *data gateway and management* for accessing/mining local data,
- *data and feature fusion* for fusing data and features,
- *distributed computing* for mining local data,
- *information retrieval and searching* for retrieving and searching data from the network,
- *distributed data mining* for mining distributed datasets,
- *combined mining* to mine for patterns consisting of sub-patterns from multiple individual data sets,
- *linkage analysis* for mining links and networks in networked data,
- *group formation* for identifying communities and groups in data, and

- *data mobility* to mine for patterns in mobile network data.
- Other techniques include *agent-based distributed data mining*, *peer-to-peer mining* and *agent-based information sharing*.

3.4.5 An example of involving network intelligence

Data mining on a peer-to-peer ($p2p$ for short) network forms a $p2p$ data mining. It utilizes the facilities of $p2p$ network including $p2p$ computing, communication, storage and human-computer interaction for data mining.

Another example of involving network intelligence for data mining is agent-based integration of multiple data mining tasks. In many networked applications, multiple data sources or multiple data mining tasks need to be involved. Some of them are heterogeneous. Multi-agents are used to implement local data mining tasks, and co-ordinate the data mining agents for task allocation, communication and message-passing, and pattern integration. Figure 3.1 illustrates the use of multi-agents for distributed data mining.

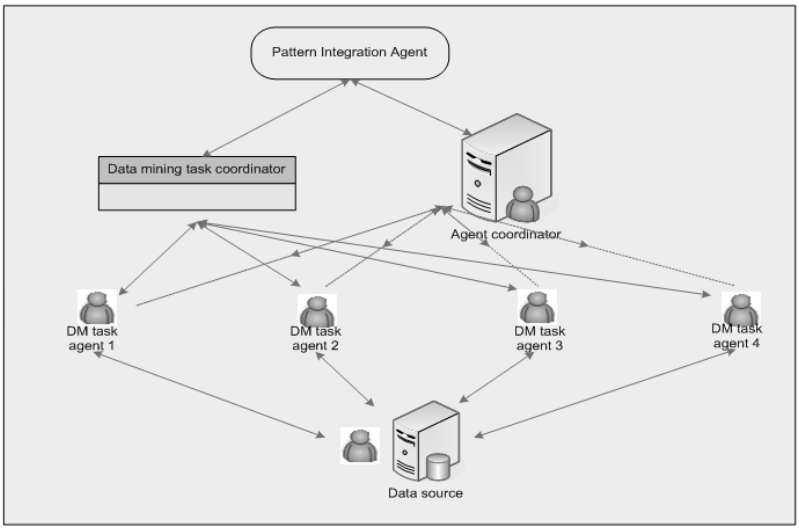

Fig. 3.1 Multi-source mining with agent-based networking

3.5 Human Intelligence

3.5.1 What is human intelligence

Definition 3.4. (Human Intelligence) refers to (1) explicit or direct involvement of human knowledge or a human as a problem-solving constituent, etc., and (2) implicit or indirect involvement of human knowledge or a human as a system component.

Explicit or direct involvement of human intelligence may consist of human empirical knowledge, belief, intention, expectation, run-time supervision, evaluation, and an individual end user or expert groups. An example of explicit human intelligence is for a domain expert to tune parameters via user interfaces. By contrast, implicit or indirect involvement of human intelligence may present as imaginary thinking, emotional intelligence, inspiration, brainstorm, reasoning inputs, and embodied cognition like convergent thinking through interaction with other members in assessing identified patterns. Examples of involving implicit human intelligence are user modeling for game behavior design, collecting opinions from an expert group for guiding model optimization, and utilizing embodied cognition for adaptive model adjustment.

In enterprise data mining, both human individuals and groups may be involved in a data mining process, which involves both intra-personal and inter-personal levels of human intelligence.The inter-personal level of human intelligence is also discussed in human social intelligence. An example is an interactive data mining system for mining and understanding critical abnormal trading behavior by engaging a group of domain experts who are familiar with relevant market models, and abnormal surveillance cases, and case management. They form an expert group in collectively tuning the models, evaluating and refining the mined trading behavior patterns. These experts sometimes discuss with each other to come up with refined parameters and models, and then tune the modeling accordingly.

3.5.2 Aims of involving human intelligence

The importance of involving humans in data mining has been widely recognized. With the systematic specification of human intelligence, we are able to convert data mining toward more human-centered, interactive, dynamic and user-friendly data mining, enhancing the capability of dealing with complex data mining issues, forming closed-loop data mining systems, and strengthening the usability of data mining.

- Human-centered data mining capability: The inclusion of human input, including individual and group knowledge, experience, preferences, cognition, thinking, reasoning etc. and more broad aspects linking to social and cultural factors (we will further expand this in social intelligence), makes it possible to utilize human intelligence to enhance data mining capability. Based on the depth and breadth of

human involvement, the cooperation of humans with data mining can be human-centered or human-assisted;

- Interactive mining capability: Human involvement takes place through interactive interfaces. This forms interactive data mining capability and systems, to effectively and sufficiently cater for human intelligence in data mining;
- Improving adaptive data mining capability: Real-life data mining applications are often dynamic. Data mining models are often pre-defined and cannot adapt to the dynamics. The involvement of human intelligence can assist with the understanding and capture of such dynamics and change, and guide the corresponding adjustment and retraining of models;
- User-friendly data mining: Catering to user preferences, characteristics, and requests in data mining will certainly make it more user-friendly;
- Dealing with complex data mining issues: Many complex issues cannot be handled very well without the involvement of domain experts. Complex knowledge discovery from complex data can benefit from inheriting and learning expert knowledge, enhancing the understanding of domain, organizational and social factors through expert guidelines, embedding domain experts into data mining systems, and so on.
- Closed-loop data mining: In general, data mining systems are open. As we learn from disciplines such as cybernetics, problem-solving systems are likely to be closed-loop in order to deal with environmental complexities and to achieve robust and dependable performance. This is the same for actionable knowledge discovery and delivery systems. The involvement of humans can essentially contribute to closed-loop data mining.
- Enhancing usability of data mining: Driven by the inclusion of human intelligence and the corresponding development and support, the usability of data mining systems can be greatly enhanced. Usability measures the quality of a user's experience when interacting with a data mining system.

3.5.3 Aspects of human intelligence

The aspects of human intelligence in AKD are embodied in many ways.

- Human empirical knowledge,
- Belief, intention, expectation,
- Sentiment, opinion,
- Run-time supervision, evaluation,
- Expert groups,
- Imaginary thinking,
- Emotional intelligence,
- Inspiration,
- Brainstorming,
- Retrospection,
- Reasoning inputs, and

- Embodied cognition like convergent thinking through interaction with other members in assessing identified patterns

3.5.4 Techniques for involving human intelligence

The involvement of the above aspects of human intelligence into data mining is challenging. Typical challenges such as dynamic involvement, cognitive emergence, group-based involvement and divergence of opinions, are important for handling complex and unclear data mining applications. To effectively involve human intelligence for AKD, fundamental studies are essential on representing, modeling, processing, analyzing and engaging human intelligence into pattern mining models, processes and systems. The following list details some of these essential techniques.

- *Dynamic user modeling* such as to capture user characteristics and inputs into data mining systems,
- *Online user interaction* such as to support online users to interact with a data mining system remotely,
- *Group decision-making* in pattern discovery such as involving a group of domain experts to evaluate and filter the identified patterns,
- *Adaptive interaction* such as for users to adapt to the pattern discovery process and for models to adapt to user thinking and decisions,
- *Distributed interaction* such as catering for multiple users to interact with models and with each other during pattern discovery and evaluation,
- *Consensus building* for a group of users to form optimal and mutually agreed findings by dealing with thinking convergence and divergence, and in particular,
- *Interactive data mining* and *human-centered interactive data mining* which deal with interface design and the major roles played by humans in pattern mining,
- For complex cases, *human-centered data mining* and *human-assisted data mining* are essential for incorporating human intelligence.

Other relevant techniques consist of interaction design, social computing, sentiment analysis, and opinion mining, which can either directly involve human intelligence or further utilize human intelligence in data mining.

3.5.5 An example

An example of tools for involving human intelligence in data mining can be seen in the development of agent services-based interfaces for users with different roles to interact with the system. In general, a data mining-based decision-support system may involves several actors with personalized user preferences. It is very important to define such actors, and understand and specify their roles, goals and properties

correspondingly. We here try to categorize system actors into four groups: *general end user (business user)*, *domain expert*, *business modeler*, and *model maintainer*.

- *General end users (business users)* are average business people, who like their business language, and prefer an interface with no technical jargon;
- *Domain experts* refer to those business people who have specific skills and knowledge that need to be applied to business modeling; they essentially co-develop a model with business modelers.
- *Business modelers* refer to those technical people (who may have a certain level of domain knowledge) who focus on modeling and adjustment; and
- *Model maintainers* refer to those who are responsible for system/model execution control and maintenance.

There are different roles, preferences and needs for these actors. It is correspondingly important to design different interaction channels, interfaces and services for them. For instance, the following agent service interfaces may be built in an AKD system to cater for different actors: *general interface*, *advanced interface*, *model interface* and *execution interface*.

- *General interfaces* support general business end users' interaction with a system, with interfaces and services supporting general functions and facilities expressed in a business language;
- *Technical interfaces* support domain experts' interaction with a system, with interfaces and services supporting advanced functions and facilities expressed in a business language;
- *Model interfaces* support business modelers and algorithm designers' interaction with a system, with interfaces and services supporting modeling and development and adjustment functions and facilities expressed in a high-level and/or low-level technical language;
- *Execution interfaces* support model maintainers' interaction with a system, with interfaces and services supporting model execution control and maintenance expressed in business language.

In practice, these interfaces are often mixed in a system. With emergent technologies such as personalized interaction, Web 2.0, semantic web and agent mining [23, 49, 51], it is possible to include them all in one system by providing personalized interfaces and services to different actor groups.

3.6 Organizational Intelligence

3.6.1 What is organizational intelligence

Definition 3.5. (Organizational Intelligence) refers to the intelligence that emerges from involving organization-oriented factors and resources into pattern mining. The

organizational intelligence is embodied through its involvement in the KDD process, modeling and systems.

3.6.2 Aims of involving organizational intelligence

In mining patterns in a complex organization, the involvement of organizational intelligence is essential in many aspects, for instance,

- Reflecting the organization's reality, needs and constraints in business modeling and finding delivery,
- Satisfying organizational goals and norms, policies, regulations and conventions,
- Considering the impact of organizational interaction and dynamics in the modeling and deliverable design,
- Catering for organizational structure and its evolution in data extraction, preparation, modeling, and delivery.

3.6.3 Aspects of organizational intelligence

Organizational intelligence consists of many aspects, for example

- Organizational structures related to key issues, such as where data comes from and who, in which branch, needs the findings
- Organizational behavior related to key issues such as understanding the business and data and finding delivery of how individuals and groups act in an organization
- Organizational evolution and dynamics related to key issues such as data and information change, affecting model/pattern/knowledge evolution and adaptability
- Organizational/business regulation and convention related to key issues such as business understanding and finding delivery, including rules, policies, protocols, norms, law, etc.
- Business process and workflow related to key issues such as data (reflecting process and workflow) and business understanding, goal and task definition, and finding delivery.
- Organizational goals related to key issues such as problem definition, goal and task definition, performance evaluation, etc.
- Organizational actors and roles related to key issues such as system actor definition, user preferences, knowledge involvement, interaction, interface and service design, delivery, etc.
- Organizational interaction related to key issues such as data and information interaction amongst sub-systems and components, data sensitivity and privacy, interaction rules applied on organizational interaction that may affect data extraction, integration and processing, pattern delivery and so on.

3.6.4 Techniques for involving organizational intelligence

In order to consider the possible needs and impacts of key organizational factors such as organizational structures, behavior, evolution, dynamics, interaction, process, workflow and actors surrounding a real-world data mining problem, many fundamental techniques may be necessary. In the following, we list a few such techniques and illustrate the benefits of utilizing them in complex data mining problems.

- *Organizational computing* benefiting the modeling, representation, analysis and design of organizational factors in social data mining software development,
- *Organizational theory* benefiting the design and management of agent-based data mining systems,
- *Organizational behavior study* benefiting the understanding of data, business and expected deliverables, as well as human-mining interaction and interface design,
- *Computer simulation* benefiting the modeling and impact analysis of involving organizational factors in social data mining software,
- *Complexity theory* benefiting the understanding of complex data mining applications and systems,
- *Swarm and collective intelligence* benefiting the optimization and self-organization that can be very important in autonomous distributed data mining,
- *Divergence and convergence of thinking* benefiting consensus building in domain expert group-assisted data mining and evaluation, and
- *Agent mining*, namely agent and data mining integration, benefiting many weak aspects that can be complemented by multi-agents to enhance data mining.

3.7 Social Intelligence

3.7.1 What is social intelligence

Definition 3.6. (Social Intelligence) refers to the intelligence that emerges from the group interactions, behaviors and corresponding regulation surrounding a data mining problem.

Social intelligence covers both *human social intelligence* and *animat/agent-based social intelligence*. *Human social intelligence* is related to aspects such as social cognition, emotional intelligence, consensus construction, and group decision. *Animat/agent-based social intelligence* involves swarm intelligence, action selection and the foraging procedure. Both sides also engage social network intelligence, collective interaction, as well as social regulation rules, law, trust and reputation for governing the emergence and use of social intelligence.

3.7.2 Aims of involving social intelligence

In mining patterns in complex data and social environments, both human social intelligence and agent-based social intelligence may play an important role, for instance,

- Enhancing the social computing capability of data mining methods and systems,
- Implementing data mining and evaluation in a social and group-based manner, under supervised or semi-supervised conditions,
- Utilizing social group thinking and intelligence emergence in complex mining problem-solving,
- Building social data mining software on the basis of software agents, to facilitate human-mining interaction, group decision-making, self-organization and autonomous action selection by data mining agents. This may benefit from multi-agent data mining and warehousing,
- Defining and evaluating social performance including trust and reputation in developing quality social data mining software, and
- Enhancing data mining project management and decision-support capabilities of the identified findings in a social environment.

3.7.3 Aspects of social intelligence

Aspects of social intelligence take multiple forms. We illustrate them from the perspective of human social intelligence and animat/agent-based social intelligence respectively.

Human social intelligence aspects consist of aspects such as *social cognition*, *emotional intelligence*, *consensus construction*, and *group decision*.

- *Social cognition* aspects relate to how a group of people process and use social information, which can inform how to involve what information into data mining,
- *Emotional intelligence* aspects relate to group emotions and feelings, which can inform interface and interaction design, performance evaluation and finding delivery for data mining.
- *Consensus construction* aspects relate to how a group of people think and how thinking evolves in a group toward a convergence, particularly in a divergent thinking situation, which can inform the conflict resolution if people from different backgrounds value different aspects in pattern selection, or if there is a conflict between technical and business interests
- *Group decision* aspects relat to strategies and methods used by a group of people in making a decision, which can inform the discussion between business modelers and end users.

Animat/agent-based social intelligence aspects consist of

- Swarm/collective intelligence aspects related to collaboration and competition of a group of agents in handling a social data mining problem, which can assist in complex data mining through multi-agent interaction, collaboration, coordination, negotiation and competition, and
- Behavior/group dynamics aspects related to group formation, change and evolution, and group behavior dynamics, which can assist in simulating and understanding the structure, behavior and impact of mining a group/community.

In addition, both human/agent social intelligence involves many common aspects, such as

- Social network intelligence,
- Collective interaction,
- Social behavior network,
- Social interaction rules, protocols, norms, etc.,
- Trust and reputation, and
- Privacy, risk, and security in a social context.

3.7.4 Techniques for involving social intelligence

The engagement of social intelligence in data mining relies on many fundamental techniques. Some of them have only recently emerged, while others may need complete investigation. We list a few here: *social computing*, *software agents*, *swarm and collective intelligence*, *social cognitive science* and *agent mining*.

- *Social computing* studying social software supporting data mining,
- *Software agent technology* developing agent-based data mining systems,
- *Swarm and collective intelligence* looking after swarm-based optimization and self-organization that can be very important in autonomous distributed data mining,
- *Social cognitive science*, in particular the *divergence and convergence of thinking* to construct consensus in social networks for mining and evaluating patterns, and
- *Agent mining* to utilize multi-agents to enhance data mining.

3.8 Involving Ubiquitous Intelligence

3.8.1 The way of involving ubiquitous intelligence

In the above sections, we state the needs and techniques for incorporating multiple forms of ubiquitous intelligence into the data mining process and systems to enhance knowledge discovery. The use of ubiquitous intelligence may take one of the

following two paths: single intelligence engagement and multi-aspect intelligence engagement.

Examples of single intelligence engagement are to involve domain knowledge in data mining, and to consider user preferences in data mining. Multi-aspect intelligence engagement aims to integrate ubiquitous intelligence as needed. It is very challenging but inevitable in mining complex enterprise applications.

It is often very difficult to have every type of intelligence well integrated in a data mining system, in addition to the challenges of modeling and involving a specific type of intelligence. New data mining methodologies and techniques need to be developed in order to involve the ubiquitous intelligence in actionable knowledge discovery and delivery.

3.8.2 Methodologies for involving ubiquitous intelligence

There are some requirements for the methodologies supporting the integration of ubiquitous intelligence, for instance,

- Facilitating the human-model interaction and interactive data mining in a data mining process-oriented, distributed, online and group-based manner;
- Supporting human-centered data mining so that a group of human experts can form the constituent of a data mining problem-solving system and infrastructure. For instance, experts call and adjust models to mine for initial patterns and further call for the next step of post-mining for more actionable patterns by supervising the pattern extraction and business-friendly deliverables;
- Suiting for developing social data mining software that caters for social interaction, group behavior, and collective intelligence in a human-involved data mining system.

Typical issues that need to be handled by the methodologies for actionable knowledge delivery consist of

- Mechanisms for acquiring and representing unstructured, ill-structured, and uncertain knowledge, such as empirical knowledge stored in domain experts' brains, and unstructured knowledge representation and brain informatics;
- Mechanisms for acquiring and representing expert thinking, such as imaginary thinking and creative thinking in group heuristic discussions;
- Mechanisms for acquiring and representing group/collective interaction behavior and impact emergence, such as behavior informatics;
- Mechanisms for modeling learning-of-learning, i.e., learning other participants' behaviors, which is the result of self-learning or ex-learning, such as learning evolution and intelligence emergence.

3.8.3 Intelligence Meta-synthesis of ubiquitous intelligence

The methodology of *Intelligence Meta-synthesis* has been proposed to study open complex giant systems [168, 173], which has been further expanded to handle open complex intelligent systems [31]. The main idea of the methodology is to develop a meta-synthesis space (m-space) supporting meta-synthetic interaction (m-interaction) and meta-synthetic computing (m-computing) [47]. M-interaction and m-computing engage and facilitate the ubiquitous intelligence through a human-centered, human-machine-cooperated problem-solving process, and the group-based interaction and decision for problem-solving in an m-space.

The performance of D^3M-based actionable knowledge discovery is highly dependent on the recognition, acquisition, representation and integration of relevant factors from human, domain, organization and society, network and web perspectives. By engaging them, we actually want to squeeze out intelligence from the above factors to strengthen the performance of knowledge discovery. Hence, we target the acquisition, integration and use of *in-depth data intelligence*, *human intelligence*, *domain intelligence*, *organizational and social intelligence*, and *network and web intelligence* in domain driven actionable knowledge delivery. To this end, a possible means is by*Intelligence Metasynthesis* [31, 74, 75, 76, 77, 79, 168, 169, 170, 171, 172, 173, 174, 175]. This also forms a key involvement and enhancement in domain driven data mining compared with traditional data-centered data mining methodologies.

The principle of intelligence meta-synthesis is helpful for involving, synthesizing and using ubiquitous intelligence surrounding actionable knowledge discovery in complex data. From a high level perspective, the methodology of intelligence meta-synthesis synthesizes data intelligence with domain-oriented social intelligence, as well as domain intelligence, human intelligence and cyberspace intelligence where appropriate. Domain driven data mining therefore is a process full of interaction and integration among multiple kinds of intelligence, as well as intelligence emergence towards actionable knowledge delivery. Fig. 3.2 outlines the basic units and their interaction in intelligence meta-synthesis for actionable knowledge discovery.

Actionable knowledge delivery through m-spaces aims to

- Acquire and represent unstructured, ill-structured and uncertain domain/human knowledge,
- Support the dynamic involvement of business experts and their knowledge/intelligence,
- Acquire and represent expert thinking such as imaginary thinking and creative thinking in group heuristic discussions during KDD modeling,
- Acquire and represent group/collective interaction behavior and impact emergence, and
- Build infrastructure supporting the involvement and synthesis of ubiquitous intelligence.

Future work will apply the theory to developing an m-space, which is able to support human-centered data mining, dynamic and distributed interaction, group-based

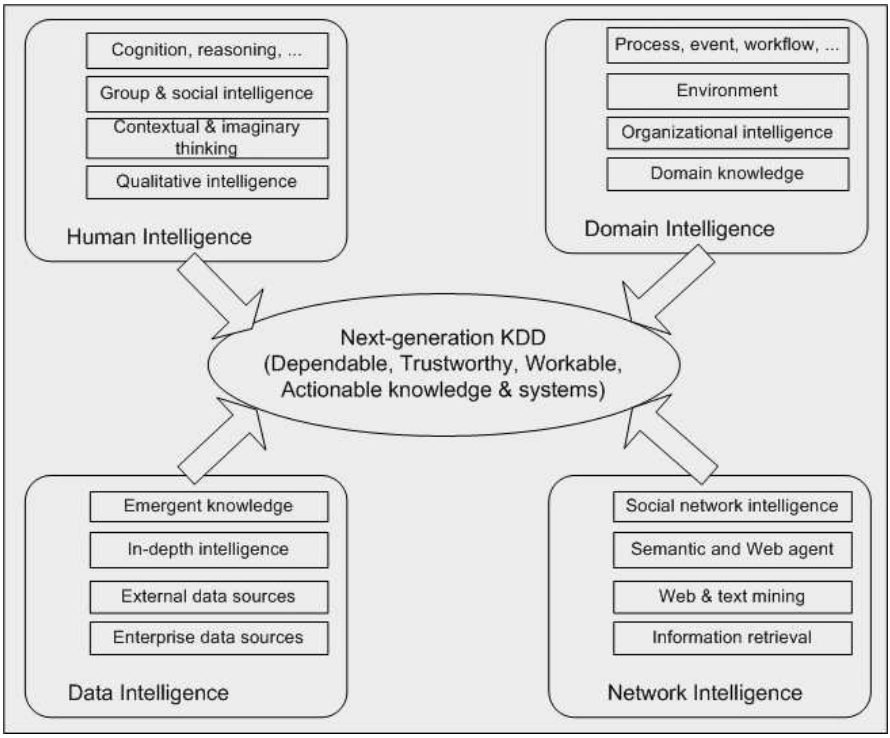

Fig. 3.2 Intelligence meta-synthesis in domain driven data mining

problem-solving, to engage human knowledge and roles in modeling and evaluation, and to satisfy domain knowledge, constraints and organizational factors.

3.9 Summary

In this chapter, we have discussed the key concepts of ubiquitous intelligence surrounding domain driven data mining in the real world. We discussed their definition, aims, aspects and techniques for involving them into data mining.

In summary, we have proposed the following ubiquitous intelligence:

- Data intelligence, which refers to both the general level and in-depth level of data intelligence from both syntactic and semantic perspectives;
- Human intelligence, which refers to both the explicit or direct and implicit or indirect involvement of human intelligence;
- Domain intelligence, which refers to both qualitative and quantitative domain intelligence;

- Network intelligence, which refers to both web intelligence and broad-based network intelligence,
- Organizational intelligence, which refers to organizational goals, structures, rules, and dynamics, and
- Social intelligence,which refers to both human social intelligence and agent/animat-based social intelligence.

Preliminary examples and case studies have been presented to illustrate the concepts. In particular, we have proposed the need to synthesize ubiquitous intelligence in domain driven actionable knowledge delivery. A methodology is intelligence meta-synthesis.

In the next chapter, we will discuss Knowledge Actionability.

Chapter 4
Knowledge Actionability

4.1 Introduction

In both Chapter 1 and Chapter 2 we emphasized the importance of engaging knowledge actionability in actionable knowledge discovery and delivery. This chapter will discuss knowledge actionability.

The goal of this chapter consists of the following aspects:

- Explaining why we need knowledge actionability in more detail,
- Summarizing the related work on knowledge actionability research and development,
- Proposing a knowledge actionability framework for actionable knowledge discovery and delivery,
- Discussing issues such as inconsistency between technical and business aspects in defining, quantifying, refining and assessing knowledge actionability,
- Providing examples and discussions for handling issues in developing and utilizing knowledge actionability,
- Finally, illustrating the use of the proposed framework and concepts in real-world data mining applications.

Correspondingly, this chapter will introduce the following contents.

- In Section 4.2, we discuss the reasons for knowledge actionability in actionable knowledge discovery and delivery;
- Section 4.3 summarizes the state-of-the-art of the related work on knowledge actionability, drawing the conclusion that current interestingness systems cannot reflect both technical and business concerns from subjective and objective perspectives;
- In Section 4.4, we discuss the need from technical significance to knowledge actionability, and then present a knowledge actionability framework composed of key concepts of objective technical interestingness, subjective technical interestingness, objective business interestingness, and subjective business interestingness, which can the weak side and satisfy the need of actionable knowledge

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discovery and delivery. Further more, we discuss the conflict of interest between technical significance and business interestingness, and illustrate the development of business interestingness in a real-world data mining application;
- Finally, Section 4.5 discusses the possibility of integrating technical significance with business interestingness, and illustrates a fuzzy aggregation approach to generate a globally ranked final pattern set balancing both technical and business performance.

4.2 Why Knowledge Actionability

As we discuss in Chapter 1, mined patterns often cannot support real user needs in taking actions in the business world [33, 42, 114, 185]. This may be due to many reasons, but in particular:

- Business interestingness is rarely considered in current pattern mining. For instance, in stock data mining, data miners normally evaluate the mined trading patterns only in terms of specific technical interestingness measures such as *correlation coefficient*. Traders, who are given these findings frequently care little about these performance measures; rather, they prefer to primarily check business-oriented metrics like *profit* and *return* of trading on those identified patterns.
- Very limited preliminary work has been conducted on developing subjective and business oriented interestingness measures, and little of this aims at a standard and general measurement.
- There is an issue in developing business interestingness metrics. Should they be more general or more specific?

With the accumulation of massive data everywhere, and with the increasingly emergent need to analyze the data, it is important to consider and quantify business expectations in pattern evaluation. Due to domain-specific characteristics [33, 42], it is difficult to capture and satisfy particular business expectations in a general manner. A more practical way is to cater for specific business expectations in a real-world mining application, and to develop business interestingness metrics. For instance, in capital markets, *profit* and *return* are often used to justify whether a trade is acceptable or not from an economic performance perspective.

Furthermore, business interestingness can also be instantiated into *objective* [106, 120] and *subjective* [140, 184] measures just like technical metrics. For example, "beat VWAP" is an empirical tool used by traders to objectively measure whether a price-oriented trading rule is confident enough to "beat" the *value-weighted average price* (VWAP) of the target market. By contrast, a stock may be subjectively evaluated in terms of certain psychoanalytic factors by a user to determine whether to go ahead with it or not. This discussion tells us that, besides technical interestingness, actionable knowledge discovery needs to consider *both objective and subjective business interestingness measures*.

Aiming at involving and satisfying business expectations in actionable knowledge discovery, this chapter discusses a *two-way significance framework*, highlighting not only technical interestingness but business interestingness in actionable knowledge discovery. The assumption is that *an extracted pattern is actionable only if it can satisfy both technical concerns and business expectations*. However, very often there is a gap or incompatibility between the values of technical and business measures for a pattern. It is not effective or feasible to simply merge two types of metrics. To this end, interestingness merger techniques such as fuzzy aggregation methods need to be developed to combine technical and business interestingness, and re-rank the mined patterns through balancing two-side concerns.

The two-way significance framework constitutes one of the essential constituents of *domain driven data mining* [33, 42, 56]. Its further development fundamentally contributes to the paradigm shift from data-driven hidden pattern mining to domain driven actionable knowledge discovery.

4.3 Related Work

The actionable capability of discovered knowledge has attracted increasing attention in retrospection on the traditional trial-and-error data mining process, and during the development of next-generation KDD methodology and infrastructure [6, 9, 33]. In fact, knowledge actionability is a critical constituent of domain driven data mining. The aims and objectives of domain driven data mining research is to push data mining research toward bridging the gap between academia and business, and to support business decision making based on the identified patterns. To this end, both technical significance and business expectations should be considered and balanced.

Limited work has been explored in developing effective knowledge actionability and business interestingness measures. Currently, preliminary efforts in developing an effective knowledge actionability framework can be roughly categorized as follows:

- Studying interestingness metrics highlighting the significance of subjective technical concerns and performance,
- Involving domain and prior knowledge in the search of actionable patterns,
- Converting and extracting actionable rules in mined results through techniques such as post-mining and post-processing, and
- Developing new theoretical frameworks catering for and measuring the actionability of extracted patterns.

The concept of actionability was initially investigated from the interestingness perspective [106, 120, 146, 158, 184, 185] to filter out pattern redundancy and "explicitly" (straightforward commonsense to business people) interesting patterns through the mining process or during post-processing [223]. A pattern is actionable if a user can obtain benefits (eg., profit [207, 209, 210, 213]) from taking actions on it, which may restore the deviation back to its norm. In particular, subjective mea-

sures such as *unexpectedness* [158, 184], *actionability* [33, 42, 184, 185, 223] (note here that actionability refers to a very specific aspect rather than a general sense of the actionable capability of a pattern), and *novelty* [203] were studied to evaluate pattern actionable capability. These research works highlight specific aspects. As a result, an unexpected pattern contradicts and hence surprises its user's expectations, while it is not necessarily actionable. On the other hand, a novel one is new but not necessarily workable to its users.

In a word, the existing actionability-oriented interestingness development mainly emphasizes technical and general interestingness, which does not necessarily address and satisfy business expectations. Even though it is straightforward, only a few researchers work on developing general business interestingness metrics [33, 42], such as profit mining [207, 209, 210, 213], to reduce the gap between pattern extraction and decision making. Furthermore, the development of general business interestingness is often limited in measuring colorful domain-specific applications and particular business concerns. Therefore, a more reasonable assumption is that business interestingness should be studied in terms of specific domain problems and should involve domain-specific intelligence [33, 42].

Recently, research on a theoretical framework of actionable knowledge discovery has emerged as a new trend. [127] proposed a high-level microeconomic framework regarding data mining as an optimization of decision "utility". [210] built a product recommender which maximizes net profit. Action rules [204] are mined through distinguishing all stable attributes from flexible ones. Other additional work includes enhancing the actionability of pattern mining in traditional data mining techniques such as association rules [146], multi-objective optimization in data mining [107], role model-based actionable pattern mining [209], cost-sensitive learning [91], and post-processing [223]. All in all, the above research mainly attempts to enhance the existing interestingness system from the technical perspective.

In contrast to the above work and traditional data-centered perspective, the proposed domain driven data mining methodology [27, 29, 33, 42] highlights actionable knowledge discovery. It involves and synthesizes domain intelligence, human intelligence and cooperation, network intelligence and in-depth data intelligence to define, measure and evaluate business interestingness and knowledge actionability. The objective is to develop theoretical and workable methodologies and techniques to convert academic outcomes directly to operationalizable business rules by reflecting business constraints, expectations and existing logics. We believe the research on domain driven actionable knowledge discovery can provide a general, concrete and practical guideline and hints on knowledge actionability research from both theoretical and practical perspectives.

4.4 Knowledge Actionability Framework

This section presents the two-way significance framework for actionable knowledge discovery. The framework measures knowledge actionability from not only a

technical perspective, but also from business perspectives. It synthesizes business expectation and traditional technical significance in justifying pattern interestingness.

4.4.1 From Technical Significance to Knowledge Actionability

The development of actionability is a progressive process in data mining. In the framework of traditional data mining, the so-called actionability ($act()$) mainly emphasizes technical significance by developing varying technical interestingness metrics. Technical interestingness metrics usually measure whether a pattern is of interest or not in terms of specific statistical significance and pattern structure corresponding to a particular data mining method.

There are two steps in technical interestingness evolution. The original focus was basically on objective technical interestingness ($tech_obj()$) [106, 120]. This focuses on capturing the complexities of pattern structure and statistical significance. Recent work appreciates subjective technical measures ($tech_sub()$) [140, 158, 184], which also recognize to what extent a pattern is of interest to a particular user. For example, probability-based belief [185] is developed to describe user confidence of unexpected rules [158]. We summarize these two phases as follows.

Let $X = x_1, x_2, ..., x_m$ be a set of items, DB be a database that consists of a set of transactions, x be an itemset in DB. Let e be interesting pattern discovered in DB through a modeling method M.

$$Phase\ 1 : \forall x \in X, \exists e : x.tech_obj(e) \rightarrow x.act(e) \tag{4.1}$$

$$Phase\ 2 : \forall x \in X, \exists e : x.tech_obj(e) \wedge x.tech_subj(e) \rightarrow x.act(e) \tag{4.2}$$

Gradually, data miners have realized that the actionability of a discovered pattern must also be assessed in terms of domain user needs. To this end, we propose the concept of *business interestingness*. Business interestingness ($biz_int()$) measures to what degree a pattern is of interest to business needs from social, economic, personal and psychoanalytic factors. It further consists of *objective business interestingness* and *subjective business interestingness*.

Recently, objective business interestingness $biz_obj()$ has been recognized by some researchers, say *profit mining* [210]. Consequently, we get Phase 3:

$$Phase\ 3 : \forall x \in X, \exists e : x.tech_obj(e) \wedge x.tech_subj(e) \wedge biz_obj() \rightarrow x.act(e) \tag{4.3}$$

In existing trading pattern mining, even though there are no of business interestingness metrics available, domain knowledge, traders' experience and suggestions, and the above-mentioned business metrics often used by traders provide the foundation for us to create business interestingness for assessing trading patterns. For

example, we use *sharpe ratio SR* as the fitness function to evaluate the performance of a mined trading rule in terms of both *return* and *risk*. If *SR* is high, the rule likely leads to high return with low risk. Where R_p is *expected portfolio return*, R_f is *risk free rate*, δ is *portfolio standard deviation*.

$$SR = \frac{(R_p - R_f)}{\delta_p} \tag{4.4}$$

Moreover, subjective business interestingness *biz_sub*() also plays an essential role in assessing *biz_int*(). For instance, empirical measures are usually used by experienced traders to roughly evaluate the business performance of a trade. As we will further discuss, the following *index return IR* is used to measure whether *trade return TR* triggered by a mined trading rule can beat *market index return IR* or not.

$$TR = \frac{\sum_{i=1}^{u} P_{a,i} \times V_{a,i} - \sum_{j=1}^{v} P_{b,j} \times V_{b,j}}{I_T} \tag{4.5}$$

$$IR = \frac{(\sum_{i=1}^{n} \frac{Index_{i+1} - Index_i}{Index_i})}{n} \tag{4.6}$$

In a specified trading period, there are n corresponding index values $Index_i$ ($i = 1, \ldots, n$) of a market. $P_{a,i}/V_{a,i}$ and $P_{b,j}/V_{b,j}$ are selling and buying price/volume at trading time t_i and t_j, which are used when executing a mined trading rule in a market. The number of sells u (at ask price and volume) is supposed to be equal to buys v (at bid price and volume). I_T is the total investment value. If $IR > TR$, a mined pattern is not actionable enough. These business measures are used in trading evidence discovery as discussed in the case studies.

Based on the above discussion, we believe knowledge actionability should highlight both academic and business concerns [42], and satisfy a two-way significance. The two-way significance indicates that actionability recognizes, from the technical perspective, the technical approach-oriented performance of an extracted pattern through satisfying nominated criteria; while, from the business aspect, it should also present justification for users to specifically react to the findings and to better service their business objectives. In this case, the satisfaction of technical interestingness may be the antecedent of checking business expectation in many cases. In a word, we view actionable knowledge as that which satisfies not only technical interestingness *tech_int*() but also user-specified business interestingness *biz_int*().

Further, knowledge actionability is instantiated in terms of *objective* and *subjective* factors from both *technical* and *business* sides. Correspondingly, we are migrating into Phase 4 for actionable knowledge discovery.

$$\begin{aligned} Phase\ 4 : \forall x \in X, \exists P : x.tech_obj(e) \wedge x.tech_subj(e) \wedge \\ biz_obj() \wedge biz_sub() \rightarrow x.act(e) \end{aligned} \tag{4.7}$$

In this case, there are two sets of interestingness measures that need to be calculated when a pattern is extracted. For instance, we say a mined association trading rule is technically interesting if it satisfies requests on *support* and *confidence*. Moreover, if it can also beat the expectation of user-specified *market index return (IR)* then it is a generally actionable rule.

The following Table 4.1 indicates the difference of interestingness systems emphasized by traditional data mining in contrast to domain driven actionable knowledge discovery.

Table 4.1 Interestingness of data-driven vs. domain driven KDD.

Interestingness		Traditional KDD	Domain Driven AKD
Technical	Objective	Objective technical *tech_obj*()	Technical objective *tech_obj*()
	Subjective	Subjective technical *tech_subj*()	Technical subjective *tech_subj*()
Business	Objective	-	Objective business *biz_obj*()
	Subjective	-	Subjective business *biz_subj*()
Integrative		-	Actionability *act*()

4.4.2 Measuring Knowledge Actionability

In order to handle the challenge of mining actionable knowledge, it is essential to specify how to measure knowledge actionability. Measuring actionability of knowledge is to recognize statistically interesting patterns permitting users to react to them to better service business objectives. The measurement of knowledge actionability should be from both objective and subjective perspectives.

Let $I = i_1, i_2, ..., i_m$ be a set of items, *DB* be a database that consists of a set of transactions, x is an itemset in *DB*. Let P be an interesting pattern discovered in *DB* through utilizing a model M. The following concepts are developed for domain driven data mining.

Definition 4.1. (Technical Interestingness) The technical interestingness ($tech_int()$) of a pattern is highly dependent on certain technical measures specified for a data mining method. Technical interestingness is further measured in terms of *objective technical measures* ($tech_obj()$) and *subjective technical measures* ($tech_sub()$).

Definition 4.2. (Objective Technical Interestingness) Objective technical interestingness ($tech_obj()$) is embodied by measures capturing the complexities of a pattern and its statistical significance. It could be a set of criteria.

For instance, the following logic formula indicates that an association rule (P) is technically interesting if it satisfies $min_support$ and $min_confidence$.

$$\forall x \in I, \exists P : x.min_support(P) \wedge x.min_confidence(P) \rightarrow x.tech_obj(P) \quad (4.8)$$

Definition 4.3. (Subjective Technical Interestingness) Subjective technical measures ($tech_subj()$) also focus and are based on technical means, and recognize to what extent a pattern is of interest to a particular technical method.

For instance, probability-based belief [158] is developed for measuring the expectedness of a pattern.

Definition 4.4. (Business Interestingness) Business interestingness ($biz_int()$) of a pattern is determined from domain-oriented personal, social, economic, user preference and/or psychoanalytic aspects. Similar to technical interestingness, business interestingness is also represented by a collection of criteria from both *objective* $biz_obj()$ and *subjective* $biz_subj()$ perspectives.

Definition 4.5. (Objective Business Interestingness) Objective business interestingness ($biz_obj()$) measures to what extent the findings satisfy business needs and user preferences based on the objective criteria.

For instance, in stock trading pattern mining, *profit* and *roi* (return on investment) are often used for objectively judging the business potential of a trading pattern. If *profit* and *roi* of a stock price predictor P are satisfied, then P is interesting to be deployed for trading.

$$\forall x \in I, \exists P : x.profit(P) \wedge x.roi(P) \rightarrow x.biz_obj(P) \tag{4.9}$$

Definition 4.6. (Subjective Business Interestingness) Subjective business interestingness ($biz_subj()$) measures business and user concerns from the subjective perspectives such as psychoanalytic factors.

For instance, in stock trading pattern mining, the psycho-index 90% may indicate that a trader thinks it is very promising for real trading purposes.

Based on the above definitions, we can define knowledge actionability as follows.

Definition 4.7. (Knowledge Actionability) Given a pattern P, its actionable capability $act()$ is described as being the degree to which can satisfy both technical interestingness and business one. If both technical and business interestingness, or a hybrid interestingness measure integrating both aspects, are satisfied, it is called an *actionable* pattern.

An actionable pattern P can be expressed in terms of the following two-way significance framework.

$$\forall x \in I, \exists P : x.tech_int(P) \wedge x.biz_int(P) \rightarrow x.act(P) \tag{4.10}$$

An actionable pattern is not only interesting to data and business modelers, but also to business users and decision-makers. A successful discovery of actionable knowledge is a collaborative work between data miners and business users, which satisfies both specific modeling approach-oriented technical interestingness measures $tech_obj()$ and $tech_subj()$, and domain-specific business interestingness

biz_obj() and *biz_subj*(), from objective and subjective perspectives. We can then further express an actionable pattern in terms of the above four aspects.

$$\forall x \in I, \exists P : x.tech_obj() \wedge x.tech_subj() \wedge x.biz_obj() \wedge x.biz_subj() \rightarrow x.act(P) \tag{4.11}$$

Table 4.2 summarizes the two-way significance/interestingness system for actionable knowledge discovery.

Table 4.2 Two-way interestingness system for AKD

T/B	S/O	Symbol
Technical	Objective	Objective technical interestingness *tech_obj*()
	Subjective	Subjective technical interestingness *tech_subj*()
Business	Objective	Objective business interestingness *biz_obj*()
	Subjective	Subjective business interestingness *biz_subj*()
Integrative		Actionability *act*()

4.4.3 Pattern Conflict of Interest

To some extent, due to the selection criteria difference, the interest gap between technical approach-oriented evaluation criteria and business performance expectation is inherent. We classify data mining projects into

- Discovery research projects, which recognize the importance of fundamental innovative research,
- Linkage research projects, which support research and development to acquire knowledge for innovation as well as economic and social benefits, and
- Commercial projects, which develop knowledge that solves business problems.

Interest gap is understood from both input and output perspectives. Input refers to the problem studied, while output mainly refers to algorithms and revenue on problem-solving. Both input and output are measured in terms of academia and business aspects. For instance, academic output of a project is mainly measured by algorithms and their associated findings, while business output may be mainly evaluated according to revenue (namely dollars).

We categorize the input and output focuses of the above projects in terms of a five-scale system, where the presence of a '+' indicates a certain extent of focus for the projects. For instance, $[+++++]$ indicates that the relevant projects fully concentrate on this aspect, while a lesser number of '+' means less focus on the target. The mark in Table 4.3 shows the gap and even conflict of problem definition and corresponding expected outcomes between business and academia.

The interest gap is embodied in terms of the interestingness satisfaction of a pattern. Unfortunately, it is often not easy to identify patterns satisfying both tech-

Table 4.3 Interest gap between academia and business

	Input			Output
	Research issues	Business prob-lems	Algorithms	Revenue
Discovery research	[+++++]	[]	[+++++]	[]
Linkage research	[++]	[+++]	[+++]	[++]
Commercial project	[]	[+++++]	[]	[+++++]

nical and business interestingness. In real-world applications, business interestingness $biz_int()$ of a pattern may differ from or conflict with the technical significance $tech_int()$ that guides the selection of a pattern. Quite often a pattern with convincing $tech_int()$ does not match with the $biz_int()$ standard. This situation happens when a pattern is selected in terms of technical significance only. Contrarily, it is usually the case that a pattern with imperfect $tech_int()$ generates interesting $biz_int()$. In this case, a tradeoff in setting technical and business performance, discussions with domain experts, and follow-up scrutinization of the patterns is necessary.

To illustrate the scenarios outlined in Table 4.3, Table 4.4 further presents examples we identified in mining activity patterns in social security data [64]. For instance, the support and confidence of the activity pattern $I,J \rightarrow \$$ (where I,J and \$ refer to activity codes in social security area) are 0.0003 and 0.0057 respectively, which is very insignificant in a statistical view, while its averaged duration amount is very high (46012.43 dollars) with a duration of around 9 business days. It shows that the business impact of this pattern is high enough, and it is worthy of taking actions on this pattern associated customers, even though it might be a rare business case. This example also shows how the business objective interestingness of a pattern can be measured, in which $\overline{d_amt}()$ and $\overline{d_dur}()$ as well as $risk_{amt}$ and $risk_{dur}$ describe the business performance of the relevant activity patterns associated with debt occurrences in governmental services.

Table 4.4 Possible inconsistency between technical and business metrics

		$tech_int()$		$biz_int()$			
Scenario	*Pattern*	*Support*	*Confidence*	$\overline{d_amt}()$ (cent)	$\overline{d_dur}()$ (min)	$risk_{amt}$	$risk_{dur}$
S_1	$I,J \rightarrow \$$	0.0003	0.0057	4601243	12781	0.505	0.203
S_2	$A,R,U \rightarrow \$$	0.386	0.757	2093	8397	0.047	0.03
S_3	$I,F,J \rightarrow \$$	0.257	0.684	639923	9478	0.185	0.094
S_4	$P,S \rightarrow \$$	0.0005	0.0013	1835	6582	0.084	0.028

In varying real-world cases, the relationship between technical and business interestingness of a pattern P may present as one of four scenarios as listed in Table 4.5. Hidden reasons for the conflict between business and academic interests may come from the neglectful business interest checking in developing models.

Table 4.5 Relationship between technical and business metrics.

Scenario	Relationship Type	Explanation
S_1	$tech_int() \Leftarrow biz_int()$	The pattern P does not satisfy business expectation but satisfies technical significance
S_2	$tech_int() \Rightarrow biz_int()$	The pattern P does not satisfy technical significance but satisfies business expectation
S_3	$tech_int() \Leftrightarrow biz_int()$	The pattern P satisfies business expectation as well as technical significance
S_4	$tech_int() \Leftrightarrow biz_int()$	The pattern P satisfies neither business expectation nor technical significance

Clearly, the scenario S_4 is of no interest to us. Even though in the business world, one may only care about the satisfaction of business expectation, actionable patterns should confirm the scenario S_3: $tech_int() \Leftrightarrow biz_int()$, which highlights two-way significance. The two-way rather than one-way significance scheme indicates that a pattern has both a solid technical foundation and a robust deployment capability in the business world. However, it is a kind of artwork to tune thresholds and balance significance and difference between *tech_int*() and *biz_int*(). Quite often a pattern with high *tech_int*() creates bad *biz_int*(). Contrarily, it is not a rare case that a pattern with low *tech_int*() generates good *biz_int*(). In this case, it is domain users who can and should determine what patterns are to be chosen by negotiation with technical modelers.

In the scenarios of S_1 and S_2, it is a kind of artwork to tune thresholds and their difference between $tech_int()$ and $biz_int()$. In real-world data mining, besides developing proper technical and business interestingness measures, there are many other things to do to reach and enhance knowledge actionability, such as tuning the roles and involvement of human intelligence, domain intelligence, organizational and social intelligence aspects specified for the problem. In particular, it is domain users and their knowledge that play essential roles in tuning the thresholds and managing the difference between $tech_int()$ and $biz_int()$.

In addition, besides the above-discussed work on developing useful technical and business interestingness measures, there are other things important for enhancing knowledge actionability, such as efforts in selecting actionability measures, testing actionability, enhancing actionability and assessing actionability in the domain driven data mining process [33].

4.4.4 Developing Business Interestingness

Business interestingness cares about business concerns and evaluation criteria. These are usually measured in terms of specific problem domains by developing corresponding business measures. There is only limited research on business interestingness development in traditional data mining. Kleinberg et al [127] presented

a framework of the microeconomic view of data mining. Profit mining [210] defined a set of past transactions and pre-selected target items, in wich a model is built for recommending target items and promotion strategies to new customers, with the goal of maximizing net profit. Cost-sensitive learning is another interesting area on modeling the error metrics of modeling and minimizing validation error.

In our work on capital market mining [22, 24], we inherit and re-define financial measures such as *profit*, *return on investment* and *sharpe ratio* to measure the business performance of a mined trading pattern in the market. In mining debt-related activity patterns in social security activity transactions, we specify business interestingness in terms of *benefit* and *risk* metrics; for instance a pattern's *debt recovery rate* and *debt recovery amount* from the benefit perspective are developed to justify the prevention benefit of an activity pattern, while *debt risk* such as *debt duration risk* and *debt amount risk* measure the impact of a debt-related activity sequence on a debt.

In the following, we take the example in Table 4.4 to illustrate how to define business interestingness. The following social security activity sequence illustrates a set of activities leading to debt, where letters A to Z represent different activities, and \$ indicates the occurrence of a debt. For instance, we may find a frequent activity pattern $ACB \rightarrow \$$. We here define how to measure the business interestingness of this pattern.

Suppose the total number of itemsets in this data set is $|D|$, where the number of the pattern ACB is $|ACB|$, then we define debt statistics in terms of the following aspects.

$$< (DABACEKB\$), (AFQCPLSWBTC\$), (PTSLD\$), \\ (QWRTE\$), (ARCZBHY) \ldots > \tag{4.12}$$

Definition 4.8. (Pattern Average Debt Amount) The total *debt amount* $d_amt()$ is the sum of all individual debt amounts d_amt_i $(i = 1, \ldots f)$ in f itemsets holding the pattern ACB. Then we get the *pattern average debt amount* $\overline{d_amt}()$ for the pattern ACB:

$$\overline{d_amt}() = \frac{\sum_1^f d_amt()}{f} \tag{4.13}$$

Definition 4.9. (Pattern Average Debt Duration) *Debt duration* $d_dur()$ for the pattern ACB is the averaged duration of all individual debt durations in f itemsets holding the pattern ACB. *Debt duration* $d_dur()$ of an activity is the number of days a debt remains valid,

$$d_dur() = d.end_date?d.start_date + 1 \tag{4.14}$$

, where $d.end_date$ is the day a debt is completed, while $d.start_date$ is the day a debt is activated.

Pattern average debt duration $\overline{d_dur}()$ is defined as:

$$\overline{d_dur}() = \frac{\sum_1^f d_dur()}{f} \tag{4.15}$$

Definition 4.10. (Pattern Debt Amount Risk) A *pattern's debt amount risk* $risk_{amt}()$ is the ratio of the total debt amount of activity itemsets containing *ACB* to the total debt amount of all itemsets in the data set, denoted by

$$risk(ACB \rightarrow \$)_{amt} \tag{4.16}$$

.

$$risk(ACB \rightarrow \$)_{amt} \in [0,1] \tag{4.17}$$

, the larger, it is the higher the risk of leading to debt.

$$risk(ACB \rightarrow \$)_{amt} = \frac{\sum_1^{|} ACB|d_amt()_i}{\sum_1^{|} D|d_amt()_i} \tag{4.18}$$

Definition 4.11. (Pattern Debt Duration Risk) A *pattern's debt duration risk* $risk_{dur}()$ is the ratio of the total debt duration of activity itemsets containing *ACB* to the total debt duration of all itemsets in the data set, denoted by

$$risk(ACB \rightarrow \$)_{dur}\,. \tag{4.19}$$

Similar to *debt amount risk*,

$$risk(ACB \rightarrow \$)_{dur} \in [0,1]\,, \tag{4.20}$$

the larger it is, the higher the risk that it will lead to debt.

$$risk(ACB \rightarrow \$)_{dur} = \frac{\sum_1^{|} ACB|d_dur()_i}{\sum_1^{|} D|d_dur()_i} \tag{4.21}$$

The above metrics can serve on business performance evaluation to measure the impact of a pattern on business outcome. Similar metrics can be developed for specific domains, for instance, in Section 4.4.1, we illustrate some business metrics for actionable trading pattern mining in stock market.

4.5 Aggregating Technical and Business Interestingness

The interest gap between academia and business as shown in scenarios S_1 and S_2 in Table 4.5 indicates the different objectives of two stakeholders. As also shown in Table 4.4, there may be inconsistency between multiple metrics belonging to the same categories. For instance, high $\overline{d_amt}$ does not mean a definitely corresponding high $risk_{dur}$. This indicates that it is necessary to develop techniques to balance

technical and business interestingness, so that a tradeoff and a final ranking can be set up to measure the global performance of a pattern set by consolidating both technical and business performance.

To fill the gap or resolve the conflict in those relevant scenarios, different kinds of action can be taken.

- First, a careful design of not only technical but also business interestingness metrics is necessary. If both ends of evaluation metrics can be designed, a full picture of the understanding of the pattern significance can be provided.
- Second, in designing technical and business interestingness metrics, it is important to consider the relationship between the two sets of metrics. It is helpful if explanatory statements can be provided by both technical and business experts for explaining inconsistent situations.
- Third, it is very helpful to involve domain knowledge and technical-business expert cooperation as much as possible during the whole mining process, in particular, during the definition and refinement of interestingness measures and their thresholds, and the filtering and pruning of the initially extracted pattern set.
- Fourth, naturally, developing a hybrid interestingness measure integrating both business and technical interestingness may reduce the burden of requiring domain users to understand those jargons and merge both-side expectations into a one-stop actionability measure.

With regard to the interestingness merge, the potential incompatibility in some cases between technical significance and business expectation makes it difficult to aggregate the two sides of metrics. A simple weight-based integration does not work due to internal inconsistency among measures. Therefore, rather than taking a conventional weighted-formula approach, we develop a fuzzy interestingness aggregation method combining $tech_int()$ and $biz_int()$ to re-rank the mined pattern set.

The idea of fuzzy aggregation of technical and business interestingness is as follows. Even though simple fuzzy aggregation of interestingness measures can be viewed as a fuzzily weighted approach, we deal with this from the pattern rather than measure perspective. Through defining fuzzy sets supervised by business users, we first fuzzify the extracted patterns into two sets of fuzzily ranked pattern sets in terms of fitness functions $tech_int()$ and $biz_int()$, respectively. We then aggregate these two fuzzy pattern sets to generate a final fuzzy ranking. This final fuzzily ranked pattern set is recommended to users for their consideration. Although this strategy is a little bit fuzzy, it combines two-side interestingness while balancing individual contributions and diversities.

In fuzzifying the pattern set in terms of specific interestingness measures, the universe of discourse of a fuzzified measure must be in $[0,1]$. In the following, we explain the fuzzy aggregation and ranking principle through a simple example. As illustrated in Figures 4.1 and 4.2, suppose we use two sets of five ascending linguistic values to fuzzify and segment the whole pattern set, we then get a fuzzily ranked technical pattern class $T_s = a,b,c,d,e$ and a fuzzy business pattern class $B_s = A,B,C,D,E$ in terms of the fuzzification of $tech_int()$ and $biz_int()$, respectively.

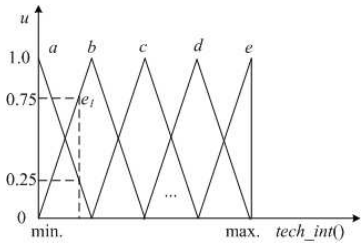

Fig. 4.1 Fuzzily ranked technical pattern class

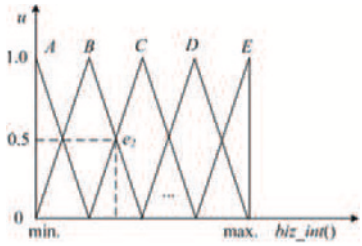

Fig. 4.2 Fuzzily ranked business pattern class

Even though classes T_s and B_s may present the same number of linguistic terms, their semantics, say b and B, are likely to vary. This means that we cannot simply aggregate the corresponding items from two classes into one integrative output through a weighted-formula approach. Instead, we develop fuzzy rules to aggregate these two fuzzy groups to generate a final recommendation list.

Definition 4.12. (Fuzzily Aggregated Pattern Ranking) Given a pattern ranked as $m-th$ in pattern class T_s, while it is ranked as $n-th$ in class B_s, its aggregated ranking is $(m+n-1)-th$.

For example, if a pattern P is technically ranked $c(3^{rd})$, while as $D(4^{th})$ from the business perspective, then its final fuzzy ranking is $6^{th}(3+4-1)$ in the final aggregated pattern set. Further, the following definition specifies the length of the final aggregated pattern set.

Definition 4.13. (Length of Final Aggregated Pattern Set) If the mined target patterns are ranked in terms of t_s levels based on technical interestingness, while they are ranked into b_s levels from business perspective, then after aggregating these two ranking classes, the length of the final aggregated ranking set is (t_s+b_s-1).

The above aggregation and ranking is based on the fuzzification of fitness and membership functions. As a result, pattern ranking presents uncertainty. For example, as shown in Figures 4.1 and 4.2, the trading pattern e_1 can be ranked into group b with membership grade $\mu = 0.75$ or a with grade $\mu = 0.25$. Similar situations may happen to the business class, the pattern e_2 can be segmented into B or C with

same grade $\mu = 0.5$. In this example, the outcomes of the fuzzily aggregated ranking present three options, namely $2^{nd}, 3^{rd} or 4^{th}$ in a total of 9 candidates. This will cause users great inconvenience.

To manage the above uncertainty emerging in fuzzy aggregation and ranking, a ranking coefficient ρ based on moment defuzzification is introduced to defuzzify a fuzzy set and to convert it into a floating point that represents the final position.

$$\rho = \frac{\sum_{l=1}^{m} \eta_l \mu_l^T \mu_l^B}{\sum_{l=1}^{m} \mu_l^T \mu_l^B} \tag{4.22}$$

Where, m refers to the number of triggered linguistic values, $l = 1, 2, \ldots, m$ corresponds to each triggered linguistic value. μ_l^T is the membership grade of No. l linguistic term relevant to the technical fitness of a pattern. μ_l^B is the membership grade of No. l linguistic term corresponding to the business interestingness of a pattern. η_l is the centroid of the No. l triggered linguistic value, it is calculated in terms of the moment and the area of each subdivision.

A real number can be calculated to measure a fuzzily aggregated pattern ranking in a relatively crisp manner. For instance, we can calculate to get $\rho = 0.125$ in the above example, which clearly indicates that the pattern is ranked as 3rd out of nine in the finally aggregated ranking set, since its membership grade is 0.75 much larger than grade 0.25 as ranked 4th.

4.6 Summary

In this chapter, we have discussed the key concept, knowledge actionability, in terms of reasons for proposing it, and a framework for defining and quantifying it. Case studies and discussions have been presented to illustrate the key concepts.

Key concepts from this chapter consist of the following:

- Knowledge actionability is essential for bridging the gap between technical approach-based and business impact-oriented expectations on patterns discovered;
- Knowledge actionability is measured in terms of technical interestingness and business interestingness from both subjective and objective perspectives;
- Often there is conflict or inconsistency between technical and business performance, and it is therefore important to develop the techniques and means to handle such situations. To this end, many aspects including domain knowledge, end user experience, organizational and social factors may be helpful, for enhancing knowledge actionability;
- The consolidation of technical and business performance toward globally ranked deliverables may make the life of end users easier in understanding findings; explanations from both technical and domain experts are helpful for understanding the possible inconsistencies.

In Chapter 5, we will discuss architectures and frameworks for actionable knowledge discovery and delivery.

Chapter 5
D^3M AKD Frameworks

5.1 Introduction

The previous chapters have constructed key foundations for domain driven data mining. From this chapter, we start to discuss techniques, means and case studies for implementing domain driven actionable knowledge discoveryAKD and delivery.

This chapter focuses on the following goals:

- Stating the AKD problem from system and micro-economy perspectives to define fundamental concepts of actionability and actionable patterns,
- Defining knowledge actionability by highlighting both technical significance and business expectations that need to be considered, balanced and/or aggregated in AKD,
- Proposing four general frameworks to facilitate AKD, and
- Demonstrating the effectiveness and flexibility of the proposed frameworks in tackling real-life AKD.

In order to present D^3M-based AKD frameworks, Table 5.1 lists key concepts and their abbreviations that will be used in this chapter.

On the basis of our experiences and case studies in several domains, such as social security [57, 64, 233] and capital markets [24, 37], we propose four AKD frameworks. The basic ideas of the four frameworks are as follows.

1. *PA-AKD*: A two-step AKD process. First, *general patterns* are mined based on technical significance; the learned patterns are then filtered and summarized in terms of business expectations, and are converted into operationalizable business rules for business people's use.
2. *UI-AKD*: AKD develops unified interestingness that aggregates and balances both technical significance and business expectation. The mined patterns are further converted into deliverables based on domain knowledge and semantics.
3. *CM-AKD*: A multi-step pattern mining on the data set in terms of a certain combination strategy. The mined patterns in a step may be fed into another mining

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Table 5.1 Key concepts

Notations	Explanations
AKD	Actionable Knowledge Discovery
PA-AKD	Post analysis-based AKD
UI-AKD	Unified interestingness-based AKD
CM-AKD	Combined mining-based AKD
MSCM-AKD	Multi-source + combined mining-based AKD
P	$P = \{p_1, \ldots, p_u\}$ is a pattern set
$\widetilde{P}$	$\widetilde{P} = \{\tilde{p_1}, \tilde{p_2}, \cdots\}$ is an actionable pattern set
$\widetilde{R}$	$\widetilde{R} = \{\tilde{r_1}, \tilde{r_2}, \cdots\}$ is a business rule set
$Int(p)$	Pattern p's interestingness
$act(p)$	Pattern p's actionability

procedure to guide its feature construction and corresponding pattern mining. Individual patterns identified from each step are then merged into final deliverables based on merger strategy, domain knowledge and/or business needs.

4. *MSCM-AKD*: Handles AKD in either multiple data sources or large quantities of data. One of the data sets is selected for mining initial patterns. Some learned patterns are then selected to guide feature construction and pattern mining on the next data set(s). The iterative mining stops when all data sets are mined, and the corresponding patterns are then merged/summarized into actionable deliverables.

This chapter is organized as follows.

- Section 5.3 discusses the work related to frameworks for actionable knowledge discovery.
- Section 5.4 presents a formal view of AKD from the system perspective.
- In Section 5.5, four types of AKD frameworks are detailed and formalized.
- Discussions about the AKD frameworks and open issues are in Section 5.7.

5.2 Why AKD Frameworks

In general, data mining (or KDD) algorithms and tools focus on the discovery of patterns satisfying expected technical significance. The identified patterns are then handed over to business people for further use. Surveys of data mining for business applications following the above paradigm in various domains [56] have shown that business people cannot effectively take over and interpret the identified patterns for business use. This may result from several aspects of challenges besides the dynamic environment enclosing constraints [17].

- There are *often many* patterns mined but they are not informative and transparent to business people who do not know which are *truly interesting and operable* for their businesses.

- A large proportion of the identified patterns may be either *commonsense or of no particular interest* to business needs. Business people feel confused by *why* and *how* they should care about those findings.
- Further, business people often do not know, and are also not informed, *how to interpret* them and *what straightforward actions can be taken* on them to support business decision-making and operation.

The above issues inform us that there is a large gap [41, 42, 61, 114] between academic deliverables and business expectations, as well as between data miners and business analysts. Therefore, it is critical to develop effective methodologies and techniques to narrow down and bridge the gap. Clearly there is a need to develop general, effective and practical methodologies for actionable knowledge discovery (AKD).

One essential way is to develop effective approaches for discovering patterns that not only are *of technical significance* [196], but *also satisfy business expectations* [41], and further *indicate the possible actions* that can be explicitly taken by business people [2, 27]. Therefore, we need to discover *actionable knowledge* that does much more than simply satisfying predefined technical interestingness thresholds. Such actionable knowledge is expected to be delivered in *operable* forms for transparent business interpretation and action-taking.

It has been increasingly recognized that traditional data mining is facing crucial problems in satisfying user preferences and business needs. For example, research work has been reported on developing *actionable* interestingness [2, 41] and subjective interestingness such as profit mining [210] to extract more interesting patterns, and on enhancing the interpretation of findings through explanation [224]. However, the nature of the existing work on actionable interestingness development is mainly technical significance-oriented, e.g., by developing *alternative* and *subjective* metrics. The critical problem to a great extent comes from the oversimplification of complex domain factors surrounding business problems, the universal focus on algorithm innovation and improvement, and the scant attention paid to enhancing KDD system infrastructure to tackle organizational and social complexities in real-world applications.

Fundamental work on AKD is therefore necessary to cater for critical elements in real-world applications such as environment, expert knowledge and operability. This is related to, but goes much beyond, algorithm innovation and performance improvement. To this end, AKD must cater for domain knowledge [227] and environmental factors, balance *technical significance* and *business expectations* from both *objective* and *subjective* perspectives [41], and support automatically *converting patterns into deliverables* in business-friendly and operable forms such as actions or rules. It is expected that the AKD deliverables will be business-friendly enough for business people to interpret, validate and action, and that they can be *seamlessly embedded* into business processes and systems. If that is the case, data mining has good potential to lead to productivity gain, as well as smarter operation and decision-making in business intelligence. Such efforts actually aim at KDD paradigm shift from traditionally *technical interestingness-oriented and data-centered hidden pattern mining*

toward *business use-oriented and domain driven actionable knowledge discovery* [57].

Relevant preliminary work on AKD mainly addresses specific algorithms and tools for the filtration, summarization and post processing [232] of learned rules. There is a need to develop general AKD frameworks that can cater for critical elements in the real world and can also be instantiated into various approaches for different domain problems. To the best of our knowledge, very limited research work has been reported in this regard.

This chapter features the definition and development of several general AKD frameworks from the system viewpoint, which follow the methodology of Domain Driven Data Mining (DDDM, or D^3M for short) [27, 41, 42, 56, 57]. Our focus is on introducing their concepts, principles and processes that are new, effective to AKD, flexible and practical. Such frameworks are necessary and useful for implementing real-world data mining processes and systems, but are often ignored in the current KDD research.

5.3 Related Work

Actionable knowledge discovery is critical in promoting and releasing the productivity of data mining and knowledge discovery for smart business operations and decision-making. Both SIGKDD and ICDM panelists identify it as one of the great challenges in developing next-generation KDD methodologies and systems [9, 102]. In recent years, some relevant work has emerged.

The term 'actionability' measures *the ability of a pattern to suggest to a user that they take some concrete actions to his/her advantage in the real world*. It mainly measures the ability to suggest business decision-making actions. Existing efforts in the development of effective interestingness metrics focus basically on developing and refining *objective technical* interestingness metrics ($t_o()$) [106, 120]. They aim to capture the *complexities of pattern structure and statistical significance*. Other work appreciates *subjective technical* measures ($t_s()$) [140, 158, 184], which also recognize *to what extent a pattern is of interest to particular user preferences*. For example, *probability-based belief* is used to describe user confidence of unexpected rules [158]. There is very limited research on developing business-oriented interestingness, for instance, profit mining [210].

The main limitations for the existing work on interestingness development lie in a number of aspects. Most work is on developing alternative interest measures focusing on technical interestingness only [155]. Emerging research on general business-oriented interestingness is isolated from technical significance. A question to be asked is "what makes interesting patterns actionable in the real world?" For that, knowledge actionability needs to pay equal attention to both *technical* and *business-oriented* interestingness from both *objective* and *subjective* perspectives [41].

With regard to AKD approaches, the existing work mainly focuses on developing post-analysis techniques to filter/prune rules [142], reduce redundancy [134] and

summarize learned rules [142], as well as on matching against expected patterns by similarity/difference [139]. In post analysis, a recent highlight is to *extract actions from learned rules* [223]. A typical effort on learning action rules is to split attributes into 'hard/soft' [223] or 'stable/flexible' [204] to extract actions that may improve the loyalty or profitability of customers. Other work is on action hierarchy [2]. Some other approaches include a combination of two or more methods, for instance, *class association rules* (or *associative classifier*) that build classifiers on association rules ($A \rightarrow C$) [124]. In [124], external databases are input into characterizing the itemsets. In [160], clustering is used to reduce the number of learned association rules. Some other work is on the transformation from data mining to knowledge discovery [101], and developing a general KDD framework to fit more factors into the KDD process [224].

Regarding the existing work, we make the following comments.

- First, existing work often stops at pattern discovery mainly based on technical significance and interestingness. As a result, the summarized 'actions' do not reflect the genuine expectations of business needs and therefore cannot support decision-making.
- Second, most of the existing post-analysis and post-mining focuses on association rules or their combination with some specific methods. This limits the actionability of learned actions and the generalization of proposed approaches for AKD.

To tackle the challenges in real-world KDD and bridge the gap, it is necessary to take a critical view of KDD, such as from micro-economic [127] and system perspectives, and to develop workable methodologies and frameworks to support AKD. To this end, D^3M [42, 57] is proposed to involve ubiquitous intelligence in the AKD process toward the delivery of operable business rules. With the D^3M, this paper proposes four types of general frameworks that can be customized to extract actionable deliverables satisfying both technical significance and business expectations. Additionally, rather than addressing the whole KDD process, as did some related works, this paper only focuses on method/algorithm frameworks towards AKD.

5.4 A System View of Actionable Knowledge Discovery

Real-world data mining is a complex problem-solving system. From the view of systems and micro-economy, the endogenous character of AKD determines that it is an optimization problem with certain objectives under a particular environment.

Let DB be a database related to business problems Ψ, $X = \{x_1, x_2, \cdots, x_L\}$ be the set of items in DB, where x_l ($l = 1, \ldots, L$) be an itemset, and the number of attributes in DB be S. Suppose $E = \{e_1, e_2, \cdots, e_K\}$ denotes the environment set, where e_k represents a particular environment setting for AKD. Further, let $M = \{m_1, m_2, \cdots, m_N\}$ be the data mining method set, where m_n ($n = 1, \ldots, N$) is a method. For method m_n,

suppose its identified pattern set $P^{m_n} = \{p_1^{m_n}, p_2^{m_n}, \cdots, p_U^{m_n}\}$ includes all patterns discovered in DB, where $p_u^{m_n}$ $(u = 1, \ldots, U)$ is a pattern discovered by m_n.

In the real world, data mining is a problem-solving process (R) from business problems Ψ (with problem status τ) to problem-solving solutions Φ:

$$R : \Psi(\tau_1) \rightarrow \Phi(\tau_2) \tag{5.1}$$

From the modeling perspective, such an AKD-based problem-solving process is a state transformation from the source data $DB(\Psi \rightarrow DB)$ to the resulting pattern set $P(\Phi \rightarrow P)$.

Definition 5.1. (Actionable Patterns) Let $\widetilde{P}^{m_n} = \{\tilde{p_1}^{m_n}, \tilde{p_2}^{m_n}, \cdots, \tilde{p_U}^{m_n}\}$ be an *Actionable Pattern Set* mined by method m_n for the given problem Ψ (its data DB), in which $\tilde{p_u}^{m_n}$ is *actionable* for the problem-solving if it satisfies the following conditions:

1.a$t_i(\tilde{p_u}) \geq t_{i,0}$; '$\geq$' indicates the pattern $\tilde{p_u}$ can beat technical interestingness t_i with threshold $t_{i,0}$;

1.b$b_i(\tilde{p_u}) \geq b_{i,0}$; '$\geq$' indicates the pattern $\tilde{p_u}$ can beat business interestingness b_i with threshold $b_{i,0}$;

1.c$R : \tau_1 \xrightarrow{A(\tilde{p_u}^{m_n})} \tau_2$; the pattern can support business problem-solving by taking action A, and correspondingly transform the problem status from the initially non-optimal status τ_1 to the greatly improved τ_2.

Definition 5.2. (Actionable Knowledge Discovery) AKD is an iterative optimization process toward the actionable pattern set $\widetilde{P}$, considering the surrounding business environment and problem states.

$$\begin{aligned} AKD^{e,\tau,m\in M} : DB &\xrightarrow{e,\tau,m_n} P^{m_n} \\ &\longrightarrow O_{p\in P}^{e,\tau,m\in M} Int(p) \\ &\longrightarrow \widetilde{P} \end{aligned} \tag{5.2}$$

where $P = P^{m_1} U P^{m_2}, \cdots, U P^{m_n}$, $Int(.)$ is the interestingness evaluation function, $O(.)$ is the optimization function to extract those $\tilde{p} \in \widetilde{P}$ when $Int(\tilde{p})$ can beat a given benchmark.

For a pattern p, $Int(p)$ can be further measured in terms of *technical interestingness* ($t_i(p)$) and *business interestingness* ($b_i(p)$) [41].

$$Int(p) = I(t_i(p), b_i(p)) \tag{5.3}$$

where $I(.)$ 'aggregates' the contributions of all particular aspects of interestingness. Further, $Int(p)$ can be described in terms of *objective* (o) and *subjective* (s) factors from both *technical* (t) and *business* (b) perspectives.

$$Int(p) = I(t_o(p), t_s(p), b_o(p), b_s(p)) \\ \rightarrow t_o(x,p) \wedge t_s(x,p) \wedge b_o(x,p) \wedge b_s(x,p) \tag{5.4}$$

where $t_o()$ is objective technical interestingness, $t_s()$ is subjective technical interestingness, $b_o()$ is objective business interestingness, and $b_s()$ is subjective business interestingness, and $I \rightarrow$ '$\wedge'$ indicates the 'aggregation'.

In general, $t_o()$, $t_s()$, $b_o()$ and $b_s()$ of practical applications can be regarded as independent of each other. With their normalization (expressed by ˆ), we can get:

$$Int(p) \rightarrow \hat{I}(\hat{t_o}(), \hat{t_s}(), \hat{b_o}(), \hat{b_s}()) \\ = \alpha\hat{t_o}() + \beta\hat{t_s}() + \gamma\hat{b_o}() + \delta\hat{b_s}() \tag{5.5}$$

where α, β, γ and δ are weights respectively.

So, the AKD optimization problem is as follows:

$$AKD^{e,\tau,m\in M} \longrightarrow O_{p\in P}(Int(p)) \\ \rightarrow O(\alpha\hat{t_o}()) \wedge O(\beta\hat{t_s}()) \wedge O(\gamma\hat{b_o}()) \wedge O(\delta\hat{b_s}()) \tag{5.6}$$

Definition 5.3. (Actionability of a Pattern) The *actionability* of a pattern p is measured by $act(p)$:

$$act(p) = O_{p\in P}(Int(p)) \\ \rightarrow O(\alpha t_o(p)) \wedge O(\beta t_s(p)) \wedge O(\gamma b_o(p)) \wedge O(\delta b_s(p)) \\ \rightarrow t_o^{act} \wedge t_s^{act} \wedge b_o^{act} \wedge b_s^{act} \rightarrow t_i^{act} \wedge b_i^{act} \tag{5.7}$$

where t_o^{act}, t_s^{act}, b_o^{act} and b_s^{act} measure the respective actionable performance.

For example, actionable frequent trading pattern mining [24, 37, 39] considers the satisfaction of *support*, *confidence* as well as business performance like *sharpe ratio*. Suppose they are independent, we then expect an actionable trading pattern to concurrently satisfy these metrics in a maximal manner.

Due to the inconsistency often existing in different aspects, we often find identified patterns only fitting in one of the following sub-sets:

$$Int(p) \rightarrow \{\{t_i^{act}, b_i^{act}\}, \{\neg t_i^{act}, b_i^{act}\}, \\ \{t_i^{act}, \neg b_i^{act}\}, \{\neg t_i^{act}, \neg b_i^{act}\}\} \tag{5.8}$$

where '$\neg$' indicates unsatisfactory.

However, in real-world mining, as we know, it is very challenging to find the most actionable patterns that are associated with both 'optimal' t_i^{act} and b_i^{act}. Quite often a pattern with significant $t_i()$ is associated with unconfident $b_i()$. Contrarily, patterns with low $t_i()$ are often associated with confident $b_i()$. Clearly, AKD targets patterns confirming the relationship $\{t_i^{act}, b_i^{act}\}$.

Therefore, it is necessary to deal with such possible conflict and uncertainty amongst respective interestingness elements. However, it is something of an art form and needs to involve domain knowledge and domain experts to tune thresholds and balance differences between $t_i()$ and $b_i()$. Another issue is to develop techniques to balance and combine all types of interestingness metrics to generate uniform, balanced and interpretable mechanisms for measuring knowledge deliverability and extracting and selecting resulting patterns. A reasonable way is to balance both sides toward an acceptable tradeoff. To this end, we need to develop interestingness aggregation methods, namely the $I - function$ (or '$\wedge$') to aggregate all elements of interestingness. In fact, each of the interestingness categories may be instantiated into more than one metric. Their 'aggregation' does not mean the essential combination into a single super measure, rather indicating the satisfaction of all respective components during the AKD process if possible. They could be checked at the same time or during the AKD processes. There could be several methods of doing the aggregation, for instance, empirical methods such as business expert-based voting, or more quantitative methods such as multi-objective optimization methods.

Besides the measurement, knowledge actionability also needs to cater for the semantic aspect of the identified actionable patterns. This is particularly important in deploying the patterns. Briefly speaking, the conversion from an identified pattern to a business rule can follow a BusinessRule Model [57] defined in OWL-S [19]. To describe a business rule, it is necessary to specify:

- Object: On what object(s) the actions are taken, with predicates to limit the range;
- Condition: Under what situations the actions can be taken on the objects, with predicates to specify the conditions;
- Operation: What actions are to be taken on the objects, with predicates to deliver the specific decision-making activities.

Subsequent to the following specification, actionable patterns are converted into business rules as a form of deliverable, which not only enhances interpretation but also indicates what actions can be taken on what objects under what conditions.

/*BusinessRule Specification*/

$< business_rule >::=< object >^{+}< condition >^{*}< operation >^{+}$

$< object >::= (All|Any|Giveni|...)$

$< condition >::= (satisfy|related|and|...)$

$< operation >::= (Alert|Action|...)$

5.5 Actionable Knowledge Discovery Frameworks

5.5.1 Post Analysis Based AKD: PA-AKD

PA-AKD is a two-step pattern extraction and refinement exercise. First, generally interesting patterns (which we call 'general patterns') are mined from data sets in terms of technical interestingness $(t_o(), t_s())$ associated with the algorithms used. Further, the mined general patterns are pruned, distilled and summarized into operable business rules (embedding actions) (which we call 'deliverables') in terms of domain specific business interestingness $(b_o(), b_s())$ and involving domain (Ω_d) and meta (Ω_m) knowledge. Fig. 5.1 illustrates the PA-AKD.

Based on the system view developed for AKD, PA-AKD is a two-step optimization problem that can be expressed as follows:

$$PA-AKD : DB \xrightarrow{e,t_i(),m_1} P \xrightarrow{e,b_i(),m_2,\Omega_d,\Omega_m} \widetilde{P}, \widetilde{R} \tag{5.9}$$

The following pseudo-code describes the PA-AKD.

FRAMEWORK 1: Post Analysis-based AKD (PA-AKD)
INPUT: target dataset DB, business problem Ψ, and thresholds ($t_{o,0}$, $t_{s,0}$, $b_{o,0}$ and $b_{s,0}$)
OUTPUT: actionable patterns $\widetilde{P}$ and operable business rules $\widetilde{R}$
Step 1: Extracting general patterns P;
  FOR $n = 1$ to N
    Develop modeling method m_n with technical
    interestingness $t_i()$ (i.e., $t_o(), t_b()$);
    Employ method m_n on DB and environment e;
    Extract the general pattern set P^{m_n};
  ENDFOR
Step 2: Extracting actionable patterns $\widetilde{P}$;
  $P = P^{m_1} U \ldots U P^{m_N}$
  FOR $j = 1$ to $(count(P))$
    Design post-analysis method m_2 by involving domain
    knowledge Ω_d and business interestingness $b_i()$;
    Employ the method m' on the pattern set P as well
    as data set DB if necessary;
    Extract the actionable pattern set $\widetilde{P}$;
  ENDFOR
Step 3: Converting pattern $\widetilde{P}$ to business rules $\widetilde{R}$.

The key point in this framework is to utilize both domain/meta knowledge and business interestingness in post-processing the learned patterns. In the real world, this framework can be further instantiated into varied mutations [139, 142, 223]. In fact, many existing methods, such as pruning redundant patterns, summarizing and aggregating patterns to reduce the quantity of patterns, and constructing actions on

top of learned patterns, can be further enhanced by expanding the PA-AKD framework and introducing business interestingness and domain/meta knowledge into the AKD process. [24] presents examples of considering domain knowledge and organizational factors in extracting actionable trading strategies in stock markets. [232] collects case studies of utilizing the PA-AKD framework for extracting effective associations.

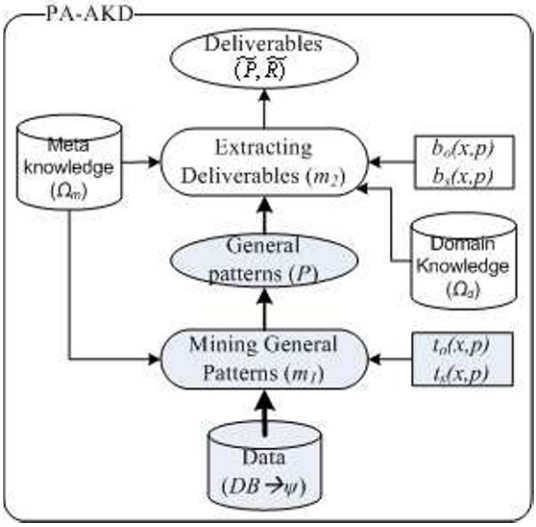

Fig. 5.1 Post analysis based AKD (PA-AKD) approach

5.5.2 Unified Interestingness Based AKD: UI-AKD

As discussed in Section 5.4, one of the essential jobs in extracting actionable knowledge is to balance the interestingness concerns of the identified patterns from both technical and business sides. To this end, a straightforward idea is to develop *unified interestingness metrics* capturing and describing both business and technical concerns, and then to extract patterns based on this unified interestingness system. Thus, UI-AKD is based on such a unified interestingness system.

Fig. 5.2 shows the framework of UI-AKD. It looks just the same as the normal data mining process except for three inherent characteristics. One is the interestingness system, which combines technical interestingness ($t_i()$) with business expectations ($b_i()$) into a unified AKD interestingness system ($i()$). This unified interestingness system is then used to extract truly interesting patterns. The second is that domain knowledge (Ω_d) and environment (e) must be considered in the data mining process. Finally the outputs are $\widetilde{P}$ and $\widetilde{R}$.

Ideally, UI-AKD can be expressed as follows:

$$UI-AKD : DB \xrightarrow{e,i(),m,\Omega_d,\Omega_m} \widetilde{P}, \widetilde{R} \tag{5.10}$$

Based on the AKD formulas addressed before, $i()$ can be further expressed as follows.

$$i() = Int() = I(t_i(), b_i()) \tag{5.11}$$

Very often $t_i()$ and $b_i()$ are not dependent, thus

$$i() \rightarrow \eta \hat{t}_i() + \varpi \hat{b}_i() \tag{5.12}$$

Weights η and ϖ reflect the interestingness balance/tradeoff negotiated between data analysts and domain experts in terms of business problem, data, environment and deliverable expectation. In some cases, both weights and aggregation can be fuzzy. In other cases, the aggregation may happen in a step-by-step manner. For each step, weights may be differentiated.

Patterns with $i()$ beating given thresholds (again, this must be mutually determined by stakeholders) come into the actionable pattern list. The pseudocode describing the UI-AKD process is as follows.

FRAMEWORK 2: Unified Interestingness-based AKD (UI-AKD)
INPUT: target dataset DB, business problem Ψ, and thresholds ($t_{O,0}$, $t_{s,0}$, $b_{O,0}$ and $b_{s,0}$)
OUTPUT: actionable patterns $\widetilde{P}$ and business rules $\widetilde{R}$
Step 1: Extracting general patterns P;
  FOR $n = 1$ to N
    Design data mining method m_n by involving
    domain knowledge Ω_d and considering
    environment e against unified interestingness $i()$;
    Employ the method m_n on DB given e and Ω_d;
    Generate pattern set P;
  ENDFOR
Step 2: Extracting deliverables $\widetilde{P}$;
Step 3: Converting $\widetilde{P}$ to business rules $\widetilde{R}$.

In practice, the combination of technical interestingness with business expectations may be implemented by various methods. An ideal situation is to generate a single formula $i()$ integrating t_i and b_i, and then to filter patterns accordingly. If such a uniform metric is not available, an alternative way is to calculate t_i and b_i for all patterns, and then rank them in terms of them respectively. A weight-based voting (weights are determined by stakeholders) can then be taken to aggregate the two ranked lists into a unified pattern set. If there is uncertainty in merging the pattern sets, fuzzy set-based aggregation and ranking may be helpful. [39] introduces fuzzy set-based aggregation of trading rules in stock markets. First, trading rules are identified through Genetic Algorithms. The identified rules are then ranked in terms of

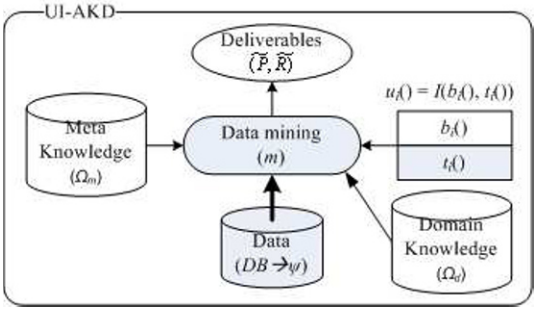

Fig. 5.2 Unified interestingness based AKD approach

technical significance and trading performance respectively, and then fuzzified into five significance levels. The two fuzzy sets are then aggregated in terms of fuzzy aggregation rules into an integrated fuzzy set. Final top-n rules are selected from this set.

As shown in Formula (5.8), in real life, potential incompatibility may exist between technical and business interestingness values for a particular pattern. The relationships between technical interestingness metrics and business ones may be linear and/or nonlinear. In addition, the simple merger of pattern sets divided on technical and business sides may cause uncertainty. These make it very challenging to develop a unified interestingness system for AKD.

5.5.3 Combined Mining Based AKD: CM-AKD

For many complex enterprise applications, one-scan mining seems unworkable for many reasons. To this end, we propose the *Combined Mining [61] based AKD* framework to progressively extract actionable knowledge. Fig. 5.3 illustrates the CM-AKD.

CM-AKD comprises multi-steps of pattern extraction and refinement on the whole data set. First, AKD is split into J steps of mining based on business understanding, data understanding, exploratory analysis and goal definition. Second, generally interesting patterns are extracted based on technical significance ($t_i()$) (or unified interestingness ($i()$)) into a pattern sub-set (P_j) in step j. Third, knowledge obtained in step j is further fed into step $j+1$ or relevantly remaining steps to guide corresponding feature construction and pattern mining (P_{j+1}). Fourth, after the completion of all individual mining procedures, all identified pattern sub-sets are merged into a final pattern set (P) based on environment (e), domain knowledge (Ω_d) and business expectations (b_i). Finally, the merged patterns are converted into business rules as final deliverables ($\widetilde{P}$, $\widetilde{R}$) that reflect business preferences and needs.

CM-AKD can be formalized as follows.

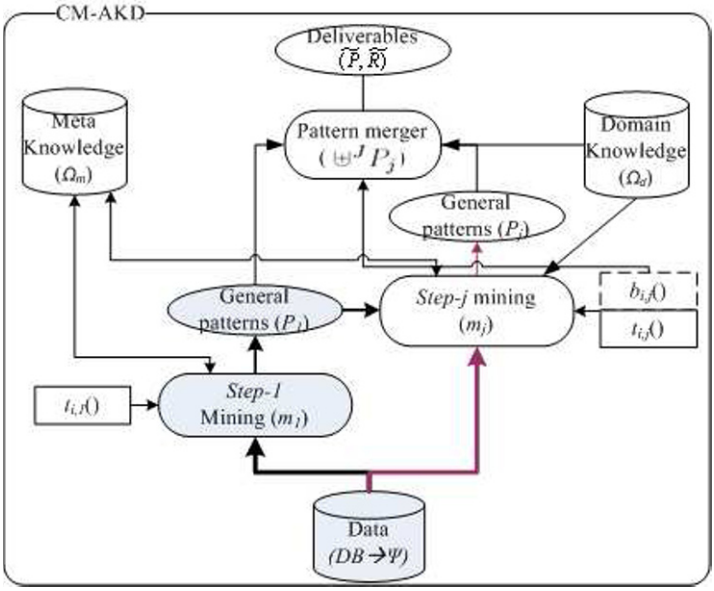

Fig. 5.3 Combined Mining based AKD (CM-ADK)

$$CM-AKD: \underbrace{DB \xrightarrow{e,t_{i,j}()[i_{i,j}()],m_j,\Omega_d,\Omega_m} \{P_j\}}_{J} \xrightarrow{e,b_{i,j}(),\uplus^J P_j,\Omega_d,\Omega_m} \widetilde{P}, \widetilde{R} \tag{5.13}$$

where $t_{i,j}$ and $b_{i,j}$ are technical and business interestingness of model m_j, and $[i_{i,j}()]$ indicates the alternative checking of unified interestingness, $\uplus^J P_j$ is the merger function, Ω_m is the meta-knowledge consisting of meta-data about patterns, features and their relationships.

The CM-AKD process can be expressed as follows.

FRAMEWORK 3: Combined Mining-based AKD (CM-AKD)
INPUT: target dataset DB, business problem Ψ, and thresholds ($t_{o,0}$, $t_{s,0}$, $b_{o,0}$ and $b_{s,0}$)
OUTPUT: actionable patterns $\widetilde{P}$ and operable business rules $\widetilde{R}$;
Step 1: AKD is split into J steps of mining;
Step 2: *Step-j mining*: Extracting general patterns P_j ($j = 1, \ldots, J$);
  FOR $j = 1$ to J
    Develop modeling method m_j with technical
    interestingness $t_{i,j}()$ (i.e., $t_o(), t_b()$) or unified $i_{i,j}()$
    Employ method m_j on the environment e and data
    DB engaging meta-knowledge Ω_m;
    Extract the general pattern set P_j;
  ENDFOR

Step 3: *Pattern merging*: Extracting actionable patterns $\widetilde{P}$;
  FOR $j = 1$ to J
    Design the pattern merger functions $\uplus^J P_j$ by
    involving domain (Ω_d) and meta (Ω_m) knowledge,
    and business interestingness $b_{i,j}()$;
    Employ the method $\uplus P_j$ on the pattern set P_j;
    Extract the actionable pattern set $\widetilde{P}$;
  ENDFOR
Step 4: Converting patterns $\widetilde{P}$ to rules $\widetilde{R}$.

This framework can be instantiated into a few mutations by employing technical and business interestingness at various stages, and by combining miscellaneous data mining models in a multi-step and iterative manner. One example is an *unsupervised + supervised learning based CM-AKD: USCM-AKD*. As shown in Fig. 5.4, the USCM-AKD first deploys an *unsupervised learning method* to mine general patterns in terms of technical interestingness $t_{i,j}()$ associated with the methods m_1. New variables triggered by the unsupervised learning process are added into the meta-knowledge base Ω_m. The original data set is then filtered, transformed and/or aggregated, guided by knowledge obtained in previous learning to generate a transformed data set for further mining. The learned patterns P_1 are then used to guide the extraction of deliverables $\widetilde{P}$ and $\widetilde{R}$ by a *supervised learning method* m_2 on the transformed data set concerning both technical ($t_i()$) and business ($b_i()$) interestingness.

An example is to develop sequential classifiers [236]. First, we mine for the most discriminative sequential patterns, in which an aggressive strategy is used to select a small set of sequential patterns. Second, pattern pruning and serial coverage tests are done on the mined patterns. Those patterns passing the serial test are used to build the sub-classifiers on the first level of the final classifier. Third, the training samples that cannot be covered are fed back to the sequential pattern mining procedure with updated parameters. This process continues until the predefined thresholds are reached or all samples are covered. Patterns generated in each loop form the sub-classifier on each level of the final classifiers.

In addition, the CM-AKD framework can be further joined with the PA-AKD approach to generate a more comprehensive framework: *Combined Mining + Post Analysis based AKD* (CMPA-AKD). In the CMPA-AKD approach, multi-step mining may be conducted by checking technical interestingness only, and leaving the checking of business interests to the post analysis component. In some other cases, multi-step mining is based on unified interestingness while pattern merging is conducted during post analysis.

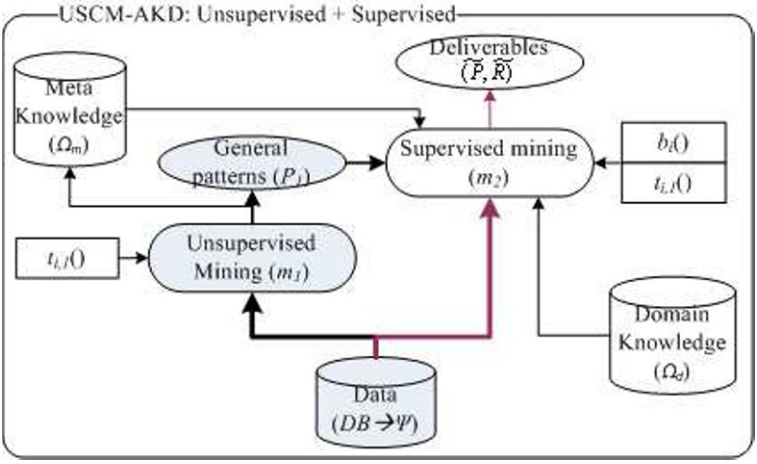

Fig. 5.4 Unsupervised + supervised learning based CM-AKD (USCM-AKD)

5.5.4 Multi-Source + Combined Mining Based AKD: MSCM-AKD

Enterprise applications often involve multiple subsystems-based and heterogeneous data sources that cannot be integrated, or are too costly to do so. Another common situation is that the data volume is so large that it is too costly to scan the whole dataset. Mining such complex and large volumes of data challenges existing data mining approaches. To this end, we propose a *Multi-source + combined mining based AKD* framework. Fig. 5.5 shows the idea of MSCM-AKD.

MSCM-AKD discovers actionable knowledge either in multiple data sets or data sub-sets $(DB_1, \ldots, DB_N)$ through partition. First, based on domain knowledge, business understanding and goal definition, one of the data sets or certain partial data (say DB_n) is selected for mining exploration (m_1). Second, the exploration results are used to guide either data partition or data set management through a *data coordinator agent* Θ_{db} (coordinating data partition and/or dataset/feature selection in terms of iterative mining processes; see more from AMII-SIG[1] regarding agents in data mining), and to design strategies for managing and conducting *parallel pattern mining* on each data set or subset and/or *combined mining* [61] on relevant remaining data sets. The deployment of method m_n, which could be either in parallel or combined, is determined by data/business understanding and objectives. Third, after the mining of all data sets, patterns P_n identified from individual data sets are merged $(\uplus^N P)$ and extracted into final deliverables $(\widetilde{P}, \widetilde{R})$.

MSCM-AKD can be expressed as follows.

[1] www.agentmining.org

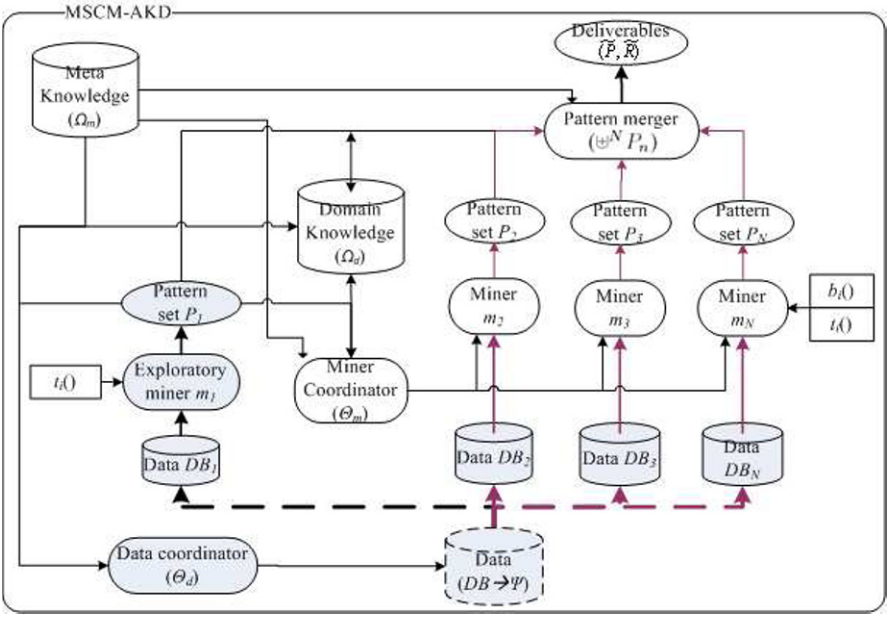

Fig. 5.5 Multi-Source + Combined Mining Based AKD

$$MSCM-AKD:$$
$$\underbrace{DB_n[DB \xrightarrow{\otimes} DB_n]^{e,t_{i,n}()[u_{i,n}()],m_n,\Omega_m} \{P_n\}}_{N}$$
$$\xrightarrow{e,b_{i,n}(),\uplus^N P_n,\Omega_d,\Omega_m} \widetilde{P},\widetilde{R} \tag{5.14}$$

where $t_{i,n}$ and $b_{i,n}$ are technical and business interestingness of model m_n on data set/subset n, and $[i_{i,n}()]$ indicates the alternative checking of unified interestingness as in UI-AKD, $\uplus^N P_n$ is the merger function, $\otimes$ indicates the data partition if the source data needs to be split.

The MSCM-AKD process is expressed as follows.

FRAMEWORK 4: Multi-Source + Combined Mining Based AKD (MSCM-AKD)
INPUT: target datasets DB, business problem Ψ, and thresholds ($t_{o,0}$, $t_{s,0}$, $b_{o,0}$ and $b_{s,0}$)
OUTPUT: actionable patterns $\widetilde{P}$ and business rules $\widetilde{R}$
Step 1: Identify or partition whole source data into N data sets DB_n ($n = 1,\ldots,N$);
Step 2: *Dataset-n mining*: Extracting general patterns P_n on data set/subset DB_n;
FOR $l = n$ to (N)
Develop modeling method m_n with technical
interestingness $t_{i,n}()$ (i.e., $t_o(), t_b()$) or unified $i_{i,n}()$
Employ method m_n on the environment e and
data DB_n engaging meta-knowledge Ω_m;

Extract the general pattern set P_n;
ENDFOR
Step 3: *Pattern merger*: Extracting actionable patterns $\widetilde{P}$;
FOR $l = n$ to N
Design the pattern merger functions $\uplus^N P_n$ to merge all patterns into $\widetilde{P}$ by involving domain and meta knowledge Ω_d and Ω_m, and business interestingness $b_i()$;
Employ the method $\uplus P_n$ on the pattern set P_n;
Extract the actionable pattern set $\widetilde{P}$;
ENDFOR
Step 4: Converting patterns $\widetilde{P}$ to business rules $\widetilde{R}$.

The MSCM-AKD framework can also be instantiated into a number of mutations. For instance, for a large volume of data, MSCM-AKD can be instantiated into *data partition + unsupervised + supervised based AKD* by integrating data partition into combined mining. An example is as follows. First, the whole data set is partitioned into several data subsets based on the data/business understanding and domain knowledge jointly by data miners and domain experts, say data sets 1 and 2. Second, an *unsupervised learning* method is used to mine one of the preference data sets, say data set 1. Some of the mined results are then used to design new variables for processing the other data set. *Supervised learning* is further conducted on data set 2 to generate actionable patterns by checking both technical and business interestingness. Finally, the individual patterns mined from both data subsets are combined into deliverables.

5.6 Case Studies

Substantial experiments show that these frameworks are effective and flexible for extracting actionable knowledge in complex real-world situations, and assist data mining practitioners with catering for their business requirements, needs and decision-making actions on the findings and deliverables in the business environment.

In particular, Chapter 9 will introduce the use of UI-based AKD framework for mining actionable market micro-structure behavior patterns. Both the technical metrics and business performance index are designed for performance evaluation in mining actionable trading strategies.

In Chapter 10, we will introduce a real-life case study in extracting deliverables for government debt prevention. The deliverables take forms of either combining arrangement activities initiated by government officers with repayment activities conducted by debt-associated customers, or combining demographics patterns with arrangement-repayment activity sequential patterns. Government officers who re-

ceive such knowledge feel more comfortable in applying them to their routine processes and rules to prevent debts.

5.7 Discussions

The D^3M views real-world AKD as a *closed optimization* system. 'Closed' indicates the problem-solving is a *closed process* starting from business problem definition and ending with operable business rules fed back into business problem-solving. 'Optimization' means that the AKD targets *optimal* solutions namely *actionable and operable* patterns and produces operable business rules. Based on the above principles, the proposed AKD frameworks present many promising characteristics.

First, the proposed AKD frameworks are *general and flexible*, and can cover many common problems and applications. Basically, they enclose many key features that are critical for offering flexibility and expandability to handle practical challenges in mining complex enterprise applications. These include catering for the organizational environment and domain knowledge (all frameworks care about domain knowledge and environment, and the interestingness system has been expanded to facilitate business concerns), mining multiple data sources [160] and large volumes of data (see MSCM-AKD), post processing the learned patterns as per business needs (see PA-AKD), and supporting multi-step and combined mining (see CM-AKD and MSCM-AKD), as well as closed data mining (by delivering operable business rules, see Section 5.5.4).

These general features, on one hand, can be instantiated into many concrete approaches. As we discussed in introducing each framework, they can be instantiated into various mutations. For instance, the post analysis-based approach can be embodied on top of association mining, clustering and classification to extract actionable associations, clusters and classes. The *unsupervised + supervised* learning process can be instantiated into approaches such as *association + classification and clustering + classification*. On the other hand, they can fit into requirements and constraints in many practical applications, for instance, analyzing rare but significant linkages isolated in multiple organizations' data, and dealing with a complex data structure that mixes heterogeneous and distributed data sources.

Second, the AKD frameworks are *effective and workable* for extracting knowledge that can be taken over by business people for instant decision-making. There are three key factors contributing to effectiveness and workability. (i) The extraction of patterns is based on both technical significance and business expectations, and as a result, they are of business interest. (ii) The frameworks support mining complex data and knowledge in the real world. (iii) The delivery of business rules as the mining deliverables make them operable and bridge the gap between the deliverables and business needs.

In addition, the deep study of D^3M-based AKD has disclosed many open issues and broad prospects in developing next-generation KDD, knowledge process-

ing and decision-support methodologies, techniques and systems for real-world applications.

- *AKD as a closed problem-solving system*: Current KDD is weak in feeding back the resulting solutions to business problems. The extraction of operable business rules presents a feasible way to achieve such an objective. Further work is necessary in defining universal representation and modeling languages for such a purpose.
- *AKD problem-solving environment*: KDD researchers increasingly recognize the significance of understanding, involving and tackling 'environment' factors in AKD modeling and presenting deliverables. Environmental factors refer to the surroundings related to human beings, business process, policies, rules, workflow, organizational factors, and networking factors [29]. Exemplary explanation can be found in [24, 37, 39]. With the system view it is necessary to develop techniques to describe, represent and involve environmental elements and to facilitate the interaction between an AKD system and its environment.
- *Ubiquitous intelligence surrounding and assisting in AKD problem-solving systemsUbiquitous intelligence*: AKD inevitably engages human intelligence, domain intelligence, network intelligence, organizational and social intelligence. Appropriate meta-synthesis [47] of such intelligence can greatly enhance the power of AKD in handling complex data and applications;
- *Representation and integration of ubiquitous intelligence in AKD*: It is necessary to develop effective mechanisms to represent, transform, map, search, coordinate and integrate such ubiquitous intelligence in AKD systems.
- *Actionability checkingActionability checking*: This involves what an actionability system is, and how to evaluate actionability. In AKD, a critical issue is what metrics and at what stage of the AKD process should actionability be checked. Appropriated combination strategies may be necessary for checking actionability from both technical and business perspectives in terms of objective and subjective aspects. In practice, identifying/pruning generally technical-interesting patterns may be conducted first, followed by checking the business interestingness of identified patterns and pruning them accordingly.
- *Status optimization and transferability*: The system status transformation by taking a decision-making action is subject to many factors and constraints such as cost and transferability. It is essential to consider them in cost-sensitive status optimization and transformation. In this instance, the corresponding mechanisms and metrics for cost-benefit analysis, for example, can be helpful.

The effectiveness, general capability, flexibility and adaptability of the proposed AKD frameworks have been tested and demonstrated in several problem domains; for example, mining social security data for debt recovery and prevention [61, 64, 233], and identifying actionable trading strategies [24], and discovering exceptional trading behavior in capital market microstructure data [24, 37, 39]. Due to space limitation, we cannot illustrate these examples one by one. However, interested readers can access more details from the references. In the following section,

we illustrate a case study using the MSCM-AKD framework for discovering actionable combined patterns in social security data. It consists of customer demographic components and customer transactional activities in social security areas to assist government officers to prevent customer debt.

5.8 Summary

This chapter has discussed an important issue in domain driven data mining, namely the development of effective, general and flexible architectures and frameworks for actionable knowledge discovery and delivery. We have proposed four such general and flexible frameworks.

The main conclusions of this chapter are as follows:

- From the viewpoint of actionable knowledge discovery systems, the development of effective, general, flexible and reusable architectures and frameworks is one of the critical issues for actionable knowledge discovery and delivery in complex enterprise data mining applications;
- Post analysis-based actionable knowledge discovery is a commonly used approach to convert initially identified results into more workable outcomes;
- With the two-way significance framework, one option of considering both technical and business interestingness is to unify technical and business interestingness during actionable knowledge discovery;
- Combined mining presents a general and useful framework for mining complex knowledge in complex data; many mutative applications can be designed on top of this framework; for instance, combined pattern mining in multiple data sources.

In the next chapter, we will further discuss the approach of *combined mining*, as an effective technique for actionable knowledge discovery.

Chapter 6
Combined Mining

6.1 Introduction

The main contributions of this chapter include the following:

- Building on existing works, generalizing the concept of combined mining that can be expanded and instantiated into many specific approaches and models for mining complex data toward more informative knowledge.
- Discussing two general frameworks, namely *multi-feature combined mining* and *multi-method combined mining*, and their paradigms and basic processes for supporting combined mining. They also contribute to *multi-source combined mining*. They are flexible for instantiation into specific needs.
- Proposing various strategies for conducting pattern interaction and combination when instantiating the above-proposed frameworks. As a result, novel combined pattern types such as incremental cluster patterns can result from combined mining, which have not previously been investigated before.
- Illustrating the corresponding interestingness metrics for evaluating certain types of combined patterns.
- Proposing a new pattern delivery method, namely *dynamic charts*, to present the evolution and interaction of a cluster of patterns and their associated impacts, which can be easily interpreted by business users and used by them to take decision-making actions.
- Finally, demonstrating the use of combined mining in discovering combined patterns in real-world e-government service data for government debt prevention in an Australian Commonwealth Government Agency.

The chapter is organized as follows. Section 6.2 discusses the needs of inventing combined mining. Section 6.3 states the problem of mining combined patterns. In Section 6.4, we introduce the basic concepts, paradigms and processes of combined mining. Section 6.5 presents the basic frameworks and procedures of the multi-feature combined mining approach. We propose several novel combined patterns including combined pattern pairs, combined pattern clusters, incremental pair patterns and cluster patterns. The corresponding interestingness measures are also

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introduced in this section. Multi-method combined mining is introduced in Section 6.6, in which we present two existing basic frameworks which are parallel and serial multi-method combined mining. In order to achieve a better interaction with multiple methods, we propose closed-loop multiple-method combined mining. As an instance of multi-method combined mining, closed-loop sequence classification is also proposed. Two case studies are demonstrated in Section 6.7, including mining multi-feature combined patterns and sequential classifiers in Centrelink e-government service data for government debt prevention. Related work is addressed in Section 6.8. The chapter is concluded in Section 6.9.

6.2 Why Combined Mining

Enterprise data mining applications inevitably involve complex data sources, for instance, multiple distributed and heterogeneous data sources with mixed structures. In these situations, business people certainly expect the discovered knowledge to present a full picture of the business settings rather than one view based on a single data source. Knowledge reflecting the full business settings is a more business-friendly, comprehensive and informative means for business decision-makers to accept the results, and to take operable actions accordingly. However, there is a challenge in mining for comprehensive, informative and actionable knowledge in complex data suited to real-life decision needs. With the accumulation of ubiquitous enterprise data, there is an increasing need to mine for complex knowledge on such data for informative decision-making.

An example is the pattern analysis of e-government service-related data. In Australia, the Australian Government accumulates a large quantity of data related to e-government services every year. For instance, the Australian Government Agency Centrelink [65] collects more than 5.5 billion transactions related to social security policies, allowances and benefits, debt[1], customer activities, and interactions between Centrelink staff and customers every year. The majority of the data is collected from online social security systems distributed in several regional data storages. The information is associated with multiple aspects such as customer demographics, online service usage, customer-officer interactions, customer earnings declaration, debts, and Centrelink arrangements and customer repayments to pay off debts. Such data consists of heterogeneous data, for instance, categorical, unordered, ordered and numerical data. Findings combining information from the relevant resources can include fruitful information and in-depth knowledge about e-government service quality, performance and impact, as well as indicators for maintaining and improving government service objectives and policies. Consequently, it can enhance e-government service intelligence and decision-making.

Similarly, many business problems involve multiple types of information, for instance, catering for user demographics, preferences, behavior, and business out-

[1] A customer debt indicates overpayment made by Centrelink to a customer who is not entitled.

comes and impacts. With the increasing demand of mining enterprise applications, including multiple and heterogeneous datasets, there is a strong business need to construct patterns consisting of components from multiple heterogeneous datasets.

Mining more informative patterns composed of multiple sources of information is not a trivial task. It actually presents many critical challenges to data mining.

- Traditionally, patterns only involve homogeneous features which come from a single source of data, for instance, frequent patterns of customer repayments. Such single patterns consist of limited information and are not very informative in business decision-making, which often involves multiple issues. If attributes from multiple aspects can be included in the pattern mining, the resulting patterns can then more closely reflect the business situation and be workable in supporting business decision-making.
- In general, it is very costly, space consuming, and sometime impossible, to join such a multiple, heterogeneous and large amount of datasets for centralized pattern analysis. As an alternative, often a single line of business data is mined, although the resulting patterns are not informative enough, and do not reflect the full picture of the business. As a result, their decision-support power is limited or weakened.
- In mining multiple heterogeneous datasets, a single method is often not powerful enough to generate results that are sufficiently sophisticated to match real-world deep and comprehensive issues.

In summary, there is a strong and challenging need to mine for more informative and comprehensive knowledge in multiple large heterogeneous datasets [150, 226] for decision-making in the real world. There is a strong need to develop general approaches to the generation of informative patterns consisting of components from respective sources in order to reflect a wider view rather than a single aspect of business, while avoiding the cost of joining large amounts of heterogeneous data together.

Typical efforts in mining for more informative knowledge in complex data mainly focus on developing specific techniques, for instance, post analysis and post mining [242] of identified results. This converts them into more useful knowledge, and enables multi-step mining of data by applying multiple methods, for instance, classification based on frequent pattern discovery [69], and the combination of association rule mining with classification to generate associative classifiers [141].

In fact, patterns consisting of components from multiple data sources or from multiple methods provides a general approach. There is a need to conceptualize such approaches, and develop general frameworks to support the discovery of more informative patterns.

To this end, in [63, 237, 240], we proposed the concepts of *combined association rules*, *combined rule pairs* and *combined rule clusters* to cater for the comprehensive aspects reflected through multiple datasets. A combined association rule is composed of multiple heterogeneous itemsets from different datasets, while combined rule pairs and combined rule clusters are built from combined association rules. For instance, a combined association rule R is in the form of $R : A_1 \cdots \wedge A_i \wedge B_1 \wedge \ldots B_j \rightarrow$

T, where $A_i \in D_i$ and $B_j \in D_j$ are itemsets in heterogeneous datasets D_i and D_j, $T \neq \emptyset$ is a target item or class and $\forall i, j, A_i \neq \emptyset, B_j \neq \emptyset, A_i \neq B_j$.

In this chapter, we further consolidate the existing works and come up with the concept of *Combined Mining*, as one of the general methods for directly analyzing complex data from multiple sources or with heterogeneous features such as covering demographics, behavior and business impacts. The aim of combined mining is to identify more informative knowledge that can provide a comprehensive presentation of problem solutions. The general ideas of combined mining are as follows.

- By involving multiple heterogeneous features, combined patterns are generated which reflect multiple aspects of concerns and characteristics in businesses;
- By mining multiple data sources, combined patterns are generated which reflect multiple aspects of nature recorded across the business lines;
- By applying multiple methods in pattern mining, combined patterns are generated which disclose a deep and comprehensive essence of the data by taking advantage of different methods; and
- By applying multiple interestingness metrics in pattern mining, patterns are generated which reflect concerns and significance from multiple perspectives.

The deliverables of combined mining are *combined patterns* such as the aforementioned combined association rules. Combined patterns consist of multiple components, a pair or cluster of atomic patterns, identified in individual sources or based on individual methods. As a result of combined mining, the delivery of combined patterns presents an in-depth and more comprehensive indication for decision-making actions, which make the patterns informative and more actionable than patterns composed of single aspects only, or identified by single method-based results.

Rather than presenting a specific algorithm for mining a particular type of combined pattern, this chapter focuses on summarizing and abstracting several general and flexible frameworks from the architecture perspective, which can foster wider implications, and in particular, can be instantiated into specific combination methods and algorithms to mine various combined patterns to tackle the above challenges.

Instantiation capability is regarded as one of the most important issues in the research of general frameworks. Several difficult issues need to be addressed when a combined mining framework is instantiated. First, the generalization capability of proposed frameworks determines their value. We therefore focus on basic concepts, paradigms and basic processes. Second, the generalization capability of proposed combined patterns is very important for producing them by different methods, from different data sources, and of different features. For this purpose, we focus on introducing combined pattern types rather than pattern merging methods. Finally, the relationship amongst the atomic patterns within a combined pattern determines how the combination is produced and measured. We therefore take the relationship analysis as one of the major tasks in constructing the framework.

Table 6.1 Customer Demographic Data

Customer ID	Gender	...
1	F	
2	F	
3	M	
4	M	

Table 6.2 Transactional Data Mixing Ordered and Unordered Data

Customer ID	Policies	Activities	Debt
1	(c_1, c_2)	$(a_1 - a_2)$	Y
2	(c_2, c_4, c_5)	$(a_1 - a_3)$	N
2	(c_2)	$(a_2 - a_3 - a_4)$	Y
2	(c_1, c_3, c_4)	$(a_1 - a_3 - a_4)$	N
3	(c_1, c_3)	$(a_2 - a_3)$	Y
3	(c_1, c_2, c_4)	$(a_1 - a_3)$	Y
4	(c_2, c_4)	$(a_1 - a_2 - a_5)$	N
4	(c_1, c_2, c_4)	$(a_1 - a_3)$	N
4	(c_1, c_2, c_5)	$(a_1 - a_3 - a_4)$	Y
4	(c_2, c_3)	$(a_2 - a_4)$	N

6.3 Problem Statement

6.3.1 An Example

In the real world, data may consist of multiple heterogeneous issues such as demographics, unordered and ordered transactions and business impact or outcomes. An example is data from social security such as the e-government allowance service-related data in Centrelink, Australia. Suppose there are several datasets, consisting of a customer demographic dataset with customer basic information (6.1), unordered government policies applied on customers, ordered customer activities, and the impact of customers on e-government service objectives, namely whether a customer results in having debts or not. For simplicity, let us assume they can be merged into one table as shown in Table 6.2 (in the case studies in Section 6.7, we do not join these datasets simply because each is too large), indicating whether a customer has a debt or not (represented by '*Y*' if yes or '*N*' for no) under various policies or activities. For instance, Centrelink has policies that customers should report their income fortnightly or irregularly, according to each different allowance. Various activities are also conducted with customers, e.g., reviewing by Centrelink, reminder letters sent to customers, and so on.

Table 6.3 Traditional Association Rules

Rules	Supp	Conf	Lift
$c_1 \rightarrow Y$	4/10	4/6	1.3
$c_1 \rightarrow N$	2/10	2/6	0.7
$c_2 \rightarrow Y$	4/10	4/8	1
$c_2 \rightarrow N$	4/10	4/8	1
$c_3 \rightarrow Y$	1/10	1/3	0.7
$c_3 \rightarrow N$	2/10	2/3	1.3
...	...	...	...

Table 6.4 Traditional Sequential Patterns

Rules	Supp	Conf	Lift
$a_1 \rightarrow Y$	3/10	3/7	0.9
$a_1 \rightarrow N$	4/10	4/7	1.1
$a_2 \rightarrow Y$	3/10	3/5	1.2
$a_2 \rightarrow N$	2/10	2/5	0.8
$a_1 - a_2 \rightarrow Y$	1/10	1/2	1
$a_1 - a_2 \rightarrow N$	1/10	1/2	1
$a_1 - a_3 \rightarrow Y$	2/10	2/5	0.8
$a_1 - a_3 \rightarrow Y$	3/10	3/5	1.2
...	...	...	...

Traditionally, such data is mined individually, by frequent pattern mining or classification conducted on the unordered policy data and ordered activity data respectively. For instance, when association mining is used to mine frequent rules, the rules as shown in Table 6.3 can be discovered from the unordered transactional dataset. Similarly, we can identify frequent sequential patterns as shown in Table 6.4.

However, such single frequent patterns are not informative for business decision-makers, because they only reflect one line of business information, do not include information of customer demographics, and are simplified and separated from real business scenarios in which unordered policies and ordered activities are closely related. As a result, the identified patterns do not reflect the full picture and reality of business, and thus are not informative. Their actionable capability is not strong enough to support business decision-making and satisfy user needs.

We now explain the use of combined mining to handle the above problem to produce more informative and actionable patterns.

First, we partition the whole population into two groups, male and female, based on the demographic data in Table 6.1, and then mine the demographic and transactional data of the two groups separately, as partially shown in Table 6.5, where *Cont* denotes the contribution of the transactional data, and I_{rule} reflects the interestingness of the combined rules. (The definitions of *Cont* and I_{rule} will be given in

Table 6.5 Combined Association Rules

Rules	Supp	Conf	Lift	*Cont*	I_{rule}
$F \wedge c_1 \rightarrow N$	2/10	1/2	1	1	1.4
$F \wedge c_2 \rightarrow Y$	2/10	2/3	1.3	1.3	1.3
$M \wedge c_2 \rightarrow N$	3/10	3/5	1.2	1.2	1.2
$M \wedge c_2 \rightarrow Y$	2/10	2/5	0.8	0.8	0.8
...	...	...	...	...	...

Table 6.6 Combined Association Rule Pairs

Pairs	Combined Rules	I_{pair}
$\mathscr{P}_1$	$M \wedge c_3 \rightarrow Y$ $M \wedge c_2 \rightarrow N$	0.55
$\mathscr{P}_2$	$F \wedge c_2 \rightarrow Y$ $M \wedge c_2 \rightarrow N$	0.63
...	... ...	...

Section 6.5.1). We can see from Table 6.5 that: (i) rules are more informative than those in Table 6.3 because they reflect multiple aspects of business, and (ii) more rules with high confidence and lift can be found by combining the rules from two separate datasets.

Second, it is more interesting to organize the rules into contrast pairs as shown in Table 6.6, where I_{pair} is the interestingness of a rule pair. For instance, $\mathscr{P}_1$ is a rule pair for the male group, and it shows that c_3 is associated with debt but c_2 is not. $\mathscr{P}_1$ is actionable in that it suggests c_2 is a preferred policy to replace c_3 and to avoid debt raised on male customers. Moreover, male customers should be excluded when initiating policy c_3. $\mathscr{P}_2$ is a rule pair with the same policy but different demographics. With the same action c_2, male customers have no debts while females tend to have debts. It suggests that c_2 is a preferable policy for male customers but an undesirable policy for female customers.

A simple way to find the rules in Table 6.5 is to join Tables 6.2 and 6.1 in a preprocessing stage, and then apply traditional association rule mining to the derived table. Unfortunately, it is often not feasible from both time and space perspectives to do so in many enterprise applications where there are multiple heterogeneous datasets with each of them consisting of hundreds of thousands of records, or more.

Third, frequent patterns combining unordered and ordered frequent patterns can be identified as shown in Table 6.7. From Table 6.5, we know that male customers under policy c_2 do not tend to have debt. However, in Table 6.7 we can see that if activities $a_1 - a_3$ are taken, male customers under policy c_2 are very likely to have debt since the interestingness I_{pair} of the pattern is as high as 2. Obviously, the ordered activity dataset provides much richer information to allow a more reasonable decision.

Table 6.7 Combined Frequent Patterns with Both Unordered and Ordered Itemsets

Rules	Supp	Conf	Lift	*Cont*	I_{rule}
$M \wedge c_2 \wedge a_1 - a_3 \rightarrow Y$	2/10	2/3	1.3	1.6	2
...	...	...	...	...	...

Table 6.8 Classification on Frequent Combined Patterns

Customer ID	Policies	Activities	Prediction
1	(c_1, c_2)	$(a_1 - a_3)$	Y
2	$(c_2, c_4$	(a_1)	N
...	...	...	

Fourth, classification can be conducted on the identified frequent pattern sets. Table 6.8 shows some examples of the frequent patterns which are further used for classification. Unlike the features in conventional classification on the demographic data, in this frequent pattern based classification, both ordered and unordered transactional data is used to make the decision. As a result, the classification is much more informative.

6.3.2 Mining Combined Patterns

The above examples show the idea of mining patterns with constituents from multiple heterogeneous sources. We make the following observations:

- There is a strong business need to mine for patterns that consist of information from related business lines and settings, to present end users with a comprehensive, or full, rather than biased or incomplete, understanding of the business.
- Patterns identified by traditional methods targeting the discovery from a single dataset or using a single method cannot reflect the full picture of business scenarios, and only indicate limited information for decision-making.
- To identify patterns involving multiple aspects of information, a common method is to join relevant tables if they are matchable, and then conduct pattern analysis on the joint table. However, this is barely workable, if not impossible, when dealing with multiple, large and heterogeneous datasets from time and space perspectives. In particular, if the datasets cannot be joined, patterns cannot be identified.
- On top of existing efforts such as associative classification and classification with frequent patterns, there is a need to develop more general and flexible methodologies to guide the identification of informative patterns through combining itemsets from multiple datasets or using multiple methods.

Combined mining is proposed to address the above issues. It is aimed to be a general approach for identifying patterns composed of components from multiple sources, or from the findings of multiple methods. The main objectives are

- As a concept and general approach for mining actionable patterns, it is necessary to define and formalize the problem of combined mining.
- There is a strong need to develop the frameworks of combined mining to fit the needs of mining complex knowledge in complex data for more informative decision-making. Although it is essential to develop specific approaches and algorithms to support combined mining, it is also important to develop corresponding general combined mining frameworks and approaches that can be customized and instantiated for various problem-solving needs.
- To make the frameworks as general as possible, the frameworks should be flexible, rather than maintaining specific methods or models. To this end, we target the introduction of basic concepts, processes, paradigms, as well as the resulting pattern types which can be instantiated and expanded for specific business needs.
- Compared to developing specific combined mining methods, the resulting general pattern types are more important since they may be identified through different methods. We therefore introduce a number of combined pattern types.
- We will illustrate the applications of some of the combined mining techniques in identifying more informative patterns in e-government service areas for government service performance analysis and quality enhancement.

Motivated by the above examples and analysis, in the remaining sections of this chapter, we present the definition, frameworks, techniques and case studies for combined mining.

6.4 The Concept of Combined Mining

6.4.1 Basic Concepts

For a given business problem (Ψ), we suppose there are the following key entities associated with it in discovering interesting knowledge for business decision-support: Data Set $\mathscr{D}$ collecting all data relevant to a business problem, Feature Set $\mathscr{F}$ including all features for data mining, Method Set $\mathscr{R}$ consisting of all data mining methods that can be used on the data $\mathscr{D}$, Interestingness Set $\mathscr{I}$ composed of all measures from all methods $\mathscr{R}$, Impact Set $\mathscr{T}$ referring to business impacts or outcomes such as fraud or non-fraud, and Pattern Set $\mathscr{P}$. They are described as follows:

- Data Set $\mathscr{D}$: $\mathscr{D} = \{\mathscr{D}_k; k = 1, \ldots, K\}$ consists of all K sub-datasets relevant to the underlying business problem, X_k is the set of all items in the dataset $\mathscr{D}_k$, $\forall k \neq j, X_k \cap X_j = \emptyset$;
- Feature Set $\mathscr{F}$: $\mathscr{F} = \{\mathscr{F}_k; k = 1, \ldots, K\}$ refers to all features used for pattern mining on K sub-datasets, $\mathscr{F}_k$ is the feature set corresponding to the dataset $\mathscr{D}_k$;
- Method Set $\mathscr{R}$: $\mathscr{R} = \{\mathscr{R}_l; l = 1, \ldots, L\}$, $\mathscr{R}_l$ is a data mining method set deployed on the dataset $\mathscr{D}_k$ involving the feature set $\mathscr{F}_k$;

- Interestingness Set $\mathscr{I}$: $\mathscr{I} = \{\mathscr{I}_{m,l}; m = 1,\ldots,M; l = 1,\ldots,L\}$, $\mathscr{I}_{m,l}$ is an interestingness metric set corresponding to a particular data mining method $\mathscr{R}_l$, which is associated with m interestingness metrics; suppose $\mathscr{I}'_k \subset \mathscr{I}$, $\mathscr{I}'_k$ is the interestingness set used by method set $\mathscr{R}'_k$;
- Impact Set $\mathscr{T}$: $\mathscr{T} = \{\mathscr{T}_j; j = 1,\ldots,J\}$ consists of the categorized business impacts associated with certain patterns; in some cases, impacts can be categorized into *impact* (T) and *non-impact* ($\bar{T}$), for instance, fraud or non-fraud. If a pattern is associated with an impact (T), represented by $X \to T$, then we call it an *impact-oriented pattern*. Similarly, if a pattern is mainly relevant to non-impact, indicated by $X \to \bar{T}$, we call it an *non-impact-oriented pattern*;
- Pattern Set $\mathscr{P}$: $\mathscr{P} = \{\mathscr{P}_{n,m,l}; n = 1,\ldots,N; m = 1,\ldots,M; l = 1,\ldots,L\}$, $\mathscr{P}_{n,m,l}$ is an atomic pattern set resulting from a data mining method $\mathscr{R}_l$ using interestingness $\mathscr{I}_{m,l}$, there are $n(n < N)$ atomic patterns in the set; Suppose $\mathscr{P}'_k \subset \mathscr{P}$, $\mathscr{P}'_k$ is the pattern set identified on dataset $\mathscr{D}_k$ using method set $\mathscr{R}'_k$ and interestingness set $\mathscr{I}'_k$.

Based on the above variables, a general pattern discovery process can be described as follows: patterns $\mathscr{P}_{n,m,l}$ are identified through data mining method $\mathscr{R}_l$ deployed on features $\mathscr{F}_k$ from a dataset $\mathscr{D}_k$ in terms of interestingness $\mathscr{I}_{m,l}$.

$$\mathscr{P}_{n,m,l} : \mathscr{R}_l(\mathscr{F}_k) \to \mathscr{I}_{m,l} \tag{6.1}$$

where $n = 1,\ldots,N; m = 1,\ldots,M; l = 1,\ldots,L$.

Combined mining is a process defined as follows.

Definition 6.1 (Combined Mining). Combined Mining is a two to multi-step data mining and post-analysis procedure, consisting of

(1) mining atomic patterns $\mathscr{P}_{n,m,l}$ as described in the Formula (6.1);
(2) merging atomic pattern sets into combined pattern set $\mathscr{P}'_k = \mathscr{G}_k(\mathscr{P}_{n,m,l})$ for each dataset $\mathscr{D}_k$ in terms of pattern merging method $\mathscr{G}_k; \mathscr{G}_k \in \mathscr{G}, \mathscr{G}$ is the pattern merging method set fitting in the characteristics of a business problem;
(3) merging dataset-specific combined patterns into the higher level of combined pattern set $\mathscr{P} = \mathscr{G}(\mathscr{P}'_k)$ if there are multiple datasets.

From a high level perspective, combined mining represents a generic framework for mining complex patterns in complex data as follows:

$$\mathscr{P} := \mathscr{G}(\mathscr{P}_{n,m,l}) \tag{6.2}$$

in which, atomic patterns $\mathscr{P}_{n,m,l}$ from either individual data sources $\mathscr{D}_k$, individual data mining methods $\mathscr{R}_l$ or particular feature sets $\mathscr{F}_k$, are combined into groups with members closely related to each other in terms of pattern similarity or difference.

In combined mining, the word 'combined' principally refers to either one or more of the following aspects on demand:

- The combination of multiple data sources ($\mathscr{D}$): combined pattern set $\mathscr{P}$ consists of multiple atomic patterns identified in several data sources respectively, namely

$\mathscr{P} = \{\mathscr{P}'_k | \mathscr{P}'_k : \mathscr{I}'_k(X_j); X_j \in \mathscr{D}_k\}$; for instance, demographic data and transactional data are two datasets that can be involved in mining for demographic-transactional patterns;

- The combination of multiple features ($\mathscr{F}$): combined pattern set $\mathscr{P}$ involves multiple features, namely $\mathscr{P} = \{\mathscr{F}_k | \mathscr{F}_k \subset \mathscr{F}, \mathscr{F}_k \in \mathscr{D}_k, \mathscr{F}_{j+k} \in \mathscr{D}_{j+k}; j,k \neq 0\}$, for instance, features of customer demographics and behavior;
- The combination of multiple methods ($\mathscr{R}$): patterns in the combined set reflect the results mined by multiple data mining methods, namely $\mathscr{P} = \{\mathscr{P}'_k | \mathscr{R}'_k \rightarrow \mathscr{P}'_k\}$, for instance, association mining and classification.

6.4.2 Basic Paradigms

In this section, we briefly introduce some basic paradigms of combined mining. This involves combined pattern types, structures formed by atomic patterns, and relationships and timeframes amongst atomic patterns.

From the pattern type perspective, combined patterns can be classified into *non-impact-oriented combined patterns* (NICP) and *impact-oriented combined patterns* (ICP), depending on whether a pattern is associated with a certain target item or business impact. For an NICP, its itemsets are associated with each other under certain interestingness metrics, while we do not bother about the impact of the pattern on business outcome.

$$\mathscr{P}_n : R_l(X_1 \wedge \cdots \wedge X_i) \rightarrow I_m, \tag{6.3}$$

$$\mathscr{P} := \mathscr{G}(P_1 \wedge \cdots \wedge P_n) \rightarrow \mathscr{I}. \tag{6.4}$$

An ICP is associated with either a *target* itemset or *resulting impact* ($T_j; T_j \subset T$, T is the target or impact set).

$$P_n : \{R_l(X_1 \wedge \cdots \wedge X_i) \rightarrow I_m\} \rightarrow T_1, \tag{6.5}$$

$$\mathscr{P} := \mathscr{G}(P_1, \ldots, P_n). \tag{6.6}$$

The *number* of constituent atomic patterns in a combined pattern can vary. For example, the following lists two kinds of general structures.

- Pair patterns: $\mathscr{P} ::= \mathscr{G}(P_1, P_2)$, two atomic patterns P_1 and P_2 are co-related to each other in terms of pattern merging method $\mathscr{G}$ into a pair. From such patterns, contrast and emerging patterns [92] can be further identified.
- Cluster patterns: $\mathscr{P} ::= \mathscr{G}(P_1, \ldots, P_n)(n > 2)$, more than two patterns are correlated to each other in terms of pattern merging method $\mathscr{G}$ into a cluster. A group of patterns such as combined association clusters [240] can be further discovered.

Further, the *structural relationships* governing constituent patterns in a combined pattern set can be multiform, and we list a few here:

- *Peer-to-Peer* relation, as illustrated by $\mathscr{P} ::= P_1 \cup P_2$, in which P_1 and P_2 take equal positions in the pair; the pattern exists due to reasons such as similarity or difference from structural or semantic relationship perspectives;
- *Master-Slave* relation, also called *Underlying-Derivative* relation, an example is $\{\mathscr{P} ::= P_1 \cup P_2, P_2 = f(P_1)\}$, in which the existence of pattern P_2 is subject to that of P_1 in terms of function f; an example is $P_2 = P_1 + \Delta P$, where ΔP is the additional part appending to P_1.
- *Hierarchy* relation, as illustrated by $\{\mathscr{P} ::= P_i \cup P_i' \cup P_j \cup P_j', P_j = \mathscr{G}(P_i), \ldots, P_j' = \mathscr{G}'(P_i)'\}$, in which some patterns are correlated in terms of relationship $\mathscr{G}$ while others with $\mathscr{G}'$ or something else.

From the *timeframe* perspective, patterns may be correlated in terms of different temporal relationships, for instance,

- *Independent* relation: as illustrated by $\{P_1 : P_2\}$, in which P_1 and P_2 occur independently from the time perspective;
- *Concurrent* relation: as illustrated by $\{P_1 | P_2\}$, in which P_1 and P_2 occur concurrently;
- *Sequential* relation: as illustrated by $\{P_1; P_2\}$, in which P_2 happens after the occurrence of P_1;
- *Hybrid* relation: as illustrated by $\{P_1 \otimes P_2 \cdots \otimes P_n; \otimes \in \{:, |, ;\}\}$, there are more than two patterns existing in $\mathscr{P}$, in which some of them happen concurrently (|) or independently (:), while others occur sequentially (;).

In Sections 6.5 and 6.6, we will illustrate some of the above pattern types and relationships.

6.4.3 Basic Process

This section discusses a general process of combined mining. In the real world, enterprise applications often involve multiple heterogeneous and distributed data sources that cannot be integrated or are too costly to integrate. Another common situation is where the data volume is so large that it cannot be handled by scanning the whole dataset. Such data has to be partitioned into either small and manageable sets, or in terms of business categories such as billing, networking and accounting data in telecommunication systems. Mining such complex data requires the handling of multi-data sources implicitly or explicitly.

Fig. 6.1 illustrates a framework for combined mining [61]. It supports the discovery of combined patterns either in multiple data sets or data sub-sets ($\mathscr{D}_1, \ldots, \mathscr{D}_K$) through data partitioning in the following manner. 1), based on domain knowledge, business understanding and goal definition, one of the data sets or certain partial data (say $\mathscr{D}_1$) is selected for mining exploration ($\mathscr{R}_1$). 2), the exploration results are used to guide either data partition or data set management through the *data coordinator*, and to design strategies for managing and conducting serial or parallel pattern mining on relevant data sets or subsets, or mining respective patterns on

relevant remaining data sets. The deployment of method $\mathscr{R}_k$ $(k = 2, \ldots, L)$, which could be either in parallel or through combination, is informed by the understanding of the data/business and objectives. If necessary, another step of pattern mining is conducted on data set $\mathscr{D}_k$ with the supervision of results from step $k-1$. 3), after finishing the mining of all data sets, patterns ($P^{\mathscr{R}_n}$) identified from individual data sets are merged ($\mathscr{G}P_n$) with the involvement of domain knowledge, and further extracted into final deliverables ($\mathscr{P}$).

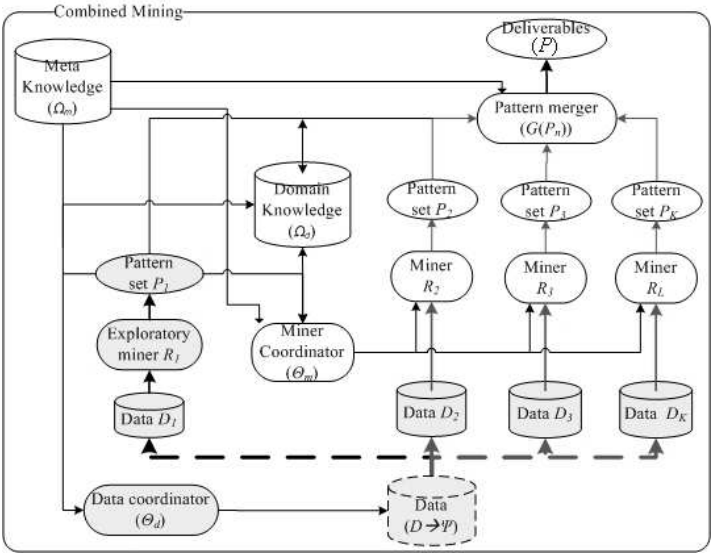

Fig. 6.1 Combined Mining for Actionable Patterns

The above combined mining process can be expressed as follows:

$$\underbrace{\mathscr{D}_k[\mathscr{D} \xrightarrow{\otimes} \mathscr{D}_k]^{e,\mathscr{I}_k,\mathscr{R}_k,\Omega_m} \{\mathscr{P}_k\}}_{K} \tag{6.7}$$

$$\xrightarrow{e,\mathscr{G}^N \mathscr{P}_k,\Omega_d,\Omega_m} \mathscr{P} \tag{6.8}$$

where $\mathscr{I}_k$ is the interestingness of data mining method $\mathscr{R}_k$ on data set/subset $\mathscr{D}_k$, $\otimes$ indicates data partition if the source data needs to be split.

For instance, if multiple data sources are involved in combined mining, the process can be further expressed as follows:

PROCESS: Multi-Source Combined Mining

INPUT: target datasets $\mathscr{D}_k$ $(k = 1, \ldots, K)$, business problem Ψ

OUTPUT: combined patterns $\mathscr{P}$

Step 1: Identify a suitable dataset or data part, say $\mathscr{D}_1$ for initial mining exploration;

Step 2: Identify the next suitable dataset for pattern mining or partition whole source data into K datasets supervised by the findings in Step 1;
Step 3: *Dataset-k mining*: Extracting atomic patterns $\mathscr{P}_k$ on dataset/subset D_k;
FOR $k = 1$ to K
Develop modeling method $\mathscr{R}_k$ with interestingness $\mathscr{I}_k$
Employ method $\mathscr{R}_k$ on the environment e and data $\mathscr{D}_k$ engaging meta-knowledge Ω_m;
Extract the atomic pattern set $\mathscr{P}_k$;
ENDFOR
Step 4: *Pattern merger*: Merging atomic patterns into combined pattern set $\mathscr{P}$;
FOR $k = 1$ to K
Design the pattern merger functions $\mathscr{G}_k$ to merge all relevant atomic patterns into $\mathscr{P}_k$ by involving domain and meta knowledge Ω_d and Ω_m, and the interestingness $\mathscr{I}$;
Employ the method $\mathscr{G}(\mathscr{P}_k)$ on the pattern set $\mathscr{P}_k$;
Generate combined patterns into set $\mathscr{P} = \mathscr{G}_k(\mathscr{P}_k)$;
ENDFOR
Step 5: Enhance pattern actionability to generate deliverables $\mathscr{P}$.
Step 6: Output the deliverables $\mathscr{P}$.

The above framework can be instantiated into a number of mutations. For instance, for a large volume of data, combined mining can be instantiated into *data partition + unsupervised + supervised combined mining* by integrating data partition into combined mining. First, the whole dataset is partitioned into several data subsets based on the data/business understanding and domain knowledge jointly by data miners and domain experts, e.g., data sets 1 and 2. Secondly, *unsupervised learning* is developed to mine one of the preference datasets, say dataset 1. Some of the mined results are then used to design new variables for processing the other dataset. *Supervised learning* is further conducted on dataset 2 to generate actionable patterns by checking both technical and business interestingness. Finally, the individual patterns mined from both datasets are combined into pattern deliverables.

6.5 Multi-Feature Combined Mining

6.5.1 Multi-Feature Combined Patterns

In multi-feature combined pattern mining, a combined pattern is composed of heterogeneous features of: 1) different data types, such as binary, categorical, ordinal and numerical; or of 2) different data categories, such as customer demographics, transactions and time series.

Definition 6.2 (Multi-Feature Combined Patterns). Assume $\mathscr{F}_k$ be the set of features in dataset $\mathscr{D}_k$, $\forall i \neq j$, $\mathscr{F}_{k,i} \cap \mathscr{F}_{k,j} = \emptyset$, based on the variables defined in Section 6.4.1, a Multi-Feature Combined Pattern (MFCP) P is in the form of

Table 6.9 Support, Confidence and Lift of pattern $X \rightarrow T$

Support	$Prob(X \wedge T)$
Confidence	$Prob(X \wedge T)/Prob(X)$
Lift	$Prob(X \wedge T)/(Prob(X) * Prob(T))$

$$\begin{aligned} \mathscr{P}_k &: \mathscr{R}_l(\mathscr{F}_1, \ldots, \mathscr{F}_k) \\ \mathscr{P} &:= \mathscr{G}_F(\mathscr{P}_k) \end{aligned} \tag{6.9}$$

where $\exists i, j, i \neq j, \mathscr{F}_i \neq \emptyset, \mathscr{F}_j \neq \emptyset$, $\mathscr{G}_F$ is the merging method for feature combination.

As shown in Section 6.3, an MFCP example is $F \wedge c_1 \wedge a_1 - a_2 \rightarrow N$. It combines one demographic component, one to many items from transactional datasets and business outcomes, e.g., whether it indicates significant impact of leading to government debt or not in Centrelink.

New evaluation metrics may be necessary to measure the interestingness of ICP. For instance, given a single combined pattern $\mathscr{P} : X_\mathrm{p} \wedge X_\mathrm{e} \rightarrow T$, the traditional *support*, *confidence* and *lift* are given in Table 6.9. Some interestingness measures can be developed on the basis of the work by [155] and [196].

In selecting actionable combined patterns, the contribution of the above traditional interestingness measures is limited. Based on traditional *support*, *confidence* and *lift*, two new metrics, *contribution* and I_rule, are designed as follows for measuring the interestingness of a single combined pattern.

Definition 6.3 (Contribution). For a multi-feature combined pattern $P : X_\mathrm{p} \wedge X_\mathrm{e} \rightarrow T$, the *contribution* of X_e to the occurrence of outcome T in rule P is

$$Cont_\mathrm{e}(X_\mathrm{p} \wedge X_\mathrm{e} \rightarrow T) = \frac{Lift(X_\mathrm{p} \wedge X_\mathrm{e} \rightarrow T)}{Lift(X_\mathrm{p} \rightarrow T)} \tag{6.10}$$

$$= \frac{Conf(X_\mathrm{p} \wedge X_\mathrm{e} \rightarrow T)}{Conf(X_\mathrm{p} \rightarrow T)} \tag{6.11}$$

$Cont_\mathrm{e}(P)$ is the lift of X_e with X_p as a precondition, which shows how much X_e contributes to the rule. *Contribution* can be taken as the increase of *lift* by appending additional items X_e to a rule. Its value falls in $[0, +\infty)$. A *contribution* greater than one means that the additional items in the rule contribute to the occurrence of the outcome, and a *contribution* less than one suggests that it incurs a reverse effect.

Based on the above definition of *contribution*, the interestingness of a single combined pattern is defined as follows.

$$I_\mathrm{rule}(X_\mathrm{p} \wedge X_\mathrm{e} \rightarrow T) = \frac{Cont_\mathrm{e}(X_\mathrm{p} \wedge X_\mathrm{e} \rightarrow T)}{Lift(X_\mathrm{e} \rightarrow T)} \tag{6.12}$$

I_rule indicates whether the *contribution* of X_p (or X_e) to the occurrence of T increases with X_e (or X_p) as a precondition. Therefore, "$I_\mathrm{rule} < 1$" suggests that

$X_p \wedge X_e \rightarrow T$ is less interesting than $X_p \rightarrow T$ and $X_e \rightarrow T$. The value of I_{rule} falls in [0,+∞). When $I_{rule} > 1$, the higher I_{rule} is, the more interesting the rule is.

6.5.2 Pair Pattern

Two atomic patterns or two patterns identified by combined mining may be able to be merged into a pair, forming a *combined pair pattern* (or simply *pair pattern*), defined as follows.

Definition 6.4 (Pair Pattern). For impact-oriented combined mining, a Pair Pattern is in the form of

$$\mathscr{P}: \begin{cases} X_1 \rightarrow T_1 \\ X_2 \rightarrow T_2 \end{cases}, \tag{6.13}$$

where 1) $X_1 \cap X_2 = X_p$ and X_p is called the *prefix* of pair $\mathscr{P}$; $X_{1,e} = X_1 \setminus X_p$ and $X_{2,e} = X_2 \setminus X_p$; 2) X_1 and X_2 are different itemsets; and 3) T_1 and T_2 are contrary to each other, or T_1 and T_2 are same but there is a big difference in their interestingness values of the two constituent patterns.

An example of a pair pattern in Section 6.3 is shown as follows.

$$\begin{array}{l} M \wedge c_3 \rightarrow Y \\ M \wedge c_2 \rightarrow N \end{array} \tag{6.14}$$

It shows that a group of male customers ($X_p = M$ may lead to different business outcomes. That is, c_2 and c_3 have different impact on business outcomes from debt to non-debt. $X_{1,e}$ and $X_{2,e}$ may show different impacts on business.

To illustrate the definition of interestingness of a pair pattern, let us define the interestingness of a combined association rule pair ($I_{pair}()$).

$$I_{pair}(\mathscr{P}) = \begin{cases} |Conf(P_1) - Conf(P_2)|, & \text{if } T_1 = T_2; \\ \sqrt{Conf(P_1)\,Conf(P_2)}, & \text{if } T_1 \text{ and } T_2 \text{ are contrary;} \\ 0, & \text{otherwise;} \end{cases} \tag{6.15}$$

where P_1 and P_2 are the two constituent patterns in the pair $\mathscr{P}$.

I_{pair} measures the contribution of the two different parts in antecedents to the occurrence of different classes in a group of customers with the same patterns. The value of I_{pair} falls in [0,1]. The larger I_{pair} is, the more interesting and actionable a pair of rules is. This kind of knowledge can help to design business campaigns and intervention strategies, and to improve business process.

6.5.3 Cluster Pattern

With combined mining, atomic patterns or combined patterns can be further organized into *clusters* by placing similar or related patterns together, which can be more informative than their constituent patterns. A *cluster pattern* is defined as follows.

Definition 6.5 (Cluster Pattern). Assume there are k atomic patterns $X_i \rightarrow T_i, (i = 1, \ldots, k)$, $k \geq 3$ and $X_1 \cap X_2 \cap \cdots \cap X_k = X_{\mathrm{p}}$, a *cluster pattern* ($\mathscr{P}$) is in the form of

$$\mathscr{P}: \begin{cases} X_1 \rightarrow T_1 \\ \cdots \\ X_k \rightarrow T_k \end{cases}, \tag{6.16}$$

where $k > 2, X_{\mathrm{p}}$ is the *prefix* of cluster $\mathscr{P}$.

Table 10.3 in Section 10.2.1.1 shows examples of cluster patterns.

With regard to the interestingness of a combined pattern cluster, let us illustrate it in terms of association rule clusters. Based on the interestingness of a pair pattern (see I_{pair} in Eqn. (10.7)), for a cluster rule $\mathscr{P}$ with k constituent patterns $P_1, P_2, \ldots, P_k$, its interestingness ($I_{\mathrm{cluster}}()$) is defined as follows.

$$I_{\mathrm{cluster}}(\mathscr{P}) = \max_{P_i, P_j \in \mathscr{C}, i \neq j} I_{\mathrm{pair}}(P_i, P_j) \tag{6.17}$$

The above definition of I_{cluster} indicates that interesting clusters are those rules including interesting rule pairs, and the other rules in the cluster provide additional information. Similar to I_{pair}, the value of I_{cluster} also falls in [0,1].

6.5.4 Incremental Pair Pattern

In some of the pair patterns, there is a certain relationship between items X_1 and X_2. One situation is $X_2 = X_1 \cup X_p, T_1 \neq T_2$, where we then have incremental pair patterns.

Definition 6.6 (Incremental Pair Pattern). An Incremental Pair Pattern is a special pair of combined patterns as follows

$$\mathscr{P}: \begin{cases} X_{\mathrm{p}} \rightarrow T_1 \\ X_{\mathrm{p}} \wedge X_{\mathrm{e}} \rightarrow T_2 \end{cases}, \tag{6.18}$$

where $X_{\mathrm{p}} \neq \emptyset$, $X_{\mathrm{e}} \neq \emptyset$ and $X_{\mathrm{p}} \cap X_{\mathrm{e}} = \emptyset$, X_{e} is a pattern increment part.

The second constituent pattern is an extension of the first, by appending pattern increment X_{e} to it, and the extension X_{e} leads to the difference between the outcomes of the constituent patterns. The relationship between X_p and X_e can be unordered or ordered.

In Section 10.2.1.4, we introduce examples of incremental pair patterns identified in social security data associated with government debt. Another example of incremental pair sequences is the *impact-reversed activity patterns* [63]. An impact-reversed activity pattern consists of an underlying activity pattern and a derivative pattern with an incremental activity sequence X_{e}. In the reversal from one pattern's impact (T_1) to the other's (T_2), the extra itemset X_{e} plays an important role. This phenomenon is of great interest to business. For instance, it can be used for improving business process, recommending activity sequences for avoiding activities or government-customer contacts that may lead to or be associated with debts.

To measure the interestingness of incremental pair patterns, we define the *conditional Piatetsky-Shapiro's ratio Cps* as follows.

Definition 6.7. *Conditional Piatetsky-Shapiro's* (P-S) *ratio Cps* measures the difference led by the occurrence of X_{e} in an incremental pair pattern, which is defined as follows.

$$Cps(X_{\mathrm{e}} \rightarrow T|X_{\mathrm{p}}) = Prob(X_{\mathrm{e}} \rightarrow T|X_{\mathrm{p}}) - Prob(X_{\mathrm{e}}|X_{\mathrm{p}}) \times Prob(T|X_{\mathrm{p}})$$

$$= \frac{Prob(X_{\mathrm{p}} \wedge X_{\mathrm{e}} \rightarrow T)}{Prob(X_{\mathrm{p}})} - \frac{Prob(X_{\mathrm{p}} \wedge X_{\mathrm{e}})}{Prob(X_{\mathrm{p}})} \times \frac{Prob(X_{\mathrm{p}} \rightarrow T)}{Prob(X_{\mathrm{p}})}$$

Cps measures the statistical or proportional significance of incremental sequence X_{e} leading to the impact reversal from T_1 to T_2.

6.5.5 Incremental Cluster Pattern

Similar to incremental pair patterns, for cluster patterns, we have *incremental cluster patterns*. We illustrate here the incremental cluster sequences.

Definition 6.8 (Incremental Cluster Sequences). An Incremental Cluster Sequence is a special cluster of combined patterns with additional items appending to every previously adjacent constituent pattern. An example is as follows.

$$\mathscr{P}: \begin{cases} X_{\mathrm{p}} \rightarrow T_1 \\ X_{\mathrm{p}} \wedge X_{\mathrm{e},1} \rightarrow T_2 \\ X_{\mathrm{p}} \wedge X_{\mathrm{e},1} \wedge X_{\mathrm{e},2} \rightarrow T_3 \\ \cdots \\ X_{\mathrm{p}} \wedge X_{\mathrm{e},1} \wedge X_{\mathrm{e},2} \wedge \cdots \wedge X_{\mathrm{e,k-1}} \rightarrow T_k \end{cases}, \tag{6.19}$$

where $\forall i$, $1 \leq i \leq k-1$, $X_{i+1} \cap X_i = X_i$ and $X_{i+1} \setminus X_i = X_{e,i} \neq \emptyset$, i.e., X_{i+1} is an *increment* of X_i. The above cluster of rules shows the impact of pattern increment on their outcomes.

In Section 10.2.1.3, we illustrate incremental cluster patterns identified in social security data associated with government debt.

Note that in extracting frequent pattern-based incremental cluster patterns, it is not necessary for all constituent patterns to have high interestingness values. For instance, combined association rule clusters don't need high *confidences*. In fact, a pattern with low confidence is also useful because it helps to judge the extent of the negative impact of the incremental part on the pattern and the extent of the positive impact of the incremental part on the next pattern.

A new metric, *impact* is designed as follows to measure the interestingness of incremental cluster sequences.

Definition 6.9 (Impact). The *impact* of X_{e} on the outcome in a cluster pattern is

$$impact_{\mathrm{e}}(P) = \begin{cases} cont_{\mathrm{e}}(P) - 1 & : \text{if } cont_{\mathrm{e}}(P) \geq 1, \\ \frac{1}{cont_{\mathrm{e}}(P)} - 1 & : otherwise. \end{cases} \tag{6.20}$$

Impact measures how much the incremental items change the outcomes, and its values fall in $[0, +\infty)$. To select interesting incremental cluster sequences, one may want to set a threshold for the minimum or the average *impact* in a cluster.

6.5.6 Procedure for Generating Multi-Feature Combined Patterns

Based on the different expectations on combined pattern types, multi-feature combined patterns may be instantiated into pairs, clusters, incremental pairs and clusters. Correspondingly, the discovery of such types of patterns can be segmented into six steps on demand. The process is as follows. First, atomic patterns P_1 are discovered in one dataset and then are used to partition another dataset. Then in a derived sub-dataset, atomic patterns P_2 are discovered. After that, P_1 and P_2 are merged into a combined pattern. Through finding common prefixes or postfixes in these patterns, interesting pair patterns are discovered by putting contrast patterns together. In addition, patterns with the same prefixes or postfixes form cluster patterns. Finally, incremental pair and cluster patterns can be further built upon the identified pattern pairs/clusters, respectively.

METHOD: Mining Multi-Feature Combined Patterns

INPUT: target datasets $\mathscr{D}_k$ $(k = 1, \ldots, K)$, business problem Ψ

OUTPUT: combined patterns $\mathscr{P}$

Step 1. Mining atomic patterns: For each dataset or partitioned dataset $\mathscr{D}_k$, mining for interesting atomic patterns $\mathscr{P}_k$ on the dataset;

Step 2. Combining atomic patterns: Merging relevant atomic patterns identified in the above step as per pattern merging method $\mathscr{G}_k$;

Step 3. Generating pair patterns: Generating pair patterns from the resulting combined patterns. For instance, those patterns with common prefixes but contrary outcomes (or same outcomes but having a big difference in interestingness) form pair patterns;

Step 4. Generating cluster patterns: For each pair pattern, add other related patterns to it to form a cluster pattern;

Step 5. Generating incremental pair patterns: For those pair patterns, if one pattern is an extension of the other, then output it as an incremental pair pattern;
Step 6. Generating incremental cluster patterns: In a cluster pattern, if there is an ordinal relation between the relevant adjacent patterns, and the latter patterns consist of additional information on top of its former ones, output them as incremental cluster patterns.

For instance, as shown in [240], interesting combined rules and rule clusters can be extracted on atomic association rules with interestingness metrics such as *support*, *confidence*, *lift*, $Cont_{\mathrm{e}}$ and I_{rule}. The learned rules with high *support* and *confidence* are also organized into clusters, and then the clusters are ranked by I_{cluster} to find actionable cluster patterns. Section 10.2.1 further illustrates these techniques.

6.6 Multi-Method Combined Mining

6.6.1 Basic Frameworks

Multi-method combined mining is another approach to discover more informative knowledge in complex data. The focus of multi-method combined mining is to combine multiple data mining algorithms as needed in order to generate more informative knowledge. In fact, the combination of multiple data mining methods has been recognized as an essential and effective strategy in dealing with complex applications.

Definition 6.10 (Multi-Method Combined Mining). Assume that there are l data mining methods $\mathscr{R}_l\,(l = 1,\ldots,L)$, their respective interestingness metrics are in the set $\mathscr{I}_m\,(m = 1,\ldots,M)$, features available for mining the dateset is $\mathscr{F}$, *multi-method combined mining* is in the form of:

$$\begin{aligned} \mathscr{P}_l &: \mathscr{R}_l(\mathscr{F}) \rightarrow \mathscr{I}_{m,l} \\ \mathscr{P} &:= \mathscr{G}_M(\mathscr{P}_l) \end{aligned} \tag{6.21}$$

where $\mathscr{G}_M$ is the merging method integrating patterns identified by multiple methods.

In dealing with complex real-world applications, the general process of multi-method combined mining is as follows.

- First, based on domain knowledge, business understanding, data analysis and goal definition, the user determines how many methods, and which methods, should be used in the framework.
- Secondly, the patterns discovered by each method are combined with the patterns by the other methods in terms of merging method $\mathscr{G}$. In reality, the merger could be through either *serial* or *parallel* combined mining.
- Finally, after mining by all methods, the combined patterns are further reshaped into more workable patterns.

In the following sections, we introduce three general frameworks of multi-method combined mining. They are *parallel multi-method combined mining*, *serial multi-method combined mining*, and *closed-loop multi-method combined mining*.

6.6.2 Parallel Multi-Method Combined Mining

One approach to involving multiple methods for combined mining is Parallel Multi-Method Combined Mining.

Definition 6.11 (Parallel Multi-Method Combined Mining). Suppose we have K data sets $\mathscr{D}_k\,(k = 1,\ldots,K)$, L data mining methods $\mathscr{R}_l\,(l = 1,\ldots,L)$ are used to mine them respectively, the *parallel multi-method combined mining* is a process as follows.

- Parallel data mining is conducted on each dataset using different data mining methods to find respective atomic pattern sets.

$$\begin{cases} \mathscr{D}_1 \xrightarrow{e,\mathscr{I}_1,\mathscr{R}_1,\Omega_m} \mathscr{P}_1 \\ \mathscr{D}_2 \xrightarrow{e,\mathscr{I}_2,\mathscr{R}_2,\Omega_m} \mathscr{P}_2 \\ \cdots \\ \mathscr{D}_K \xrightarrow{e,\mathscr{I}_l,\mathscr{R}_l,\Omega_m} \mathscr{P}_n \end{cases} \tag{6.22}$$

- The atomic patterns identified by individual methods are merged into combined patterns in terms of merging method $\mathscr{G}$:

$$\mathscr{P} := \mathscr{G}(\mathscr{P}_1, \mathscr{P}_2, \ldots, \mathscr{P}_n) \tag{6.23}$$

In parallel multi-method combined mining, multiple methods are implemented on multiple data sources or partitioned data sets. The resulting patterns are the combination of the outputs of individual methods on particular data sources.

An example of parallel multi-method combined mining is to mine for demographic patterns on customer demographic data using association rule mining, and at the same time to discover event classes on transactional datasets by a decision tree. The identified results from association rule mining and decision tree-based classification are then merged to form combined patterns: *frequent demographic pattern - event class*. The combined patterns may show that customers with certain frequent demographic characteristics are likely to be further associated with the occurrences of particular types of events.

6.6.3 Serial Multi-Method Combined Mining

The second type of approach to involving multiple methods into combined mining is *serial multi-method combined mining*, which is described as follows.

Definition 6.12 (Serial Multi-Method Combined Mining). Suppose we have L data mining methods $\mathscr{R}_l\,(l = 1,\ldots,L)$, the *serial multi-method combined mining* is a gradual process as follows.

- Based on the understanding of domain knowledge, data, business environment and meta knowledge, select a suitable method (say $\mathscr{R}_1$) on the dataset $\mathscr{D}$. Consequently, we obtain the resulting pattern set $\mathscr{P}_1$:

$$\mathscr{D} \xrightarrow{e,\mathscr{R}_l,\mathscr{F}_l,\mathscr{I}_l,\Omega_m} \mathscr{P}_1, or \tag{6.24}$$

$$\{\mathscr{R}_1,\mathscr{F}_1,\mathscr{I}_1\} \xrightarrow{e,\mathscr{D},\Omega_m} \mathscr{P}_1 \tag{6.25}$$

- Supervised by the resulting patterns $\mathscr{P}_1$ and deeper understanding of the business and data during mining P_1, select the second appropriate data mining methods $\mathscr{R}_2$ to mine $\mathscr{D}$ for pattern set P_2:

$$\{\mathscr{R}_2,\mathscr{F}_2,\mathscr{I}_2\} \xrightarrow{e,\mathscr{D},\Omega_m,\mathscr{P}_1} \mathscr{P}_2 \tag{6.26}$$

  where, $\mathscr{P}_1$ involves and contributes to the discovery of $\mathscr{P}_2$.
- Similarly, select the next data mining method to mine the data with supervision of the corresponding patterns from the previous stages; repeat this process until the data mining objective is met, and we get eventual pattern set P.

$$\{\mathscr{R}_L,\mathscr{F}_L,\mathscr{I}_L\} \rightarrow \mathscr{P} \tag{6.27}$$

In serial multi-method combined mining, data mining methods are used one by one according to specific arrangements. That is, a method is selected and used based on the output of the previous methods. Such serial combination of data mining methods is often very useful for mining complex datasets.

An example is associative classification through gradual deployment of association rule mining and classification [141]. In other cases, association mining can be deployed on top of the results of clustering for rarity mining [164], or vice versa to discover more interesting patterns as in [133, 237, 239]. More examples include the combination of sequential pattern mining and classification [135], classification and clustering [228] and regression and association rule mining [157].

6.6.4 Closed-Loop Multi-Method Combined Mining

Most current combinations of multiple data mining methods are either in parallel or serial. In these two approaches, we generally do not bother about the impact of

one method on the other. For example, in serial multi-method combined mining, a previously applied method R_j in general has no impact on another method R_i's resulting patterns and performance, even though R_j follows R_i. This is actually a common issue in open loop combination.

In practice, the feedback from a latter method's results to its previous methods may assist with the pattern refinement in combination, and enhance the deliverable performance and the efficiency of the data mining process. To this end, we propose the concept of *closed-loop multi-method combined mining*. Its general idea is as follows.

Definition 6.13 (Closed-Loop Multi-Method Combined Mining). Suppose we have dataset $\mathscr{D}$, L data mining methods $\mathscr{R}_l\,(l = 1,\ldots,L)$ are used to mine $\mathscr{D}$, if multiple data mining methods are serially applied, we then conduct *closed-loop multi-method combined mining* through multiple loops of pattern discovery processes as follows.

- Loop 1, follow the process of serial multi-method combined mining to generate pattern set $\mathscr{P}$, through a progressive pattern formation process as shown in the following formula 6.28. During each step of extracting pattern set $\mathscr{P}^1$, there are some samples that cannot be properly identified. This may not indicate that such samples are not identifiable, rather that this is due to the constraints and conditions applied on the respective methods.

$$\{\mathscr{R}_1,\mathscr{F}_1,\mathscr{I}_1\} \rightarrow \{\mathscr{R}_2,\mathscr{F}_2,\mathscr{I}_2\} \rightarrow \ldots\{\mathscr{R}_L,\mathscr{F}_L,\mathscr{I}_L\} \rightarrow \mathscr{P} \tag{6.28}$$

- Loop 2, the patterns identified by data mining methods $\mathscr{R}_l(l = 1,\ldots,L)$ are further checked to see whether the identified patterns are valid to all samples in the dataset $\mathscr{D}$. Those samples on which patterns are not valid, form a dataset $\mathscr{D}^1$ from the dataset $\mathscr{D}$. They are called *exceptional itemsets*. The exceptional itemsets are further fed back to another loop of mining by re-using methods from $\mathscr{R}_1$ through $\mathscr{R}_L$ as needed and with refinement of parameters etc. Suppose we then get another resulting pattern set $\mathscr{P}^2$.
- Repeat the process of Loop 2 as needed. Suppose Z loops are needed, in order that final remaining exceptional itemsets $\mathscr{D}^Z$ that cannot be covered by patterns are within an acceptable level. We correspondingly obtain Z pattern sets in the whole process, namely $\{\mathscr{P}^1,\ldots,\mathscr{P}^Z\}$.
- Merge the identified Z pattern sets to generate final combined patterns.

$$\mathscr{P} := \mathscr{G}_C(\mathscr{P}^1,\mathscr{P}^2,\ldots,\mathscr{P}^Z) \tag{6.29}$$

  where $\mathscr{G}_C$ represents merging methods for closed-loop multi-method combined mining.

Fig.6.2 further illustrates the process of the closed-loop multi-method combined mining. In the closed-loop combination, whether a pattern is interesting or not does not only depend on a particular method that extracts the pattern, but also on the other

methods used in the system. Hence, the performance and efficiency of the system could be much improved by using the same interestingness measures.

In Section 6.6.5, we introduce an example of closed-loop multi-method combined mining, namely *closed-loop sequence classification*.

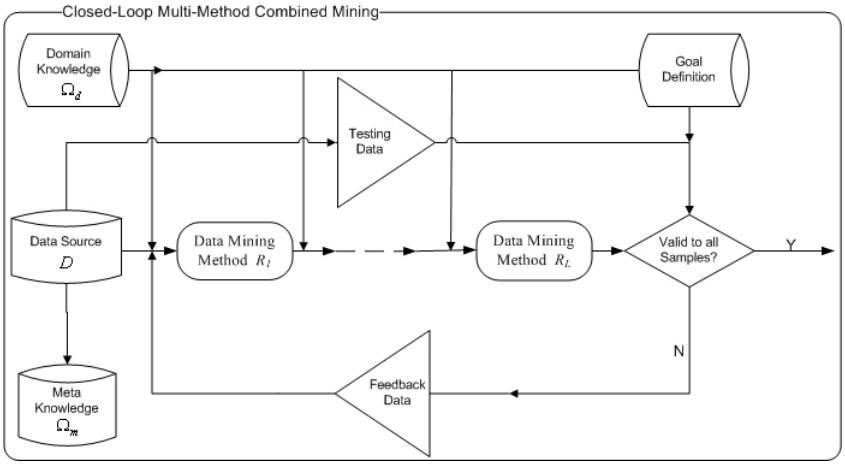

Fig. 6.2 Closed-Loop Multi-Method Combined Mining

6.6.5 Closed-Loop Sequence Classification

In recent years, sequence classification has been recognized as a challenging data mining issue. It has a wide range of applications such as bioinformatics [182, 190] and customer behavior predictions [97]. Most existing sequence classification algorithms follow the theory of serial multi-method combined mining, in which classification follows sequence pattern mining. It is known that efficiency is a key problem in sequential pattern mining even though many algorithms have been proposed to improve efficiency. In sequential pattern mining, time order has to be taken into account to find frequent sub-sequences. Hence, a huge number of candidates have to be checked in the algorithm. The sequential pattern mining may take weeks or even months if all the candidates are generated and processed. On the other hand, in order to build sequential classifiers, a number of processes such as *significance test* and *coverage test* have to be conducted on the sequential pattern set. If the sequential pattern set contains huge amounts of sequential patterns, the classifier building can also be extremely time-consuming. Therefore, in sequence classification, the efficiency problem exists not only in sequential pattern mining but also in classifier building.

In fact, in rule-based classification, the most important task is not to find the complete rule set but to unearth the most discriminative rules [69, 116]. In [69], experimental results show that "redundant and non-discriminating patterns often over-fit the model and deteriorate the classification accuracy". To solve such issues, we propose a novel closed-loop sequence classification method as follows. First, a small set of the most discriminating sequential patterns are mined. These patterns are then used for coverage test on the training dataset. If the sequential pattern set is small enough, there must be some samples that have not been covered by the mined patterns. These uncovered samples are further fed back to the next loop of sequential pattern mining. Again, a coverage test is implemented on the newly mined patterns. The remaining samples that still cannot be covered are fed back for sequential pattern mining until the predefined thresholds are reached or all samples are covered.

6.6.5.1 Discriminating Measures

In order to discover a small set of discriminating patterns in each loop, we use Chi-Square test and Class Correlation Ratio (*CCR*) [205] as the principal interestingness measures. The *CCR* can be defined given a contingency table shown in Table 6.10.

Table 6.10 2 by 2 Feature-Class Contingency Table

	x	$\neg x$	$\sum rows$
C	a	b	$a+b$
$\neg C$	c	d	$c+d$
$\sum cols$	$a+c$	$b+d$	$n=a+b+c+d$

CCR measures how a correlated sequence X is with the impact T compared to non-impact $\bar{T}$. *CCR* is defined as follows.

$$CCR(X \rightarrow T) = \frac{\hat{corr}(X \rightarrow T)}{\hat{corr}(X \rightarrow \bar{T})} = \frac{a \cdot (c+d)}{c \cdot (a+b)} \tag{6.30}$$

Here, $\hat{corr}$ is the correlation between X and T.

$$\hat{corr}(X \rightarrow T) = \frac{sup(X \cup T)}{sup(X) \cdot sup(T)} = \frac{a \cdot n}{(a+c) \cdot (a+b)}, \tag{6.31}$$

CCR measures to what extent the antecedent is correlated with the class (e.g., impact T or non-impact) it predicts $\bar{T}$. *CCR* falls in $[0, +\infty)$. $CCR = 1$ means the antecedent is independent of the class. $CCR < 1$ means the antecedent is negatively correlated with the class. $CCR > 1$ means the antecedent is positively correlated with the class.

6.6.5.2 Algorithm Outline

The closed-loop sequence classification algorithm is outlined as follows. Since an aggressive pattern mining strategy is used in the closed-loop sequence classification algorithm, only a very small set of sequential patterns, rather than the complete sequential patterns, can be mined in each loop. The number of sequential patterns increases in each loop so that the total sequential pattern number is much less than the complete sequential pattern set.

Algorithm: Mining closed-loop sequential classifiers
INPUT: Transactional data
OUTPUT: Sequential classifiers
(1). Calculate the *frequency* of each one-event sequence and the corresponding *CCR*. Only the events with $CCR > 1+m_1$ or $CCR < 1-m_2$ (m_1 and m_2 are margins) are extracted into a sequential pattern set. The pattern growth is also based on this sequential pattern set. With this greedy strategy, only a small set of sequential patterns are mined.
(2). Calculate the *frequency*, *chi-square* value and *CCR* of each sequence, and only those where sequences meet *support*, *significance* and *CCR* criteria are output into the resulting sequential pattern set.
(3). After all sequential patterns are extracted in the above steps, pattern pruning is implemented on the mined sequences. We follow the pattern pruning algorithm in [147]. The only difference is, in our algorithm, *CCR* instead of *confidence* is used as the measure for pruning.
(4). Conduct coverage test following the ideas in [141] and [147]. Since greedy pattern mining strategy is used in this algorithm, a large number of training samples cannot be covered by the mined sequential patterns.
(5). These training samples are fed back to Step (1). With updated parameters, sequential patterns are mined again. After pattern pruning and coverage test, those uncovered training samples are fed back to (1) for further sequential pattern mining by updating parameters. The process iterates until the predefined thresholds are reached or all samples are covered.

We use two strategies to build the sequence classifier as follows.

- Highest weighted score ($CCR_{Highest}$). Given a sequence instance s, the class label corresponding to the classifiable sequential pattern with highest weighted score is assigned to s; and
- Multiple weighted scores (CCR_{Multi}). Given one sequence instance s, all the classifiable sequential patterns on one level covered s are extracted. It is not difficult to compute the sum of the weighted score corresponding to each target class. The class label corresponding to the largest weighted score sum is assigned to s.

6.7 Case Study: Mining Combined Patterns in E-Government Service Data

In Chapter 10, we introduce some results in mining combined patterns on e-government service data in the Australian Commonwealth Government Agency, Centrelink. Centrelink is responsible for delivering government social security policies and services to one third of Australians. Every year, Centrelink accumulates many customer debts for various reasons. A key purpose of combined mining is to identify debt-related patterns that indicate causes and effects of Centrelink customers and officer-customer interactions.

Social security data such as that in Australian Commonwealth Government Agency Centrelink is widely seen in welfare states. The data consists of customer demographics, debt information, activities [63] such as government arrangements for debtors' payback agreed by both parties, and debtors' repayment information. Such data encloses important information about the experience and performance of government service objectives and social security policies, and may include evidence and indicators for recovering, detecting, preventing and predicting debt occurrences. The case studies illustrate the discovery of multi-feature combined patterns and sequential classifiers on social security data for debt recovery and prevention. In mining the combined patterns, we consider the effects of domain factors, customer behaviors, interactions between customers and officers, and government policies and so on, which make the deliverables more informative and actionable for business problem-solving.

6.8 Related Work

Generally speaking, approaches for mining more informative and actionable knowledge in complex data can be categorized as follows: (1) post-analysis and post-mining of learned patterns, (2) involving extra features from other datasets, (3) integrating multiple methods, (4) joining multiple relational tables, and (5) direct mining by inventing effective approaches.

Direct mining for discriminative patterns has recently been highlighted in Harmony [208], DDPMine [70], and model-based search tree [100]. Our combined mining methods, multi-feature and multi-method combined mining, belong to this category. We now focus on introducing the work related to the first four approaches, and explain their difference from combined mining.

Post-analysis and post-mining of learned patterns is a commonly used approach [242]. Many existing works focus mainly on developing post-analysis techniques to prune rules [142], reduce redundancy [134], summarize learned rules [142], and to match expected patterns by similarity difference [139]. A recent highlight was the extraction of actions from learned rules [223]. A typical approach to learning action rules is to split attributes into either 'hard/soft' [223] or 'stable/flexible' [178, 204]

to extract actions that may improve the loyalty or profitability of customers. In this case, an action is reported to be the conversion of valuable customers from a likely attrition status to a loyal status. Some other work has been undertaken on action hierarchy [2].

Different to the post-analysis-based methods, the combined patterns introduced in this chapter do not rely on post-analysis. We actually conduct direct mining of combined patterns, and add on post-mining if necessary. For example, the multi-feature combined mining approach considers features from multiple datasets during the generation of more informative patterns. This is related to multi-data source mining as well. For instance, [63, 240] mine for combined activity patterns consisting of heterogeneous features from either multiple data sources or multiple partitioned data sets to mine for patterns consisting of multiple aspects of information.

Further taking the cluster patterns as an example, in our work they are not generated by pattern summarization. Cluster patterns are mined through method as discussed in Section 4.3. The patterns in a cluster have the same prefix, but the remaining items in the patterns make the results different. The unique aspect of our method is that it can generate incremental and decremental combined clusters as well as pairs. However, the current methods mainly target contrast patterns, emerging patterns, etc. which are much simpler than ours.

The integration of multiple data mining methods is widely used to mine more informative knowledge. There have been several methods combining multiple data mining algorithms. For instance, the combination of association rule mining and classification, namely the associative classification [141], is reported to have not only high classification accuracy, but also strong flexibility in handling unstructured data. The combination of clustering and association rules can be used for rarity mining [164] if clustering is implemented on the dataset followed by association rule mining. When clustering is used after association rule mining, more interesting patterns and more actionable knowledge can be mined [239]. [157] combined regression with association rule mining. [92] proposes the concept of mining emerging contrast patterns. [147] mines association rules from a class distribution-associated FP-tree. [225] combines associative classification and FOIL-based rule generation. [131] proposes tree and cyclic patterns for structure-based graph classification. In addition, methods from different areas may be combined, for instance, the integration of boosting with associative classifiers [193]. In this chapter, we summarize the main paradigms of integrating multiple methods for complex pattern mining, and propose the approach of closed-loop multi-method combined mining, which can lead to more effective and efficient mining of discriminative patterns.

A recent trend is for classification to be conducted on top of frequent pattern mining, which has attracted more and more research interest because of its wide applications. [135] proposed an algorithm for sequence classification using frequent sequential patterns [162, 221] as features in the classifier. In their algorithm, subsequences are extracted and transformed into sets of features. After feature extraction, general classification algorithms such as Naïve Bayes, SVM or neural network can be used for classification. Their algorithm is the first attempt on the combination of classification and sequential pattern mining. However, a huge amount of sequen-

tial patterns are mined in the sequential mining procedure. Although the pruning algorithm is used for post-processing, a large amount of sequential patterns still construct the feature space. Their algorithm does not tackle some important problems such as how to efficiently and effectively select discriminative features from the large feature space. This issue can be handled in our closed-loop sequence classification method.

[218] studied the problem of early prediction using sequence classifiers. The prefix of a sequence as short as possible is used to make prediction reasonably accurate. They proposed a sequential classification rule to mine sequential classification rules, which are then selected by an early-prediction utility measure. Based on the selected rules, a generalized sequential decision tree is used to build a classification model with the divide-and-conquer strategy. Tseng and Lee [202] proposed a CBS algorithm to combine sequential pattern mining and classification. In their paper, two algorithms, CBS_Class and CBS_All were proposed. CBS_Class used selected discriminative sequential patterns while CBS_All used all mined sequential patterns. The experimental results in their paper show that the performance of CBS_Class is better than the performance of CBS_All. Exarchos [99] proposed to combine sequential pattern mining and classification followed by an optimization algorithm. The accuracy of their algorithm is higher than that of CBS. However, optimization is a very time-consuming procedure. The above algorithms are combined in a serial way, and there are two main shortcomings with these algorithms. First, a large amount of patterns are mined from the combined mining algorithms; however, without the interaction of different methods, it is very difficult to discover the most discriminating patterns. Secondly, in order to keep as many useful patterns as possibly for analysis at later stages, each method has to mine as many patterns as possible, in which a large number of patterns are uninteresting. The efficiency of this combination is very low. In our closed-loop combination method, both effectiveness and efficiency can be significantly improved because only discriminating patterns are mined in the algorithm.

Table joining is widely used in order to mine patterns from multiple relational tables by putting relevant features from individual tables into a consolidated one. As a result, a pattern may consist of features from multiple tables. This method is suitable for mining multiple relational databases, and in particular for small datasets. However, enterprise applications often involve multiple heterogeneous datasets consisting of large volumes of records. It is too costly in terms of time and space, thus sometimes impossible, to join multiple sources of data. It is not feasible for general multiple sources of data. Combined mining can identify such compound patterns while handling large datasets.

In addition, multi-relational data mining [95] and multi-database mining [226] has been intensively studied. The main difference between them and our combined mining approach is that our method is not aimed at multi-database only, rather it is a general approach for mining complex knowledge in complex data. As the multi-source combined mining has shown, combined mining does not take place through joining related tables. The resulting patterns of multi-source combined mining can

consist of pair or cluster patterns with components from multiple datasets, which is new to multi-relational mining, to the best of our knowledge.

The most important difference between our combined mining and other existing methods is the combined pattern types. Combined mining can produce new pattern types such as incremental/decremental cluster patterns that have not previously been identified by the above method. Such combined patterns are not discoverable through post-analysis or table joining, but rather by direct combined mining.

Another interesting finding from this research is the dynamic chart for presenting pattern mining deliverables. Dynamic charts provide a straightforward view for describing the evolution of customer behavior changes, the interaction between actions, and the impact and impact dynamics of customer activities. Cluster patterns presented in dynamic charts can inform business users with the dynamics and impact of customer behavior and interaction, and support them in taking suitable actions.

6.9 Summary

Mining complex data for complex knowledge is a problem in data mining research and development. Typical examples involve multiple distributed and heterogeneous features and data sources with large quantities, catering for user demographics, preferences, behavior, business appearance, service usage, and business impact. There is an increasing need to mine for patterns consisting of multiple aspects of information from multiple relevant business lines, so as to reflect comprehensive business scenarios, and present patterns that can inform decision-making actions. This challenges existing data mining methods such as post-analysis and table joining based analysis, and the traditional deliverables from data mining that are not informative and actionable for real-world use.

Even though many efforts are under investigation for the above purpose, by building on existing works, this chapter has presented a comprehensive and general approach named *combined mining* for handling multiple large heterogeneous data sources targeting more informative and actionable knowledge. We focus on providing general frameworks and approaches to handle multi-feature, multi-source and multi-method issues and requirements.

We have addressed challenging problems in combined mining, and summarized and proposed effective pattern merging and interaction paradigms, combined pattern types such as pair patterns and cluster patterns, interestingness measures, and effective tools such as dynamic charts for presenting complex patterns in a business-friendly manner. To the best of our knowledge, new pattern types such as incremental/decremental cluster patterns resulting from combined mining and pattern deliverable methods dynamic charts are new in the current body of knowledge, and cannot be produced by simple post-analysis and table joining based approaches, even if targeting multiple relational tables.

Case studies have been conducted on real-world e-government service data from the Australian Commonwealth Government Agency Centrelink, discovering single

combined patterns, pair patterns, cluster patterns, incremental pair and cluster patterns, and sequential classifiers on customer demographic data, government debt arrangement and repayment activity data, and debt information for government debt prevention. Combined patterns are further presented in terms of dynamic charts, a novel pattern presentation method reflecting the evolution and impact change of a cluster of patterns. Experiments have shown that the identified combined patterns are more informative and actionable than any single patterns identified in the traditional way.

In fact, combined mining provides a general framework for discovering more informative knowledge in complex data. Typical challenges such as mining heterogeneous data sources can benefit from combined mining. The proposed frameworks can be instantiated and expanded to cater for other complex situations in which single line, table joining, and one-step mining cannot be handled very well, or in which their outcomes cannot satisfy real business needs. To this end, more efforts will be made to develop effective paradigms, combined pattern types, combined mining methods, pattern merging methods and interestingness measures for large and multiple sources of data.

Chapter 7
Agent-Driven Data Mining

7.1 Introduction

This chapter discusses a new technique – *agent-driven data mining* for domain driven data mining (D^3M). We focus on introduce the basic concept, driving forces, technical means, research issues and case studies of agent-driven data mining.

The goals of this chapter consist of the following aspects:

- An overview of agent mining – the integration of multi-agent technology with data mining;
- The potential of using agents for data mining; and
- Case studies of agent-driven data mining.

Correspondingly, we address the following aspects:

- Section 7.2 discusses the complementation between agents and data mining, disclosing the need of integrating agents and data mining.
- In Section 7.3, the field of agent mining is briefly introduced. Agent mining fulfills the respective strengths of both agents and data mining to handle either critical challenges in an individual party or mutual issues.
- In Section 7.4 and Section 7.5, we discuss why agents for data mining, and what agents can do for data mining, respectively.
- Agents for distributed data mining is introduced in Section 7.6.
- Section 7.7 lists some of research issues in agent-driven data mining.
- Sections 7.8 and 7.10 illustrates the use of agent-driven data mining for solving real-world prolems.

7.2 Complementation between Agents and Data Mining

In many cases, both data mining and multi-agent systems (MAS) involve ubiquitous intelligence, as we discussed in Chapter 3. This actually discloses many mutual

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challenges faced by both areas. In dealing with such challenges, besides the techniques that need to be innovated from either end, a promising way is to marry both parties and let them complement each other.

Why is the interaction and integration between agents and data mining (agent mining [22, 23, 48, 51, 235] for short) important? There are both explicit and implicit reasons. Explicit reasons may include the following system complexities.

- Explicit limitations and challenges in pure agent systems, as addressed in [26, 44, 54], can be complemented by data mining, for instance, data mining driving agent learning, user modeling and information analysis.
- Explicit limitations and challenges in pure data mining systems, as discussed in [26, 44, 54], can be better serviced by agent technology, for instance, agent-based data mining infrastructure, agents for data management and preparation, agent-based service provision.
- Agent mining has the potential to result in new strengths and advantages that cannot be delivered by any single side, for instance, leading to more intelligent agent-mining symbiont fusing capabilities of in-depth perception, learning, adaptation, discovery, reasoning, and decision.

Implicit driving forces for including the above mutual issues are equally significant.

- Agent-mining symbionts are substantially essential for dealing with complicated intelligence phenomena and system complexities in complex intelligent systems. Simple intelligent systems and other issues that can be tackled using one side of these technologies, for instance, an agent-based data integration system, may not necessarily involve both sides.
- The emergence of intelligence in agent mining may massively strengthen the problem-solving capability of an intelligent system, which cannot be carried out by either part.
- Implicit roles need to be discovered through interdisciplinary studies, which may extensively promote either one side or the whole of an agent-mining integrative system, once the roles are disclosed and properly developed.
- New research issues, opportunities, techniques and systems may be triggered in the agent mining community.

It is arguable that agents and data mining are complementary. Agent mining can enhance both sides considerably through introducing new approaches and techniques to solve those domain-specific challenges that cannot be tackled well by either methods. Some typical benefits and roles in agent and mining areas that can be achieved through agent mining.

- Enhancing agents through data mining. Agent-mining interaction was originally initiated by data mining driven agent learning in 1991 [18, 183]. Data mining has the potential to enhance agent technology through introducing and improving the learning and reasoning capability of agents. Agents can be enhanced through involving data mining in broad aspects, in particular, agent learning, agent coordination and planning, user modeling and servicing, and network servicing.

- Promoting data mining through agents. Sometime around 1993, another effort was started on agent-based data mining [84, 85, 96], namely to utilize agent technology to enhance data mining. The enhancement may be embodied in terms of varying aspects, for instance, agent-based KDD infrastructure, agent-based distributed processing, agent-based interactive data mining, and agent-based data warehouse.
- Building super intelligent symbionts. As evidenced by the agent service-based trading support system F-Trade [38], the use of agent mining can lead to more intelligent systems that can best fuse the strengths of agents in building intelligent systems as well as the beauty of data mining in processing deep knowledge.

7.3 The Field of Agent Mining

Agents and data mining interaction and integration forms a promising research area [1], or agent mining for short [23, 48, 51, 61]. As an emerging scientific field, *agent mining* studies the methodologies, principles, techniques and applications of the integration and interaction between agents and data mining, as well as the community that focuses on the study of agent mining. The interaction and integration between agents and data mining are comprehensive, multiple dimensional, and interdisciplinary.

On the basis of complementation between agents and data mining, agent mining fosters a synergy between them from different dimensions, for instance, *resource*, *infrastructure*, *learning*, *knowledge*, *interaction*, *interface*, *social*, *application* and *performance*. As shown in Fig. 7.1, we briefly discuss these dimensions.

- Resource layer – interaction and integration may happen on data and information levels;
- Infrastructure layer – interaction and integration may be on infrastructure, architecture and process sides;
- Knowledge layer – interaction and integration may be based on knowledge, including domain knowledge, human expert knowledge, meta-knowledge, and knowledge retrieved, extracted or discovered in resources;
- Learning layer – interaction and integration may be on learning methods, learning capabilities and performance perspectives;
- Interaction layer – interaction and integration may be on coordination, cooperation, negotiation, communication perspectives;
- Interface layer – interaction and integration may be on human-system interface, user modeling and interface design;
- Social layer – interaction and integration may be on social and organizational factors, for instance, human roles;
- Application layer – interaction and integration may be on applications and domain problems;

[1] www.agentmining.org

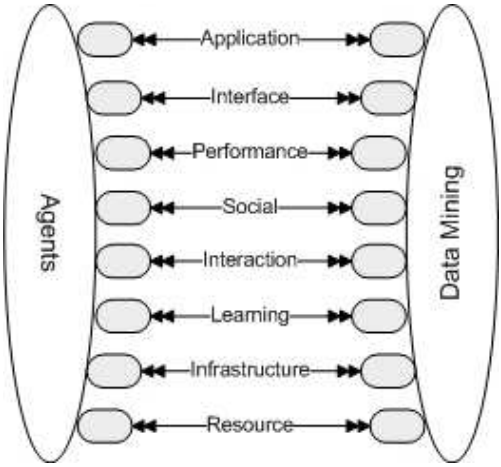

Fig. 7.1 Multi-Dimensional Agent-Mining Synergy.

- Performance layer – interaction and integration may be on the performance enhancement of one side of the technologies or the coupling system.

From these dimensions, many fundamental research issues/problems in agent mining emerge. Correspondingly, we can generate a high-level research map of agent mining as a disciplinary area. Fig. 7.2 shows such a framework, which consists of the following research components: *agent mining foundations*, *agent-driven data processing*, *agent-driven knowledge discovery*, *mining-driven multi-agent systems*, *agent-driven information processing*, *mutual issues in agent mining*, *agent mining systems*, *agent mining applications*, *agent mining knowledge management*, and *agent mining performance evaluation*. We briefly discuss them below.

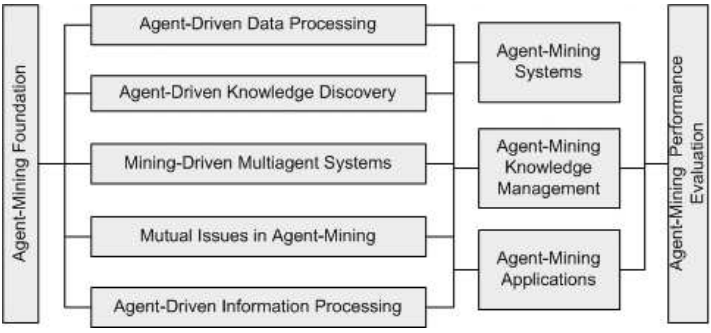

Fig. 7.2 Agent-Mining Disciplinary Framework.

- *Agent mining foundations* studies issues such as the challenges and prospects, research map and theoretical underpinnings, theoretical foundations, formal methods, and frameworks, approaches and tools;
- *Agent-driven data processing* studies issues including multi-agent data coordination, multi-agent data extraction, multi-agent data integration, multi-agent data management, multi-agent data monitoring, multi-agent data processing and preparation, multi-agent data query and multi-agent data warehousing;
- *Agent-driven knowledge discovery* studies problems like multi-agent data mining infrastructure and architecture, multi-agent data mining process modeling and management, multi-agent data mining project management, multi-agent interactive data mining infrastructure, multi-agent automated data learning, multi-agent cloud computing, multi-agent distributed data mining, multi-agent dynamic mining, multi-agent grid computing, multi-agent interactive data mining, multi-agent online mining, multi-agent mobility mining, multi-agent multiple data source mining, multi-agent ontology mining, multi-agent parallel data mining, multi-agent peer-to-peer mining, multi-agent self-organizing mining, multi-agent text mining, multi-agent visual data mining, and multi-agent web mining;
- *Mining-driven multi-agent systems (MAS)* studies issues such as data mining-driven MAS adaptation, data mining-driven MAS behavior analysis, data mining-driven MAS communication, data mining-driven MAS coordination, data mining-driven MAS dispatching, data mining-driven MAS distributed learning, data mining-driven MAS evolution, data mining-driven MAS learning, data mining-driven MAS negotiation, data mining-driven MAS optimization, data mining-driven MAS planning, data mining-driven MAS reasoning, data mining-driven MAS recommendation, data mining-driven MAS reputation/risk/trust analysis, data mining-driven self-organized and self-learning MAS, data mining-driven user modeling and servicing, and semi-supervised MAS learning;
- *Agent-driven information processing:* multi-agent domain intelligence involvement, multi-agent human-mining cooperation, multi-agent enterprise application integration, multi-agent information gathering/retrieval, multi-agent message passing and sharing, multi-agent pattern analysis, and multi-agent service-oriented computing;
- *Mutual issues in agent mining* including issues such as actionable capability, constraints, domain knowledge and intelligence, dynamic, online and ad-hoc issues, human role and intelligence, human-system interaction, infrastructure and architecture problems, intelligence metasynthesis, knowledge management, lifecycle and process management, networking and connection, nonfunctional issues, ontology and semantic issues, organizational factors, reliability, reputation, risk, privacy, security and trust, services, social factors, and ubiquitous intelligence;
- *Agent mining knowledge management:* knowledge management is essential for both agents and data mining, as well as for agent mining. This involves the representation, management and use of ontologies, domain knowledge, human empirical knowledge, meta-data and meta-knowledge, organizational and social factors, and resources in the agent-mining symbionts. In this, formal methods and tools are necessary for modeling, representing and managing knowledge. Such tech-

niques also need to cater for identifying and distributing knowledge, knowledge evolution in agents, and enabling knowledge use.

- *Agent mining performance evaluation* researches on methodologies, frameworks, tools and testbeds for evaluating the performance of agent mining, and performance benchmarking and metrics. Besides technical performance such as accuracy and statistical significance, business-oriented performance such as cost, benefit and risk are also important in evaluating agent mining. Other aspects such as mobility, reliability, dependability, trust, privacy and reputation, etc., are also important in agent mining.
- *Agent mining systems:* this research component studies the formation of systems, including techniques for the frameworks, modeling, design and software engineering of agent-mining systems. It provides agents and data mining technologies as basic resources, thus they can perform as parts of the system. Techniques and tools for engineering and constructing agent-mining systems are important for handling various kinds of applications. A specific agent-mining system may be extended to fit into some applications based on the provided features of the system. With regards to integrated systems, applications can be constructed from the pre-defined framework of a particular problem domain or fully tailored to serve the purpose of the applications. Either way, system implementation and deployment are to be further investigated since development of the two technologies has been done in parallel. It is hard to generate dedicated platforms for agent mining systems.
- *Agent mining applications* This refers to any real-world applications and domain problems that can be better handled by agent mining technologies. Based on the need from particular applications, any issues discussed in the above topics may be engaged here. For instance, in some cases, an agent mining simulation system needs to be built for us to understand the working mechanism and potential optimization of a complex social network. In other cases, the enhancement of learning capability is the main task, and appropriate learning tools need to be used on demand.

7.4 Why Agent-Driven Data Mining

Multiagent technology is good at user interaction, autonomous computing, self-organization, coordination, cooperation, communication, negotiation, peer-to-peer computing, mobile computing, collective intelligence and intelligence emergence. These main strengths of multi-agent technology can greatly complement data mining in particular complex data mining problems in aspects such as data processing, information processing, pattern mining, user modeling and interaction, infrastructure and services. One of its main tasks is multi-agent-based data mining (otherwise known as multi-agent-driven data mining, multi-agent data mining, in this book, we use the term “agent-driven data mining”).

Agent-driven data mining (ADDM) refers to the contributions made by multi-agents for enhancing data mining tasks. ADDM can contribute to the problem solving of many data mining issues, eg., agent-based data mining infrastructure and architecture, agent-based interactive mining, agent-based user interaction, automated pattern mining, agent-based distributed data mining, multi-agent dynamic mining, multi-agent mobility mining, agent-based multiple data source mining, agent-based peer-to-peer data mining, and multi-agent web mining. In the following, we discuss the unique roles of agents in supporting distributed data mining (DDM).

In enterprise applications, data is distributed in heterogeneous sources coupling in either a tight or loose manner. Distributed data sources associated with a business line are often complex, for instance, some is of high frequency or density, mixing static and dynamic data, mixing multiple structures of data. In some cases, multiple sources of data are stored in parallel storage systems. Local data sources can be of restricted availability due to privacy, their commercial value, etc, which also prevents, in many cases, its centralized processing even in a collaborative mode. For this data, data integration and data matching are difficult to conduct. It is not possible to store them in centralized storage and not feasible for processing in a centralized manner. To mine the data, the infrastructure and architecture weaknesses of existing distributed data mining systems requires more flexible, intelligent and scalable support.

Agent technology can help with these challenges by involving autonomy, interaction, dynamic selection and gathering, scalability, multi-strategy and collaboration. Other reasons include privacy, mobility, time constraint (stream data, it is too late to extract and then mine), and computational costs and performance requests. In particular, multi-agent technology can complement distributed data mining in many aspects, for instance,

- Distributed and multiple data sources are often isolated from each other. For in-depth understanding of a business problem, it is essential to bring relevant data together through centralized integration or localized communication. From this, agent planning and collaboration, mobile agents, agent communication and negotiation can benefit.
- Data and device mobility requires the perception and action of data mining algorithms on a mobile basis. Mobile agents can adapt to mobility very well.
- Pro-actively assisting agent is necessary to drastically limit how much the user has to supervise and interfere with running the data mining process.
- In changing and open distributed environment, KDD agents may be applied to adaptively select data sources according to given criteria such as the expected amount, type and quality at the considered source, actual network and KDD server load. Agents may be used, for example, to dynamically control and manage the process of data gathering.
- Some data distributed in different storages is dependent on time, e.g. time difference.
- For some complex application settings an appropriate combination of multiple data mining techniques may be more beneficial than applying a particular one. KDD agents may learn in, due course which of their deliberative actions

to choose, depending on the type of data retrieved from different sites and the mining tasks to be pursued.

- KDD agents may operate independently on data gathered at local sites and then combine their respective models. Alternatively, they may agree to share potential knowledge as it is discovered, in order to benefit from the additional options of other agents.
- Distributed local data is not allowed to be extracted and integrated with other sources directly, due to privacy issues. A KDD agent with authority to access and process the data locally can dispatch identified local patterns for further engagement with findings from other sources.
- In some organizations, business logic, process and work-flow determine the order of data storage and access. This, therefore, augments the complexity of DDM. Agents located in each storage area can communicate with each other and dispatch the DDM algorithm agents instantly, once the response is over.

In fact, ADDM provides a unique approach to involve, represent and tackle

- data intelligence such as agent-based distributed data access and collaboration,
- human intelligence such as through user-agent interaction, user modeling and servicing,
- domain and organizational factors such as through multi-agent swarm intelligence, collective intelligence, and intelligence emergence,
- network intelligence such as through mobile agents and multi-agent coordination and communication, and
- social intelligence such as building multi-agent social cognition and interaction to involve a group of experts into mining process.

7.5 What Can Agents Do for Data Mining?

Data mining faces many challenges when it is deployed to real world problem-solving, in particular, in handling complex data and applications. We list here a few aspects that can be improved by agent technology. These include enterprise data mining infrastructure, involving domain and human intelligence, supporting parallel and distributed mining, data fusion and preparation, adaptive learning, and interactive mining.

- *Enterprise data mining infrastructure* The development of data mining systems supporting real-world enterprise applications are challenging. The challenge may arise from many aspects, for instance, integrating or mining multiple data sources, accessing distributed applications, interacting with varying business users, and communicating with multiple applications. In particular, it has been a grand challenge and a longstanding issue to build up a distributed, flexible, adaptive and efficient platform supporting interactive mining in real-world data.

- *Involving domain and human intelligence* Another grand challenge of existing data mining methodologies and techniques are the roles and involvement of domain intelligence and human intelligence in data mining. With respect to domain intelligence, how to involve, represent, link and confirm to components such as domain knowledge, prior knowledge, business process, and business logics in data mining systems is a research problem. Regarding human intelligence, we need to distinguish the role of humans in specific applications, and further build up system support to model human behavior, interact with humans, bridge the communication gap between data mining systems and humans, and most importantly incorporate human knowledge and supervision into the system.
- *Supporting parallel and distributed mining* One of the major efforts of data mining research is to enhance the performance of data mining algorithms. This is usually conducted through designing efficient data structures and computational methods to reduce computational complexities. In many cases, computational performance can be greatly improved through developing parallel algorithms. In other cases, distributed computing is necessary such as dealing with distributed data sources or applications, or peer-to-peer computing is required. However, how to design effective and efficient parallel and distributed algorithms is an issue.
- *Data fusion and preparation* In the real world, data is getting more and more complex, in particular, sparse and heterogeneous data distributed in multiple places. To access and fuse such data needs intelligent techniques and methods. On the other hand, today's data preparation research is facing new challenges such as processing high frequency time series data stream, unbalanced data distribution, rare but significant evidence extraction from dispersed data sets, linking multiple data sources, accessing dynamic data. Such situations expect new data preparation techniques.
- *Adaptive learning* In general, data mining algorithms are predefined to scan data sets. In real-world cases, it is expected that data mining models and algorithms can adapt to dynamic situations in changing data based on their self-learning and self-organizing capability. As a result, models and algorithms can automatically extract patterns in changing data. However, this is a very challenging area, since existing data mining methodologies and techniques are basically non-automatic and unadaptable. To enhance the automated and adaptive capability of data mining algorithms and methods, we need to search for support from external disciplines that are related to automated and adaptive intelligent techniques.
- *Interactive mining* Controversies regarding either automatic or interactive data mining have been raised in the past. A clear trend for this problem is that interaction between humans and data mining systems plays an irreplaceable role in domain driven data mining situations. In developing interactive mining, one should study issues such as user modeling, behavior simulation, situation analysis, user interface design, user knowledge management, algorithm/model input setting by users, mining process control and monitor, outcome refinement and tuning. However, many of these tasks cannot be handled by existing data mining approaches.

7.6 Agent-Driven Distributed Data Mining

This section particularly discusses the state-of-the-art of agent-driven knowledge discovery [49]. As discussed in the above, agent-driven knowledge discovery forms a big area for agent mining. It is actually the mostly addressed area in agent mining.

7.6.1 The Challenges of Distributed Data Mining

Data mining and machine learning currently forms a mature field of artificial intelligence supported by many various approaches, algorithms and software tools. However, modern requirements in data mining and machine learning inspired by emerging applications and information technologies and the peculiarities of data sources are becoming increasingly tough. The critical features of data sources determining such requirements are as follows:

- In enterprise applications, data is distributed over many heterogeneous sources coupling in either a tight or loose manner;
- Distributed data sources associated with a business line are often complex, for instance, some is of high frequency or density, mixing static and dynamic data, mixing multiple structures of data;
- Data integration and data matching are difficult to conduct; it is not possible to store them in centralized storage and it is not feasible to process them in a centralized manner;
- In some cases, multiple sources of data are stored in parallel storage systems;
- Local data sources can be of restricted availability due to privacy, their commercial value, etc., which in many cases also prevents its centralized processing, even in a collaborative mode;
- In many cases, distributed data spread across global storage systems is often associated with time difference;
- Availability of data sources in a mobile environment depends on time;
- The infrastructure and architecture weaknesses of existing distributed data mining systems requires more flexible, intelligent and scalable support.

These and some other peculiarities require the development of new approaches and technologies of data mining to identify patterns in distributed data. Distributed data mining (DDM), in particular, Peer-to-Peer (P2P) data mining, and multi-agent technology are two responses to the above challenges.

7.6.2 What Can Agents Do for Distributed Data Mining?

The practical implementation of distributed and P2P data mining and machine learning creates many new challenges. While analyzing these challenges, [128] argues

why agent technology is best able to cope with them in terms of autonomy, interaction, dynamic selection and gathering, scalability, multi-strategy and collaboration. Other reasons include privacy, mobility, time constraints (stream data which is too late to extract and then mine), and computational costs and performance requests.

- Isolation of data sources. Distributed and multiple data sources are often isolated from each other. For in-depth understanding of a business problem, it is essential to bring relevant data together through centralized integration or localized communication. From this, agent planning and collaboration, mobile agents, agent communication and negotiation can benefit.
- Mobility of source data and computational devices. Data and device mobility requires the perception and action of data mining algorithms on a mobile basis. Mobile agents can adapt to mobility very well.
- Interactive DDM. Pro-actively assisting agent is necessary to drastically limit how much the user has to supervise and interfere with running the data mining process.
- Dynamic selection of sources and data gathering. One challenge for an intelligent data mining agent acting in an open distributed environment, in which to pursue the DM tasks, for example, where the availability of data sites and their content may change at any time, is to discover and select relevant sources. In these settings, DM agents may be applied to adaptively select data sources according to given criteria such as the expected amount, type and quality at the considered source, actual network and DM server load. Agents may be used, for example, to dynamically control and manage the process of data gathering.
- Time constraints on distributed data sources. Some data distributed in different storages is dependent on time, e.g., time differences.
- Multi-strategy DDM. For some complex application settings, an appropriate combination of multiple data mining techniques may be more beneficial than the application of a particular one. DM agents may learn in, due course which of their deliberative actions to choose, depending on the type of data retrieved from different sites and the mining tasks to be pursued.
- Collaborative DDM . DM agents may operate independently on data they have gathered at local sites and then combine their respective models. Alternatively, they may agree to share potential knowledge as it is discovered, in order to benefit from the additional options of other DM agents.
- Privacy of source data. Distributed local data is not allowed to be extracted and integrated with other sources directly, due to privacy issues. A DM agent with authority to access and process the data locally can dispatch identified local patterns for further engagement with findings from other sources.
- Organizational constraint on distributed data sources. In some organizations, business logic, process and work-flow determine the order of data storage and access. This, therefore, augments the complexity of DDM. Agents located in each storage area can communicate with each other and dispatch the DDM algorithm agents instantly, once the response is over.

7.6.3 Related Work

There are quite a few research works on ADDM [2]. [159] provides a survey on agent technology for data mining. [86] is one of the first works attracting attention to ADDM arguing its advantages in mining vast amount of data stored in network and using collaborative capabilities of DM agents. It studies agent-based approach to distributed knowledge discovery using Inductive Logic Programming (ILP) approach and provides for some experiment results of application of the agent-based approach to DDM.

The paper [123] proposes PADMA (Parallel Data Mining Agents) system addressing use of agent architecture to cope large scale and distributed nature of data sources as applied to hierarchical clustering. It is intended to handle numeric and textual data with the focus on the latter. Agency of DDM system consists of DM agents that are responsible for local data access and extraction of high level useful information, agent-facilitator coordinating the DM agents operation while handling their SQL queries and presenting them to user interface. Agent-facilitator gets "conceptual graphs"from DM agents, combines them and passes the results to user. The focus of the paper is to show the benefit of agent-based parallel data mining.

JAM (Java Agents for Meta-learning) system proposed in [192] is an agent based system supporting the launching of learning, classifier and meta-learning agents over distributed database sites. It uses parallelism and distributed nature of meta-learning and its possibility to share meta-information without direct access to distributed data sets. JAM [8] in constituted of distributed learning and classification programs operating in parallel on JAM sites that are linked to a network. In turn, JAM site contains one or more local data base; one or more learning agents (machine learning programs); one or more meta-learning agents, intended for combing decisions produced by local classifier agents; a repository of decisions computed locally and imported by local and meta classifier agents; a local user configuration file and graphical user interface. Once the local and meta-classifiers are generated the user manages the execution of the above modules to classify new (unlabelled) data. A peculiarity of JAM system operation is that each local site may import decisions of remote classifiers from peer JAM sites and combine these decisions with own local classifier decisions using local meta-learning agent. JAM sites are operating simultaneously and independently. Administration of JAM site local activity is performed by the user via local user configuration file. Details of the JAM system architecture and implementation can be found in [8].

The paper [211] is motivated by the desire to attack increasingly difficult problems and application domains which often require to process very large amounts of data or data collected at different geographical locations that cannot be processed by sequential and centralised systems. It is also motivated by capabilities of MAS to provide for computing processes with robustness, fault tolerance, scalability, and

[2] Part of the following content comes from the work "Agent-Based Distributed Data Mining: A Survey", in the book "Data Mining and Multi-Agent Integration" by Springer, edited by L. Cao, with author permission.

speed-up as well as by maturity of the computer and network technology supporting implementation of parallel and distributed information processing. Based on multi-agent learning, target system can be built as self improving their performance. In particular, the paper considers job assignment problem, the core of many scheduling tasks, aiming to find reasonable solution in a reasonable time. The nodes of partially ordered set of jobs are considered as active entities or agents and the jobs are considered as passive entities or resources to be used by the agents. The agents interact in order to find a solution meeting some predefined criteria. The solution search procedure implemented by agents is based on distributed reinforcement learning starting from an initial solution that is performed using low-level communication and coordination among agents. The paper experimentally proves some advantages of the developed multi-agent model for implementation of parallel and distributed machine learning in the problem in question.

[11] describes the developed Papyrus system that is Java-based and intended for DDM handling with clusters and meta-clusters distributed over heterogeneous data sites. Mobile DM agents of the system are capable to move data, intermediate results and models between clusters for local processing thus reducing network load. Papyrus supports several techniques for exchanging and combining locally mined decisions (predictive models) and meta-data that are necessary to describe the above models specified in terms of a special mark-up language.

[129] studies advantages and added value of using ADDM, reviews and classifies existing agent-based approaches to DDM and proposes agent-oriented implementation of a distributed clustering system. The paper explicitly formulates why agent-based approach is very perspective one for DDM (autonomy, interactivity, capability to dynamically select data sources in changeable environment, etc.). The proposed KDEC scheme addressing computing statistical density estimation and information theoretic sampling to minimise communication between sites is implemented on the basis of agent technology as distributed data clustering system. In addition, one of its distinctive features is that it preserves the local data privacy.

[80] emphasises the possible synergy between MAS and DDM technologies. It particularly focuses on distributed clustering, having every increasing application domains, e.g. in sensor networks deployed in hostile and difficult to access locations like battle fields where sensors are measuring vibration, reflectance, temperature, and audio signals; in sensor networks for monitoring a terrain, smart home, and many other domains. In these domains analysing, e.g. DDM task, are non-trivial problems due to many constraints such as limited bandwidth of wireless communication channels, peer-to-peer mode of communication and the necessity of interaction in asynchronous network, data privacy, coordination of distributed computing, non-trivial decomposability, formidable number of data network nodes, limited computing resources, e.g. due to limited power supply, etc. The authors state that the traditional framework for centralised data analysis and DM algorithms does not really scale very well in such distributed applications. In contrast, this distributed problem solving can be very well coped with the multi-agent framework supporting semi-autonomous behaviour, collaboration and reasoning, among other perspective MAS properties. From DDM domain perspective, the paper focuses on clustering in

sensor networks that offers many aforementioned challenges. This paper suggests that traditional centralised data mining techniques may not work well in the above domains and underscores, among others, and that DDM algorithms integrated with MAS architecture may offer novel very perspective synergetic information technology for these domains.

Recently some efforts were paid to development of technological issues of multi-agent data mining that can be evaluated as a sign of increasing maturity. [109] states that the core problem of distributed data mining and machine learning design does not concern particular data mining techniques. Instead of this, its core problem is development of an infrastructure and protocols supporting coherent collaborative operations of distributed software components (agents) performing distributed learning. The paper proposes a multi-agent architecture of an information fusion system possessing of DDM and machine learning capabilities. It also proposes a design technology, which core is constituted by a number of specialised agent interaction protocols supporting distributed agent operations in various use cases (scenarios) including, in particular, DDM protocol. Further development of DDM system design technology is given in [110, 111].

[200] proposes a framework (an abstract architecture) for agent-based distributed machine learning and data mining. The proposed framework , as it is motivated by the authors, is based of the observation that "despite the autonomy and self-directedness of learning agents, many of such systems exhibit a sufficient overlap in terms of individual learning goals so that beneficial operation might be possible if a model for flexible interaction between autonomous learners was available that allowed agents to (i) exchange information about different aspects of their own learning mechanism at different levels of detail without being forced to reveal private information that should not be disclosed, (ii) decide to what extent they want to share information about their own learning processes and utilise information provided by other learners, and (iii) reason about how this information can best be used to improve their own learning performance."

The idea underlying the proposed framework is that each agent is capable to maintain meta-description of own learning processes in a form that makes it admissible, due to privacy issue, to exchange meta-information with other agents and reason about it rationally, i.e. to reason in a way providing for improving of their own learning results. The authors state that this possibility, for learning agents, is a hypothesis they intend to justify experimentally within a proposed formal framework. Actually this paper presents a preliminary research results and there are a lot of efforts to be done to reach reliable evaluation of the hypothesis used.

Besides those role-model systems presented earlier, there are more ADDM related works which the reader is encouraged to review: [13, 14, 15, 151, 177, 152, 163, 200, 230].

7.7 Research Issues in Agent Driven Data Mining

There are many open issues in the research direction of agent driven data mining. In establishing an agent-based enterprise data mining infrastructure, one may study organization and society-oriented study system analysis and design techniques for large-scale agent systems. Correspondingly, solutions for agent service based application integration, distributed data preparation, distributed agent coordination and parallel agent computing should be considered. In many cases of data mining, people should study algorithms that can adapt to dynamic data changes, dynamic user requests. To this end, it has the potential for agents to detect and reason such changes. Automated and adaptive data mining algorithms should be studied. The following is a list of some research open issues and promising areas.

- Activity modeling and mining
- Agent-based enterprise data mining
- Agent-based data mining infrastructure
- Agent-based data warehouse
- Agent-based mining process and project management
- Agent-based distributed data mining
- Agent-based distributed learning
- Agent-based grid computing
- Agent-based human mining cooperation
- Agent-based link mining
- Agent-based multi-data source mining
- Agent-based interactive data mining
- Agent-enriched ontology mining
- Agent-based parallel data mining
- Agent-based web mining
- Agent-based text mining
- Agent-based ubiquitous data mining
- Agent knowledge management in distributed data mining
- Agent for data mining data preparation
- Agent-human-cooperated data mining
- Agent networks in distributed knowledge discovery and servicing
- Agent service-based KDD infrastructure
- Agent-supported domain knowledge involvement in KDD
- Agent system providing data mining services
- Automated data mining learning
- Autonomous learning
- Distributed agent-based data preprocessing
- Distributed learning
- Domain intelligence in agent-based data mining
- Mobile agent-based knowledge discovery
- Protocols for agent-based data mining
- Self-organizing data mining learning.

7.8 Case Study 1: F-Trade – An Agent-Mining Symbiont for Financial Services

F-Trade [38] [3] is the acronym of Financial Trading Rules Automated Development and Evaluation, a web-based automated enterprise infrastructure for trading strategies and data mining on stock/capital markets. The system offers data connection, management and processing services. F-Trade supports online automated plug and play, and automatic input/output interface construction for trading signals/rules and data mining algorithms, data sources, and system components. It provides powerful and flexible supports for online backtesting, training/test, optimization and evaluation of trading strategies and data mining algorithms. Users can plugin, subscribe, supervise and optimize trading strategies and data mining algorithms in a human-machine cooperated manner.

F-Trade is built in Java agent services on top of Windows/Linux/Unix. XML is used for system configuration and metadata management. A super-server functions as the application server, another one acts as the data warehouse. It is constructed with online connectivity to distributed data sources as well as user-specific data sources.

Major roles played by agents in F-Trade consist of agent service-based architecture, agent-driven human interaction, agent for data source management, data collection and dispatch agents roaming to remote data sources, agentized trading strategies and data mining algorithms, agent and service recommender providing optimum algorithms and rules to users, and so on. Data mining assists the system in aspects such as data mining-driven trading rule/algorithm recommender agents, data mining-driven user services, data mining-driven trading agent optimizers, mining actionable trading rules in generic trading pattern set, parameter tuning of algorithm agents through data mining, etc. Mutual issues involve ontology-based domain knowledge representation and transformation to problem-solving terminology, human involvement and agent-based human interaction with algorithms and the system for algorithm supervision, optimization and evaluation, among others.

With this infrastructure, financial traders and researchers, and financial data miners can plug their algorithms onto the system and concentrate on improving the performance of their algorithms with iterative evaluation on a large amount of real stock data from international markets. The system services include (i) trading services support, (ii) mining services support, (iii) data services support, (iv) algorithm services support, and (v) system services support. In order to support all these services, soft plug-and-play is essential in the F-TRADE. Each system components interact through XML schema which specifies details of the components and allows agents to examine and use.

[3] www.f-trade.info

7.9 Case Study 2: Agent-based Multi-source Data Mining

Enterprise data mining applications often involve multiple data sources, which are likely distributed, heterogeneous and costly to be integrated. Multiple data mining agents, with each of them powered with specific algorithms focuses on mining local patterns on an individual data source, and other coordinator agents coordinate and communicate with each other to arrange for the distributed pattern mining, and additional pattern merging agents organize the aggregation of local patterns into global patterns.

An example is the application of the above idea onto the multi-source combined mining framework introduced in Section 5.5.4. Fig. 5.5 illustrates the principle of multi-source combined mining. The system can be agentized, as shown in Fig. 7.3. In the system, we could build up the following agents.

- Data source management agents: data monitors (DMonitor) and data coordinators (DCoordinator), to be responsible for managing data set connections, or data partition etc., and the dispatching of data mining tasks on data sets; in adaptive learning, data change monitors are deployed to monitor significant data change;
- Local data mining agents: data miners (DMiner) m_1 to m_N, to fulfill the data mining functions on data sets DB_1 to DB_N respectively;
- Data mining coordination agents: data mining coordinators (DMCoordinator), to coordinate the scheduling of local pattern mining by DMiners on each data set in terms of a certain protocol, either a predefined order or a negotiated decision made based on dynamic detection; another function for a DMCoordinator is to monitor the execution status of one DMiner and then notify the other to start the pattern mining;
- Local pattern set management agents: local pattern management agents (LPMAgent) aim to manage the local patterns produced by local data mining agents DMiners, for instance, pattern filtering according to filtering rules;
- Pattern merging agents: pattern merging agents (PMAgent) are to merge local patterns from corresponding DMiners into global pattern sets; PMAgents are composed of pattern merging methods, for instance, pattern clustering methods as discussed in Section 6.5.5 to generate a cluster of patterns consisting of local components from relevant local pattern sets;
- Knowledge management agents: knowledge management agents (KMAgent) are instantiated to manage knowledge acquired from different sources such as domain knowledge (Ω_m) and meta-knowledge (Ω_d);

In practice, the above-mentioned agents are instantiated into specific functional agents to fulfill particular tasks. For instance, DMiners may be customized into association rule miners, negative frequent sequence miners etc. DMCoordinators may be customized into data mining execution coordinators looking after the local dataset mining scheduling, message passing from one DMiner to another etc.

Many other types of agents may be involved. For instance, user interface agents (UIAgent) are created to interact with domain experts collecting domain knowledge

from them and further pass it over to KMAgent for further representing the knowledge. Negotiation agents may be applied as well. For example, in a trading agent system, as discussed in [24, 28], multiple trading strategy agents may negotiate with each other for maximizing self benefit and rewards, and eventually maximizing the global benefits for the trading agent coordinated by a global coordinator agent.

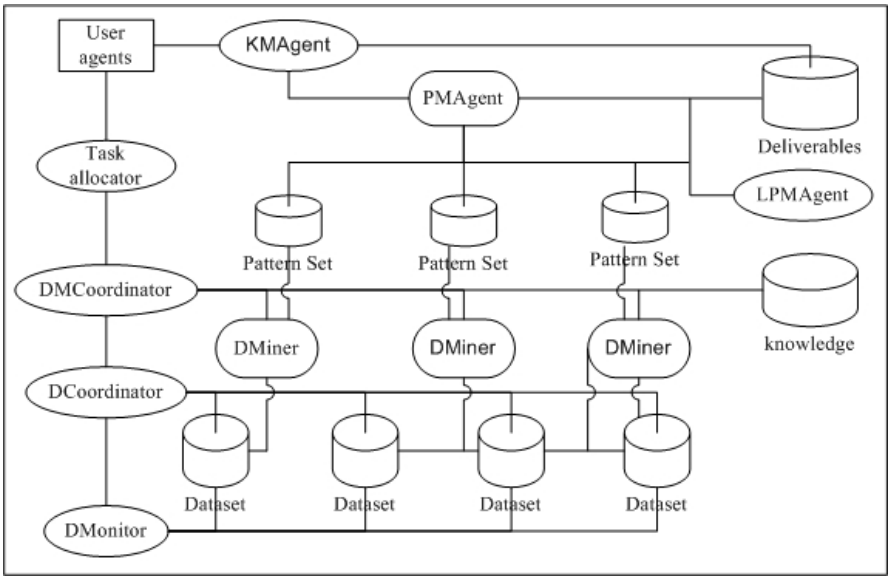

Fig. 7.3 Agentized Multi-source Combined Mining

Another example of agent-based distributed data mining is introduced in Section 3.4.5. Fig. 3.1 illustrates the working mechanism of multiple distributed agent coordination in undertaking respective data mining tasks and for generating global patterns.

7.10 Case Study 3: Agent-based Adaptive Behavior Pattern Mining by HMM

7.10.1 System Framework

In mining abnormal trading behavior in capital markets, agents are used for adaptive detection of abnormal trading behavior in dynamic and multiple activity streams. This section presents the framework of this system.

7.10.1.1 Basic process

Technically, the above business problem and observations can be converted into the following problem. Suppose there are J activity streams, each of them consists of F, G to H itemsets respectively.

$$\{e_{11}, e_{12}, \ldots, e_{1F}\} \tag{7.1}$$
$$\{e_{21}, e_{22}, \ldots, e_{2G}\} \tag{7.2}$$
$$\ldots \tag{7.3}$$
$$\{e_{J1}, e_{J2}, \ldots, e_{JH}\} \tag{7.4}$$

Itemsets (for instance, $e_{ab}, \ldots, e_{cd}$) in different streams are associated with each other for some relations $f(\cdot)$ such as ordering and timing.

$$f(\cdot) := f(e_{ab}, \ldots, e_{cd}) \tag{7.5}$$

where $a \neq c$ and $b \neq d$.

To deal with the above data, coupled Hidden Markov Model (CHMM) is used.

$$\lambda^{CHMM} = (A, B, C, \pi) \tag{7.6}$$

The CHMM models multiple activity streams as multiple processes, reflecting the relationships amongst the streams, and the transitions between hidden states and observations. This makes it possible for us to observe the state change of individual stream, as well as the correlation amongst relevant streams.

Furthermore, we use agent technology to construct the system. The system framework is shown in Fig. 7.4.

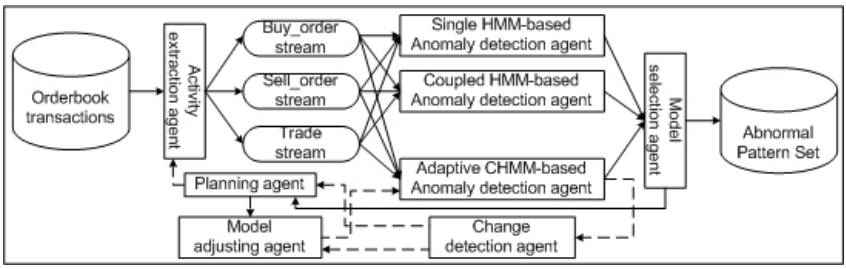

Fig. 7.4 Framework of agent-based abnormal pattern discovery by HMM

The system consists of the following agents: Activity Extraction Agent, Anomaly Detection Agent, Change Detection Agent, Model Adjusting Agent, and Planning Agent. They collaborate with each other to find out the best training model, and then deploy this best one for activity pattern discovery. The basic working process of the system is as follows.

PROCESS: Agent-based abnormal behavior pattern discovery by HMM
INPUT: Orderbook transactions $\mathscr{D}$, domain knowledge Ψ
OUTPUT: Abnormal behavior patterns $\mathscr{P}$
Step 1: Activity Extraction Agent splits orderbook transactions into activity streams $\mathscr{D}_k$ ($k = 1,\ldots,K$) in terms of market microstructure theory (Ψ). In our case, we obtain Buy_order stream D_B, Sell_order stream D_S, and Trade stream D_T;
Step 2: Training HMM-based model $\mathscr{R}_k$ on stream D_k respectively, or on selected streams $\{D_i,\ldots,D_k\}$;
Step 3: Model Selection Agent selects the best model;
Step 4: Testing the best model;
Step 5: Exporting final pattern set $\mathscr{P}$.

Furthermore, we propose an adaptive CHMM driven by agents. The pattern Change Detection Agent detects changes in the output of CHMM, and then the Planning Agent triggers the adjustment and retraining of the CHMM model to adapt to the source data dynamics.

7.10.1.2 Agent modeling

Each agent carries respective goals and roles, and follows certain interaction rules to collaborate with other agents for fulfilling the tasks. The basic goals and roles of all agents are described as follows.

AGENT: Activity Extraction Agent
GOAL: Extracting activity sequences from source data;
ROLE: ActivityExtractor
– Understanding activity types in source data;
– Splitting source data in terms of activity types;
– Extracting activity sequences.

AGENT: HMM-based Anomaly Detection Agent
GOAL: Detecting anomalies in activity streams by given HMM-based models
ROLE: AnomalyDetector
– Training the given HMM-based models;
– Testing the given HMM-based models;
– Exporting resulting abnormal patterns.

AGENT: Model Selection Agent
GOAL: Selecting the best models from given model list
ROLE: ModelSelector
– Collecting performance data in terms of given metrics for each model;

– Comparing performance of each model;
– Determining the best model.

AGENT: Planning Agent
GOAL: Coordinating the action planning of agents in terms of inputs
ROLE: Planner
– Receiving inputs from Change Detection Agent and Model Selection Agent;
– Sending response request to the corresponding agents;
– Triggering the actions of corresponding agents.

Each agent follows some protocols in conducting their roles to achieve their goals. For example, the Planning Agent follows an interaction protocol, named RetrainModelRequest. It guides the successful retraining of the best model.

PROTOCOL: RetrainModelRequest
Requester: PlanningAgent
Responder: HMM-based AnomalyDetectionAgent
Input: the best model with $model_id$
Rule: $model_id$.Training.Fulfilled()
Output: $retrain.Successful() = True$

7.10.2 Agent-Based Adaptive CHMM

The agent-based adaptive CHMM (ACHMM) is an extended version of CHMM. In this section, we introduce the working process and agent coordination in achieving the adaptation to data dynamics.

7.10.2.1 The process of ACHMM

ACHMM is implemented through a closed loop system, which consists of CHMM, Change Detection Agent, Model Adjusting Agent and Planning Agent. The adaptation is mainly based on the detection of change between the identified patterns and the benchmark ones. The benchmark patterns are those updated most recently which reflect the pattern change. Its basic idea is as follows.

The working process of ACHMM is as follows.

PROCESS: Agent-based abnormal behavior pattern discovery by ACHMM
INPUT: Orderbook transactions $\mathscr{D}$, domain knowledge Ψ
OUTPUT: Abnormal behavior patterns $\mathscr{P}$
Step 1: Stream Extraction Agent splits orderbook transactions into sequences $\mathscr{D}_k$ $(k = 1, \ldots, K)$ in

terms of market microstructure theory (Ψ). In our case, we get Buy_order stream D_B, Sell_order stream D_S, and Trade stream D_T;
Step 2: Training CHMM-based model $\mathscr{R}_k$ on stream D_k respectively, or on the selected streams $\{D_i, \ldots, D_k\}$;
Step 3: Change Detection Agent detects the change of the identified patterns compared with the benchmarks;
IF there is any significant change;
Informing the Planning Agent;
Triggering Model Adjusting Agent to adjust CHMM models;
Re-training CHMM models, re-extracting streams if necessary;
Testing CHMM-based models;
ENDFOR
Step 5: Retraining the CHMM model;
Step 6: Testing the CHMM model;
Step 7: Exporting final pattern set $\mathscr{P}$.

The key process is for the Change Detection Agent to detect change happening in the identified patterns compared with that of the benchmark patterns. If there is a significant pattern difference between those identified at previous time t_1 compared to those of the current moment t_2, the Planning Agent will receive a notice of that, and then trigger the Model Adjusting Agent to adjust the CHMM model, and then retrain the model.

7.10.2.2 Agent coordination

In fulfilling the adaptive model adjustment, multiple agents collaborate with each other. In particular, the Planning Agent coordinates with the Change Detection Agent to obtain indicators about the pattern change. If there is significant change, it then sends an adjustment request to the Model Adjusting Agent. The Model Adjusting Agent then calls the CHMM agent to retrain the model.

In conducting the coordination, the Planning Agent follows some interaction protocols. For instance, one of interaction protocols is AdjustModelRequest, which is described as follows.

PROTOCOL: AdjustModelRequest
Requester: PlanningAgent
Responder: ModelAdjustingAgent
Input: change_signal
Rule: $change_signal_{t1} \neq change_signal_{t2}$
Output: adjust_rates[]

We further show the interleaving protocol fulfilled by Change Detection agent, Planning Agent, and Model Adjusting Agent. Fig. 7.5 shows the interleaving protocol.

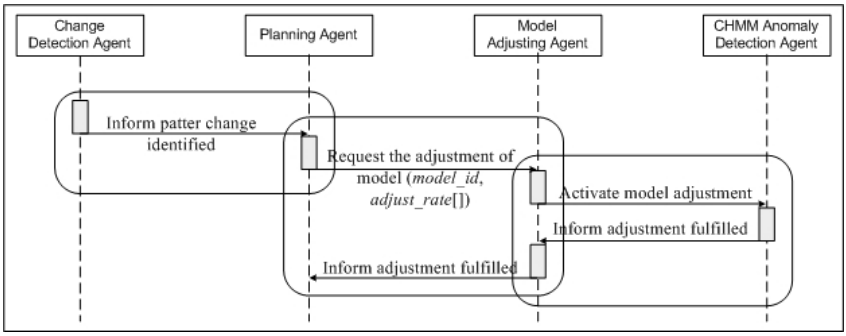

Fig. 7.5 Interleaving protocol for adjusting CHMM model

Its working mechanism is as follows. After detecting a significant pattern change, the Change Detection Agent informs the Planning Agent, the Planning Agent requests the model adjustment to be coordinated by the Model Adjusting Agent with *model_id* and adjustment indicators *adjust_rate*[]. The Model Adjusting Agent further activates the CHMM Anomaly Detection Agent to retrain the model on the updated data. After the model adjustment, it responds the Model Adjusting Agent with the status of adjustment, and this message is further passed to the Planning Agent.

7.11 Research Resources on Agent Mining

7.11.1 The AMII Special Interest Group

With regards to the symbiosis, interaction and integration of the two researches have attracted the communities' attention. Many research bodies have listed agent and data mining interaction and integration (AMII) as a special interest.

Agent mining communities are formed through emerging efforts in both AAMAS and KDD communities, as evidenced by continuous acceptance of papers on agent mining issues by prestigious conferences such as AAMAS, SIGKDD and ICDM. A Special Interest Group on Agent Mining Interaction and Integration (AMII-SIG) has been formed [4], which provides related resources, such as list of research topics, activities, workshops and conferences, links, related publications, research groups, etc.. AMII-SIG shares information covering many issues, and has also led the annual workshop series on Agent and Data Mining Interaction (ADMI) since 2006, moving globally and attracting researchers from many different communities. Another biennial workshop series - Autonomous Intelligent Systems - Agents and Data Min-

[4] AMII-SIG: www.agentmining.org

ing (AIS-ADM) has existed since 2005. Other events include special journal issues [51, 60], tutorials [21, 195], an edited book [23, 48] and monographs [194].

7.11.2 Related References

For the general knowledge and area overview of agent mining, we here introduce a few papers. The paper [44] presents a general overview of agent and data mining interaction, including research issues and field development in agent mining. The latest paper [49] presents an expert opinion of the field of agent mining, including disciplinary framework, agent-driven data mining, data mining-driven agents, and mutual issues in both areas.

Two recent edited books [23, 48] collect around 35 papers addressing different angles of agent mining, from fundamental issues to specific techniques and means, and to agent mining applications. In particular, Chapter 1 ”Introduction to Agent Mining Interaction and Integration” in [23] presents an overall picture of agent mining from the perspective of positioning it as an emerging area. It summarizes the main driving forces, complementary essence, disciplinary framework, applications, case studies, and trends and directions, as well as brief observation on agent-driven data mining, data mining-driven agents, and mutual issues in agent mining. The author draws the following conclusions:

- agent mining emerges as a new area in the scientific family,
- both agent technology and data mining can greatly benefit from agent mining,
- it is very promising to result in additional advancement in intelligent information processing and systems.
- However, as a new open area, there are many issues waiting for research and development from theoretical, technological and practical perspectives.

7.12 Summary

This chapter has discussed the basic concept and overall picture of agent-driven data mining for D^3M. This is through a progressive procedure: introduction of agent mining, agent-driven data mining, and agent-driven distributed data mining.

Conclusions from this chapter consist of:

- Agents and data mining, as two of most active techniques in IT area, can complement each other toward individual complementation and mutual enhancement. A new field, named agent mining, has formed with the dialog amongst all relevant fields.
- In the agent mining field, agent-driven data mining is very interesting and promising through the involvement of agents into handling challenges in data mining.

- In agent-driven data mining, agent-driven distributed data mining has attracted special attention.

In Chapter 8, we will discuss another technique, namely post-analysis and post-mining for domain driven data mining.

Chapter 8
Post Mining

8.1 Introduction

Data mining is widely used in many areas, such as retail, telecommunication, finance, etc. However, many data miners often face the following problems: How to read and understand discovered patterns, which are often in thousands or more? What are the most interesting ones? Is the model accurate and what does the model tell us? How to use the rules, patterns and models? To answer the above questions and present useful knowledge to users, it is necessary to do post mining to further analyse the learned patterns, evaluate the built models, refine and polish the built models and discovered rules, summarize them, and use visualisation techniques to make them easy to read and understand [242]. The function of post-mining in knowledge discovery process is shown in Figure 8.1, which bridges the gap between the patterns discovered by data mining techniques and the useful knowledge desired by end users.

Post mining is an important step in knowledge discovery process to refine discovered patterns and learned models and present useful and applicable knowledge to users. This chapter presents an introduction into recent advances of techniques for post mining, including interestingness measures, pruning and redundancy reduction, filtering and selection, summarization, visualisation and presentation, post-analysis and maintenance of discovered patterns.

This chapter is organized as follows. Section 8.2 presents interestingness measures. The filtering, selection, pruning and redundancy reduction of patterns are discussed in Section 8.3. The visualization and presentation of patterns are given in Section 8.4. Section 8.5 presents the techniques for summarizing discovered patterns and rules. The post-analysis of rules are discussed in Section 8.6. Section 8.7 describes the maintenance of models and patterns. The last section concludes this chapter.

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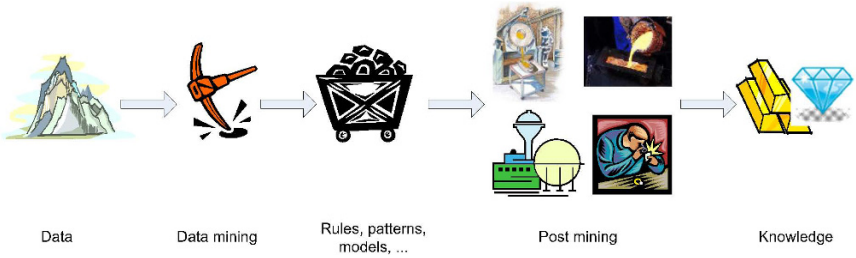

Fig. 8.1 Post Mining in the Process of Knowledge Discovery

8.2 Interestingness Measures

In data mining applications, the results are often some rules or patterns, which can be either interesting or simply common sense. Which rules or patterns are the most interesting ones? One way to tackle the above problem is to rank the discovered rules or patterns with interestingness measures, such as *lift*, *utility* [64], *profit* [210], *actionability* [41], etc.

To select interesting and useful rules, various interestingness measures have been designed. The measures of rule interestingness fall into two categories, subjective and objective [106, 185]. Objective measures, such as *lift*, *odds ratio* and *conviction*, are often data-driven and give the interestingness in terms of statistics or information theory. Subjective (user-driven) measures, e.g., *unexpectedness* and *actionability*, focus on finding interesting patterns by matching against a given set of user beliefs.

Support and confidence are the most widely used objective measures to select interesting rules. In addition to support and confidence, there are many other objective measures introduced by Tan et al. [196], such as ϕ*-coefficient*, *odds ratio*, *kappa*, *mutual information*, *J-measure*, *Gini index*, *laplace*, *conviction*, *interest* and *cosine*. Their study shows that different measures have different intrinsic properties and there is no measure that is better than others in all application domains. In addition, any-confidence, all-confidence and bond, are designed by Omiecinski [155]. *Utility* is used by Chan et al. [66] to find top-k objective-directed rules. *Unexpected Confidence Interestingness* and *Isolated Interestingness* are designed by Dong and Li [92] by considering its unexpectedness in terms of other association rules in its neighbourhood.

Unexpectedness and *actionability* are two main categories of subjective measures [184]. A pattern is unexpected if it is new to a user or contradicts the user's experience or domain knowledge. A widely-accepted definition of the actionability in knowledge discovery is that: a pattern is actionable if the user can do something with it to his/her advantage [184, 146]. Yang et al. [223] proposed to extract, from decision trees, actionable knowledge which "suggest actions to change customers from an undesired status to a desired one" while maximizing the expected net profit. Ras and Wieczorkowska [178] designed action-rules which show "what actions should be taken to improve the profitability of customers". The attributes are grouped into

"hard attributes" which cannot be changed and "soft attributes" which are possible to change with reasonable costs. The status of customers can be moved from one to another by changing the values of soft ones.

Liu and Hsu [139] proposed to rank learned rules by matching against expected patterns provided by the user. *Rule_Similarity* and *Rule_Difference* are defined to compare the difference between two rules based on their conditions and consequents, and *Set_Similarity* and *Set_Difference* are defined to measure the similarity between two sets of rules. The learned rules are ranked by the above similarity and difference, and then it is up to the user to identify interesting patterns.

Boettcher et al. [16] presented a unified view on assessing rule interestingness with the combination of rule change mining and relevance feedback. Rule change mining extends standard association rule mining by generating potentially interesting time-dependent features for an association rule during post-mining, and the existing textual description of a rule and those newly derived objective features are combined by using relevance feedback methods from information retrieval. Their idea is to find changes in rules by post analysing association rules at different time points. At first, a timestamped dataset is partitioned into intervals along the time axis, and then association rule mining is applied to each subset to find associations in every timeframe. The supports and confidences of each rule along history make up sequences, and the time-dependent features of association rules are discovered by post-mining those association rules and their history supports and confidences. Then the rules with a change pattern are analysed and evaluated based on objective interestingness. At last, A user's feedback about the rules are collected and used to obtain a new subjective interestingness ranking. A user expresses what he considers to be relevant by manually marking the rules, and the annotated rules can be taken as a representation of his notion of relevance. After each feedback cycle, the remaining rules are compared to the set of annotated rules and a new relevance score is calculated. Moreover, the user's knowledge beliefs and assumptions may change as he sees more rules, so Boettcher et al. introduced a decay of a relevant or non-relevant rule's importance over time, which gives a higher priority to a user's latest choices over the first ones.

Rezende et al. [180] present a new methodology for combining data-driven (objective) and user-driven (subjective) evaluation measures to identify interesting rules. With the proposed new methodology, data-driven measures can be used to select some potentially interesting rules for the user's evaluation, and then the rules and the knowledge obtained during the evaluation can be employed to calculate user-driven measures for identifying interesting rules. Their methodology is that, the objective measures are fist used to filter the rule set, and then subjective measures are used to assist the user in analysing the rules according to his knowledge and goals. The process are composed of four phases: objective analysis, user's knowledge and interest extraction, evaluation processing and subjective analysis. The first phase, objective analysis, uses objective measures to select a subset of the rules (named PIR: potentially interesting rules) generated by an association mining algorithm. In the second phase, the selected rules are ordered and short rules are presented to user first. The user evaluates the rules and makes them as unexpected knowledge, useful

knowledge, obvious knowledge, previous knowledge, and irrelevant knowledge. In the third phase, rules similar with "irrelevant" rules are removed, and then the subjective measures, conforming, unexpected antecedent, unexpected consequent and both-side unexpected, are computed for the rules no in PIR. And in the last phase, the user can explore the rules based on the subjective measures to find rules interesting to him.

Besides the above interestingness measures, there are also many others. A comprehensive review on interestingness measures of association rules and how to select the right ones can be found in the work of Silberschatz and Tuzhilin [185] and Tan et al. [196], Omiecinski [155] and Wu et al. [216].

8.3 Filtering and Pruning

There are often too many association rules discovered from a dataset and it is necessary to remove redundant rules before a user is able to study the rules and identify interesting ones from them. There are many techniques proposed to filter and prune the learned association rules and some of them are introduced in this section.

Toivonen et al. [201] proposed *rule covers* to prune discovered rules, where a rule cover is a subset of discovered rule base such that for each row in the relation there is an applicable rule in the cover if and only if there is an applicable rule in the original rule base. Rule covers turned out to produce useful short descriptions of large sets of rules.

Liu et al. [145] presented a technique on post-processing for rule reduction using closed set. Superfluous rules are filtered out from rule base in a post-processing manner. With dependent relation discovered by closed set mining technique, redundant rules can be eliminated efficiently. Their method is composed of three stages: 1) generating a transaction-rule contest from a given database and its rule-sets; 2) building closed rule-sets using closed set mining; and 3) pruning the original rule set by virtue of the obtained closed rule-sets.

In associative classification, it is very important to prune discovered rules and build highly accurate classifiers with a small set of association rules. Baralis and Chiusano [12] proposed *essential rule set* to yield a minimal rule subset, which compactly encodes without information loss the classification knowledge available in a classification rule set. It includes the rules that are essential for classification purposes, so it can replace the complete rule set. The essential rule set is a general purpose compact rule set, and it can be exploited to build many different associative classifiers. Their technique is particularly effective in dense datasets, where tradition extraction techniques may generate huge rule sets.

Antonie et al. [4] discussed the techniques for associative classification and described the three phases for building associative classifiers: 1) mining the training data for association rules and keeping only those that can classify instances, 2) pruning the mined rules to weed out irrelevant or noisy rules, and 3) selecting and combining the rules to classify unknown items. They discussed mining data sets with

re-occurring items and using the number of occurrences of an attribute in rule generation and in classification. They argued that negative association rules, which use negated value or imply a negative classification, can capture relationships that would be missed by positive only rule-based associative classifiers. Then, they designed a technique to evaluate each rule by using the rule set to re-classify the training dataset and measuring each rule's number of correct and incorrect classifications. The measurements are then plotted on a graph, based on which the rules are categorized and prioritized for pruning. They also presented a system, ARC-UI, that allows a user to interactively test and improve the results of classifying an item using an associative classifier.

Liu et al. [141] proposed to mine for *class association rules* and build a classifier based on the rules. With their rule generator, the rule with the highest confidence is chosen for all the rules having the same conditions but different consequents.

Zaïane and Antonie [229] studied strategies for pruning classification rule to build associative classifiers. Their method is selecting rules with high accuracy based on the plot of correct/incorrect classification for each rule on the training set. The above technique is designed for building an associative classifier with high accuracy, and cannot work to find the proposed actionable combined patterns.

Chiusano and Garza [72] discussed the selection of high quality rules in associative classification. They presented a comparative study of five well-known rule pruning methods that are frequently used to prune associative classification rules and select the rules for classification model generation. The five methods are database coverage, redundancy removal, chi-square test, pessimistic error rate based pruning and confidence-based pruning. Database coverage selects the minimal set of high-precedence rules needed to cover the training data and discards from the original set the rules that either do not classify any training data or cause incorrect classifications only. With redundancy removal, a rule set is split into two parts, a set of essential rules and a set of redundant rules, based on an order relation, and the redundant set is removed. Chi-square (χ^2) test is used to check the correlation between the antecedent and the consequent of a rule, to state whether a rule is statistically significant, and a rule is pruned when its antecedent and consequent are not correlated according to the chi-square test. With pessimistic error rate (PER) based pruning, the PER is used to evaluate the expected classification error of the rules. A rule is pruned if its PER value is higher than the PER value of at least one of its general rules, because it is expected to be less accurate than its general rules. Confidence-base pruning is the main traditional approach used for association rule pruning. It operates by selecting rule with confidence above a given threshold *minconf* and is performed a-posteriori, after having extracted the rules. Their experimental results show that the database coverage, either applied alone or together with other techniques, achieves on average the highest accuracy and the most compact and human-interpretable classification model. However, the results also show that the database coverage prunes on average more potentially useful rules than other approaches.

8.4 Visualisation

Visualization through visual imagery is an effective way to aid in understanding and exploring of a large number of complex patterns. Yang [223] designed a technique to use parallel coordinates for visualising frequent itemsets and many-to-many association rules. An association rule is visualised as a continuous polynomial curve connecting items in the rule, with one item on each parallel coordinate. The closure properties among frequent itemsets and association rules are embedded in the visualisation. When there is an item taxonomy, each coordinate is used to visualise an item taxonomy tree. The leaf nodes in the tree are items and non-leaf nodes are item categories. The tree is expandable and collapsible by user interaction. His approach is capable of visualising a large number of many-to-many rules and itemsets with many items. A user can interactively select and visualise a subset of itemsets or association rules with the above technique.

Yahia et al. [219] presented two meta-knowledge based approaches for an interactive visualisation of large amounts of association rules. Different from traditional methods of association rule visualisation where association rule extraction and visualisation are treated separately in a one-way process, the two proposed approaches that use meta-knowledge to guide the user during the mining process in an integrated framework covering both steps of the data mining process. The first one builds a roadmap of compact representation of association rules from which the user can explore generic bases of association rules and derive, if desired, redundant ones without information loss. The second approach clusters the set of association rules or its generic bases, and uses a fish-eye view technique to help the user during the mining of association rules. Their experimental results show that thousands of rules can be displayed in few seconds. With a merging representation which exploits both a 2D representation and a fish-eye view, the clustering-based visualisation is efficient to handle a large number of association rules. Their techniques make an interactive and intuitive approach that allows the expert-user to be able to visualise the whole set of generated rules and consequently to shed light on relevant ones. However, the work of selecting relevant rules is not covered by their methods.

Yamamoto et al. [220] also discussed the visualisation techniques to assist the generation and exploration of association rules. It presents an overview of the many approaches on using visual representations and information visualisation techniques to assist association rule mining. A classification of the different approaches that rely on visual representations is introduced, based on the role played by the visualisation technique in the exploration of rule sets. A methodology that supports visually assisted selective generation of association rules based on identifying clusters of similar itemsets is also presented. They developed an I_2E system (Interactive Itemset Extraction System) for visual exploration of itemsets and selective extraction and exploration of association rules. In the system, a projection-based graph visualisation of frequent itemsets is constructed by applying a multi-dimensional projection technique to feature vectors extracted form the itemsets, in order to obtain a two-dimensional layout. Then, clustering is used to identify clusters of similar itemsets and itemset filtering is performed on each cluster to make sure that rules

representative of different groups of similar itemsets will be extracted and displayed. After that, a pairwise comparison of rules using relations is conducted to identify rules that bear some relation with a given rule of interest and decide which rules are worth saving for knowledge acquisition. A parallel coordinates visualisation of rule interestingness measures assists the comparison. Each axis maps one interest measure and each rule is represented by a polyline that intersects the axes on the points corresponding to its measure values. The system employs visualisation to assist exploration of mining results and user-driven interactive rule extraction and exploration.

8.5 Summarization and Representation

In association rule mining, many researchers work on summarising a large number of rules to present users with a small set of rules which are easier to interpret and understand, or representing the discovered rule base with a small subset without losing much information.

Liu et al. [142] proposed a technique for pruning and summarising discovered association by first removing those insignificant associations and then forming direction setting (DS) rules, a summary of the discovered associations. The chi-square (χ^2) test is used to measure the significance of rules and insignificant ones are pruned. The test is then used again to remove the rules with "expected directions", that is, the rules which are combinations of direction setting rules. Their method is to provide a brief summary of discovered association rules, "analogous to summarization of a text article" [142]. A user is supposed to view the summarized DS rules first, and then focus on the interesting aspects and study the non-DS rules to get more details.

Pasquier [161] presented frequent closed itemset based condensed representations for association rules. Many applications of association rules to data from different domains have shown that techniques for filtering irrelevant and useless association rules are required to simplify their interpretation by the end-user. A *condensed representation* is a reduced set of association rules that summarizes a set of strong association rules. A condensed representation is called *a basis of an association rule set* if each rule in the basis must contain information not deducible from other rules in the basis and all strong association rules can be deduced from the basis. Condensed representations and bases present to the end-user of a very small set of rules selected among the most relevant association rules according to objective criteria.

Lent et al. [134] proposed to reduce the number of learned association rules by clustering. Using two-dimensional clustering, rules are clustered by merging numeric items to generate more general rules. It is useful when the user desires to segment the data. A *clustered association rule* is defined as a rule formed by combining similar, "adjacent" assocition rules. An example of a clustered rule is "$40 \leq$ age $< 42 \Rightarrow$ own_home=yes" could be formed from two rules "age=40 $\Rightarrow$ own_home=yes" and "age=41 $\Rightarrow$ own_home=yes". Clustered association rules are

helpful in reducing the large number of rules that are typically computed by existing algorithm, thereby rending the clustered rules much easier to interpret and visualise.

8.6 Post-Analysis

With post-analysis, the discovered patterns are further analysed in the relationship between patterns and the temporal features of patterns.

Liu et al. [143] proposed to analyse association rules from the temporal dimension. Give the history of the association rules and their support, confidence, etc., the rules are devided into three groups: *semi-stable rules*, *stable rules* and *trend rules*. A rule is semi-stable if non of its confidences (or supports) in the time periods is statistically below *minconf* (or *minsupp*). A stable rule is a semi-stable rule and its confidences (or supports) over time do not vary a great deal, i.e., they are homogeneous. The stable rules and trend rules help users to understand the behavior of rules over time.

Cherfi et al. [71] presented a new technique to combine data mining and semantic techniques for post-mining and selection of association rules. To focus on the result interpretation and discover new knowledge units, they introduce an original approach to classify association rules according to their conformity with a background knowledge model. Its successful application on text mining shows the benefits of taking into account a knowledge domain model of the data.

In the case of stream data, the post-mining of associations is more challenging. Thakkar et al. [199] presented a Stream Mill Miner (SMM) prototype for continuous post-mining of association rules in a data stream management system. In their system, the candidate rules are passed to the Rule Post-Miner (RPM) model that evaluates them by various semantic criteria and by comparing them against a historical database of rules previously characterised by an analyst. As a result, the current deployed rules are revised and the historical rule repository is updated. The analyst continuously monitors the system and provides critical feedback to control and optimise the post-mining process by querying and revising the historical rule repository.

The Receiver Operating Characteristics (ROC) graph is a popular way of assessing the performance of classification rules, but they are inappropriate to evaluate the quality of association rules, as there is no class in association rule mining and the consequent part of two different association rules might not have any correlation at all. Prati [166] presented a novel technique of QROC (Quality ROC), a variation of ROC space to analyse itemset costs/benefits in association rules. In a QROC graph, the axes are re-scaled so that points in QROC space can be directly compared regardless of the consequent. Therefore, it is especially suitable when there is a strong skew ratio between the marginal probabilities in a contingency table. It can be used to prune uninteresting rules from a large set of rule base and can be used to help analysts to evaluate the relative interestingness among different association rules in different cost scenarios.

Zhao et al. [239, 240] proposed to post analyse association rules by finding rule pairs and rule clusters, which can provide more useful knowledge than presenting separate rules. Moreover, they also proposed growing sequential patterns which can show the changes of patterns as time goes on. Several novel interestingness measures, *Contribution*, I_{rule}, I_{pair} and I_{cluster}, are designed by considering information from different types of data in combined rules and combined patterns. *Contribution* measures how much a part of a pattern contributes to the appearance of a target outcome. Based on *Contribution*, the other three measures, I_{rule}, I_{pair} and I_{cluster}, give respectively the interestingness of a single rule, a pair of rules and a cluster of rules.

8.7 Maintenance

The data in most real life applications are not static. Instead, new data keeps coming as there are always new transactions and new customers every day. As a result, the patterns may evolve in response to data updates. Therefore, the maintenance of discovered knowledge on updated databases is very important.

Feng et al. [104] studied the techniques for the maintenance of frequent patterns. The frequent pattern maintenance problem is summarized with a study on how the space of frequent patterns evolves in response to data updates. Focusing on incremental and decremental maintenance, four major types of maintenance algorithms are introduced, which are apriori-based algorithms, partition-based algorithms, prefix tree based algorithms and concise representation based algorithms. The advantages and limitations of these algorithms are studied from both the theoretical and experimental perspectives. Their experimental evaluation shows that algorithms that involve multiple data scans suffer from high I/O overhead and low efficiency, which can be solved by employing a prefix tree, such as FP-tree, to summarize and store the dataset. Their experiments also show that CanTree (Canonical-order Tree) [137], a prefix tree based algorithm, is the most effective one for incremental maintenance and TRUM [103], which maintains the equivalence classes of frequent patterns, is the most effective one for decremental maintenance.

Wu et al. [215] designed an adaptive model for debt detection in social security by adjusting the sequence classification model based on new data. At first, sequence patterns are discovered from training data and then a set of discriminative sequential patterns are selected and used to build a classification model. After that, the model is applied to new data to predict debt occurrences. Based on the correct and wrong predictions on the new data, the ranks and weights of the sequential patterns are adjusted accordingly. A pattern which classify more patterns correctly than wrongly gets increased rank and weight. A pattern performed badly on new data are decreased in its rank and weight. Patterns giving no correct predictions but only wrong predictions are removed from the model, since they become out-of-date in new data. The model is proved to be effective for debt detection when the patterns evolve overtime.

8.8 Summary

This chapter presents a brief introduction of recent advances in post mining, which does further analysis after patterns are discovered by data mining algorithms. It discusses interesting measures, pruning, selection, summarization, visualisation and maintenance of patterns. Besides the techniques presented in this chapter, there are some other topics that might be related to post mining, such as interactive analysis, validation and evaluation of built models and discovered patterns.

The next chapter will discuss some case studies in which domain driven data mining has been applied to develop actionable trading strategies for smart trading, and mining for actionable market microstructure behavior patterns for deeply understanding of exceptional trading behavior on capital market data.

Chapter 9
Mining Actionable Knowledge on Capital Market Data

9.1 Case Study 1: Extracting Actionable Trading Strategies

9.1.1 Related Work

In financial markets, traders always pursue profitable trading strategies (also called trading rules) to make good return on investment. For instance, an experienced trader may use an appropriate pairs trading strategy, namely taking a long position in the stock of one company and shorting the stock of another in the same sector. A long position reflects the view that the stock price will rise; shorting reflects the opposite. This strategy may statistically increase profit while decreasing risk compared to a naive strategy of putting all money on one stock. In practice, an actionable strategy can not only maximize the profit or return, but also result in the proper management of risk and trading costs.

An artificial financial market [5, 67, 98, 212] provides an economic, convenient and effective electronic marketplace (also called e-market) for the development and back-testing of actionable strategies taken by trading agents without losing a cent. A typical simulation drive is the Trading Agent Competition [197, 212], for instance, auction-oriented protocol and strategy design [82], bidding strategy [83], design tradeoffs [121], and multi-attribute dynamic pricing [81] of trading agents. However, existing trading agent research presents a prevailing atmosphere of academia. This is embodied in aspects such as artificial data, abstract trading strategy and market mechanism design, and simple evaluation metrics. In addition, little research has been done on strategy optimization in continuous e-markets. We know continuous markets form the major market form [5, 67].

The above atmosphere has led to a big gap between research and business expectation. As a result, the developed techniques are not necessarily of business interest or cannot support business decision-making. In fact, the development of actionable strategies is a non-trivial task due to domain knowledge, constraints and expectations in the market [41]. Very few studies on continuous e-markets have been conducted for actionable trading strategies in the above constrained practical scenarios

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[53, 138, 38]. Therefore, it is a very practical challenge and driving force to narrow the gap towards workable trading strategies for action-taking to business advantage.

An actionable trading strategy can assist trading agents in determining right actions at right time with right price and volume on right instruments to maximize the profit while minimizing the risk [36, 24]. The development of actionable strategies targets an appropriate combination or optimization of relevant attributes such as target market, timing, actions, pricing, sizing and traded objects based on proper business and technical measurement. This combination and optimization in actionable trading strategy development should consider certain market microstructure and dynamics, domain knowledge and justification, as well as investors' aims and expectations. These form the constrained environment in developing actionable strategies for trading agents.

In this section, we discuss lessons learned in actionable trading strategy development in continuous e-markets based on our experience of research and practical development. We (1) discuss real-life constraints that need to be taken into account in trading agent research, (2) propose an actionable trading strategy framework, and (3) investigate a series of approaches to actionable strategy development.

We first identify some important institutional features and constraints on designing trading strategies for trading agents: for instance, varying combinations of organizational factors form different market microstructure. The basic ideas include the notion that the development of actionable strategies in constrained scenarios is an iteratively in-depth strategy discovery and refinement process where the involvement of domain knowledge and human-agent cooperation are essential. The involvement of domain experts and their knowledge can assist in understanding, analyzing and developing highly practical trading strategies. In-depth strategy discovery, refinement and parallel supports can effectively improve the actionable capabilities of an identified strategy.

Following the above ideas, we study a few effective techniques for developing actionable trading strategies in a continuous e-market context. These include designing and discovering quality trading strategies, and enhancing the actionable performance of a trading strategy through analyzing its relationship with target stocks. In addition, parallel computing is also imposed on efficiently mining actionable strategies. All of these methods are simulated and back-tested in an agent service-based artificial financial market F-Trade [38] with connections to multiple market data.

9.1.2 What Is Actionable Trading Strategy?

9.1.2.1 Trading Strategies

Intelligent agent technology is very useful and is increasingly used for developing, back-testing and evaluating automated trading techniques and program trading strategies in e-market places [212] without market costs and risks before they are deployed into the business world [82]. In fact, with the involvement of business

lines in agent-based computational finance and economics studies, a trading agent has the potential to be customized for financial market requirements. The idea is to extend and integrate the concepts of trading agents, agent-based financial and economic computation and data mining with trading strategy development in finance to design and discover appropriate trading strategies for trading agents. Classic agent intelligence such as autonomy, adaptation, collaboration and computation is also encouraged in aspects such as automated trading and trading agent collaboration for strategy integration. In this way, trading agents can be dedicated to the development of financial trading strategies for market use. A trading strategy indicates when a trading agent can take which trading actions under certain market situation. For instance, the following illustrates a general Moving Average (MA) based trading strategy.

EXAMPLE 1. (MA Trading Strategy). An MA trading strategy is based on the calculation of moving average of security prices over a specified period of time. Let n be the length (i.e. number of prices) of MA in a time period, $P_{I,i}$ (or for short P_i) be the price of an instrument I at the time of No. i ($i < n$) price occurrence. An MA at time t (which corresponds to No. n price) is calculated as $MA_t(n)$:

$$MA_t(n) = \frac{1}{n}\sum_{i=0}^{n-1} P_i \tag{9.1}$$

A simple MA strategy is to compare the current price P_t of security I with its MA value: MA_t. Based on the conditions met, a MA strategy generates 'sell' (denoted by 1), 'buy' (denoted by 1) or 'hold' (denoted by 0) trading signal at time t. If P_t rises above $MA_t(n)$, a buy signal is triggered, the security is then bought and held until the price falls below MA, at which time a *sell* signal is generated and the security is sold. For any other cases, a *hold* signal is triggered.

The pseudo code of MA strategy for generating trading signal sequence S is represented as follows.

$$\begin{cases} S = 1 & \text{if} P_t > MA_t(n) \text{and} P_i < MA_t(i), \forall i \in 1,\ldots,n-1 \\ S = -1 & \text{if} P_t < MA_t(n) \text{and} P_i > MA_t(i), \forall i \in 1,\ldots,n-1 \\ S = 0 & \text{otherwise} \end{cases} \tag{9.2}$$

This strategy is usually not workable when it is employed in the real world. To satisfy real-life needs, market organizational factors, domain knowledge, constraints, trader preference and business expectation are some key factors that must be considered in developing actionable trading strategies for workable trading agents.

9.1.2.2 Actionable Trading Strategies

As discussed above, searching actionable trading strategies for trading agents is a process to identify trading patterns that can reflect the 'most appropriate' combination of purchase timing, position, pricing, sizing and objects to be traded under certain market situations and interest-driving forces [206]. To this end, trading agents

may cooperate with each other to either search the 'optimal' solutions from a huge amount of searchable strategy space denoted by a trading pattern. In some other cases, they collaborate to synthesize multiple trading strategy fragments favored by individual agents into an integrative strategy satisfying the general concerns of each agent as well as global expectation representing the trader's interest.

In addition, data mining can play a critical role in actionable strategy searching and trading pattern identification. Data mining in finance has the potential to identify not only trading signals, but also patterns indicating either iterative or repeatable occurrences. Therefore, developing actionable trading strategies for trading agents [81, 105] is an interaction and collaboration process between agents and data mining [149]. The aim and objective of this process is to develop smart strategies for trading agents to take actions in the market that can satisfy a trader's expectation within a certain market environment.

Definition Trading Strategy. A trading strategy actually represents a set of individual instances, the rule set Ω is a tuple defined as follows.

$$\begin{aligned}\Omega &= \{r_1, r_2, \ldots, r_m\} \\ &= \{(t, b, p, v, i) | t \in T, b \in B, p \in P, v \in V, i \in I)\}\end{aligned} \tag{9.3}$$

where r_1 to r_m are instantiated individual trading rules, each of them is represented by instantiated parameters of t, b, p, v and an instrument i to be traded. $T = \{t_1, t_2, \ldots, t_m\}$ is a set of appropriate time trading signals to be triggered. $B = \{$buy, sell, hold$\}$ is the set of possible behaviors (i.e., trading actions) executed by trading participants. $P = \{p_1, p_2, \ldots, p_m\}$ and $V = \{v_1, v_2, \ldots, v_m\}$ are the sets of trading price and volume matching with corresponding trading time. $I = \{i_1, i_2, \ldots, i_m\}$ is a set of target instruments to be traded.

With the consideration of environment complexities and the trader's favorite strategy, the optimization of trading strategies is to search an 'appropriate' combination set Ω' in the whole rule candidate set Ω, in order to achieve both user-preferred technical (tech_int()) and business (biz_int()) interestingness in an 'optimal' or 'sub-optimal' manner. Here 'optimal' refers to the maximal/minimal (in some cases, smaller is better) values of technical and business interestingness metrics under certain market conditions and user preferences. In some situations, it is impossible or too costly to obtain 'optimal' results. For such cases, a certain level of 'sub-optimal' results are also acceptable. Therefore, the sub-set Ω' indicates 'appropriate' parameters of trading strategies that can support a trading participant a ($a \in A$, A is market participant set) to take actions to his/her advantage. As a result, in some sense, trading strategy optimization aims to extract actionable rules with multiple attributes towards multi-objective optimization in a constrained market environment.

Definition Actionable Trading Strategy. An optimal and actionable trading strategy set Ω' is to achieve the following objectives:

$$\begin{aligned}tech_int() &\rightarrow max\{tech_int()\} \\ biz_int() &\rightarrow max\{biz_int()\}\end{aligned} \tag{9.4}$$

while satisfying the following conditions:

$$\begin{aligned} \Omega' &= \{e_1, e_2, \ldots, e_n\} \\ \Omega' &\subset \Omega \\ m &> n \end{aligned} \tag{9.5}$$

where tech_int() and biz_int() are general technical and business interestingness metrics, respectively. As the main optimization objectives of identifying 'appropriate' trading strategies, the performance of trading rules and their actionable capability are encouraged to satisfy expected technical interestingness and business expectations under multi-attribute constraints. In Formula 9.4, we only show the maximal situations of objective optimization. As we have pointed out, this could be in a minimal situation. The ideal aim of actionable trading strategy discovery is to identify trading patterns and signals, in terms of certain background market microstructure and dynamics, so that they can assist traders in taking the right actions at the right time with the right price and volume on the right instruments. As a result of trading decisions directed by the identified evidence, benefits are maximized while costs are minimized.

For instance, the very common trading strategy Moving Average is a function of attributes trading price, start date/time, and the size of the sliding window for calculating the average. In market trading, the moving average can be formulated into many different forms of trading strategies, for instance, some MA(x, y) or MA(x, y, z). On the other hand, a general model of moving average-based trading strategy can be parameterized into various instances. For example, MA(x, y) may be replaced by MA(5, 20) and MA(13, 26).

The task of optimizing a moving average based strategy, say MA(x, y), is to search appropriate x and y to reach the best of the expected business performance. This effort generates a subset of moving average strategies of the general model.

9.1.3 Constraints on Actionable Trading Strategy Development

Typically, actionable trading strategy development must be based on a good understanding of organizational factors associated with a market, otherwise it is not possible to accurately evaluate actionability. In real-world actionable pattern mining, the underlying environment is more or less constrained [197]. Constraints may be broadly embodied in terms of data, domain, interestingness and deployment aspects [82].

9.1.3.1 Domain Constraint

Typically, actionable trading rule optimization must be based on a good understanding of organizational factors hidden in the mined market and data, otherwise it is

not possible to accurately evaluate the dependable capability of the identified trading rules. The actionable capability of optimized trading rules is highly dependent on the mining environment where the rule is extracted and applied. In real-world actionable rule extraction, the underlying environment is more or less constrained. Constraints may be broadly embodied in terms of data, domain, interestingness and deployment aspects. Here we attempt to explain domain and deployment constraints surrounding actionable trading rule discovery. In Section 9.1.3.3, interestingness constraint will be discussed.

Market organization factors relevant to trading rule discovery consist of the following fundamental entities: $M = \{I, A, O, T, R, E\}$. Table 9.1 briefly explains these entities and their impact on trading rule actionability. In particular, the entity $O = \{(t, b, p, v) | t \in T, b \in B, p \in P, v \in V\}$ is further represented by attributes T, B, P and V, which are attributes of trading rule set Ω. The elements in M form the constrained market environment of trading rule optimization. In the strategy and system design of trading rule optimization, we need to give proper consideration to these factors.

In practice, any particular actionable trading rule needs to be identified in an instantiated market niche m ($m \in M$) enclosing the above organization factors. This market niche specifies particular constraints, which are embodied through the elements in Ω and M, on rule definition, representation, parameterization, searching, evaluation and deployment. The consideration of a specific market niche in trading rule extraction can narrow the search space and rule space in trading rule optimization. In addition, there are other constraints such as data constraints D that are not addressed here due to limited space. Comprehensive constraints greatly impact the development and performance of extracting trading strategies.

Table 9.1 Market Organizational Factors and Their Impact on Rule Actionability

Organizational factors	Impact on actionability
Traded instruments I, such as a stock or derivative, I={stock, option, feature, ...}	Varying instruments determine different data, analytical methods and objectives
Market participants A, A = {broker, market maker, mutual funds, ...}	Trading agents have the final right to evaluate and deploy discovered trading rules to their advantage
Order book forms O, O = {limit, market, quote, block, stop}	Order type determines what data set (eg., order book) is to be mined, as well as particular business interestingness
Trading session, whether including a call market session or continuous session, is indicated by time frame T	Setting up the focusing session can greatly prune order transactions
Market rules R, e.g., restrictions on order execution defined by exchange	Determine pattern validity of discovered trading rules when deployed
Execution system E, e.g., a trading engine is order-driven or quote-driven	Limit pattern type and deployment manner after migration to real trading system

Constraints surrounding the development and performance of actionable trading rule set Ω' in a particular market data set form a constraint set:

$$\Sigma = \{\delta_i^k | c_i \in C, k \in N\}$$

where δ_i^k stands for the k-th constraint attribute of a constraint type c_i. $C = \{M, D\}$ is a constraint type set covering all types of constraints in market microstructure M and data D in the searching niche. N is the number of constraint attributes for a specific type i.

Correspondingly, actionable trading rule set Ω' is a conditional function of Σ, which is described as

$$\Omega' = \{(\omega, \delta) | \omega \in \Omega, \delta \in \{(\delta_i^k, a) | \delta_i^k \in \Sigma, a \in A\}\}$$

where ω is an 'optimal' trading pattern instance, δ indicates specific constraints on the discovered pattern that is recommended to a trading agent a.

Let us explain this again through the illustration of Moving Average-based trading strategies. For the rule MA(l), let it be deployed to trade in the order-driven Australian Securities eXchange (ASX) market by a broker. The staff member instantiates it into a form of MA(5), which is a five-transaction moving average, and sets a benchmark $\tau = AU\$25.890$ in trading BHP (BHP Billiton Limited) on 24 January 2007. In this situation, he/she believes the instantiated rule is one of the most dependable MA(l). Here M is instantiated into {stock, broker, market order, continuous session, order-driven}. In Section 9.1.4.1, we will discuss how genetic algorithms can play a role in working out an appropriate parameter combination such as {l, start date, τ} to profitably trade a given stock in a specific constrained market.

To work out the actionable set Ω' efforts on many levels are essential. In particular, the concerns and expectations of market investors play inevitable roles. Section 9.1.3.3 will briefly discuss the relationship and balance between technical significance and business expectations in trading rule optimization.

9.1.3.2 Data Constraint

The second type of constraint on trading strategy development is data constraint. Huge quantities of historical data can play an especially important role in strategy modeling. We model strategies using data mining to discover interesting and actionable trading patterns. Data constraint set D consists of the following factors, where $D = \{quantity, attribute, location, format, frequency, privacy, compression\}$. An exchange intraday data stream normally presents characteristics such as high quantities, high frequency and multiple attributes of data. Some exchanges specify user-defined data format or follow a standard format such as FIX protocol [121]. Data may be distributed in multiple clusters with compression for large exchanges. Data constraint seriously affects the development and performance of trading strategies. For instance, the efficiency of complex strategy simulation and modeling may in-

volve parallel supports on multiple sources, parallel I/O, parallel algorithms and memory storage.

9.1.3.3 Deliverable Constraint

Often deliverable trading strategies are not actionable in the real market even though they are functional in research. This gap may be due to the interestingness gaps between academia and business. What makes this trading strategy more interesting than the other? This is determined by deliverable constraint, which is embodied through interestingness, namely a set $Int = Tint, Bint$. Trading strategy interestingness covers both technical interestingness $Tint$ and business interestingness $Bint$. In the real world, simply emphasizing technical interestingness such as statistical measures of validity is not adequate for designing strategies. Social and economic interestingness (we refer to business interestingness), for instance, profit, return and return on investment, should be considered in assessing whether a strategy is actionable or not. Integrative interestingness measures which integrate both business and technical interestingness are expected. Satisfying the integrative interestingness can benefit the actionability of trading agents in the real world.

There may be other types of constraints such as dimension/level constraints. These ubiquitous constraints form a multi-constraint scenario, namely $M, D, Int, \ldots$, for actionable strategy design. We think that actionable trading strategy development and optimization in continuous e-markets should be studied in the above constrained scenario.

9.1.3.4 Human-Agent Cooperation

The constraint-based context and actionable requirement of trading strategies determine that the development process is more likely to be human involved rather than automated. Human involvement is embodied through cooperation between humans (including investors and financial traders) and trading agents. This is achieved through the compensation between human intelligence (e.g., domain knowledge and experience) and agent intelligence. Therefore, trading strategy development likely presents as a human-agent-cooperated interactive discovery process.

The role of humans may be embodied in the full period of development from market microstructure design, problem definition, data preprocessing, feature selection, simulation modeling, strategy modeling and learning to the evaluation, refinement and interpretation of discovered strategies and resulting outcomes. For instance, the experience and meta-knowledge of domain experts can guide or assist with the selection of features and strategy modeling, adding trading factors into the modeling, designing interestingness measures by injecting traders' concerns, and quickly evaluating results. This can largely improve the effectiveness of designing trading strategies.

To support human involvement, human agent cooperation is essential. Interaction often takes explicit form, for instance, setting up interaction interfaces to tune the parameters of trading strategies. Interaction interfaces may take various forms, such as visual interfaces, virtual reality technique, multi-modal and mobile agents. On the other hand, interaction may also follow implicit mechanisms, such as accessing a knowledge base or communicating with a user assistant agent. In interactive trading strategy optimization, the performance of the discovered strategies relies highly on interaction quality in terms of the representability of domain knowledge, and the flexibility, user-friendliness and run-time capability of interfaces.

9.1.4 Methods for Developing Actionable Trading Strategies

Following the ideas introduced in Section 9.1.2, this section illustrates some approaches for developing actionable trading strategies. They consist of optimizing trading strategies, extracting in-depth trading strategies, discovering trading strategies, developing trading strategy correlated with instruments, and integrating multiple strategies. Their promising business performance is demonstrated in tick-by-tick data.

9.1.4.1 Optimizing Trading Strategies

There are often huge quantities of variations and modifications of a generic trading strategy by parameterization. For instance, *MA(2, 50, 0.01)* and *MA(10, 50, 0.01)* refer to two different strategies. However, it is not clear to a trader which specific rule is actionable for his or her particular investment situation. In this case, trading strategy optimization may generate an optimal trading rule from the generic rule set.

Optimizing trading strategies aims to find trading strategies with better target performance. This can be achieved through developing varying optimization methods. Genetic Algorithm (GA) is a valid optimization technique, which can be used for searching combinations of trading strategy parameters satisfying user-specified performance [138]. However, a simple use of GA may not necessarily lead to trading strategies of business interest. To this end, domain knowledge must be considered in fitness function design, search space and speed design, and so on. The fitness function we used for strategy optimization is Sharpe Ratio (*SR*).

$$SR = \frac{R_p - R_f}{\delta_p} \tag{9.6}$$

R_p is the expected portfolio return, R_f is the risk free rate, and δ_p is the portfolio standard deviation. When *SR* is higher, it indicates higher return but lower risk.

Fig. 9.1 illustrates some results of GA-based trading strategy optimization. The trading strategy is *Filter Rule Base*, namely $FR(\delta)$. It indicates a generic class of correlated trading strategies, which necessitates going long on the day that the price

rises by $\delta\%$ and holding until the price falls $\delta\%$, subsequently closing out and going short

Algorithm *Trading Strategy 1*

1. A generic strategy $FR(\delta)$
2. At time point t, get $high(t)$ and $low(t)$
3. IF $price(t-1) > high(t-1)$
4. $high(t) = price(t-1)$

   ELSE
5. $high(t) = high(t-1)$
6. IF $price(t-1) < low(t-1)$
7. $low(t) = price(t-1)$
8. ELSE
9. $low(t) = low(t-1)$
10. Generate trading signals
11. IF $price(t) < high(t) * (1-\delta)$
12. Generate SELL signal
13. IF $price(t) > low(t) * (1+\delta)$
14. Generate BUY signal
15.

In this rule, there is only one parameter δ, which can be used for optimization because δ is hard to manage well in a real-life market. Fig. 9.1 shows the optimization results of the stock Australian Commonwealth Bank (CBA) in the Australian Stock Exchange (ASX) in 2003$\tilde{2}$004. It shows that from 14 July 2003, the cumulative payoff with $\delta = 0.04$ always beats other δ.

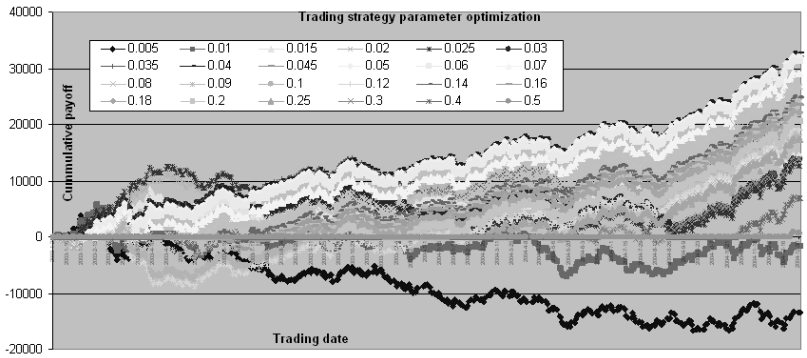

Fig. 9.1 Some results of GA-based trading strategy optimization.

9.1.4.2 Extracting In-depth Trading Strategies

In many real-life cases, a given trading strategy may not work well due to the failure to consider some organizational factors and constraints. To this end, we need to enhance a trading strategy by involving real-life constraints and factors. For instance, the above rule $FR(\delta)$ does not consider the noise impact of false trading signals and dynamic difference between high and low sides. These aspects can be reflected in the rule by introducing new parameters.

Extracting in-depth trading strategies is not a trivial task. It is necessary to consider domain knowledge and expert advice, conduct massive back-testing in historical data, and mine hidden trading patterns in market data, otherwise a developed strategy is unlikely to make sense to business. For instance, we create an Enhanced Filter Rule $FR(t, \delta_H, \delta_L, h, d)$ as follows.

Algorithm *Trading Strategy* 2A

n enhanced $FR(t, \delta_H, \delta_L, h, d)$
At time point t, get $high(t)$ and $low(t)$
IF $price(t-1) > high(t-1)$
    $high(t) = price(t-1)$
ELSE
    $high(t) = high(t-1)$
IF $price(t-1) < low(t-1)$
    $low(t) = price(t-1)$
ELSE
    $low(t) = low(t-1)$
Generate trading signals
    IF $price(t) < high(t) * (1-\delta_H)$
        Generate SELL signal
            IF $position(t-1) <> 0$ & $hold(t-1) = h$
                $position(t) = 1$
    IF $price(t) > low(t) * (1+\delta_L)$
        Generate BUY signal
            IF $position(t-1) <> 0$ & $hold(t-1) = h$
                $position(t) = -1$

This enhanced version considers the following domain-specific aspects, which make it more adaptive to real market dynamics compared with the generic rule $MA(sr, lr, \delta)$.

- More filters are imposed on the generic FR to filter out false trading signals which would result in losses, say fixed percentage band filter δ_H and δ_L for high and low price movement respectively, and time hold filter h;
- The fixed band filter δ_H (or δ_L) requires the buy or sell signal to exceed high or low by a fixed multiplicative band δ_H (or δ_L);

- The time hold filter h requires the buy or sell signal to hold the long or short position for a pre-specified number of transactions or time h to effectively ignore all other signals generated during that time.

Fig. 9.2 shows the trading results of a trading agent taking the above strategy in ASX data 2003-2004.

Fig. 9.3 further shows the performance difference between a base rule and its enhanced version. It indicates that the involvement of domain knowledge and organizational constraints can to a great extent enhance the business performance (cumulative payoff in our case) of trading strategies.

9.1.4.3 Discovering Trading Strategies

Another method of trading strategy development is through mining trading patterns in stock data. For instance, based on the domain assumption that some instruments are associated with each other, we can discover trading patterns effective on multiple correlated instruments. The following illustrates an effective pair trading strategy. It indicates that a trading agent can go long with one instrument while short on another.

Algorithm *Trading Strategy 3*A
1. pair trading strategy PTS, T
2. C1. Calculating the $tech_int()$, e.g., coefficient ρ, of two stocks S and T considering market index;

	A	B	C	D	E	F	G	H	I	J	K	L
4	Trade Date	Trade Price	Maximum Price	Minimum Price	Sell Cond Met	Buy Cond Met	Trade Signal Indicator	Trades	Position Hold Days	Positions	Payoff ($)	Payoff($) Cumulative
5	1/2/2003	3027	0	0	0	0	0	0	0	0	0	0
6	1/3/2003	3066	0	0	0	0	0	0	0	0	0	0
7	1/6/2003	3079	0	0	0	0	0	0	0	0	0	0
8	1/7/2003	3060	0	0	0	0	0	0	0	0	0	0
9	1/8/2003	3072	0	0	0	0	0	0	0	0	0	0
10	1/9/2003	3065	3088	3027	0	0	0	0	0	0	0	0
11	1/10/2003	3066	3088	3065	0	0	0	0	0	0	0	0
12	1/13/2003	3063	3088	3066	-1	0	-1	-1	0	0	0	0
13	1/14/2003	3077	3088	3063	0	0	-1	0	1	-1	-350	-350
14	1/15/2003	3075	3077	3063	0	0	-1	0	2	-1	50	-300
15	1/16/2003	3062	3077	3063	-1	0	-1	0	3	-1	325	25
16	1/17/2003	3055	3077	3062	-1	0	-1	0	4	-1	175	200
17	1/20/2003	3059	3077	3055	0	0	-1	0	5	-1	-75	125
18	1/21/2003	3054	3077	3055	-1	0	-1	0	6	-1	100	225
19	1/22/2003	3016	3075	3054	-1	0	-1	0	7	-1	950	1175
20	1/23/2003	3016	3062	3016	0	0	-1	0	8	-1	0	1175
21	1/24/2003	3021	3059	3016	0	0	-1	0	9	-1	-125	1050
22	1/27/2003	3001	3059	3016	-1	0	-1	0	10	-1	500	1550
23	1/28/2003	2962	3054	3001	-1	0	-1	0	11	-1	1225	2775
24	1/29/2003	2924	3021	2962	-1	0	-1	0	12	-1	700	3475
25	1/30/2003	2947	3021	2924	0	0	-1	0	13	-1	-575	2900
26	1/31/2003	2943	3021	2924	0	0	-1	0	14	-1	100	3000
27	2/3/2003	2933	3001	2924	0	0	-1	0	15	-1	250	3250
28	2/4/2003	2934	2962	2924	0	0	-1	0	0	0	0	3250
29	2/5/2003	2892	2947	2924	-1	0	-1	0	0	0	0	3250
30	2/6/2003	2881	2947	2892	-1	0	-1	0	0	0	0	3250
31	2/7/2003	2887	2943	2881	0	0	-1	0	0	0	0	3250
32	2/10/2003	2870	2934	2881	-1	0	-1	0	0	0	0	3250
33	2/11/2003	2871	2934	2870	0	0	-1	0	0	0	0	3250
34	2/12/2003	2874	2892	2870	0	0	-1	0	0	0	0	3250
35	2/13/2003	2818	2887	2870	-1	0	-1	0	0	0	0	3250
36	2/14/2003	2812	2887	2818	-1	0	-1	0	0	0	0	3250
37	2/17/2003	2839	2874	2812	0	0	-1	0	0	0	0	3250
38	2/18/2003	2846	2874	2812	0	0	-1	0	0	0	0	3250
39	2/19/2003	2864	2874	2812	0	0	-1	0	0	0	0	3250
40	2/20/2003	2803	2864	2812	-1	0	-1	0	0	0	0	3250
41	2/21/2003	2810	2864	2803	0	0	-1	0	0	0	0	3250
42	2/24/2003	2846	2864	2803	0	0	-1	0	0	0	0	3250
43	2/26/2003	2790	2864	2803	-1	0	-1	0	0	0	0	3250
44	2/26/2003	2825	2864	2790	0	0	-1	0	0	0	0	3250
45	2/27/2003	2783	2846	2790	-1	0	-1	0	0	0	0	3250
46	2/28/2003	2794	2846	2783	0	0	-1	0	0	0	0	3250
47	3/3/2003	2816	2846	2783	0	0	-1	0	0	0	0	3250
48	3/4/2003	2803	2825	2783	0	0	-1	0	0	0	0	3250
49	3/5/2003	2771	2825	2783	-1	0	-1	0	0	0	0	3250
50	3/6/2003	2760	2816	2771	-1	0	-1	0	0	0	0	3250
51	3/7/2003	2732	2816	2769	-1	0	-1	0	0	0	0	3250
52	3/10/2003	2733	2816	2732	0	0	-1	0	0	0	0	3250
53	3/11/2003	2713	2803	2732	-1	0	-1	0	0	0	0	3250
54	3/12/2003	2710	2771	2713	-1	0	-1	0	0	0	0	3250

Fig. 9.2 Some results of enhanced trading strategy FR.

3. C2. Determining stock pairs according to $tech_int()$ and $biz_int()$ defined through cooperation with traders, market aspects such as market sectors, volatility, liquidity and index are considered;
4. C3. Designing trading strategy to trade stock S and T alternatively by training it in in-sample data:
5. IF $PS - (*PT >= d_0$, THEN buy T and sell S
6. IF $PS - (*PT <= -d_0$, THEN sell T while buy S
7. Where P_S and P_T are prices of S and T, weight (and distance d_0 are business factors that are optimized and tuned by the system with user guide
8. C4. Generating trading signals by checking the above rules in out-of-sample data,
9. C5. Iteratively evaluating and refining the strategy by calculating $biz_int()$, e.g., return in our case, and considering impacts of volatility, liquidity and index
10. C6. Deploying the refined strategy to the market

Fig. 9.4 shows pair trading signals generated in trading Australia stock pairs CBA and GMF in the ASX market. Fig. 9.5 further shows the impact of business factors distance d_0 and weight ϕ on return and the number of triggered signals.

In discovering pair trading strategy for trading agents on ASX Top 32 stocks from January 1997 to June 2002 in F-Trade [53, 55, 38], the trading agent made the following findings.

- Pair relationship between stocks and the combination of the above four factors interesting to trading cannot just be determined by technical measures such as coefficient . They are also highly affected by stock movement such as volatility and liquidity. High volatility improves return while high liquidity balances the market impact on return.

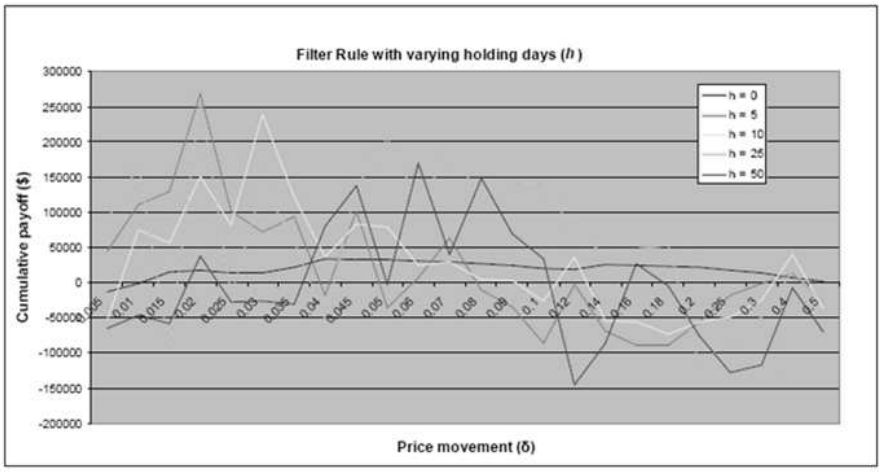

Fig. 9.3 Performance comparison between base and enhanced trading strategies.

- All 13 correlated stocks mined in the ASX Top 32 come from different sectors. This finding means that pairs from the same sector are not necessary, as presumed by financial researchers. The pairs trading strategies discovered have been deployed into the famous exchange surveillance system SMARTS [188].

9.1.4.4 Developing Trading Strategies Correlated with Instruments

In a real-life market, some trading rules are tested to be more profitable to trade a class of stocks, while others are more suitable for different stocks. This triggers another method for improving trading strategies, through analyzing the relationship between a trading strategy and its tradable instruments. We developed the following algorithms to discover the correlations between trading strategies and instruments, and then let a trading agent trade those identified associated instruments only.

Algorithm *Trading Strategy 4*I
1. mproving trading strategies by analyzing correlation between strategies and stocks
2. Mining actionable rules for an individual stock;
3. Mining highly correlated rule-stock pairs by high dimension reduction;

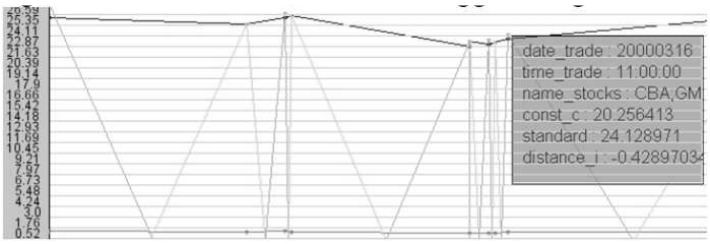

Fig. 9.4 Some results of discovered trading strategy (CBA and GMF, ASX data from 1 Jan 2000 to 20 Jun 2000).

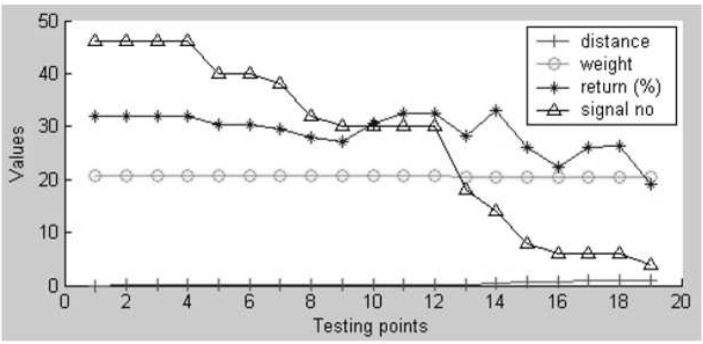

Fig. 9.5 Relation between d_0, ϕ, return and signal number (ASX intraday data, from 1 Jan 2000 to 20 Jun 2000).

4. Evaluating and refining the rule-stock pairs by considering traders' concerns;
5. Recommending actionable rule-stock pairs.

In discovering trading strategy-stock pairs, traders were invited to make suggestions on designing features, interestingness measure and parameter optimization. They also helped us design mechanisms for evaluating and refining rule-stock pairs. Taking the ASX as an instance, six types of trading rules such as MA and Channel Breakout and 27 ASX stocks such as ANZ and TEL were chosen for the experiments in the intraday training data from 1 January 2001 to 31 January 2001 and the testing set from 1 February 2001 to 28 February 2001. Five different investment plans were conducted on the above rules and stocks. In organizing pairs, we ranked them based on *return*, and generated 5% pair, 10% pair, and so forth from the whole pair set. The *5% pair* means that the return of trading these pairs is the top 5% in the whole pair set.

Fig. 9.6 illustrates returns for different investment plans on different pair groups (where the digital legend below the graph refers to pair percentage, e.g., 5 means top 5% pairs). These results show that trading top pairs can achieve a high return but with concomitant high risk, while trading middle-level pairs can obtain a lower return with lower risk. Thus, massively trading a basket of stocks with a lower ranking pair may be an investment strategy more interesting to investors who have big money. As a result, they can quite probably obtain superior profit with low risk.

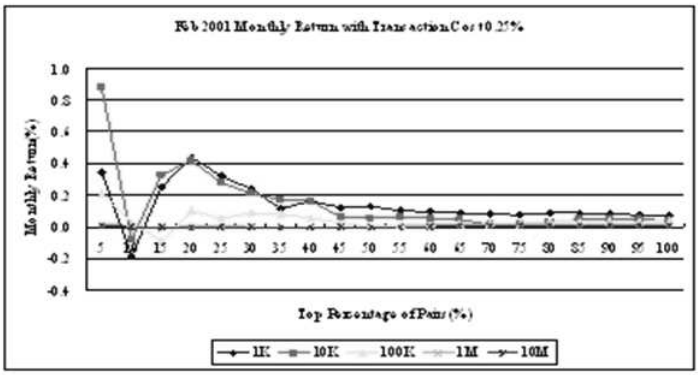

Fig. 9.6 Return on investment of trading strategy-stock pairs.

9.2 Case Study 2: Mining Actionable Market Microstructure Behavior Patterns

9.2.1 Market Microstructure Behavior in Capital Markets

In capital markets, trading behavior refers to actions and operations conducted by investors and recorded into trading engine systems. Basic trading actions consist of buy (B), sell (S) and hold (H). Operations conducted by investors include the interaction and intervention activities between investors and trading engines, for instance, placing an order or withdrawing an order. Such trading behavior indicates significant information about the investment intention and corresponding actions taken by investors to achieve that intention, for instance, manipulating orders to take advantage of the resulting manipulated market movement.

In capital markets, the above trading behavior is governed by the theory of *market microstructure* [119, 149]. Consequently, trading engine systems record relevant information such as orders and trades into transactions, which consist of *market microstructure data* [37]. In general, market microstructure data consists of basic data categories including orders, trades, indices and market data. It involves several dimensions such as time, value and trade actions.

The existing research in capital markets, from either finance such as behavior finance [165] or information technology areas like insider trading analysis [94], mainly rely on data related to market and price movements. To the best of our knowledge, little research has been done on analyzing genuine trading behavior to disclose interior driving forces and causes of market movement and related investment performance.

In fact, transactions in market microstructure data reflect the results of investors' trading behavior in a market, which is controlled by certain trading models following market microstructure theory. We call such trading behavior monitored in terms of market microstructure theory *market microstructure behavior*. Though such transactions are usually managed in terms of entities such as orders and trades, we can actually scrutinize market microstructure behavior hidden in the transactions by squeezing out behavioral elements, and then make them explicit to form behavioral data for behavior-oriented analysis. This is to cause the *shift from behavior implication in transactional space to explication in behavioral data*. For this, behavior modeling is necessary for extracting and representing market microstructure behavior in market transactions.

9.2.2 Modeling Market Microstructure Behavior to Construct Microstructure Behavioral Data

Normal orderbook transactions available from stock markets consist of attributes from related entities such as *orders* and *trades*. Table 9.2 illustrates a data sample of

stock transactional data. Such data indicates an investor's actions, such as whether a trade is a buy (B) or a sell (S), while it does not explicitly exhibit an investor's intention and corresponding behavior in the market. For such a purpose, we have to convert the transactional data into microstructure behavioral data.

Table 9.2 Trade order sequences related to the order O100

Serial_ID	Date	Time	Account_ID	Security	Action	Price	Volume
O100	28/06/2005	09:54:07	A123	S123	B	10.00	1000
O078	28/06/2005	09:59:52	A348	S123	S	10.10	500
O102	28/06/2005	09:59:52	A980	S123	B	10.10	500
O067	28/06/2005	09:59:56	A690	S123	S	10.00	200
O100	28/06/2005	09:59:56	A123	S123	B	10.00	200
O089	28/06/2005	10:07:49	A531	S123	S	10.00	300
O100	28/06/2005	10:07:49	A123	S123	B	10.00	300

Following the vector-based behavior model [45], to model market microstructure behavior, we build vector-based microstructure behavior sequences. Considering the fact that every order follows market microstructure theory and indicates information about the order holder's intention, the proper representation of a trade behavior instance should reflect the trader's intention and actions associated with the corresponding order and trade's lifecycles.

In constructing the trading behavior sequences in terms of the abstract behavioral model $\overrightarrow{\gamma}$, we only consider the ordinal relations existing amongst trading actions. Therefore, elements such as *time* are excluded in the vector. In addition, *context* and *place* refer to the market and its orderbook which are ignored in the vector, *price* and *volume* are corresponding elements of *goal* and *belief* in the behavior model respectively, *plan* refers to the probability of trading, *status* refers to trading status, and finally *associate* refers to the number of follow-up trading actions. In this example, we ignore the behavior attributes *plan* and *constraint*, while the *impact* (f) of a trading behavior will be further modeled in microstructure behavior pattern analysis in Section 9.2.3. As a result, the generic abstract behavior model $\overrightarrow{\gamma}$ is instantiated into the following microstructure trading behavior vector.

$$\begin{aligned}\overrightarrow{\gamma} &= \{s,o,g,b,a,l,f,u,m\}\\ &= \{account_id, security, price, volume,\\ &trading_action, trading_probability, impact, trading_status,\\ &followup_actions\}\end{aligned} \tag{9.7}$$

These elements in the microstructure behavior vector are explained as follows.

- s refers to an investor indicated by *account_id*;
- o refers to the *security* traded by an investor;
- g refers to *price* an investor wants to reach;

- b refers to order size (*volume*) conducted by a trading behavior instance, $b \in \{b_H, b_M, b_L\}$ represents *large*, *medium* and *small* orders respectively;
- a reflects the *action* of a trading behavior associated with an order, $a \in \{B, S, K, B_i, A_s\}$, where B_i and A_s represent *bid* and *ask* respectively;
- l stands for the probability that an order is traded by the current trading behavior, $l \in \{l_H, l_M, l_L\}$, where l_H, l_M and l_L represent *high*, *medium* and *low* probabilities of an order to be traded respectively;
- f reflects the impact of a trading behavior as measured in terms of business interestingness (see Section 9.2.3);
- u reflects the trading status (say balance of an order) associated with a behavior sequence at the end of a trading period, $u \in \{u_0, u_{-1}, u_1\}$ represents the sequences of trading behavior leading to a *completely traded order*, an *outstanding order* or a *deleted order*;
- m represents the number of associated trading actions following the creation of an order, $m \in \{m_{-1}, m_0, m_1\}$ represents *none*, *one* or *many* trading actions followed.

The example we show here is to mine for exceptional trading behavior that may indicate abnormal trading in the market. We are particularly interested in the following attributes: trade_action, trading_probability, trading_status, followup_actions, because they are the main features related to microstructure behavior. We then get the following sub-vector of $\overrightarrow{\gamma}'$.

$$\begin{aligned} \overrightarrow{\gamma}' &= \{b, a, l, u, m\} \\ &= \{order_size, trading_action, trading_probability, \\ &\quad trading_status, followup_actions\} \end{aligned} \tag{9.8}$$

Based on this microstructure behavior vector ($\overrightarrow{\gamma}'$), market trading behavior can be modeled into vector-based microstructure behavior sequences ($\overrightarrow{\Gamma}'$). A vector-based microstructure behavioral sequence $\overrightarrow{\Gamma}$ consists of sequences of behavioral vectors ($\overrightarrow{\gamma}$) for an investor within a certain trading period. Such behavioral sequences systematically reflect an investor's intention, trading activities, relationships with other associated behavior instances, and behavior procedure (lifecycle) in a market.

Correspondingly, vector-based microstructure behavior sequences ($\overrightarrow{\Gamma}'$) can be constructed for an investor as follows.

$$\begin{aligned} \overrightarrow{\Gamma}' &= \{\overrightarrow{\gamma}'_1, \overrightarrow{\gamma}'_2, \ldots, \overrightarrow{\gamma}'_j, \ldots\} \\ &= \{\overrightarrow{\gamma}'_1(b^1, a^1, l^1, u^1, m^1), \\ &\quad \overrightarrow{\gamma}'_2(b^2, a^2, l^2, u^2, m^2), \\ &\quad \ldots \\ &\quad \overrightarrow{\gamma}'_j(b^j, a^j, l^j, u^j, m^j), \\ &\quad \ldots\} \end{aligned} \tag{9.9}$$

With the above microstructure behavior vector, we can convert behavior-related orderbook transactions into vector-based microstructure behavior sequences. For that, we also need to consider investor data such as *account_id* that are needed for filling in vector attributes and constructing sequences. As a result of data conversion from transactions to vector-based behavior sequences, the microstructure orderbook is then transformed into microstructure behavioral sequences that are more suitable for explicit behavior pattern and impact analysis.

9.2.3 Mining Microstructure Behavior Patterns

With vector-based microstructure behavior sequences, microstructure behavior patterns can be identified. In order to discover behavior patterns, we first define microstructure behavior impact (f) (refer to business interestingness $b_i()$) and technical interestingness ($t_i()$) for pattern analysis.

Definition 9.1. *(Microstructure Behavior Impact (f))* f in the microstructure behavior model indicates the impact of microstructure behavior in the market. One way to model the impact is through the calculation of *Abnormal Return* (*AR*) of all actions in a behavior sequence within a trading period Δt, which reflects the return volatility of the behavior in the market.

$$AR = \frac{std(\ln \frac{p_t}{p_{t-1}})}{\sqrt{t}} \tag{9.10}$$

where std is the standard division, p_t is the volume-weighted average price on trading a security at time t.

$$p_t = \frac{\sum g * b}{\sum b} \tag{9.11}$$

g and b refer to the price and volume of all orders placed in the relevant time period.

Two more metrics are defined to measure the technical interestingness of a microstructure behavior pattern. They are *intentional interestingness* (I_i) and *exceptional interestingness* (I_e).

Definition 9.2. *(Intentional Interestingness (I_i))* I_i reflects an investor's intention of deploying a series of trading actions ($\overrightarrow{\gamma}$) toward expected impact and goal.

$$I_i = Supp_t * \frac{|\overrightarrow{\gamma}|}{AvgL_t} \tag{9.12}$$

where $Supp_t$ is the support of behavior sequence $\overrightarrow{\gamma}$ in the behavioral data within a period Δt, $|\overrightarrow{\gamma}|$ is the sequence length, and $AvgL_t$ is the weighted average length of behavior sequences $\overrightarrow{\gamma}$.

Definition 9.3. *(Exceptional Interestingness (I_e))* I_e reflects how exceptionally a pattern presents in a target time period rather than in the benchmark one.

$$I_e = \frac{\frac{Supp_t}{AvgL_t} * \sum_{j=1}^{m} \omega_j}{\sum_{j=1}^{m} (\frac{Supp_j}{AvgL_j} * \omega_j)} \tag{9.13}$$

where ω_j is the weight for the benchmark period j, $Supp_j$ is the support of behavior sequence $\overrightarrow{\gamma}$ in the behavioral data within a period j which is the benchmark period, and $AvgL_j$ is the weighted average length of behavior sequence $\overrightarrow{\gamma}$ in benchmark period j, there are in total m benchmark periods.

Definition 9.4. *(Exceptional Microstructure Behavior Patterns (P))* P is an exceptional microstructure behavior in the market if it satisfies the following conditions:

$$P(\overrightarrow{\gamma}) \in \overrightarrow{\Gamma}; \tag{9.14}$$

$$I_i \geq I_{i0}; \tag{9.15}$$

$$I_e \geq I_{e0}; \tag{9.16}$$

where I_{i0} and I_{e0} are the respective thresholds defined by domain experts.

The algorithm for identifying such exceptional trading behavior patterns is as follows.

METHOD 1: Mining Exceptional Microstructure Behavior in Market Microstructure Behavior Data
INPUT: Microstructure behavior data $\overrightarrow{\Gamma}$, thresholds m, I_{i0} and I_{e0}
OUTPUT: Microstructure behavior patterns $\widetilde{P}$

Step 1: Mining general patterns P;
  FOR $j = 1$ to m
    Construct the behavior sequences $\overrightarrow{\Gamma_j}$;
    Extract behavior pattern set P_j on $\overrightarrow{\Gamma_j}$;
  ENDFOR
Step 2: Extracting deliverables $\widetilde{P}$;
  FOR $j = 1$ to $(count(P))$
    IF satisfy conditions of I_i and I_e as defined in Definition 9.4;
      Extract the behavior pattern set $\widetilde{P}$;
    ENDIF
  ENDFOR
Step 3: Converting the patterns $\widetilde{P}$ into business alert rules $\widetilde{R}$;

Table 9.3 Selected Exceptional Microstructure Behavior Patterns

Date	Exceptional Microstructure Patterns	I_i	I_e	AR (%)
18/01/2005	$\{(b_S,S,l_M,u_1,m_{-1}),(b_S,S,l_M,u_1,m_{-1}),(b_S,S,l_M,u_1,m_{-1}),(b_S,S,l_M,u_0,m_0)\}$	0.026	10.8	1.85
01/02/2005	$\{(b_S,B,l_H,u_1,m_{-1}),(b_S,B,l_H,u_1,m_{-1}),(b_S,B,l_H,u_1,m_{-1})\}$	0.030	5.2	0.93
24/05/2005	$\{(b_S,S,l_M,u_0,m_0),(b_S,S,l_M,u_0,m_0)\}$	0.054	11.2	6.38
13/07/2005	$\{(b_S,B,l_H,u_{-1},m_{-1})\}$	0.028	5.6	9.12
15/09/2005	$\{(b_S,B,l_H,u_{-1},m_{-1}),(b_S,B,l_H,u_{-1},m_{-1})\}$	0.035	8.8	3.49
05/12/2005	$\{(b_S,S,l_M,u_0,m_0),(b_S,S,l_M,u_0,m_0)\}$	0.035	8.6	3.41

9.2.4 Experiments

Substantial experiments have been conducted on orderbook data provided by the Shanghai Stock Exchange in China for a collaborative research project. One of the datasets consists of 240 trading days from 2005 to 2006 for a security, which include 213,898 orders and 228,156 trades. In terms of the microstructure behavior vector model, the orderbook data is converted into vector-based microstructure behavior sequences. For instance, two consecutive transactions that happened on July 16, 2004 are converted into the following vector-based behavioral sequences.

$$\begin{aligned}\overrightarrow{\Gamma}' &= \{\overrightarrow{\gamma_1}', \overrightarrow{\gamma_2}'\} \\ &= \{(b_L,B,l_L,u_{-1},m_1), \\ &\quad (b_M,S,l_H,u_1,m_0)\}\end{aligned} \tag{9.17}$$

Patterns are further mined on the vector-based behavioral sequences. Table 9.3 illustrates some of the exceptional microstructure behavior patterns identified in the behavioral data. These patterns are interesting because

(1) they are of high intentional and exceptional interest as indicated by I_i and I_e, which reflect the investor's intention and goal in achieving exceptional performance; and
(2) they lead to high *abnormal return* which indicates exceptionally high return as a result of such trading behavior compared to other normal trading.

Compared to price-centered exceptional trading pattern analysis, for the first time as we know, such microstructure behavior patterns disclose much more informative and deeper knowledge about interior trading activities and process. It also has the ability to inform the resulting status and investment impact of conducting them in the market. Such information can assist market surveillance officers to deeply understand market price movement and macroeconomic dynamics with in-depth driving forces and the process of related investors.

Chapter 10
Mining Actionable Knowledge on Social Security Data

10.1 Case Study: Mining Actionable Combined Associations

10.1.1 Overview

Social security data is widely seen in welfare states. The data consists of customer demographics, government overpayment (debt) information, activities such as government arrangements for debtors' payback agreed by both parties, and debtors' repayment information. Such data encloses important information about the experience and performance of government service objectives and social security policies, and may include evidence and indicators for recovering, detecting, preventing and predicting debt occurrences.

We have developed activity mining [62, 64] methods based on AKD frameworks to identify patterns of high impact-oriented customer behavior and customer-government officer contacts that are likely correlated to debts. Some of the methods are based on a CI-AKD framework by developing interestingness metrics measuring both the technical significance and business impact of the patterns, for instance, in identifying *sequential impact-contrasted activity patterns* where P is frequently associated with both patterns $P \rightarrow T$ and $P \rightarrow \bar{T}$ in separated data sets, and *sequential impact-reversed activity patterns* in which both $P \rightarrow T$ and $PQ \rightarrow \bar{T}$ are frequent.

This section offers several examples of utilizing the MSCM-AKD framework in identifying combined associations and combined association clusters for debt prevention.

10.1.2 Combined Associations and Association Clusters

Based on the MSCM-AKD framework, combined associations and association clusters are composed of items from heterogeneous data. They are defined as follows.

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Definition 10.1. *(Combined Associations)* A combined association rule r is in the form of $u + v \rightarrow c$, where u and v are different itemsets from two heterogeneous datasets, and c is a target item or class.

For example, u is a demographic itemset, v is a transactional itemset, and c is the class of a customer.

Further, the combined association rules can be organized into rule clusters by putting similar or related rules together, which can provide more useful knowledge than separated rules.

Definition 10.2. *(Combined Association Rule Cluster)* A combined association rule cluster R is a set of combined association rules with the same u (or v) but different v (or u).

Examples of combined association rules and rule clusters are shown in Formula (10.1). R_1 and R_2 are two combined rule clusters. The rules in cluster R_1 have the same u but different v, which makes them associated with various results c. On the contrary, the rules in R_2 have the same v but different u.

$$R_1: \begin{cases} u + v_1 \rightarrow c_{k_1} \\ u + v_2 \rightarrow c_{k_2} \\ \cdots \\ u + v_m \rightarrow c_{k_m} \end{cases}, \quad R_2: \begin{cases} u_1 + v \rightarrow c_{l_1} \\ u_2 + v \rightarrow c_{l_2} \\ \cdots \\ u_n + v \rightarrow c_{l_n} \end{cases} \tag{10.1}$$

The word "combined" means: 1) each single rule consists of itemsets from different datasets, namely demographic data and transactional data including repayments, arrangements and debt transactions; 2) each rule cluster is composed of two or more related local rules identified in individual datasets. The process of mining combined associations and association clusters in social security data is shown in Fig. 10.1, where $DB1$ is demographic data, $DB2$ is transactional data.

Mining Combined Associations and Association Clusters
INPUT: target datasets $DB1$ and $DB2$
OUTPUT: Combined association rules and association clusters
Step-1 mining: Mining frequent patterns on the whole population in dataset $DB1$. Select the top m frequent demographic patterns discovered P_i $(i = 1, 2, ..., m)$;
Step-2 partition: Partitioning the dataset $DB2$ into n sub-datasets DB_{2n} $(n = 1, \ldots, m)$ based on the identified top demographic patterns.
Step-3 mining: Mining associations on dataset DB_{2n} in terms of the groups of people identified in *Step-1*. Identifying $\{Q_n = v_{nj} \rightarrow c_{nj}\}$ $(n = 1, \ldots, N;$ $j = 1, \ldots, J)$, the top J frequent patterns discovered from DB_{2n}.
Step-4 merging patterns: Merging results discovered in the relevant groups into combined patterns like:

1) $u_1 + v_1 \rightarrow c_1$ and $u_1 + v_2 \rightarrow c_2$, and 2) $u_1 + v_1 \rightarrow c_1$ and $u_2 + v_1 \rightarrow c_3$.

In utilizing the MSCM-AKD framework, two critical steps are data partition and pattern merging. In this exercise, the partition of debt-related transactional data is driven by domain knowledge and top m frequent demographic patterns recognized by domain experts. Being recognized by business analysts, these top m frequent demographic patterns actually represent m groups of customers. As a result, each sub-dataset in $DB2$ is associated with a particular group of people in the whole population.

Patterns identified in respective datasets are finally merged in terms of combined associations and combined association clusters. A combined pattern combines one demographic component, namely one of the m frequent demographic groups, and one to many items from the transactional datasets based on inputs of business analysts. Business inputs represent the expectations of domain experts including whether a pattern makes sense in business or not, and whether it indicates significant business impacts (namely business interestingness).

In mining social security data, *risk* and *cost* are defined as subjective and objective business interestingness to measure the impact of a pattern.

Definition 10.3. The *risk* of pattern $P \rightarrow T$, is defined as

$$Risk(P \rightarrow T) = \frac{Cost(P \rightarrow T)}{TotalCost(P)},$$

where $Cost(P \rightarrow T)$ is the sum of the cost associated with $P \rightarrow T$ and $TotalCost(P)$ is the total cost associated with P. The *average cost* of pattern $P \rightarrow T$ is defined as

$$AvgCost(P \rightarrow T) = \frac{Cost(P \rightarrow T)}{Cnt(P \rightarrow T)}.$$

Here *risk* and *cost* are instantiated into $Risk_{amt}$, $Risk_{dur}$, $AvgAmt$ and $AvgDur$, which stand for debt-amount risk, debt-duration risk, average debt amount and average debt duration, respectively. These metrics measure the impact of a pattern on governmental debt.

Finally, a combined pattern is interesting only if it satisfies the interestingness of a corresponding combined pattern type as defined in the following sections.

10.1.3 Selecting Interesting Combined Associations and Association Clusters

To support the discovery of combined associations and association clusters, we first develop corresponding interestingness metrics: 1) interestingness for combined associations, and 2) interestingness for combined association cluster.

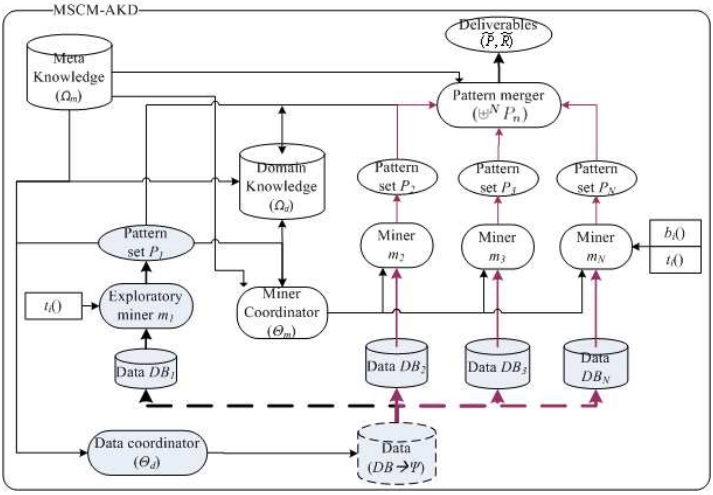

Fig. 10.1 MSCM-AKD-based framework for mining combined associations and association clusters.

For a combined association rule $u+v \rightarrow c$, the traditional interestingness metrics such as supports, confidences and lifts contribute little to selecting actionable combined association rules. To measure the interestingness of a combined association rule, we define two new lifts based on traditional supports, confidences and lifts.

$$Lift_u(u+v \rightarrow c) = \frac{Conf(u+v \rightarrow c)}{Conf(v \rightarrow c)} \tag{10.2}$$

$$Lift_v(u+v \rightarrow c) = \frac{Conf(u+v \rightarrow c)}{Conf(u \rightarrow c)} \tag{10.3}$$

$Lift_u(u+v \rightarrow c)$ is the lift of u with v as a precondition, which shows how much u contributes to the rule. Similarly, $Lift_v(u+v \rightarrow c)$ gives the contribution of v in the rule. The follows can be derived from the above formulas.

$$Lift_u(u+v \rightarrow c) = \frac{Lift(u+v \rightarrow c)}{Lift(v \rightarrow c)} \tag{10.4}$$

$$Lift_v(u+v \rightarrow c) = \frac{Lift(u+v \rightarrow c)}{Lift(u \rightarrow c)} \tag{10.5}$$

Based on the above lifts, the interestingness of a single combined association rule is defined as follows.

$$I_{\text{rule}}(u+v \rightarrow c) = \frac{Lift_u(u+v \rightarrow c)}{Lift(u \rightarrow c)} = \frac{Lift_v(u+v \rightarrow c)}{Lift(v \rightarrow c)} \tag{10.6}$$

I_{rule} indicates whether the contribution of u (or v) to the occurrence of c increases with v (or u) as a precondition. Therefore, "$I_{\text{rule}} < 1$" suggests that $u+v \rightarrow c$ is less interesting than $u \rightarrow c$ and $v \rightarrow c$. The value of I_{rule} falls in [0,+∞). When $I_{\text{rule}} > 1$, the higher I_{rule} is, the more interesting the rule is.

Furthermore, we define the interestingness of a combined association cluster. First, the interestingness of a pair of combined association rules is defined as follows. Suppose p_1 and p_2 are a pair of combined association rules with different consequents within a single rule cluster, say, $p_1 = (u_1 + v_1 \rightarrow c_1)$ and $p_2 = (u_2 + v_2 \rightarrow c_2)$ where $u_1 = u_2$, $v_1 \neq v_2$ and $c_1 \neq c_2$ (or $u_1 \neq u_2$, $v_1 = v_2$ and $c_1 \neq c_2$), the interestingness of the rule pair $\{p_1, p_2\}$ is defined as:

$$I_{\text{pair}}(p_1, p_2) = Conf(p_1)\, Conf(p_2) \tag{10.7}$$

I_{pair} measures the contribution of the two different parts in antecedents to the occurrence of different classes in a group of customers with the same demographics or the same transaction patterns. Such knowledge can help to design business campaigns and improve business process. The value of I_{pair} falls in [0,1]. The larger I_{pair} is, the more interesting and actionable a pair of rules are.

For an association cluster R with J combined associations $p_1, p_2, \ldots, p_J$, its interestingness is defined as follows:

$$I_{\text{cluster}}(R) = \max_{p_x, p_y \in R, x \neq y, c_x \neq c_y} \{I_{\text{pair}}(p_x, p_y)\} \tag{10.8}$$

The above definition of I_{cluster} indicates that interesting clusters are the rule clusters with interesting rule pairs, and the other rules in the cluster provide additional information. Similar to I_{pair}, the value of I_{cluster} also falls in [0,1].

With the above interestingness and traditional metrics: *support*, *confidence*, *lift*, $Lift_u$, $Lift_v$ and I_{rule}, we can select interesting combined associations from the learned rules. Learned rules with high *support* and *confidence* are further merged into association clusters ranked in terms of I_{cluster}.

10.2 Experiments: Mining Actionable Combined Patterns

In this section, we introduce two case studies in the same application, namely mining combined patterns for governmental overpayment prevention in e-government service data in the Australian Commonwealth Government Agency Centrelink. The first is on mining multi-feature combined patterns, and the second is on discovering sequential classifiers.

Table 10.1 Arrangement Rules "$v \rightarrow c$"

v		c	$Conf(\%)$	$Count$	$Lift$
Arrangements	Repayments	Class			
irregular	cash or post office	A	82.4	4088	1.8
withholding	cash or post office	A	87.6	13354	1.9
withholding & irregular	cash or post office	A	72.4	894	1.6
withholding & irregular	cash or post office & withholding	B	60.4	1422	1.7

10.2.1 Mining Multi-Feature Combined Patterns

This section illustrates case studies of identifying single combined patterns, cluster patterns, and incremental pair and cluster patterns, which combine demographics, arrangement and repayment activities. More relevant information can be found from the public papers [63, 237, 239, 240].

10.2.1.1 Mining Single Combined Patterns and Cluster Patterns

The case study centers on Centrelink social security data with debts raised in the calendar year 2006 and the corresponding customers, debt arrangement and repayments. The cleaned sample data contains 355,800 customers with their demographic attributes, arrangement and repayment activities. There are 7,711 traditional associations mined (see [240] for more information). These association rules are illustrated in Table 10.1.

Based on the methods of mining multi-feature combined patterns in Section 6.5, some examples of discovered single combined patterns and cluster patterns are shown in Tables 10.2 and 10.3, respectively. In the two tables, columns $Cont_p$ and $Cont_e$ stand for the *contributions* of X_p and X_e in the pattern, respectively. I_r and I_c are the interestingness of the corresponding patterns and pattern clusters.

Compared with the single associations identified in respective datasets, the single combined patterns and cluster patterns are much more informative than single rules presented in the traditional way. They contain much richer information from multiple business aspects rather than a single one, or from a collection of separated single rules. For instance, the following combined pattern shows that customers aged 65 or more, whose arrangement method is of 'withholding' plus 'irregular', and who actually repay in the approach of 'withholding', can be classified into class 'C'. It combines information regarding the specific group of the debtor's demographic, repayment and arrangement methods that cannot be identified in a traditional way given the large quantity of respective datasets to be joined.

$$\{X_\mathrm{p} = age : 65+, X_\mathrm{e} = withholding \ \& \ irregular + \\ withholding \rightarrow T = C\} \tag{10.9}$$

Table 10.2 An Excerpt of Single Combined Patterns

Rules	X_p	X_e		T	Cnt	$Conf$	I_r	$Lift$	$Cont_p$	$Cont_e$	L_p [1]	L_e [2]
	Demographics	Arrangements	Repayments	Class		(%)						
P_1	age:65+	withholding & irregular	withholding	C	50	63.3	2.91	3.40	2.47	4.01	0.85	1.38
P_2	income:0 & remote:Y & marital:sep & gender:F	withholding	cash or post & withholding	B	20	69.0	1.47	1.95	1.34	2.15	0.91	1.46
P_3	income:0 & age:65+	withholding	cash or post & withholding	A	1123	62.3	1.38	1.35	1.72	1.09	1.24	0.79
P_4	income:0 & gender:F & benefit:P	withholding	cash or post	A	469	93.8	1.36	2.04	1.07	2.59	0.79	1.90

Moreover, the pattern clusters shown in Table 10.3 present further information, which hasn't been reported before, to the best of our knowledge. For example, pattern cluster $\mathscr{P}_1$ shows that, for single women on benefit "N", the best way to get their debts repayed as quickly as possible is through repayments "cash or post" with "irregular" or "withholding" arrangements. An actionable policy is to push them to pay through these arrangements, instead of those given in patterns p_7 to p_{10}.

In summary, combined patterns are more business-friendly and indicate much more straightforward decision-making actions to be taken by business analysts in the business world, while this cannot be reached by traditional methods targeting individual patterns from single datasets.

To support business-friendly delivery of identified combined patterns, we further transform the combined patterns into business rules. Business rules-based pattern representation is more business-friendly, and indicates direct actions for business decision-making. For instance, for the above combined patterns, it actually connects key business elements with segmented customer characteristics, and we can generate a business rule as shown in Fig.10.2. The business rules are deliverables presented to business people. They are convenient and operable for clients to embed into their routine business processes and operational systems for automatically monitoring the debt recovery process in order to recover government debt.

10.2.1.2 Mining Incremental Pair Patterns

A case study of incremental pair patterns on Centrelink debt-related activity data is given as follows. The data involves four data sources, which are activity files recording activity details, debt files logging debt details, customer files enclosing customer circumstances, and earnings files storing earnings details. To analyze the relationship between activity and debt, the data from activity files and debt files are extracted. The activity data used to test the proposed approaches is Centrelink activity data from Jan. 1st to Mar. 31st 2006. We extract activity data including 15,932,832 activity records recording government-customer contacts with 495,891

Table 10.3 Sample Combined Association Clusters

Clusters	Rules	u	v		c	Cnt	$Conf$ (%)	I_r	I_c	$Lift$	L_u	L_v	$Lift$ of $u \rightarrow c$	$Lift$ of $v \rightarrow c$
		demographics	arrangements	repayments										
R_1	p_5	marital:sin	irregular	cash or post	A	400	83.0	1.12	0.67	1.80	1.01	2.00	0.90	1.79
	p_6	&gender:F	withhold	cash or post	A	520	78.4	1.00		1.70	0.89	1.89	0.90	1.90
	p_7	&benefit:N	withhold & irregular	cash or post & withhold	B	119	80.4	1.21		2.28	1.33	2.06	1.10	1.71
	p_8		withhold	cash or post & withhold	B	643	61.2	1.07		1.73	1.19	1.57	1.10	1.46
	p_9		withhold & vol. deduct	withhold & direct debit	B	237	60.6	0.97		1.72	1.07	1.55	1.10	1.60
	p_{10}		cash	agent	C	33	60.0	1.12		3.23	1.18	3.07	1.05	2.74
R_2	p_{11}	age:65+	withhold	cash or post	A	1980	93.3	0.86	0.59	2.02	1.06	1.63	1.24	1.90
	p_{12}		irregular	cash or post	A	462	88.7	0.87		1.92	1.08	1.55	1.24	1.79
	p_{13}		withhold & irregular	cash or post	A	132	85.7	0.96		1.86	1.18	1.50	1.24	1.57
	p_{14}		withhold & irregular	withhold	C	50	63.3	2.91		3.40	2.47	4.01	0.85	1.38
R_3	p_{15}	benefit:Y	irregular	cash or post	A	218	79.6	1.15	0.52	1.73	0.97	2.06	0.84	1.79
	p_{16}	&age:22-25	cash	cash or post	C	483	65.6	0.78		3.53	1.38	1.99	1.78	2.56
R_4	p_{17}	income:0	irregular	cash or post	A	191	76.7	1.03	0.48	1.66	0.93	1.85	0.90	1.79
	p_{18}	&age:22-25	cash	cash or post	C	440	62.1	1.08		3.34	1.31	2.76	1.21	2.56

DELIVERING BUSINESS RULES: Customer Demographic-Arrangement-Repayment combination business rules

For All *customer i (i ∈ I is the number of valid customers)*

Condition:

satisfies *S/he is a debtor aged 65 or plus*;

relates

S/he is under arrangement of 'withholding' and 'irregularly',

and

His/her favorite Repayment method is 'withholding';

Operation:

Alert = "*S/he has 'High' risk of paying off debt in a very long timeframe.*"

Action = "*Try other arrangements and repayments in R_2, such as trying to persuade her/him to repay under 'irregular' arrangement with 'cash or post'.*"

End-All

Fig. 10.2 An Example of Business Rule

customers, which are associated with 30,546 debts in the first three months of 2006. After data pre-processing and transformation, there are 454,934 activity sequences: 16,540 (3.6%) activity sequences associated with debts and 438,394 (96.4%) sequences with non-debts. Labels T and $\bar{T}$ denote the occurrences of debt and non-debt respectively, and code a_i represents an activity.

Table 10.4 shows an excerpt of single sequential activity patterns and pairs of contrast patterns. One is the underlying pattern $X_p \rightarrow T_1$, the other is derived pattern $X_p \wedge X_e \rightarrow T_2$, where T_1 is opposite to T_2, and X_e is a derived activity or sequence. The last two columns show respectively the local supports of underlying patterns

and reverse patterns. For example, the first row shows an incremental pair pattern as follows.

$$\begin{cases} a_{14} \rightarrow \bar{T} \\ a_{14}, a_4 \rightarrow T \end{cases}. \tag{10.10}$$

As shown in Table 10.4, the local support of $a_{14} \rightarrow \bar{T}$ on non-debt data is 0.684, and the local support of $a_{14}, a_4 \rightarrow T$ on debt data is 0.428. $Cont_e$ shows the impact of the derived activity on the outcome when an underlying pattern happens. Cps denotes *conditional P-S ratio*. Both $Cont_e$ and Cps show to what extent the impact can be reversed through adding additional sequences. The first row shows that the appearance of a_4 tends to change the impact from $\bar{T}$ to T when a_{14} happens first. The following analysis will help to better understand the meaning of the pair of patterns $a_{14} \rightarrow \bar{T}$ and $a_{14}, a_4 \rightarrow T$. The local supports of $a_{14} \rightarrow T$ and $a_{14} \rightarrow \bar{T}$ are 0.903 and 0.684 respectively, so the ratio of the two values is 0.903/0.684= 1.3. The local supports of $a_{14}, a_4 \rightarrow T$ and $a_{14}, a_4 \rightarrow \bar{T}$ are 0.428 and 0.119 respectively, so the ratio of the two values are 0.428/0.119 = 3.6. The above two ratios indicate that, when a_{14} occurs first, the appearance of a_4 makes it more likely to become a debt. These kinds of pair patterns help show what effect an additional activity may have on the impact of the patterns. In practice, such patterns are much more informative and actionable than any single form of patterns for conducting effective intervention and business campaigns.

Table 10.4 An Excerpt of Incremental Pair Patterns

X_p	T_1	X_e	T_2	$Cont_e$	Cps	Local support of $X_p \rightarrow T_1$	Local support of $X_p \wedge X_e \rightarrow T_2$
a_{14}	$\bar{T}$	a_4	T	2.5	0.013	0.684	0.428
a_{16}	$\bar{T}$	a_4	T	2.2	0.005	0.597	0.147
a_{14}	$\bar{T}$	a_5	T	2.0	0.007	0.684	0.292
a_{16}	$\bar{T}$	a_7	T	1.8	0.004	0.597	0.156
a_{14}	$\bar{T}$	a_7	T	1.7	0.005	0.684	0.243
a_{15}	$\bar{T}$	a_5	T	1.7	0.007	0.567	0.262
a_{14}, a_{14}	$\bar{T}$	a_4	T	2.3	0.016	0.474	0.367
a_{14}, a_{16}	$\bar{T}$	a_5	T	2.0	0.005	0.393	0.118
a_{15}, a_{14}	$\bar{T}$	a_5	T	1.7	0.007	0.381	0.179
a_{14}, a_{16}, a_{14}	$\bar{T}$	a_{15}	T	1.2	0.005	0.248	0.188

10.2.1.3 Mining Incremental Cluster Patterns

A case study of incremental cluster patterns is given as follows (see [241] for more information). The data used are a sample of debt data and activity transaction data

from July 2007 to February 2008. After data preprocessing, 15,931 sequences are constructed. Minimum support is set to 0.05, that is, 797 out of 15,931 sequences. There are 2,173,691 patterns generated and the longest pattern has 16 activities.

An example of discovered incremental cluster pattern is

$$\begin{cases} PLN \rightarrow T \\ PLN, DOC \rightarrow T \\ PLN, DOC, DOC \rightarrow T \\ PLN, DOC, DOC, DOC \rightarrow T \\ PLN, DOC, DOC, DOC, REA \rightarrow T \\ PLN, DOC, DOC, DOC, REA, IES \rightarrow T \end{cases}, \tag{10.11}$$

where T stands for a debt.

10.2.1.4 Dynamic Charts

To more explicitly disclose the dynamics and impact of combined sequences, we create a *dynamic chart*. Dynamic charts present the dynamics of sequential patterns, activity interaction, and impact change, and the formation of associated pairs and clusters in terms of pattern interestingness. They are selected based on the support, confidence, lift, contribution and impact defined for combined patterns. We rank all combined patterns based on the above measures and show the top ones, because they supposedly show us which activities have big impacts on the outcomes.

The above incremental cluster patterns are also shown in Fig.10.3 in terms of dynamic charts. This figure illustrates the impact of a cluster of patterns on the occurrence of debt. There are four charts in the figure: the upper-left chart shows the changes in confidence (i.e., the likelihood of debt occurrences) and in lift (i.e., the improvement in the likelihood of debt occurrences); the upper-right chart shows the changes in support, that is, the portion of sequences supporting the above association between activity patterns and debts; the two charts at the bottom show respectively the contribution and impact of items. Each point in the curve gives the value for the sequential pattern from the first activity to the corresponding activity.

The upper-left chart shows that pattern "PLN, DOC, DOC, DOC, REA" is associated with debt occurrence, and the lift is 1.1. However, an additional IES following the sequence will dramatically reduce the likelihood of debt occurrence. The bottom-left chart shows that PLN and IES are negatively associated with debt occurrence, and DOC is slightly positively associated with debt. The bottom-right chart suggests that IES has a big impact on the outcome, and PLN also has some impact on debt, and the impact of DOC is relatively minor.

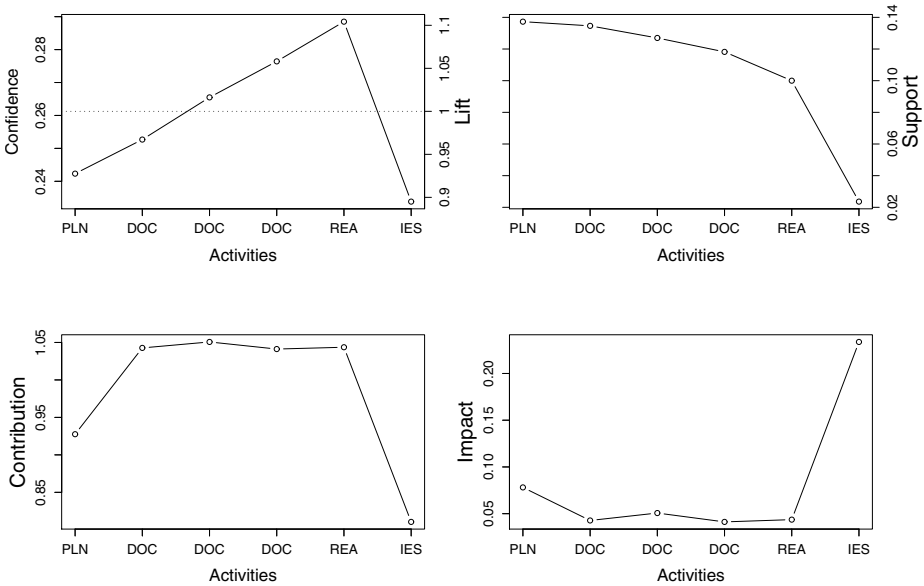

Fig. 10.3 Dynamic Charts Showing Dynamics of Incremental Cluster Patterns

10.2.2 Mining Closed-Loop Sequence Classifiers

The proposed closed-loop sequence classification algorithm is tested on Centrelink e-governmental service data. The data involves a range of Commonwealth services to the Australian community. We combine sequential pattern mining with classification to build a sequential classifier that can predict whether a customer tends to have debt or not. Here, a debt indicates overpayment made by Centrelink to a customer who is not entitled.

We use the same dataset as the one in Section 10.2.1.3. After data cleaning, there are 15,931 activity sequences including 849,831 activity records used in this exercise. 60% out of the whole dataset is extracted as training set while the remaining as testing set. By keeping the ratio of training and testing sets but randomly dividing them, we test the built classifier five times. The average classification performance is shown in Table 10.5.

Table 10.5 The performance of our sequence classifier

Predict	Actual	Count	Percent (%)
Debt	Debt	1303	20.4
Debt	No_debt	1203	19.0
No_debt	Debt	370	5.8
No_debt	No_debt	3496	54.8

In our algorithm, we set $min_sup = 1\%$ on both debt data and non_debt data. The number of the mined patterns are $39,320$. In order to compare the searching space of our algorithm and conventional algorithm, we also implement the standard sequence mining algorithm using SPADE [231]. SPADE cannot work in our case study if $min_sup < 5\%$. When $min_sup = 5\%$, the number of the mined patterns is $2,173,691$. After sequence pattern mining is done, we use a strategy similar to CMAR [147], and the classification result is as shown in Table 10.6.

Table 10.6 The performance of standard sequence classifier

Predict	Actual	Count	Percent (%)
Debt	Debt	1300	20.4
Debt	No_debt	1585	24.8
No_debt	Debt	332	5.2
No_debt	No_debt	3155	49.5

In this experiment, we can see that, in the proposed algorithm the searching space is much smaller than in the conventional algorithm ($39,220 \ll 112,212,436$). However, our algorithm achieves an accuracy of 75.2% while the accuracy of the conventional algorithm is 69.9%. From the above test, we can see that our algorithm outperforms the conventional algorithm on both efficiency and accuracy. One of the possible reasons for the accuracy improvement is that less redundant sequence patterns are used in our algorithm.

In order to evaluate the accuracy of the proposed algorithm, we implement the algorithm CCR_{CMAR} which is the same as CMAR [147] but replacing confidence with CCR in ranking. The CCR_{CMAR} is also implemented in our hierarchical framework. On each level, the sub-classifier is trained using a weighted χ^2 on multiple patterns. We compare the accuracy of CCR_{CMAR} to our proposed algorithms $CCR_{Highest}$ and CCR_{Multi} at difference min_sup levels. The results are shown in Table 10.7. In all our experiments, 60% of the whole dataset is extracted as a training set while the remaining data put into the testing set. By keeping the ratio of training and testing sets but randomly dividing them, we test the built classifiers.

In Table 10.7, at all min_sup levels, CCR_{Multi} outforms $CCR_{Highest}$. It verifies again that the classifier using the highest ranking pattern for one instance suffers from over-fitting. Between the two algorithms using multiple patterns for one instance, CCR_{Multi} and CCR_{CMAR}, CCR_{Multi} outperforms CCR_{CMAR} at all min_sup levels. When min_sup becomes greater, the difference between the two algorithms increases, which means our algorithm is more robust than CCR_{CMAR} when fewer patterns are discovered for classification.

Table 10.7 The Performance of Different Sequence Classification Algorithms

min_sup	$No.Pattern$	CCR_{CMAR}	$CCR_{Highest}$	CCR_{Multi}
1%	39,220	75.0%	72.7%	75.2%
2%	10,254	74.4%	71.8%	74.9%
5%	1,116	69.4%	70.9%	72.4%
10%	208	64.2%	61.0%	66.7%

10.3 Summary

This chapter has illustrated the use of domain driven data mining on mining actionable knowledge on social security data for better understanding government service quality, causes and effects of government service problems, customer behavior and demographics, and government officer-customer interactions. The identified combined patterns, in particular, combined associations, combined association clusters, combined patterns combining multiple features and through closed-loop sequence classification, have been shown to be much more informative than traditional patterns consisting of a single line of business information. The considerations of domain factors and business delivery make the findings much more actionable in social security government debt prevention, recovery and intervention.

In Chapter 11, we will summarize the main open issues surrounding domain driven data mining, as well as the trends and prospects of domain driven data mining.

Chapter 11
Open Issues and Prospects

11.1 Open Issues

While D^3M opens opportunities for the paradigm shift from data-centered hidden pattern discovery to domain driven actionable knowledge delivery, there are fundamental problems that require investigation.

- How to involve domain knowledge in the data mining modeling process?
- How to involve human qualitative intelligence such as imaginary thinking in interactive data mining?
- How to involve expert group in complex data mining applications?
- How to involve network resources in active and run-time mining?
- How to support social interaction and cognition in data mining?
- How to measure knowledge actionability, and balance technical significance with business expectation?
- How to make data mining operable, repeatable, trustful and business-friendly?
- How to build an intelligent data mining infrastructure that can synthesize ubiquitous intelligence?

Accordingly, many open issues await further investigation. For instance, to effectively synthesize the ubiquitous intelligence in actionable knowledge discovery, many research issues need to be studied or revisited.

- Typical research issues and techniques in *Data Intelligence* include mining in-depth data patterns, and structured knowledge in mix-structured data.
- Typical research issues and techniques in *Domain Intelligence* consist of representation, modeling and involvement of domain knowledge, constraints, organizational factors, and business interestingness into data mining.
- Typical research issues and techniques in *Network Intelligence* include involving information retrieval, text mining, web mining, semantic web, ontological engineering techniques, and web knowledge management into data mining process and systems.

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- Typical research issues and techniques in *Human Intelligence* include human-machine interaction, representation and involvement of empirical and implicit knowledge into data mining process and models.
- Typical research issues and techniques in *Social Intelligence* include collective intelligence, social network analysis, and social cognition interaction into data mining systems.
- Typical issues in *intelligence meta-synthesis* consist of building meta-synthetic interaction as working mechanism, and meta-synthetic space (m-space) as a data mining-based problem-solving system.
- The integration of agents with data mining for involving and employing ubiquitous intelligence through agent-based AKD systems and m-spaces.
- The next-generation data mining methodologies, frameworks and processes towards actionable knowledge discovery.
- Effective principles and techniques for acquiring, representing, modeling and engaging intelligence in real-world data mining through automated, human-centered and/or human-machine-cooperated means.
- Workable and operable tools and systems balancing technical significance and business concerns, and delivering actionable knowledge expressed as operable business rules seamlessly engaging business processes and systems.
- Project management methodologies for governing domain-driven AKD projects.
- Techniques supporting dynamic mining, evolutionary mining, real-time stream mining, and domain adaptation.
- Techniques for enhancing reliability, trust, cost, risk, privacy, utility and other organizational and social issues.
- Techniques for handling inconsistencies between the mined knowledge and the existing domain knowledge.

11.2 Trends and Prospects

- Methodologies and techniques incorporating ubiquitous intelligence .
- Data Intelligence, for instance, deep knowledge in complex data structure; mining in-depth data patterns, and mining structured & informative knowledge in complex data, etc.
- Domain Intelligence, for instance, domain & prior knowledge, business processes/logics/workflow, constraints, and business interestingness; representation, modeling and involvement of domain knowledge, constraints, organizational factors, and business interestingness into data mining in KDD, etc.
- Network Intelligence, for instance, network-based data, knowledge, communities and resources; information retrieval, text mining, web mining, semantic web, ontological engineering techniques, and web knowledge management, etc. into data mining process and systems
- Human Intelligence, for instance, empirical and implicit knowledge, expert knowledge and thoughts, group/collective intelligence; human-machine interac-

tion, representation and involvement of human intelligence, etc. into data mining process and models

- Social Intelligence, for instance, organizational/social factors, laws/policies/protocols, trust/utility/benefit-cost; collective intelligence, social network analysis, and social cognition interaction, etc.
- Intelligence Metasynthesis, for instance, synthesize ubiquitous intelligence in KDD; metasynthetic interaction (m-interaction) as working mechanism, and metasynthetic space (m-space) as an AKD-based problem-solving system, etc. for complex problems
- Post-analysis and post-mining for actionable knowledge delivery
- Architectures and frameworks supporting D^3M
- The integration of agents with data mining for involving and employing ubiquitous intelligence through agent-based AKD systems and m-spaces.
- Evaluation system for actionable knowledge delivery by balancing technical and business concerns
- Delivery of actionable and operable decision-support findings
- Project management methodologies supporting D^3M

Chapter 12
Reading Materials

This section lists some materials related to domain driven data mining, for the convenience of readers who are interested in studying this interesting, challenging and very practical area. Section 12.1 lists some of professional activities happened so far, Section 12.4 reads papers initiating the concept and discussing the research issues of domain driven data mining, as well as applications of domain driven data mining in various business domains. Section 12.3 collects readings on agent-driven data mining, Section 12.4 selects some papers regarding post analysis and post mining. Certainly, we cannot list all materials related to domain driven data mining, in particular, the research outputs from our colleagues who either work on this area, or contribute to this problem. The book references hopefully cover many of such publications.

12.1 Activities on D^3M

- Workshops:
  - Domain-driven data mining 2009, joint with ICDM2009
  - Domain-driven data mining 2008, joint with ICDM2008.
  - Domain-driven data mining 2007, joint with SIGKDD2007.
- Tutorial:
  - Cao, L. Domain-driven data mining: Empowering Actionable Knowledge Delivery, PAKDD2009.
- Special issues:
  - Domain-driven data mining, IEEE Trans. Knowledge and Data Engineering, 2010.
  - Domain-driven, actionable knowledge discovery, IEEE Intelligent Systems, Department, 22(4): 78-89, 2007.

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  - Cao, L. Yu, P.S., Zhang, C. and Zhang, H.(Eds.) Data Mining for Business Applications, Springer, 2008.
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  - Cao, L. Domain Driven Data Mining: Challenges and Prospects, IEEE Trans. on Knowledge and Data Engineering, 2009.
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Glossary

Actionability measures the ability of a pattern to suggest a user to take some concrete actions to his/her advantage in the real world. The pattern satisfies both technical and business performance needs from both objective and subjective perspectives. It particularly measures the ability to suggest business decision-making actions.

Actionable knowledge discovery is an iterative optimization process toward the actionable pattern set, considering the surrounding business environment and problem states. It is a loop-closed and iterative refinement process, multiple feedbacks, iterations and refinement are involved in the understanding of data, resources, the roles and utilization of relevant intelligence, the presentation of patterns, the delivery specification, and knowledge validation.

Actionable knowledge delivery aims to deliver knowledge that is of solid foundation, business-friendly, and can be taken over by business people for decision-making seamlessly. During the the process and iterations of actionable knowledge discovery, understanding and deliverables are progressively improved and enhanced toward the final deliverables satisfying user and business needs and supporting direct decision-making action-taking. Its main objective is to enhance the actionability of identified patterns for smart problem-solving.

Actionable pattern satisfies both technical and business interestingness needs, is business-friendly and understandable, reflects user preferences and business needs, and can be seamlessly taken over by business people for decision-making action-taking. Actionable patterns can support business problem-solving by taking actions recommended by the pattern, and correspondingly transform the problem status from an initially non-optimal status to a greatly improved one.

Agent-driven data mining (ADDM) refers to the contributions made by multi-agents for enhancing data mining tasks. ADDM can contribute to the problem solving of many data mining issues, eg., agent-based data mining infrastructure and architecture, agent-based interactive mining, agent-based user interaction, automated pattern mining, agent-based distributed data mining, multi-agent dynamic mining,

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multi-agent mobility mining, agent-based multiple data source mining, agent-based peer-to-peer data mining, and multi-agent web mining.

Agent mining namely agents and data mining interaction and integration, is a new research area that fulfills the respective strengths of both agents and data mining to handle either critical challenges in an individual party or mutual issues. Agent mining studies the methodologies, principles, techniques and applications of the integration and interaction between agents and data mining, as well as the community that focuses on the study of agent mining. The interaction and integration between agents and data mining are comprehensive, multiple dimensional, and inter-disciplinary.

Business interestingness Business interestingness of a pattern is determined from domain-oriented personal, social, economic, user preference and/or psychoanalytic aspects. It consists of both subjective and objective aspects.

Closed-loop mining The discovery of patterns is through a process is with closed-loop feedback and iterations. Actionable knowledge discovery in a constraint-based context is more likely to be a closed-loop rather than open process. A closed-loop process indicates that the outputs of data mining are fed back to change relevant parameters or factors in particular stages. The feedback and change effect may be embodied through analyzing and adjusting the relationships between outputs and particular parameters and factors, and eventually tuning the parameters and factors accordingly.

Cluster pattern more than two patterns are correlated to each other in terms of pattern merging method G into a cluster. Atomic patterns are combined in terms of certain relationships from the structural (for instance, Peer-to-Peer relation, Master-Slave relation) or timeframe (for example Independent relation, Concurrent relation or Sequential relation, or Hybrid relation) perspectives.

Constraint refers to conditions applied on or involved in the process of actionable knowledge discovery and delivery, including domain constraints, data constraints, interestingness constraints, and deliverable constraints.

Contrast pattern results from the mining process in which one considers the mining of patterns/models that contrast two or more datasets, classes, conditions, time periods, and so forth. It captures the situations or contexts (the conditional contrast bases) where small changes in patterns to the base make big differences in matching datasets.

Coupled sequence refers to multiple sequences of itemsets, which are coupled with each other in terms of certain relationships. An example is the trade sequence, buy sequence and sell sequence in stock markets, in which they are coupled in terms of trading mechanisms, trading rules and investment intention etc.

Combined association rule consists of association rules identified from multiple datasets, which are combined into one combined pattern in terms of a certain relationship.

Combined association cluster is a set of combined association rules based on a combined rule pair, where the rules in the cluster share a same underlying pattern but have different additional pattern increments on the left side.

Combined association pair consists of a pair of association rules.

Combined mining is a two to multi-step data mining and post-analysis procedure, consisting of mining atomic patterns, merging atomic pattern sets into combined pattern set, or merging dataset-specific combined patterns into the higher level of combined pattern set. It directly analyzes complex data from multiple sources or with heterogeneous features such as covering demographics, behavior and business impacts. The aim of combined mining is to identify more informative knowledge that can provide an informative and comprehensive presentation of problem solutions. The deliverables of combined mining are combined patterns.

Combined pattern consists of multiple components, a pair or cluster of atomic patterns, identified in individual sources or based on individual methods. As a result of combined mining, the delivery of combined patterns presents an in-depth and more comprehensive indication for taking decision-making actions, which make the patterns informative and more actionable than patterns composed of single aspects only, or identifying by single method-based results.

Data constraint Constraints on particular data, may be embodied in terms of aspects such as very large volume, ill-structure, multimedia, diversity, high dimensionality, high frequency and density, distribution and privacy, dynamics and changes.

Data intelligence reveals interesting stories and/or indicators hidden in data about a business problem. The intelligence of data emerges in the form of interesting patterns and actionable knowledge. It consists of multi-level of data intelligence, namely explicit intelligence, implicit intelligence, syntactic intelligence, and semantic intelligence.

Decremental cluster pattern also called decremental pattern cluster, is a special cluster of combined patterns, within which a former atomic pattern has an additional pattern increment compared to its next adjacent constituent pattern.

Decremental pair pattern also called decremental pattern pair, is a pair of combined patterns which are paired in terms of certain relationship, within which the first atomic pattern has a pattern increment part compared to the second constituent.

Deliverable constraint refers to conditions on deliverables such as business rules, processes, information flow, presentation, etc. may need to be integrated into the domain environment. For instance, learned patterns can be converted into operationalizable business rules for business peoples use.

Derivative pattern is a derived pattern on top of an underlying pattern, namely by appending additional pattern components on to the base pattern. When it is applied to the impact-oriented pattern, the extension leads to the difference between the

outcomes of the constituent patterns. The derivative relationship can be unordered or ordered.

Discriminative pattern or discriminating patterns, refers to those patterns drawing distinctions from other candidates, usually taken in consideration based on class, category, significance, or impact difference etc. Its opposite form is often called indiscriminative pattern.

Domain constraint includes the domain and characteristics of a problem, domain terminology, specific business process, policies and regulations, particular user profiling and favorite deliverables.

Domain driven data mining also called Domain Driven Actionable Knowledge Delivery, is on top of the traditional data-centered pattern mining framework, refers to the set of methodologies, frameworks, approaches, techniques, tools and systems that cater for human, domain, organizational and social, and network and web factors in the environment, for the discovery and delivery of actionable knowledge.

Domain factor consists of the involvement of domain knowledge and experts, the consideration of constraints, and the development of in-depth patterns, which are essential for filtering subtle concerns while capturing incisive issues.

Domain intelligence refers to the intelligence that emerges from the involvement of domain factors and resources in pattern mining, which wrap not only a problem but its target data and environment. The intelligence of domain is embodied through the involvement into KDD process, modeling and systems. It consists of qualitative and quantitative domain intelligence.

Dynamic chart is a pattern presentation method, which presents the dynamics of sequential patterns, activity interaction, and impact change, and the formation of associated pairs and clusters in terms of pattern interestingness.

Emerging pattern are sets of items whose frequency changes significantly from one dataset to another. It describes significant changes (differences or trends) between two classes of data.

General pattern refers to the pattern mined based on technical significance associated with the algorithm used.

Human intelligence refers to (1) explicit or direct involvement of human knowledge or a human as a problem-solving constituent, etc., and (2) implicit or indirect involvement of human knowledge or a human as a system component.

Impact-oriented pattern An impact-oriented pattern consists of two components, namely the left-hand itemsets and the right-hand target impact associated with the left-hand itemsets. It means that the occurrence of the left-hand itemsets likely results in the impact defined on the right hand side.

Impact-reversed pattern An impact-reversed pattern consists of an underlying activity pattern and a derivative pattern with an incremental component. In the reversal

from one patterns impact (T_1) to the others (T_2), the extra itemset plays an important role.

Incremental cluster pattern also called incremental pattern cluster, is a cluster of combined patterns coupled in terms of certain relationships, within which additional pattern increments are appended to every previously adjacent constituent patterns.

Incremental pair pattern also called incremental pattern pair, is a pair of combined patterns which are paired in terms of certain relationship, within which the second atomic pattern has an additional pattern increment part compared to the first constituent. For instance, a contrast pattern consisting of an underlying pattern and a derivative pattern.

In-depth pattern also called deep pattern, uncovers not only appearance dynamics and rules but also inside driving forces, reflects not only technical concerns but also business expectations, and discloses not only generic knowledge but also something that can support straightforward decision-making actions. It is a pattern actionable in the business world. In-depth pattern is either filtered and summarized in terms of business expectations on top of general pattern(s), or itself discloses deep data intelligence. In-depth pattern mining discovers more interesting and actionable patterns from a domain-specific perspective.

Interestingness measures the significance of a pattern learned on a dataset through a certain method. The pattern interestingness is specified in terms of technical interestingness and business interestingness, from both objective and subjective perspectives.

Interestingness constraint determines what makes a rule, pattern and finding more interesting than the other.

Intelligence meta-synthesis involves, synthesizes and uses ubiquitous intelligence surrounding actionable knowledge discovery and delivery in complex data and environment.

Knowledge actionability Given a pattern P, its actionable capability is described as being the degree to which can satisfy both technical interestingness and business one. If both technical and business interestingness, or a hybrid interestingness measure integrating both aspects, are satisfied, it is called an actionable pattern.

Market microstructure data refers to the data acquired in capital markets, which is produced in terms of the theory of market microstructure and trading rules. Market microstructure data presents special data complexities, such as high frequency, high density, massive quantity, data stream, time series, mutliple coupled sequences etc.

Market microstructure pattern refers to the pattern learned on market microstructure data.

Multi-feature combined mining is a kind of combined mining which learns patterns by involving multiple feature sets, usually heterogeneous. For instance, a com-

bined pattern may consist of demographic features, business policy-related features, and customer behavioral data.

Multi-method combined mining is a kind of combined mining which learns by involving multiple data mining methods. It consists of serial multi-method combined mining, parallel multi-method combined mining, and closed-loop multi-method combined mining.

Multi-source combined mining is a kind of combined mining which learns patterns by involving multiple data sets, usually distributed and heterogeneous.

Network intelligence refers to the intelligence that emerges from both web and broad-based network information, facilities, services and processing surrounding a data mining problem and system. It involves both web intelligence and broad-based network intelligence.

Objective technical interestingness measures to what extent the findings satisfy business needs and user preferences based on the objective criteria.

Objective technical interestingness is embodied by measures capturing the complexities of a pattern and its statistical significance. It could be a set of criteria.

Organizational factor refers to many aspects existing in an organization, such as organizational goals, actors, roles, structures, behavior, evolution, dynamics, interaction, process, organizational/business regulation and convention, workflow and actors surrounding a real-world data mining problem,

Organizational intelligence refers to the intelligence that emerges from involving organization-oriented factors and resources into pattern mining. The organizational intelligence is embodied through its involvement in the KDD process, modeling and systems.

Pair pattern consists of two atomic patterns that are co-related to each other in terms of a pattern merging method into a pair.

Pattern summarization is a process of data mining, which summarizes learned patterns into higher level of patterns.

Pattern merging is a process of data mining, which merges multiple relevant patterns into one or a set of combined patterns. For instance, local patterns from corresponding data miners are merged into global pattern sets, merging atomic pattern sets into combined pattern set, or merging dataset-specific combined patterns into the higher level of combined pattern set.

Pattern increment refers to the additional component on top of an underlying pattern (a prefix or postfix) to form a derivative pattern, or an incremental pattern. For instance, with an underlying pattern U, different pattern increment V_1, V_2, $\ldots$, V_n may be added to U, to form into different derivative pattern U, V_1, U, V_2, $\ldots$, U, V_n.

Pattern interaction refers to the process and protocol in which patterns are interacted with each other to form into certain new patterns. Cluster patterns and pair

patterns may be resulted from pattern interaction. Many pattern interaction mechanisms can be created, for instance, pattern clustering, classification of patterns etc.

Pattern impact refers to the business impact associated with a pattern or a set of patterns. For instance, a frequent sequence is likely associated with the occurrence of government debt, here government debt is the impact.

Post analysis refers to techniques that are used to post-process learned patterns, for instance, to prune rules, reduce redundancy, summarize learned rules, merge patterns, match expected patterns by similarity difference, and the extraction of actions from learned rules.

Post mining refers to the pattern mining on learned patterns, or on learned patterns combined with additional data. The main difference between post analysis and post mining is whether another round of pattern mining process is conducted on the learned pattern set or not.

Reverse pattern also called impact-reserved pattern, is a pattern corresponding to another pattern, which triggers the impact change from one to another, usually opposite impact.

Social factor refers to aspects related to human social intelligence such as social cognition, emotional intelligence, consensus construction, and group decision; animat/agent-based social intelligence aspects such as swarm/collective intelligence aspects, behavior/group dynamics aspects, as well as many common aspects such as collective interaction, social behavior network, social interaction rules, protocols, norms, trust and reputation, and privacy, risk, and security in a social context, etc.

Social intelligence refers to the intelligence that emerges from the group interactions, behaviors and corresponding regulation surrounding a data mining problem. Social intelligence covers both human social intelligence and animat/agent-based social intelligence.

Subjective business interestingness measures business and user concerns from the subjective perspectives such as psychoanalytic factors.

Subjective technical interestingness focuses and is based on technical means, and recognize to what extent a pattern is of interest to a particular technical method.

Technical interestingness The technical interestingness of a pattern is highly dependent on certain technical measures specified for a data mining method. Technical interestingness is further measured in terms of objective technical measures and subjective technical measures.

Ubiquitous intelligence refers to the emergence of intelligence from many related aspects surrounding a data mining task, such as in-depth data intelligence, domain intelligence, human intelligence, network and web intelligence, and/or organizational and social intelligence. Real-world data mining applications often involve multiple aspects of intelligence, a key task for actionable knowledge discovery and delivery is to synthesize such ubiquitous intelligence. For this, methodologies, tech-

niques and tools for intelligence meta-synthesis in domain driven data mining is necessary. The theory of M-computing, M-interaction and M-space provide a solution for this purpose.

Underlying pattern also called base pattern, a base of a combined pattern, on top of which new pattern(s) is(are) generated. An underlying pattern may be taken as prefix or postfix of a derivative pattern.

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